Eastern Economy Edition

Food Processing and Preservation

B. Sivasankar

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FOOD PROCESSING AND PRESERVATION
by B. Sivasankar

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Preface

The subject of food science including the processing and preservation aspects is largely an interdisciplinary one involving topics from chemistry, microbiology, engineering operations and technological processes. These topics need an interactive approach to enable one to gain a proper perspective and insight into the subject of food processing and preservation. This book, an outgrowth of my long experience in teaching the subject at the postgraduate level at Anna University, Chennai, is an attempt in this direction.

The book provides an exhaustive coverage on all the major aspects of food science in an easy, accessible style. The initial chapters present topics related to the chemical constituents of foods, i.e. nutrients, and their functional roles, the changes that the nutrients undergo during processing and preservation, and the chemical reactions responsible for spoiling the food as well as maintaining the organoleptic properties of foods. The use of various chemical additives for specific purposes during food processing is discussed at length in a separate chapter (Ch. 8). The next four chapters (Chs. 9–12) are devoted to different types of microorganism associated with food, their beneficial roles in producing new and modified foods by fermentation and their detrimental effects in causing food borne diseases and food spoilage. The unit operations practised in food industry for processing and conversion of food are dealt with in Chs. 13 and 14. Topics that are discussed next include the various operations for food preservation by the use of high and low temperatures, evaporation and drying, and irradiation, which are detailed in Chs. 15–18. The remaining chapters (19–26) present the major plant and animal foods and their processing and preservation techniques. These include various foodstuffs such as dairy products, vegetable and fruit products, beverages, cereal products, meat products, confectionery items and spices, all of which have been discussed separately in different chapters. Finally, the assessment of food quality and safety has been outlined in the last chapter. The book with its student friendly features presents the complex subject

of food science and technology in a manner which even a student who needs an introduction to the subject would find easy to comprehend.

This is an ideal textbook for undergraduate students of science and engineering and a useful reference for postgraduate students offering courses in food science. Besides, researchers and professionals in the food processing industry would also find this study extremely rewarding.

Many people have contributed either directly or indirectly to make this book a reality. I take this opportunity to express my deep sense of gratitude to (Late) Prof. B. Jagannadhaswamy and Prof. C.M. Lakshmanan, former Directors of A.C. College of Technology, Chennai for initiating me into this area and giving me encouragement and guidance. The wholehearted support extended by Prof. K. Rengaraj and Dr. V. Sadasivam of Department of Chemistry, Anna University is also gratefully acknowledged. Special thanks are due to the colleagues and authorities of Anna University for providing with the ideal academic environment while writing this book.

I wish to express my gratitude towards my family members for their understanding and patience during the preparation of the manuscript. Finally, I convey my sincere thanks to the publishers, Prentice-Hall of India for the meticulous processing of the manuscript, both during editorial and production stages.

B. SIVASANKAR

CHAPTER

1

Introduction

1.1 WORLD FOOD REQUIREMENT

Food is a basic requirement of man than even shelter and clothing. It is the source of nutrients and provides the energy required for all activities of a human body, such as growth, repair of the damaged tissues, reproduction and sustenance. Food is consumed to satisfy primarily hunger, which is an unpleasant sensation. Hunger is a physiological condition associated with the contraction of the stomach, and leads to other sensations such as weakness, irritability, tenseness and even nausea. Food is also sought to satisfy the appetite, a pleasant sensation causing a person to satisfy the craving of the body for food. It is more of a psychological state usually felt in the mouth depending on odour and flavour as well as the memory of a pleasant food. Satiety is the feeling of having had enough of the desired food, after eating it.

Apart from being a source of nutrients and energy, food also gives a sense of security in the sense that individuals, communities and nations feel secure when enough food is stored to meet the requirements during scarcity periods. Food is a source of power. If we go back to history which is replete with references of blockading food supplies during times of war, we can easily understand that the only way to make the enemy weak those days was to disrupt the supply of food. The variety of foods consumed by someone is more of a status symbol because foods consumed by rich people are generally exotic and unaffordable to the poor. Food is a symbol of hospitality as guests are customarily offered food and drink. Food is also an outlet for emotion.

The world food requirement is directly related to population and population growth. The crop pattern, demand for semi-processed and processed foods and food consumption pattern keep changing. Unfortunately, the availability of food in different parts of the world is not equitable with many regions having undernourished people who do not get sufficient quantity of food. A more important aspect than the quantity of

food is the nutritional quality and wholesomeness of food. It is imperative to achieve the food production target and ensure the quality standard to combat hunger and food related diseases the world over.

1.2 FOOD SAFETY FOR THE CONSUMER

Food safety is a global challenge to meet the requirements of consumers with respect to hygienic and nutritional qualities. The food and beverage products have to satisfy stringent requirements with respect to human safety, freedom from microbiological and physicochemical contamination, organoleptic and nutritional quality, aesthetic appeal, total compliance with food laws and environmental safety regulations.

The fast emerging food biotechnology involving the use of genetically modified foods (GM foods) and food products produced by genetically modified organisms (GMOs) has become controversial even though the use of microorganisms and microbial products in food processing has been in practice for centuries. It is necessary that the controversies be resolved satisfactorily to ensure consumer safety.

1.3 AIMS OF FOOD SCIENCE AND TECHNOLOGY

Food requirements for the global population and consumer safety have necessitated defining the objectives of food science and technology. These include the following steps:

(i) Make available wholesome, nutritious and appetizing food at economical rates.

(ii) Improve nutritive value and minimize loss of essential nutrients during processing and preservation.

(iii) Ensure long-term storage stability even at elevated temperatures.

(iv) Market the processed foods of high caloric density in compact and easily reconstitutable form, in light weight and easily openable containers.

(v) Prevent food poisoning, contamination or adulteration.

(vi) Cater to the special dietary requirements of astronauts, patients, sports persons and children.

(vii) Improvise mechanical processing operations such as harvesting to replace or minimize labour.

(viii) Develop new varieties of instant or convenience foods.

1.4 INTERDISCIPLINARY APPROACH

The stated objectives of food science and technology may be achieved by adopting an interdisciplinary approach to food processing and preservation. Food science essentially involves the study of (i) problems of the dietary needs of people of different places (regions and countries), (ii) composition of food for an adequate diet and nutritional characteristics of various types of foods, (iii) causes of food spoilage, food poisoning and diseases and (iv) storing and preservation of different types of foods with minimum loss of nutrients. Food technology exploits the information obtained from food science so that food of better quality is grown, harvested, stored, processed, preserved and transported on a large scale.

The different areas of study in food science and technology include the following:

1. *Food chemistry* deals with the constituents of food and their composition, structure and properties and the chemistry of changes occurring during processing and utilization; developing principles, methods and techniques for physical and chemical analyses of foods and food products which are to be related to the standards and regulations of food processing.
2. *Food microbiology* involves the study of the relationship of habitat to the occurrence of microorganisms in foods; microbiological action in relation to food spoilage and food manufacture; physical, chemical and biological destruction of microorganisms in foods; microbiological examination of foodstuffs and public health.
3. *Food engineering* is concerned with the engineering aspects and unit operations of food processing including mechanics, heat and mass transfer, rate processes and process control instrumentation.
4. *Food processing technology* is about the general characteristics of raw food materials, harvesting, biotechnological methods of food processing and preservation, processing objectives including factors influencing food acceptability and preferences, packaging, waste disposal and sanitation.

The following chapters aim at providing information on the principles and factors involved in food processing and preservation.

1.5 CONSTITUENTS OF FOOD

1.5.1 Nutrients

Food may be defined as substances, which when eaten and absorbed by the body, maintain life and growth, i.e. supply, build and repair tissues. The chemical components that perform these functions are called nutrients. A

food must contain at least one nutrient. Nutritious foods such as milk contain a variety of nutrients and can fulfil all the requirements of the body, while glucose contains only one nutrient. Six types of nutrients have been identified in foods. They are carbohydrates, fats, proteins, minerals, vitamins and water. Lack of the necessary minimum amounts of these nutrients leads to general deficiency and undernourishment while lack of any one particular nutrient in sufficient quantity leads to malnutrition. The chemical composition of each group of nutrients and its function in the body are interrelated. The nutrients may be classified into three main types based on their biological functions.

1. The nutrients required for energy include carbohydrates, fats and proteins.
2. The nutrients for control of body processes include proteins, minerals, vitamins and water.
3. The nutrients required for the growth of tissues and their maintenance include proteins, minerals and water.

In addition to the major nutrients, foods contain natural or added minor constituents such as colourants, flavours, preservatives, toxins, etc. The additives in processed foods may be intentional to achieve certain aims during processing such as preservation, emulsification or flavour enhancement, or unintentional/accidental additives such as metallic contaminants, and fertilizer and pesticide residues.

The nutrient contents of a few typical foods are shown in Table 1.1.

TABLE 1.1 Amounts of Nutrients in Different Foods (g/100 g)

Food	*Water*	*Carbohydrates*	*Fats*	*Proteins*	*Minerals and vitamins*
Milk	88	4.8	3.8	3.3	0.1
Cheese	37	0	34	25	1.0
Egg	74	0.9	11	12	0.1
Beef	59	0	18	16	1.0
Bread	38	52	2	7	1.0
Apples	84	12	0	0.3	0.1
Cabbage	90	4	0	3	0.1
Jam	30	69	0	0.6	0.2
Chocolate	0	53	38	8	0.8

1.5.2 Food as a Source of Energy

Plants are autotropic and produce food in the form of carbohydrates utilizing sunlight, water, minerals and carbon dioxide by photosynthetic

process. Animals and humans cannot synthesize food in their body and therefore depend on plants and other animals for food. Energy is provided in the human body by metabolic process involving the breakdown of carbohydrate, fat and protein constituents of food that is digested and absorbed.

The energy value of food is measured in heat units called calories. A calorie is defined as the amount of heat required to raise the temperature of 1 g of water through 1°C. A kilo calorie is one thousand calories and is equal to 4.19 kilo Joules. The energy yield of different foods is compared by determining the amount of energy given when the substance is completely oxidised by igniting it in an oxygen filled chamber under pressure and expressed as heat of combustion of food. However, the available energy from a food in the human body is slightly less than the corresponding heat of combustion value due to loss within the body. The energy yield of the main nutrients of food is given in Table 1.2.

TABLE 1.2 Heats of Combustion and Available Energy of Nutrients

Nutrient	*Heat of combustion* (k.cal/g)	*Available energy* (k.cal/g)
Carbohydrates	4.1	4
Fats	9.4	9
Proteins	5.7	4

The energy values of some foods are given in Table 1.3.

TABLE 1.3 Energy Value of Some Foods

Food	*Energy value* (k.cal/100 g)	*Food*	*Energy value* (k.cal/100 g)
Butter	735	Fish	77
Cheese	406	Milk	65
Sugar	394	Apples	46
Jam	260	Tomatoes	15
Bread	230	Cabbage	25
Eggs	147	Potatoes	74

Energy requirement in the human body. Energy is required in the body for (i) basal metabolism, (ii) thermogenesis, (iii) growth and (iv) muscular activity. Basal metabolism or metabolic rate (BMR) is the energy required to sustain the body at complete rest. BMR varies with age. BMR values are about 1600 k.cal. for an average man and 1500 k.cal. for an average woman. These values account for about two-thirds of the total energy requirement of the body per day.

Thermogenesis or diet induced thermogenesis refers to the increase in the BMR after eating food. The increase is proportional to the energy content of the meal regardless of the nutrient composition of the food. Thermogenesis appears as heat.

Energy is required for the growth of tissue and body, e.g. during pregnancy about 30,000 k.cal. of energy is required to increase the size of placenta and the womb, create additional store of fat in the mother, meet the energy requirements of the baby in the womb and give birth to the baby. Once the baby is born, lactation starts and the energy requirement during this period is 750 k.cal. per day. In new born babies and up to three months of age, 25% of food energy is utilized for growth. The energy requirement for growth decreases to about 6% in a one-year old and to 2% in a five-year old child. In general, one gram of tissue formation requires about 5 k.cal. of food energy.

Muscular activity for doing external work requires energy. The energy requirement varies with the nature of physical activity. For example, sleeping requires energy equivalent of 1.0 BMR, while sitting, walking and heavy work such as mining or farm work require energy of 1.2, 3.2 and 6.0 BMR equivalents respectively.

1.6 FOOD, HEALTH AND DISEASE

Health is the attainment and maintenance of the highest state of mental and bodily vigour of which any individual is capable. No single food provides sufficient amounts of all the essential nutrients to keep a human being healthy. Different foods have widely differing nutritional contents and in order to maintain health it is necessary to select foods that fulfill the requirements of a healthy diet. The nutritional requirements vary according to the age, size and level of activity of a consumer. The recommended amounts of different nutrients are specified in terms of recommended daily amounts (RDAs). For some nutrients, RDAs are not specified because of their wide distribution in foods. A balanced diet is one which provides sufficient amounts of all the essential nutrients. It will prevent malnutrition, undernutrition and starvation as well as occurrence deficiency diseases. Such balanced diets can be achieved by eating animal foods such as meat, fish, eggs, milk, cheese and butter which are relatively rich in proteins and fats together with vegetables and fruits as a source of vitamins and minerals. It is essential that consuming food should be functional as well as fun.

Food is related to diseases just as it is related to health. Deficiency diseases such as scurvy, rickets and beriberi have been found to occur due to lack of sufficient amounts of vitamins. These diseases have almost been eradicated with the availability of better processed and fortified foods. Other food related diseases include obesity, slimming disease, coeliac disease, food intolerance and diseases of affluence.

Obesity is the state in which a person accumulates an excessive amount of body fat. Normal fat content of an adult man is about 12% while in an adult woman it is about 25%. If the body fat content exceeds 20% in a male and about 35% in a female the person is considered as obese. Obesity is considered as a major threat to health as it has an adverse effect on the cardiovascular system. Obese people are likely to suffer from diabetes, kidney failure, blindness, osteoarithritis, cancer of colon, rectum and prostrate, infertility in women, difficulty in childbirth and cancer of the breast, cervix and uterus.

Slimming disease (*Anorexia nervosa*) is a diet related disease which is caused probably more due to psychological problems. Due to loss of appetite, a severe loss of weight takes place and problems with menstrual periods are associated with the disease.

Coeliac disease is associated with malabsorption of food resulting in loss of weight and deficiency in vitamins and minerals. Coeliacs are sensitive to gliadin protein present in wheat and rice. The disease normally originates in childhood when cereals are introduced in the diet. Children with coeliac disease do not grow properly, have a characteristic pot belly and suffer from anemia and rickets. Gluten-free foods are provided as diet for such children.

Food intolerance is due to hypersensitivity exhibited by some people. Food intolerance may be caused by specific substances present in the food (allergens) which cause food allergy, e.g. tomatoes and strawberries may release histamine in some people producing irritating rashes, caffeine present in tea and coffee or amines in wine, cheese, yeast extracts and bananas can cause unpleasant effect. Similarly, certain additives present in the food such as tartrazine dye and preservatives such as benzoic acid and salicylates may cause food allergy. Food intolerance may also arise due to inability of some people to digest certain foods, e.g. children and adults lacking lactase cannot digest milk.

Diseases of affluence include coronary heart disease which is attributed to a variety of risk factors such as smoking, high blood pressure and high blood cholesterol, the last two being related to nature of diet. Cholesterol is an essential component of all membranes in animal tissues but is absent in plants. It is a precursor of steroid hormones, Vitamin D and bile acids. Esters of cholesterol predominate in plasma and adrenals whereas in nervous tissue and brain, cholesterol exists in free form. Since cholesterol is synthesized by the body it is not a dietary essential. It is the principal constituent of deposits or plaques which form in the wall of arteries, narrowing them and thereby restricting the blood flow (a condition called atheroscelerosis) which may lead to formation of blood clot and subsequently to a heart attack.

Cholesterol is transported in the bloodstream as lipoprotein compounds. Low density lipoproteins (LDL) and very low density lipoproteins (VLDL) carry cholesterol from the digestive system to tissues, and during this transport, can form plaques along the artery walls. Thus,

these two types of lipoproteins are referred to as 'bad' cholesterol. On the other hand, high density lipoprotein (HDL) or 'good' cholesterol scavenges cholesterol molecules from blood and tissues and delivers them to the liver where they are processed for excretion.

Daily intake of cholesterol in diet accounts for only 15–20% of the total cholesterol in the body and is not solely responsible for the risk of heart attack. The major portion of body cholesterol (about 80%) is biosynthesized and acetyl-coA is the major precursor. Biosynthesis of cholesterol depends on the nature of diet. High inputs of saturated fatty acids facilitate the formation of excess of acetyl-coA which enhances the plasma cholesterol level. However, if the saturated fats are replaced by polyunsaturated fatty acids (PUFAs) in the diet, plasma cholesterol level decreases as the PUFAs in the plasma lipoproteins decrease the capacity of the lipoproteins to carry cholesterol. The normal plasma cholesterol level is in the range 140–235 mg/100 ml of which HDL content is 45–85 mg with the ratio of HDL to total cholesterol as 0.22 to 0.45. It is suggested that a ratio greater than 0.2 is desirable.

Maintaining a low level of blood cholesterol through dietary changes seems to be effective in lowering the risk of heart attack. It is recommended that intake of animal fat, particularly egg and butter, be restricted in favour of vegetable fat because such substitution increases the polyunsaturated fatty acids in the diet thereby decreasing the cholesterol level. However, other risk factors such as smoking, stress and hypertension need to be addressed in the prevention of coronary heart diseases.

Questions

1. Write a note on food requirements of the world population.
2. What are the aims of food science?
3. What are nutrients and what are their functions?
4. How does food function as a source of energy?
5. Give an account of the energy needs of human beings.
6. What is basal metabolic rate (BMR)? What are the factors that influence BMR?
7. What are deficiency diseases? Give examples.
8. Write a note on the relationship between diet and coronary heart disease.

CHAPTER

2

Water

2.1 ROLE AND FUNCTION IN BIOLOGICAL SYSTEMS

Water is essential to sustain life in every living organism. Living species contain water as much as 60 to 95% of their total weights. In the case of human beings, about two-thirds weight of the body is water. All the organs, tissues and fluids contain water as a constituent. Only a few parts of the body such as teeth, bones and hair contain very little water.

Water, unlike other nutrients, does not undergo any chemical change in the body. It functions mainly as a carrier of nutrients and waste products, as a solvent to dissolve solutes, as a liquid medium for biochemical reactions and to form colloidal suspensions, as a stabiliser of biopolymer conformation and as a determinant of protein activity.

2.2 DIETARY REQUIREMENTS

Water is continuously lost from the body in the form of sweat, urine and faeces. Hence replacement of water is necessary continuously. The main sources of water are food and drink. Some nutrients on oxidation in the body produce water, e.g. one kg of glucose gives about half-a-litre of water. The amounts of intake and loss of water per day in an adult are shown in Table 2.1.

TABLE 2.1 Water Intake and Loss in Human Body

Source	*Intake* (l/d)	*Source*	*Loss* (l/d)
Food	1.12	Urine	1.30
Drink	1.18	Lungs	0.30
Oxidation of nutrients	0.28	Skin	0.92
		Faecus	0.06
Total	2.58	Total	2.58

2.3 DIETARY SOURCES

Water is the main constituent of most foods. The presence of water in optimal amount is necessary for viability of biological systems and preserving the quality of foods. Each food has its own characteristic water content. The water contents of a few types of foods are shown in Table 2.2.

TABLE 2.2 Water Content of Various Foods

Animal foods	*Water content* (%)	*Plant foods*	*Water content* (%)
Lean meat, raw	60–65	White bread	35–40
Pork, raw	55–60	Apples, peaches,	
Beef	50–70	Oranges and grapefruits	85–90
Chicken	70–75	Strawberries and tomatoes	90–95
Fish	65–80	Berries, cherries and pears	80–85
Cheese	35–40	Peas(green) and bananas	75–80
Honey	20–25	Carrots and potatoes	80–90
Milk	85–90	Beans, cabbage	
Butter	16–20	and cauliflower	90–95
Milk powder	4–5	Fruit juices and beer	85–90

Such a large content of water necessitates effective methods of preservation of food. Preservation methods such as dehydration alter the native properties of food. Hence a basic knowledge of the physical characteristics of water in foods is essential for practising proper food preservation.

2.4 PHYSICAL PROPERTIES OF WATER

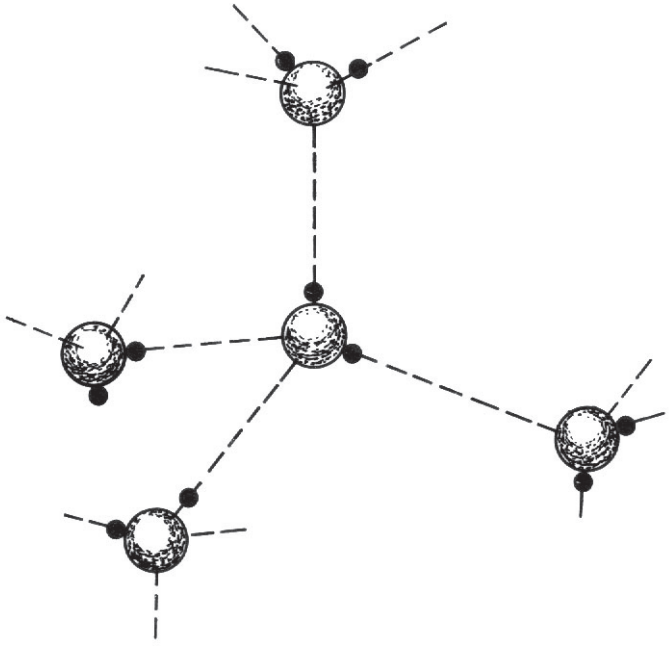
The physical constants of water of vital importance to food processing and preservation, include the following:

1. Melting point, boiling point, surface tension, dielectric constant, heat capacity and heat of fusion, vapourization and sublimation—all of which have unusually high values.
2. Normal viscosity and a moderately low density that has an unusually maximum value at 3.98°C and unusual characteristic of expanding on solidification.
3. A large thermal conductivity, (1.43 cal/sec cm^2 °C/cm at 20°C) compared to other liquids and a large thermal conductivity of ice compared to other nonmetallic solids. The thermal conductivity of ice at 0°C is about four times that of water at the same temperature (5.35 cal/sec cm^2 °C/cm)—thus ice conducts heat at a much faster rate than immobilized water in tissues.

4. Thermal diffusivity (the rate at which a substance undergoes changes in temperature) of ice (~0.011 cm^2/sec) is nine times that of water (~0.0014 cm^2/sec). This implies that tissues freeze more rapidly than they thaw due to differences in thermal conductivity and diffusivity values of water and ice.

2.5 STRUCTURE OF PURE WATER AND ICE

The unusual properties of water are attributed to hydrogen bonding. The V-shaped structure of water molecule and the polarized nature of the O-H bond result in association of water molecules through intermolecular H-bonding. Each water molecule is H-bonded with a maximum of four other water molecules in a three-dimensional tetrahedral arrangement, as shown in Figure 2.1. Extra energy is thus needed to break the intermolecular bonding resulting in larger values of physical constants. Water, with its tetrahedrally directed forces, crystallizes in an HCP structure. Ice exists in 11 crystalline polymorphic and also in an amorphous or vitreous state. However, the hexagonal ice is stable under normal pressure at 0°C.



Oxygen - ◯; Hydrogen - ●; H - bonding ------

Figure 2.1 Structure of H-bonded water molecules.

2.6 SOLUTE-WATER INTERACTIONS

The interactions of food components (solutes) with water are of relevance with respect to the processing and preserving foods. The normal structure of pure water based on a H-bonded tetrahedral arrangement is disrupted by the addition of solutes. Ionic solutes containing small ions such as lithium, sodium, hydrogen, calcium, barium, magnesium, aluminium, fluoride and hydroxide have strong electric field and bind water molecules forming net structures (solvation). The bound water is less mobile and more compressed than the water molecules in pure water. Ionic solutes of large sized ions such as potassium, rubidium, cesium, ammonium, halides (except fluoride), nitrate, iodate, bromate, perchlorate, etc. have weak electric field and function as net-structure breakers.

Solutes such as sugars capable of H-bonding have a profound influence on the normal structure of water and hinder freezing. The water binding and water retaining (humectancy) capacity of sugars in foods have been attributed to saccharide-water interactions. Similarly, the texture characteristics of starchy foods and the stability of gels in fruit preserves and jams are essentially due to the interaction of polysaccharide components of food and water.

Inert solutes such as rare gases, hydrocarbons and non-polar groups of such compounds as fatty acids, amino acids and proteins have a structure forming action. They position themselves in the H-bonded clusters of water molecules giving rise to extended array of water molecules and the solute is surrounded by water with an enhanced structure, e.g. protein-water interactions and clathrate hydrates. The stability and solubility of protein aggregates in seeds and emulsifying characteristics of proteins are attributed to protein-water interactions.

2.7 CONCEPT OF WATER ACTIVITY

Though water is abundant in biological systems and foods, its free or ready availability is more important for the growth and activity of microorganisms. The 'free availability' of water may be quantitatively expressed by the use of the concept of activity, a_w. The water activity of a solution (or even a food stuff) is defined as the ratio of the vapour pressure (P) of the solution or water in the food to the vapour pressure pure water (P_0) at the same temperature.

$$a_w = P/P_0 \tag{2.1}$$

The activity of pure water is 1.0 and that for a 1.0 M solution of the ideal solute it would be 0.9823.

Water activity of a food would be in equilibrium with relative humidity

(RH) of the atmosphere surrounding the food. Water activity is directly related to RH as given by Equation 2.2.

$$a_w = RH/100 \tag{2.2}$$

A relative humidity about a food corresponding to a water activity lower than that of food would tend to dry the surface of the food. Conversely, if the relative humidity is higher than the water activity, the latter would be increased on surface of the food due to condensation of moisture. The relationship between water activity and relative humidity at equilibrium and the moisture content of a food can be precisely described by its moisture sorption isotherm as shown in Figure 2.2.

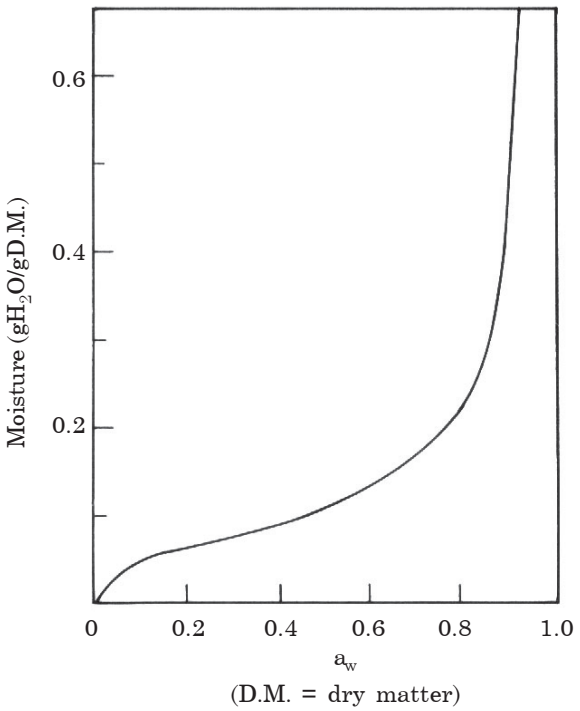


Figure 2.2 Water adsorption isotherm.

(Source: Food Chemistry, O.R. Fennema (Ed.), Marcel Dekker, New York, 1976.)

The practical importance of controlling water activity in foods is primarily related to the minimization of microbial deterioration of foods. By lowering the water activity by dehydration, salting or sugaring, microbial deterioration of the food can be reduced and even eliminated. Water activity also influences the rates of chemical and enzymatic reactions in foods.

Most fresh foods have water activity of 0.99. The water content and the water activity of a few typical foods are listed in Table 2.3.

TABLE 2.3 Water Content and Water Activity of Foods

Food	*Water content* (%)	*Water activity* (a_w)
Fresh meat	65	0.98
Cheese	40	0.97
Jam	33	0.88
Dried fruits	18	0.76
Honey	20	0.75

2.7.1 Water Binding in Foods

Water in food and biological systems exists in three types or degrees of boundness. Pure water with full activity can be considered as 'free water' (Type IV) and is not found in biological systems.

Entrapped water or Type III water involves physically entrapped water in tissues and membranes and is similar in its properties to water in dilute solutions. The freezing point is reduced only to a slight extent and the normal solvent capacity is exhibited by entrapped water. It is easily removed during evaporation, concentration or drying operation. This type of water constitutes the major proportion of water in plant and animal food tissues and is readily available for chemical reactions and the growth and activity of microorganisms. As it is removed the remaining water gradually assumes a lower activity. When all the Type III water is removed the moisture content is about 12 to 25% by weight and the water activity is lowered to 0.8. The rates of chemical reactions such as sugar-amine reactions (Maillard reaction) increase as the water activity is lowered to about 0.8 and less.

Multilayer water or Type II water involves water that is H-bonded in water-solute and water-water clusters, and water in micro-capillaries. It has limited solvent capacity and does not freeze completely even at −40°C. It is more difficult to remove this type of water than Type III water. Partial removal of Type II water eliminates the last possibility of microbial growth and greatly reduces most kind of chemical reactions. Complete removal of Type II water leaves 3 to 7% of moisture level in the food (water activity is about 0.25) and corresponds approximately to optimum stability of dry products that contain significant amount of oxidizable lipids. This small amount of water is required to inhibit oxidative rancidity through several mechanisms. The small amount of water facilitates destruction of free radicals, H-bonds to hydroperoxides and slows the rate of their conversion to other products, and hydrates or coordinates with metals, thereby reducing their ability to catalyze oxidation.

Monolayer, vicinal water or Type I water involves water adsorbed to solutes as a monolayer and water in chemical hydrates. This corresponds to a moisture content in the range of 0 to 0.7% by weight and can be partially removed by dehydration but not by freezing even at −40°C. Type I water is tightly bound and is considered to be true bound water. It is in this region where acceleration of lipid phase reactions (oxidative rancidity) occurs as lipids become more exposed. However, the rates of enzyme catalyzed reactions such as hydrolysis of lipids or proteins decrease.

Figure 2.3 shows the effect of moisture content and water activity of foods on the growth rates of microorganisms and of various chemical reactions that occur in foods. It also indicates the shelf life or storage stability of foods as a function of water activity.

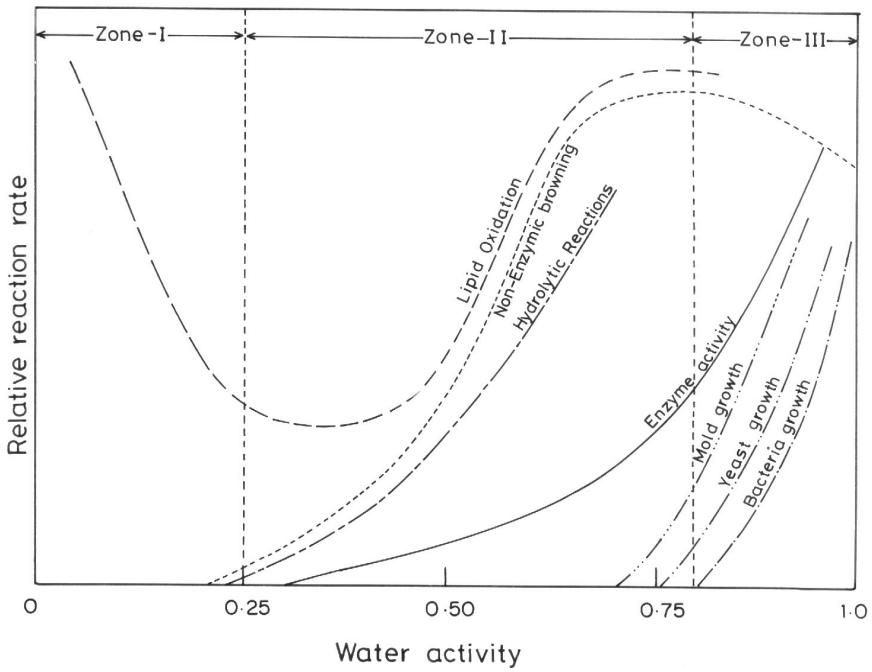


Figure 2.3 Rates of chemical and enzymatic reactions and microbial growth as a function of water activity in foods.
(Source: Food Chemistry, O.R. Fennema (Ed.), Marcel Dekker, New York, 1976.)

The moisture content—water activity relationship is temperature dependent and a decrease in temperature shifts the boundaries of zones I and II to points of greater moisture and lower water activity. Water adsorption and desorption isotherms do not coincide, i.e. they exhibit hysterisis as shown in Figure 2.4. At any given moisture content, the water activity during desorption is less than that during adsorption and at any given water activity the content during desorption is greater than that during adsorption.

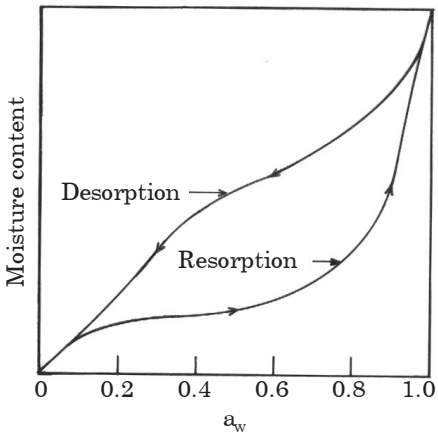


Figure 2.4 Hysteresis of moisture sorption isotherm.
(Source: Food Chemistry, O.R. Fennema (Ed.), Marcel Dekker, New York, 1976.)

2.7.2 Water Activity and Activity of Microorganisms

Most food spoilage bacteria do not grow below a water activity of 0.91. The minimum water activity requirements for the growth of microorganisms are listed in Table 2.4.

TABLE 2.4 Water Activity Requirements of Microorganisms

Microorganism	*Minimum a_w required for growth*
General	
Normal bacteria	0.91
Normal yeasts	0.88
Normal moulds	0.80
Halophilic bacteria (salt loving)	0.75
Xerophilic moulds (prefer dry conditions)	0.65
Osmophilic yeasts (prefer high osmotic pressure)	0.60
Specific organisms (food poisoning organisms)	
Clostridium botulinum Type E	0.97
Pseudomonas spp.	0.97
Escherichia coli	0.96
Enterobacter aerogenes	0.95
B. subtilis	0.95
Clostridium botulinum Types A and B	0.94
Staphylococcus aureus	0.86
Aspergillus conicus	0.70
Sacharomyces rouxii	0.62

Each microorganism has a maximal, optimal and minimal water requirement for its growth. This range depends on a variety of factors which are discussed below:

(i) *Type of solute employed to reduce the water activity.* For moulds, the lowest water activity for growth is practically independent of the type of solute used. Other microorganisms have limiting values of water activity with some solutes compared to other solutes, e.g. potassium chloride is less toxic to bacteria than sodium chloride which in turn is less toxic than sodium sulphate.

(ii) *Nutritive value of culture medium.* The better the medium, the lower the minimum water activity value required for the growth of the microorganism.

(iii) *Temperature.* Most microorganisms have the greatest tolerance to low water activity at about optimal temperature of their growth.

(iv) *Oxygen supply.* Aerobes have a lower limiting water activity value in the presence of air than in its absence. The reverse is true for anaerobes.

(v) *pH value.* Most microorganisms tolerate low water activity at neutral pH than in acid or alkaline pH range.

(vi) *Inhibitors.* The presence of inhibitors narrows the range water activity required for the growth of microorganisms.

2.7.3 Controlling Water Activity in Foods

Water is made unavailable for the growth of the microorganisms in foods by (i) addition of solutes, (ii) formation of gels, (iii) lowering of temperature and (iv) dehydration.

Addition of ionic solutes ties-up water in solution. Thus higher concentration of salts and sugars, in effect, dries the food tying-up water molecules. In addition, water inside the microbial cells tends to leave by osmosis when there is a higher concentration of solutes outside the cell in the surrounding food killing the microorganisms ultimately. The formation of hydrophilic gels (e.g. 3–4% agar) makes water unavailable in the medium, preventing bacterial growth. Lowering the temperature tends to crystallize water into ice which is not useful for the growth of the microorganisms. The water activity of water-ice mixtures (ratio of vapour pressure ice to that of water) decreases with decreasing temperature below 0°C. Thus water activity of water at 0°C is 1.0 and it decreases to 0.953 at −5°C, 0.907 at −10° C, 0.846 at −15°C and 0.823 at −20°C. In foods, as more ice is formed, the concentration of solutes in the unfrozen water increases further and this also contributes to the lowering of water activity. Dehydration of foods effected by drying decreases the moisture content and thereby lowers water activity.

2.7.4 Determination of Water Content in Foods

Any one of the three established methods may be used to determine the water content of a food. The gravimetric method is based on the difference in the weights of the sample food prior to and after complete drying. The volumetric method uses Dean and Stark apparatus shown in Figure 2.5.

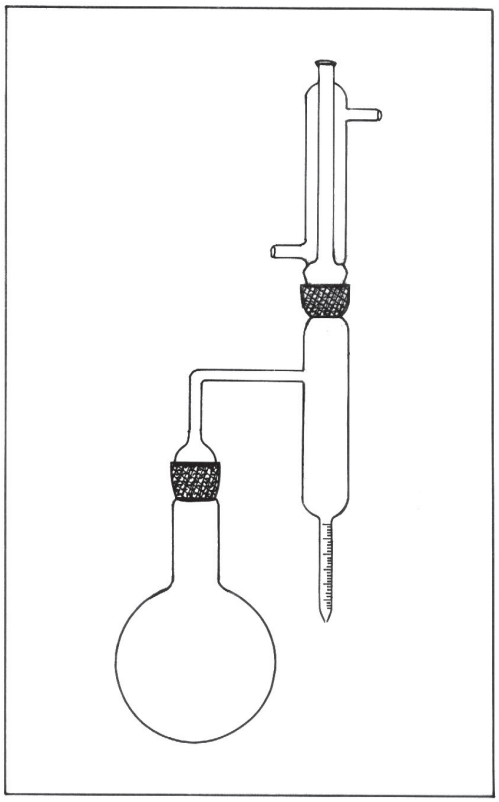


Figure 2.5 Dean and Stark apparatus.

The weighed sample of food such as cereal grains is subjected to reflux distillation in the apparatus using toluene as solvent. The volume of water collected in the graduated collecting tube gives the water/moisture content of the sample food. Both the gravimetric and volumetric methods are useful for samples of higher water/moisture content. The more sensitive chemical method called Karl Fischer titration is useful to determine the water content of a variety foods such as meat, fruits and vegetables and also in foods where water is relatively tightly bound. The weighed food sample is dispersed in water-free sulphur dioxide-pyridine-methanol

reagent and then titrated with standard iodine in water-free methanol. The reaction may be represented as

$$SO_2 + I_2 + 2H_2O \longrightarrow H_2SO_4 + 2HI \quad (2.3)$$

The appearance of free excess iodine from the titrant may be detected visually or amperometrically and taken as the end point of the titration.

Near infrared spectrophotometry (NIR) is also useful in the determination of water content in flour, cereal grains and similar relatively dry foods. The absorbance at 1.94 μm is measured and compared with that at 2.08 μm to determine the moisture content. Nuclear magnetic resonance (NMR) spectrometry is also useful in determination of moisture content. Both these instrumental methods are fast, sensitive and non-destructive.

2.7.5 Determination of Water Activity

Water activity of a food is determined based on the relationship between water activity and relative humidity at equilibrium conditions. The tendency of the food sample to gain or lose water when exposed to air at a constant temperature and relative humidity is measured by placing the sample along with a solution of known water activity in a closed chamber for equilibration and determining the weights at regular intervals. Saturated solutions of salts have known water activity and may be used in the experiment. The water activity at 25°C is known for saturated solutions of potassium dichromate (0.980), potassium citrate (0.925), barium chloride (0.902), sodium benzoate (0.880), potassium chloride (0.845), ammonium sulphate (0.790), sodium chloride (0.755) and ammonium nitrate (0.618).

Questions

1. What is the role of water in biological systems?
2. Explain the concept of water activity and its relevance in food preservation.
3. What physical properties of water are of importance in food processing and preservation?
4. What is a sorption isotherm? How is water sorption isotherm related to the moisture content in a food?
5. Write a note on the different types of bound water in foods and their role in chemical and enzymatic reactions during processing.
6. How is moisture content of a food raw material determined?
7. What methods are adopted to control water activity in a food?

CHAPTER

3

Carbohydrates

Carbohydrates are widely distributed organic compounds that have a central role in the metabolism of animals and plants. Photosynthesis of carbohydrates by green plants, in fact, supports the existence of all other organisms. They constitute the major portion of our daily food. The economic availability of fairly large amounts of carbohydrates, their nutritional, physiological effects and functional properties are of relevance to food quality and acceptance. Carbohydrates undergo changes during milling, cooking, drying, freezing and storage, and it is essential to understand the effects of these changes during processing, particularly on the nutritional and aesthetic values of food. Physical properties of carbohydrates such as association of polar molecules through hydrogen bonding, ionic interactions and substituent effects influence their solubility, diffusion, osmosis, viscosity and plasticity. These in turn influence the texture and acceptance of processed foods particularly in the manufacture of confectionery, pastries, gels, coatings, frozen foods and dehydrated foods which have been reconstituted. In addition, chemical properties such as thermal degradation reactions and chelation of metal ions influence flavour production, retention of natural flavour and colour in processed foods.

3.1 CHEMICAL COMPOSITION AND NOMENCLATURE

The empirical formula of carbohydrates is $(CH_2O)_n$. Carbohydrates are classified into three major groups. The simplest carbohydrates are monosaccharides with three to eight carbon atoms, but only those with five or six carbon atoms are common which include the hexoses such as glucose, fructose, galactose, mannose and rhamnose and the pentoses such as arabinose, ribose, xylose and lyxose. Most of them are referred to

as sugars. Oligosaccharides contain two to ten monosaccharide units linked by a glycosidic group. Disaccharides are the most important oligosaccharides and contain two monosaccharide units. Examples of disaccharides include sucrose, lactose, maltose and cellobiose. Polysaccharides are polymers of monosaccharides containing several (more than ten) monosaccharide units. Examples of polysaccharides are starch, glycogen, cellulose and pectin.

3.2 STRUCTURE

Monosaccharide hexoses such as glucose and fructose have been initially assigned a straight chain structure containing free carbonyl group. The free carbonyl group is responsible for reducing properties exhibited by monosaccharides, such as the reduction of cupric ions in alkaline medium (Fehlings' reaction) giving the name *reducing sugars* for monosaccharides. But the straight chain structural formula does not satisfactorily explain many of the properties of monosaccharides. The open chain structure has been shown to exist in equilibrium with a ring structure in aqueous solution of monosaccharides. Only 0.02% of D-glucose molecules have the open chain structure in aqueous solution. The ring structure is formed by condensation of the carbonyl group with a hydroxyl group at far end of the chain to give a hemiacetal. There are four possible ring structures for D-glucose, differing in the configuration of the new asymmetric centre created by hemiacetal formation and the size of the ring. A six membered ring is referred to as *pyranose*, and a five membered ring, as *furanose*. The two isomeric forms known as anomers designated as α- and β- are generated due to the ring structure. The anomers differ in optical rotation. α-D-glucopyranose obtained by crystallization from aqueous solution has a specific rotation of +112 whereas the β-D-glucopyranose obtained by crystallization from pyridine solution has a specific rotation +19. When crystals of any of the anomers are dissolved in water, the specific rotation of the solution changes to +52 irrespective of the anomer used to prepare the solution. This phenomenon is known as *mutarotation* and the transition of one anomer to another occurs via the open chain aldehydic structure.

The thermodynamically favourable arrangement in most of the six-membered aliphatic ring structures is the chair form. Theoretical studies have shown that the favoured conformation for β-D-glucose will be the one with greatest number of bulky substituents such as the -OH and CH_2OH in equatorial positions and neighbouring hydroxyl groups are as far apart as possible. The universal role of glucose in living systems, both as a metabolic intermediate and a structural element is that all its hydroxyl

groups are in the favoured equatorial positions. The conformation and anomer proportions of a particular sugar in solution are not only a function of the intramolecular non-bonding interactions but also dependent on the interactions between the sugar molecule and the solvent, usually water in foodstuffs. In water at 25°C, the α-anomer of D-glucose is present to the extent of 37%.

The condensation of the hemiacetal or the reducing group of a monosaccharide with a hydroxyl group of another monosaccharide molecule through an acetal or glycosidic link gives a disaccharide. The glycosidic link of carbohydrates has the effect of fixing the hemiacetal in either the α- or the β-configuration. Two important food sugars are the disaccharides—lactose (milk sugar), β-D-galactopyranosyl-(1,4)-α-D-glucopyranose, a *reducing sugar* and sucrose (sugar from cane or beet), α-D-glucopyranosyl-(1,2)-β-D-fructofuranose, a *non-reducing sugar*. The glycosidic link is stable under ordinary conditions but gets readily hydrolyzed in acid solutions or in the presence of appropriate hydrolytic enzymes. Thus, under mildly acidic conditions or by the action of enzyme invertase sucrose is hydrolysed to invert sugar, a mixture of glucose and fructose. This hydrolysis is termed *inversion* because the optical rotation of sucrose changes from +66.5 to +19 because of the formation of glucose with optical rotation of +52.7 and fructose with optical rotation of −92.4. Other examples of reducing disaccharides include maltose, isomaltose and cellobiose while trehalose is a non-reducing sugar.

Polysaccharides are long chains of glycosidic sugar units containing a few types of hexoses, modified hexoses and pentoses. Starch, hemicellulose, pectin and cellulose are some of the important polysaccharides, of which starch is the major food carbohydrate. Starch consists of linear polymer amylose and a highly branched polymer amylopectin. Both are homoglycans of D-glucose. Amylose consists of the disacchardie maltose as the repeat unit linked by α-1,4-glycosidic linkages while amylopectin contains both α-1,4- as well as a small percentage of α-1,6-glycosidic linkages giving rise to a branched structure.

3.3 DIETARY CARBOHYDRATES AND THEIR SOURCES

Carbohydrates occur in food as sugars and starch, which are the major sources of energy in the diet and as cellulose, the main component of dietary fibre. Plants produce sugars through photosynthesis. These are converted into polysaccharides. Common carbohydrates and their sources are listed in Table 3.1.

TABLE 3.1 Common Carbohydrates and Their Sources

Carbohydrates	*Sources*
Monosaccharides	
glucose and fructose	honey, fruits
Oligosaccharides	
(i) raffinose, stachyose	cereals, onion, malt, tubers
(ii) fructosyl sucrose and sucrose	sugarcane, fruits, vegetables
(iii) lactose	milk and other dairy products
(iv) maltose, isomaltose	starch syrup, honey, malt
Polysaccharides	
(i) starch and dextrins	cereals, roots, tubers and legumes
(ii) cellulose	plant cell wall and fibres
(iii) glycogen	liver and animal tissue
(iv) hemicellulose	plant cell walls, cereals, nuts, flour and bran
(v) pentoses and pectic substances	fruits, citrus fruits, apples, vegetables

The carbohydrate composition in foods influences flavour, texture and colour and thereby the acceptability of foods. Moreover, information of the carbohydrate composition in raw food materials is useful for determining the process conditions to be employed. In general, cereals are rich in starch content and poor in sugars while fruits are rich in free sugars and poor in starch content. The natural changes in carbohydrate composition due to hydrolysis of oligo- and polysaccharides giving reducing sugars that occur during maturation or ripening of plant foods is of considerable importance in determining the time of harvest and the type of post harvest treatment. For example, citrus fruits are allowed to ripen on the tree as the starch is converted to sugars and they do not undergo any further change in their carbohydrate composition after harvest. In contrast, fruits such as apples, bananas and pears are harvested before they completely ripen and the starch is converted to sugars as ripening proceeds during storage. Green peas, green beans and sweet corn are harvested before maturity to obtain succulent textures and sweetness. As maturation progresses, the sugars are converted into polysaccharides, water is lost and tough textures develop. But crops such as soybeans are allowed to mature completely before harvesting so that starch is depleted as sucrose and galactosylsucrose (raffinose and stachyose) are formed.

The reducing sugar content of fruits and vegetables that are to be dehydrated or processed by heat, indicates the extent of non-enzymic browning that will occur during processing. During sun drying of grapes and dates, sucrose is converted to glucose and fructose by invertase activity and non-enzymic browning of these reducing sugars deepens the colour of the dried products. Non-enzymic browning in fried potato wafers

depends on the reducing sugar content of potatoes which normally increases during cold storage. Enzymic hydrolysis of glycosides, thioglucosides and cyanogenetic glycosides in vegetables produce toxic substances due to improper processing and preservation methods.

In animal foods, postmortem activity of enzymes is of relevance to carbohydrate composition. The glycogen of animal tissues, particularly of liver is rapidly depolymerized to glucose after slaughter and hence immediate deep freezing is required to preserve glycogen.

3.4 CARBOHYDRATE DERIVATIVES

The important derivatives of monosaccharides include the uronic acids and sugar alcohols (alditols). Uronic acids are oxidation products of monosaccharides obtained by the oxidation of the hydroxyl group at C-6 of the aldose sugars to give carboxylic acids, e.g. glucuronic and galacturonic acid. These uronic acids are important constituents of polysaccharides such as pectins and alginates. On the other hand, oxidation of the carbonyl group of the monosaccharides gives onic acids. For example, the enzyme glucose oxidase catalyzes the oxidation of the carbonyl group of glucose to form gluconolactone which hydrolyzes to form gluconic acid and this reaction forms the basis of glucose determination in food materials. The reaction is also used for the removal of traces of glucose from liquid egg used in bakeries to prevent Maillard reaction. Reduction of the carbonyl group of the sugar to a hydroxyl gives sugar alcohols such as xylitol, maltitol and sorbitol. Sorbitol is found in nature in fruits such as pears, apples and plums. Inositol, a cyclic alditol is found as its hexaphosphate (phytic acid) or as calcium or magnesium salt (phytin) in plants. These sugar alcohols are not absorbed by the human intestine. Hence these sugar alcohols are used as sweeteners to replace sugars in calorie-reduced and special foods for diabetics. Sugar alcohols are also used to decrease water activity in intermediate moisture foods, as softeners, as crystallization inhibitors and for improving rehydration characteristics of dehydrated foods.

Though any compound containing a glycosidic link is a glycoside, the term is normally reserved for a class of compounds that occurs naturally in plants. These glycosides have a sugar component linked to a non-sugar component, such as a flavanoid or a steroid, generally called the aglycone. Anthocyanin pigments of plants such as amygdalin, present in bitter almonds, which gives rise to traces of HCN in some foods, are important examples of glycosides.

Amino sugars (e.g. glucosamine, galactosamine, etc.) have an amino group substituting one of the hydroxyl groups in monosaccharides.

3.5 FUNCTIONAL PROPERTIES OF DIETARY CARBOHYDRATES

Carbohydrate constituents of the food undergo a variety of chemical changes during mechanical, heat or cold processing of foods. The main chemical reactions of carbohydrates of relevance to food processing include hydrolysis of oligosaccharides and polysaccharides, thermal decomposition reactions of sugars and their condenzation to form disaccharides, oligosaccharides and dextrins, enolization, isomerization and tautomerization. Other constituents of the food, particularly proteins, interact with carbohydrates, resulting in the loss of certain essential amino acids and thereby reducing the nutritional quality of the food. The chemical changes affect the various functional characteristics of carbohydrates and contribute immensely to the changes in texture, colour, flavour and organoleptic properties of food.

The functional properties of carbohydrates in food include: (i) nutrition, (ii) flavour and colour production, (iii) sweetening, (iv) texturing and (v) plasticizing action and humectancy.

3.5.1 Nutritional Aspects

Carbohydrates are the main and the cheapest source of energy in the human diet providing about 4 k.cal/g. Though fat provides energy of 9 k.cal/g, digestible carbohydrates supply energy at lower cost, promote the utilization of fats and reduce the wastage of proteins. Proteins are the costliest source of energy and use of proteins for energy deprives the body of essential building blocks of enzymes, proteins and other biomolecules. Apart from their nutritional and economic values, carbohydrates contribute to the appetizing and pleasurable aspects of food such as sweetness and flavour due to caramelization.

Glucose, sucrose, maltose, lactose, galactose, fructose and ribose are digestible readily in human metabolism. Melibiose, mannose, sorbose, arabinose and xylose either remain undigested or poorly absorbed in human metabolism. Among the oligosaccharides, galactose derivatives of sucrose, viz. raffinose and stachyose which occur in legume seeds such as peas, beans and soybeans, are neither hydrolyzed nor absorbed by human digestive system. However, a meal containing large quantities of these beans are utilized by *E.coli* bacteria in the large intestines resulting in the production of large quantities of hydrogen and some carbon dioxide as flatus or fermentation gases giving the discomforts of flatulence. Lactose of milk consumed by individuals lacking the enzyme lactase causes stomach upsets.

From the nutritional point of view, starch is the major plant polysaccharide that is readily digested in the intestine and thus serves as a source of carbohydrates. The major portion of energy requirements of

humans is met by starch of cereal grains and tubers such as potatoes. Other polysaccharides in foods are pentosans, pectic substances, cellulose and hemicellulose, collectively called as fibres. Fibrous foods are low fat foods not easily digestible but have an essential role in the functioning of the large intestine. They reduce the incidence of bowel cancer by reducing the time that potential carcinogens derived from the foods spend in the bowel.

Glycogen (animal starch) acts as an energy/carbohydrate reserve in the liver and muscles. Other animal polysaccharides include chitin (a polymer of N-acetylglucosamine, the fibrous component of anthropod exoskeletons), polyglycans and mucopolysaccharides have specialized roles (structural role) in connective tissues.

Overindulgence of foods rich in sugars leads to obesity and underconsumption of other foods containing essential nutrients. If more monosaccharides are produced in the body than is required, the excess is converted into fat and stored in fat depots.

Digestion of carbohydrates starts in the mouth where salivary amylose hydrolyzes starch to maltose. The hydrolysis continues in the stomach till the food is acidified with gastric juice. The food passes from the stomach to the small intestine, where amylase continues the conversion of starch to maltose. The enzymes maltase, lactase and sucrase present in the intestinal juice hydrolyze maltose, lactose and sucrose, respectively to glucose, galactose and fructose. These monosaccharides pass into the blood stream and are carried to the liver, where fructose and galactose are converted to glucose. A part of the glucose is transported to muscles and the balance stored in liver. In the liver and muscles, the glucose is converted to glycogen which serves as the reserve of carbohydrate in the body.

The energy requirement of body is met by the oxidation of glucose in about 30 steps. The two main stages involve the formation of pyruvic acid by anaerobic glycolysis releasing about 8 k.cal of energy followed by the oxidation of pyruvic acid to carbon dioxide and water producing about 30 k.cal. The energy from glucose oxidation is stored in the form ATP.

3.5.2 Flavour and Colour Production by Carbohydrates

The chemical reactions of carbohydrates such as hydrolysis, enolization, isomerization and browning, occurring during sterilizing, cooking and dehydration processes affect food quality and acceptance. In neutral or weakly alkaline media, cyclic hemiacetals rapidly change to acyclic carbonyl forms of reducing sugars. The reactive carbonyl groups enolize and begin a series of decomposition reactions. In acidic media, the carbonyl groups undergo acid catalyzed dehydration after enolization to produce objectionable furfurals and dark brown colours. The reaction is known as Lobry de Bruyn-Alberda van Eckenstein transformation and

requires an acid or base catalyst. The enediol is dehydrated in neutral or acidic environment to give hydroxymethyl furfural (HMF) and other cyclic compounds. HMF can be readily detected in sugar based food products that have been heated, such as confectionery. It also serves as a means of detecting adulteration of honey with invert sugar.

The extent of hydrolysis of carbohydrates during processing depends primarily on the acidity of the medium, the anomeric form and position of the interglycosidic linkages, the ring form of sugar units, the extent of H-bonded association between molecules and the rate of inactivation of native glycosyl hydrolases. Hydrolysis of sucrose followed by non-enzymic browning of the released reducing sugars results in dark colours.

Flavour and colour reactions of carbohydrate constituents of foods may be classified into (i) enzymic browning reactions, (ii) caramelization, (iii) Maillard reaction and (iv) Strecker degradation. The last three reactions are non-enzymic browning reactions which are responsible for flavour and colour development in foods.

Enzymic browning reactions. Browning reactions are a part of the natural process of decay and consist of a series of downhill exothermic reactions that proceed spontaneously, e.g. drying and decomposition of vegetation to humus, peat and coal. When the ordered life-supporting substances are deprived of the protection of water they get oxidized and/or chemically dehydrated to reactive intermediates, which polymerize to dull brown humic substances. The browning reactions that occur in foods may be classified into two types: (i) oxidative browning and (ii) non-oxidative browning. Both the types may be catalyzed by enzymes.

In oxidative browning, oxidases (e.g. phenolase, catechol oxidase, ascorbic acid oxidase, lipoxidase) are active in the first step of conversion of phenol, enediol or conjugated diene functional groups to reactive carbonyl compounds which undergo further non-enzymic oxidative browning reactions. Ultraviolet light and ionizing radiations also induce oxidation reactions.

Non-oxidative browning also involves enzymes in the first step in which glycosyl hydrolases initiate the so-called non-oxidative browning in natural products by the release of reducing sugars. This is an important step in colour and flavour development in dates, honey, maple syrup, chocolate and vanilla processing. The reducing sugars undergo a series of reactions involving isomerization, enolization and dehydration even in the absence of oxygen or other oxidants to yield reactive carbonyl compounds and ultimately brown polymers (melanoidins) and flavour compounds.

Caramelization. Decomposition reactions may be induced by melting dry sugars and to a limited extent by heating their syrups at temperatures above 100°C, e.g. in candy making. Heat hastens the inversion of sucrose to yield reducing sugars which condense to higher oligosaccharides and dextrins. When amino compounds do not participate in the heat induced reactions, the complex colour and flavour forming reactions of

carbohydrates are called caramelization. The brown pigments that characterize caramel and other foods are due to HMF and its precursors. The pigment molecules have very high molecular weight and are not absorbed in the intestine. Caramelization is done in the industry with different catalysts to produce either flavour or colour. For flavour purposes, sucrose is caramelized in concentrated syrup. The sugar fragmentation reactions are promoted by neutralizing and buffering the acids that are formed by the addition of basic salts. The formation of humic substances is limited by the basic salts and thereby bitter and astringent tastes are avoided. The caramel aroma is mainly due to a group of cyclic alkylenolones, dihydrofuranones and pyrones.

Caramel colouring, for use in beverages, confections and baked goods, is prepared by heating glucose syrup with dilute sulphuric acid and partial neutralization with ammonia to yield brown coloured polymers. The sugar is dehydrated and polymerized to form soluble polymers of high tinctorial power. Humic substances are filtered off to yield an acidic, very dark, viscous syrup that further polymerizes in storage as the browning reactions continue. The addition of sulphite forms strong anionic sulphonic groups in the caramel colloid, which stabilize them in the presence of tannins. The nitrogen content of caramel colouring should be at the lowest possible level because of the possibility of formation of toxic 4-methylimidazole.

Flavours produced by non-enzymic browning reactions are essential for recognition and taste acceptance of many processed foods. For example, coffee beans, peanuts and cashew nuts develop distinctive flavours only after roasting. Roasted aromas are derived from mixtures of many volatile Maillard reaction products. Aromas from coffee, peanuts, popcorn, freshly baked bread, toasted marsh mallows and barbeque all contain many compounds from browning reactions. Thermal decomposition of sugar produces furans, pyrones, lactones, aldehydes, ketones, acids, esters, pyrazines, pyrroles and pyridines. Two important character compounds in processed foods are caramel aroma compounds and nutty or bready aroma compounds.

Both caramelization and Maillard reactions are important colour and flavour producing reactions of non-enzymic nature. Caramel flavour is not acceptable in some foods such as orange juice, dehydrated potato and dried milk.

Maillard reaction. In the presence of amino compounds the browning of sugars occurs more rapidly, particularly in neutral or alkaline conditions in a sequence of reactions called Maillard reaction. Sugar-amine reactions result in amorphous brown to black polymers called melanoidins. The interaction of amino compounds with monosaccharides results in the formation of an intermediate imine which cyclizes to a glycosylamine (N-glycoside). The N-glycosides are widely distributed in nature (e.g. nucleic acids, NAD^+, coenzyme A, etc.). They are generated in food whenever a reducing sugar occurs together with a compound

containing an amino group such as amines, free amino acids, peptides and proteins. Aldosylamine undergoes Amadori rearrangement to yield a 1-amino-1-deoxyketose and a ketosylamine yields 2-amino-2-deoxyaldose via a Heyns rearrangement. Both reactions correspond to alkali catalyzed isomerization of aldoses and ketoses. The amadori compounds are found in nature in various foods such as dried fruits, dehydrated vegetables, milk powder or liver extract. The glycosylamines and amadori compounds are only intermediate compounds in the Maillard sequence of reaction which undergo further acid or alkali catalyzed reactions at higher reaction rates to yield brown pigments known as melanoidins. Melanoidins are nitrogenous polymers and copolymers with variable contents of nitrogen and differing molecular weight and solubilities in water. In addition, volatile compounds with strong odour intensity are formed. Browning and aroma formation during cooking, frying, baking or roasting are essentially caused by the Maillard reaction. In addition to the amadori compounds, reactive dicarbonyl compounds are also formed during the sugar-amine reactions. The disintegration of dicarbonyl compounds and others such as maltol and isomaltol (formed from amadori compounds) gives rise to flavours of cooked foodstuffs. The browning of bread crust and the undesired discolouration and off-flavour development in powdered forms of eggs and milk are due to heating the sugars in the presence of proteins.

Maillard reaction results in a loss of amino acids. Lysine is the most reactive of amino acids followed by methionine, arginine, tryptophan and histidine. Since lysine, histidine and tryptophan are essential amino acids (these cannot be synthesized by the body and hence must be supplied with the diet), it is important to retard the Maillard reaction in foods. Inhibition of Maillard reaction is accomplished by keeping the pH below the isoelectric points of amino acids, use of low temperatures and by diluting with water.

Strecker degradation. An important class of volatile flavours arises by an interaction of α-dicarbonyl compounds (which are end products or intermediate products of other degradation reactions) with α-amino acids at elevated temperatures called the Strecker degradation reaction which involves transmination and decarboxylation reactions. The degradation reaction yields aldehydes (often called Strecker aldehydes), aminoketones and carbon dioxide. The Strecker aldehydes (e.g. formaldehyde, acetaldehyde, methyl propanal, 2- and 3-methyl butanals, 2-phenyl ethanal, etc.) contribute to the fruity, ester like, sharp penetrating green, malty green or honey like aromas in foods. The aminoketone formed undergoes further reactions to yield pyrazine derivatives which are also powerful aroma constituents. These compounds together with products obtained from dimerization of sugar residues make a major contribution to attractive and characteristic aroma of baking.

3.5.3 Sweetness of Sugars

The most important nutritive sugars include sucrose and glucose. Most sugars are much less sweet than sucrose and some are not sweet at all. The sweet characteristic of sugars and other sweeteners has been attributed to the presence of a saporous unit represented as A,H-B system (vide Figure 3.1). A and B represent electronegative atoms (usually oxygen) and A,H hydrogen bonding capability. For sweetness to be recognized by the tongue, distance between the atom B of the sweet molecule (sugar) and the H atom on A (i.e. on the neighbouring atom) should be 0.3 nm. It is assumed that the A,H-B system interacts via H-bonding with a similar structure in the sweet sensitive protein in the epithelium of the taste bud. In sugars 3,4-α-glycol structure is considered to be the primary A,H-B of the aldohexoses. In a ketohexose such as β-D-fructopyranose, the anomeric hydroxyl and the hydroxy-methylene oxygen have this role. The conformation of the sugar ring is of crucial importance to exhibit sweetness. For the α-glycol A,H-B system to have correct dimensions the two -OH groups must be in skewed or guache form. In eclipsed conformation the two groups are too close to each other and form intramolecular hydrogen bond, while in the staggered form, they are far apart to interact with the receptor protein. The CI conformation of β-D-glucopyranose has all the -OH groups attached directly to the ring (C1, C2, C3 and C4) in the guache conformation and are potential saporous units. In contrast, the IC conformation has all the α-glycol structure in anti-conformation.

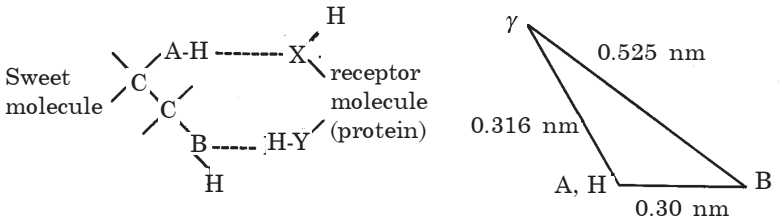


Figure 3.1 A,H-B system of a sweet molecule, its interaction with receptor protein of taste bud and the rigid cyclic structure of lipophilic site and A,H-B system.

In addition to the presence of an A, H-B system of the correct dimensions, the relative orientation of the other hydroxyl groups and the presence of a non-polar lipophilic binding site (γ-site) are also important for the molecule to exhibit sweetness. The axial C4 hydroxyl of D-galactose is able to form an intramolecular hydrogen bond with the ring oxygen because of its orientation, and this has been attributed to the fact that D-galactose is less sweet than glucose. Another factor is that though a sugar molecule may have more than one saporous unit, only one at a

time actually binds with the receptor. Thus trehalose with twice the number of saporous units of glucose is not twice as sweet as the latter. The sweetest substance with the A,H-B system and the non-polar lipophilic site forming a rigid cyclic structure binds tightest to the receptor site indicating the inflexibility of binding site on the receptor protein. High molecular weight polymers of glucose, e.g. starch, are without taste.

The structures of a few sweet carbohydrate molecules are shown below in Figure 3.2.

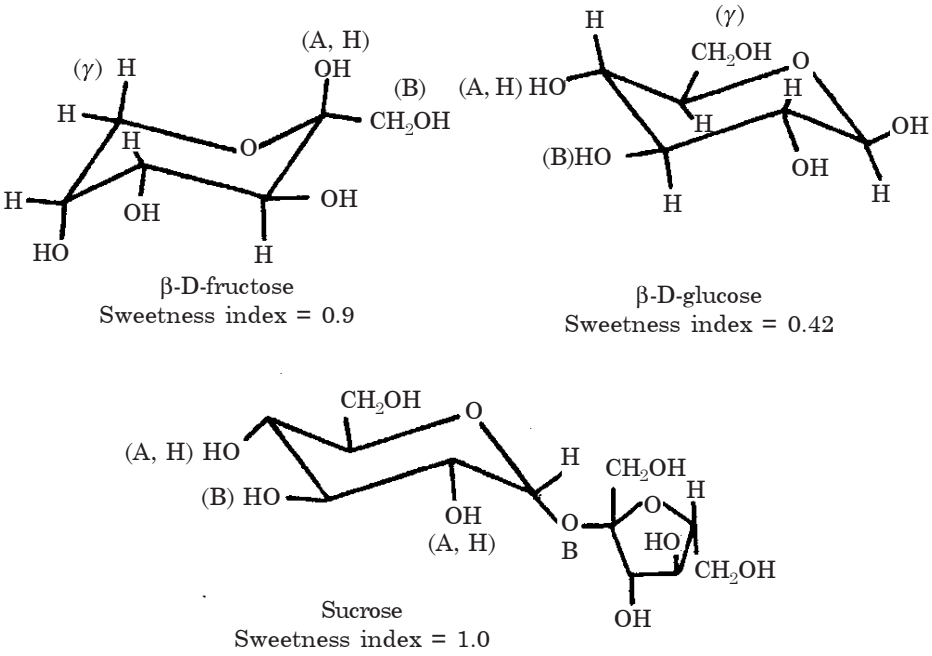


Figure 3.2 Sweet carbohydrate molecules.

The A, H-B concept has been extended to many non-sugar sweeteners, such as chloroform, cyclamate, saccharin and aspartame. Non-sugar sweeteners are discussed in Chapter 7 (page 99).

3.5.4 Texturing Characteristics of Carbohydrates

Monosaccharides, oligosaccharides and particularly, polysaccharides immensely influence textural characteristics of foods. Sugars in their crystalline state contribute to the appearance and texture of food products, particularly, the confectionery, biscuits and cakes. The granulated sugar and powdered form for icing are commercially available. The relative proportions of undissolved sugar crystals and saturated sugar solution (syrup) controls the texture of the cream centres of

chocolates. The crystalline forms of most sugars are anhydrous. However, monohydrates of sugars are also known. D-lactose crystallizes as the monohydrate of the α-anomer. These crystals are very slow to dissolve and occasionally form a gritty deposit in evaporated milk. The texture of sweetened condensed milk is also highly influenced by the size of the lactose monohydrate crystals.

Sugar. Sugars form super saturated syrups when their aqueous solutions are concentrated. The syrups of reducing sugars are very resistant to crystallization. This fact is taken advantage of by industry, in the production of hard-boiled sweets, in cake recipes to retain moisture. Very high dissolved sugar concentrations are an essential feature of jam and other preserves. Fruits are canned in concentrated sugar syrup to give them a texture. In preserves, water activity is reduced by hydrogen bonding with sugar molecules thereby preventing the growth of microorganisms. The concentrated syrups withdraw water from the fruit by osmosis and sugar molecules enter the cell to form complexes with cell wall polysaccharides thereby strengthening it and giving a firm texture. In home preparation of preserves and jellies, the acidity of the fruit and the extended boiling times to evaporate water facilitate extensive hydrolysis of sucrose and pectin methoxyl groups resulting in gelling failures.

Polysaccharides contribute immensely to the texture of foods through their rheological and physical characteristics. Starch is the most important polysaccharide with reference to the texture of many foods. Other polysaccharides obtained from various sources find use in food industry as texturing agents. These include pectin, agar, alginates, carrageenan, gum arabic, guaran gum, tamarind flour, xanthan gum and dextran.

Starch. The food's texture in relation to molecular structure of starch is quite interesting. Starch occurs in abundance in the form of irregular round shaped aggregates or granules, ranging in size from 2–100 μm. The shape and size of the granules are the peculiarity of the species of the plant and can be used to identify the origin of starch or flour. Starch consists of two types of glucose polymers, amylose (linear) and amylopectin (highly branched). Most starches have 20–25% amylose. However, pea starch is 60% amylose. Waxy varieties of maize and other cereals have very little or no amylose. The arrangement of amylose and amylopectin molecules in the granules of starch is not yet clearly understood. When viewed under a microscope, starch granules are birefringent (show a Maltese cross pattern). This implies that there is a high degree molecular orientation. X-ray diffraction patterns show that root starch granules are more crystalline than those of cereals. Crystallinity is attributed to amylopectin.

Raw uncooked starchy foods are not easy to digest because starch granules are contained within the cell walls of the plant which cannot be

penetrated easily by the digestive juices. Cooking softens the cell walls and allows water to enter starch granules causing them to disintegrate and gelatinize. Overcooking may cause the cell walls to disintegrate completely to give an unpalatable stuff.

Whole undamaged starch granules are insoluble in cold water due to the collective strength of hydrogen bonding of the chains of amylose and amylopectin. As the temperature is raised, water begins to be imbibed, starch granules swell and form a sticky suspension which consists of highly swollen starch granules dispersed in a solution of free starch molecules that are leached out from the granules during swelling. The temperature range at which the swelling occurs with the simultaneous destruction of the internal structure of the granules is called the gelatinization temperature. This trend is characteristic of particular starch, e.g. grains of sorghum starch gelatinizes at higher temperature compared to corn starch. The gelatinization temperature is usually in the range of 55–70°C. The birefringence of starch granules is lost as water is imbibed. Swelling occurs and the viscosity of the solution rises steeply. If heating is continued, the viscosity of the solution soon begins to fall, as the integrity of the granules is lost. On cooling, the viscosity rises again and a gel is obtained. When starch solutions or pastes are allowed to stand for a few hours they begin to show changes in their rheological properties. Dilute solutions lose viscosity but concentrated pastes and gels become rubbery and exude water (syneresis). This phenomenon is called *retrogradation* and involves amylose molecules. It involves an irreversible transition from the solubilized or dispersed state to an insoluble microcrystalline state due to the realignment of amylose molecules. Retrogradation is regarded as a normal progression in the firming of a starch gel. The rate and extent of retrogradation are influenced by temperature, size and shape of starch granules, concentration of the starch solution and other ingredients present in the solution or paste. It occurs more rapidly at temperatures close to 0°C. Retrogradation is the cause of a number of undesirable changes in food products. For example, starch is used to thicken the juice or gravy of commercial fruit or meat pies. After manufacture, if the pie is frozen, retrogradation sets in and on thawing, the paste becomes completely liquid, revealing how much or how little of the filling was actually fruit or meat. Retrogradation in such products is minimized or eliminated by using starch from waxy varieties of maize with very little amylose content. Such pastes or soft gels survive freezing and thawing operations.

The behaviour of starch during manufacture of bread and bakery products is very important in determining the texture of the finished products. When wheat flour is first mixed with water the damaged (due to flour milling) starch granules absorb some water. During bread making, the kneading and proving stages prior to baking allow time for the α- and β-amylases (naturally present in the flour) to breakdown a small portion of the starch to maltose and other sugars. These are

fermented by the yeast added to the mix to give out carbon dioxide which leavens the dough. Other baked goods rely on the carbon dioxide emanating from the baking powder added, together with air whipped into the mixture for the same purpose. Once in the oven, the starch granules gelatinize and undergo varying degree of disruption and dispersion. The α-amylase activity of flour persists till about 75°C and some enzymatic fragmentation of starch granules also occurs in bread. The factors that influence the degree of break down of starch include: (i) availability of water and (ii) the presence of fat. Butter and sugar are added to the flour to limit the access of water towards starch. After baking, the granular structure of starch persists giving the springy oven fresh bread.

Cake with low fat content and bread become stale in about two days after manufacture due to retrogradation of amylose. This gives the extra whiteness to bread crumbs, and increased rigidity of the gelatinized starch causes the lack of 'spring' in the crumb texture. Retrograded amylose can be returned to solution only by heating. Hence the stale bread is moistened and returned to the oven for a few minutes to revive it.

Starch with a high proportion of amylose can be obtained by fractionation of common starch. Addition of 1-butanol to a hot slurry of starch granules allows separation of amylose-complex on cooling, which can be recovered by filtration or centrifugation. Alternatively, starch can be dissolved in DMSO to weaken the H-bonds, prior to precipitation with butanol. Pure amylose is obtained by redissolving the butanol complex and the subsequent removal of butanol by distillation. High amylose starch is used when rapid setting is required for the manufacture of gums and impervious films. Strong, flexible films can be cast from pure amylose. The films have unusual properties of being water soluble and edible. Foods can be coated with films of amylose to improve water retention and to decrease surface stickiness of dehydrated fruits, while extenders may be added to starch to modify the physical properties of starch pastes. Sodium chloride reduces swelling and break down of starch granules whereas sodium hydroxide accelerates swelling. Sugars limit swelling by competing for H-bonding and fats inhibit swelling. Pregelatinized starch is obtained by heating a starch suspension above its gelatinization temperature, followed by drying. It contains mainly physically damaged starch which is soluble in cold water and forms gels readily. It is used in instant foods, puddings and in bakery products. Thin boiling starch is a partially acid hydrolyzed starch product that is readily soluble in boiling water giving a low viscous solution. On cooling, it remains a liquid and exhibits low retrogradation. It is used as a thickener and for forming protective films.

Chemically modified starch prepared by incorporating artificial crosslinks or chemical groups (e.g. $POCl_3$, adipic or succinic anhydride) available commercially, forms a gel readily with water or milk and is used for preparing instant desserts, toppings, thickeners, stabilizers and whippings. Reaction of starch with monochloroacetic acid in an alkaline solution gives carboxymethyl starch which can be used as a thickener and gelling agent.

Pectin. Pectin (polygalactouronic acid) occurs in the middle lamella of plant tissues and in the primary cell wall. It is the structural polysaccharide of soft tissues in soft fruits and fleshy roots. It is produced from peels of citrus fruits and from apple pomace by extraction at pH 1.5–3.0 at 60–100°C. The process is carefully controlled to prevent hydrolysis of glycosidic and ester linkages. The extract is concentrated and spray or drum dried into a powdery product. Pectin forms gels that give the texture of jam and fruit preserves. These gels consist largely of water and yet are stable and retain the shape of moulds. This is attributed to the stable three dimensional structure of the polysaccharide. The polymer chains in a gel interact in regions of ordered structure. These regions are called junction zones and are essentially crystalline. For making jam, pectin solubilization by boiling the fruit alone is not enough. The pectin must be esterified to at least 70%. The acid present naturally in the fruit or added must maintain the pH below 3.5 and total sugar content must be at least 50% w/v to stabilize the jam.

Agar. Agar is obtained from seaweed (red algae *Rhodophyceae*) by hot water extraction. The main components of the polysaccharide are β-D-galactopyranose and 3,6-anhydro-α-L-galactopyranose. Agar is insoluble in cold water but may be solubilized by precipitating it from a warm aqueous dispersion by the addition of ethanol. Agar is a good gelling agent and the gel setting and stability depend on the concentration and molecular weight. For example, a 1.5% solution of agar sets to a gel at 32–38° C with high stability as it melts only at 60–95°C. The large difference in gelling and melting temperatures, due to hysteresis, is a distinct feature of agar. Agar finds use as a stabilizer and emulsifier in frozen desserts of fruit juice, ice creams, yoghurt, cheese, candy and bakery products. It retards the staling of bread and also provides the desired texture in canned meat and poultry products. It is also used for preparing nutritive media in microbiology.

Alginates. Alginates occur as skeletal components of cell walls of brown algae (*Phaeophyceae*). Alginates are extracted with alkali and precipitated by acids or calcium salts. Alginate contains α-L-guluronic acid and β-D-mannuronic acid. The polysaccharide is water soluble as alkali metal or magnesium salts, giving solutions with high viscosity depending on the molecular weight. Adding Ca^{2+} to sodium alginate solution forms gels, fibres or films. Alginate is used in food industry as thickening, stabilizing and gel forming agent. At 0.2–0.5% concentration, it stabilizes the consistency of filling in baked products, salad dressings and milk chocolates. It prevents formation of large sized ice crystals in ice creams. As a gelling agent it is used in fruit gels, puddings and dessert gels.

Carrageenan. Carrageenan is obtained by hot water extraction of red seaweeds (*Chondrus, Eucheuma, Iridaea* and *Gigartina species*) under mild alkaline conditions followed by drying or precipitation. Carrageenans

are a complex mixture of polysaccharides mainly based on D-galactose and 3,6-anhydro-D-galactose. The polysaccharide finds use in food industry because of its ability to form gels (in desserts and canned meat), increase viscosity, stabilize emulsions and dispersions (in chocolate milk, cocoa suspensions, cheese and in dough used for baking bread).

Gum arabic. Gum arabic obtained from the bark of *Acacia* trees is a mixture of polysaccharides derived from L-arabinose, L-rhamnose, D-galactose and D-glucuronic acid. It is highly soluble in water and the solution viscosity increases rather steeply at higher concentrations. Gum arabic is used as an emulsifier and stabilizer in baked products, for retarding sugar crystallization, fat separation in confectionery products and formation of large sized ice crystal in ice creams and as a foam stabilizer in beverages. It is also used as a flavour fixative in the production of encapsulated, powdered aroma concentrates, particularly of essential oils to protect them against oxidation and other undesirable degradative changes.

Guaran gum. Guar flour of the seed endosperm of the tropical leguminous plant *Cyamopsis tetragonoloba* contains the polysaccharide chain of β-D-mannopyranosyl units. The gum is used as a thickening agent and a stabilizer in salad dressings and ice creams. It also finds use in paper, cosmetic and pharmaceutical industries.

Tamarind flour. The flour of the seeds of the tree *Tamarindus indica* is rich in polysaccharide which can be extracted with hot water and dried into a powder. The polysaccharide consists of D-galactose, D-xylose, D-glucose and L-arabinose units. The polysaccharide forms a stable gel over a wide pH range with a low syneresis. The gel is stabilized with relatively smaller amounts of sugar compared to pectin gels. The polysaccharide is used in the production of marmalades and jellies and as a thickening agent and stabilizer in ice cream and mayonnaise.

Xanthan gum. The polysaccharide is produced by fermentation using *Xanthomonas campestris* as an extracellular product. It is precipitated out from the fermentation broth by the addition of isopropanol in the presence of potassium chloride. It consists of D-glucose, D-mannose and D-glucuronic acid. It is highly soluble in water giving highly viscous solutions exhibiting a pseudoplastic behaviour (i.e. viscosity of the solution depends on shear rate). Xanthan gum has a high thermal stability and contributes the freeze-thaw stability of solutions, gels and emulsions. It is used as a thickening agent in canned foods, salad dressings and instant puddings and jellies.

Dextran. Dextran is produced by *Leuconostoc mesenteroide* and *Streptobacterium dextranicum*. It is a polysaccharide made of glucose and highly soluble in water. It is used as a thickening and stabilizing agent in bakery and confectionery products. It also finds use as a blood substitute for medical applications.

Cellulose. Cellulose is the main constituent of plant cell walls along with hemicellulose, pectin and lignin. These are not digested and constitute the dietary fibre of the human food. Microcrystalline cellulose is used in low calorie food products and salad dressings, desserts and ice creams along with carboxymethyl cellulose.

3.5.5 Plasticizing Action and Humectancy of Carbohydrates

Carbohydrates are hydrophilic to different degrees, depending on their structures which governs their plasticizing action and humectancy (ability to retain water). The rate and extent of water absorption by sugars in the dry state depends on their absolute purity, anomeric purity and homogeneity of crystal structure. The presence of a small percent of malto-oligosaccharides in β-maltose monohydrate (not removed even by repeated recrystallization) significantly increases the hygroscopicity of maltose. The hygroscopicity of sugars, dextrins and their mixtures is an important factor affecting the acceptance of confections, bakery toppings, whiteners and instantly reconstitutable powders or granular foods that must not become sticky. Maltose and lactose with their limited uptake of moisture are useful in such products. In contrast, the more hygroscopic liquid sugars (e.g. invert sugar, glucose syrup) help retain moisture in baked goods, plastic candies and fillings that should not become brittle. If the sugar in such products crystallizes out, water is given up and plasticizing effect is lost. To retain plasticity and inhibit crystallization, starch hyrolysates containing non-crystallizing malto-oligosaccharides and dextrins or sorbitol are used as additives.

Questions

1. List the dietary carbohydrates and their sources.
2. Write a note on the structure of monosaccharides.
3. Give a detailed account on the nutritional characteristics of carbohydrates.
4. What are dietary fibres? What is their use?
5. Discuss in detail the role of carbohydrates in flavour production.
6. What is browning reaction? What are its implications?
7. What is caramelization? How is it carried out?
8. How is caramel prepared for colouring foods?
9. What is the significance of Maillard reaction?

10. What is A,H-B concept? How does it explain the sweetness of a molecule?
11. How does starch contribute to the texture of bread?
12. Write a note on the contribution of polysaccharides other than starch to the texture of foods.
13. What are chemically modified starches? What are their applications?
14. Write a note on plasticizing action and humetancy of carbohydrates.

CHAPTER

4

Lipids

4.1 DEFINITION AND CLASSIFICATION

Lipids are a group of substances which are insoluble in water but soluble in nonpolar solvents. The majority of lipids are oils and fats which are derivatives of long chain fatty acids. Other lipids include phospholipids and terpenes. The fatty acids are usually monocarboxylic acids with straight chains containing even number of carbon atoms. Both saturated and unsaturated fatty acids are constituents of lipids. Lipids are of three main types: (a) simple lipids, (b) compound lipids and (c) derived lipids.

Simple lipids (also called neutral lipids) are esters of fatty acids with alcohols. These include fats, which are esters of fatty acids with glycerol and waxes, which are esters of fatty acids with alcohols other than glycerol.

Compound lipids contain other groups in addition to the esters. Some example of compound lipids are: phosphatides (phospholipids) containing phosphoric acid and a nitrogen moiety, cerebrosides (glycolipids) containing fatty acids, carbohydrates and a nitrogen moiety but not phosphoric acid, sphingolipids and sulpholipids.

Derived lipids are substances derived from neutral or compound lipids and have general properties of lipids. These include fatty acids, higher alcohols and sterols, and hydrocarbons.

4.2 FATS AND OILS

Fats and oils represent the most prevalent category of lipids and together with phospholipids are the most important food constituents. Liquid fats are called oils, e.g. soybean oil, olive oil, cotton seed oil, sesame seed (gingely) oil, groundnut oil, mustard oil, etc. These are mainly of plant origin. Lard (pig fat) and tallow (bovine fat) are examples of non-liquid fats of animal origin and horse oil is a liquid fat.

4.2.1 Fat Group Classification

Fats and oils can be classified according to their group characteristics into five fat groups: (i) milk fat group, (ii) lauric acid group, (iii) oleic-linoleic acid group, (iv) linolenic acid group and (v) animal depot fat group.

Milk fat group is essentially from the milk of cow, buffalo, goat and sheep. This group is characterized by the presence of oleic acid (30–40%), palmitic acid (25–32%), stearic acid (10–15%) and butyric acid (3–15%).

The lauric acid group fats contain high percentage of lauric acid (40–50%) which is a C_{12} acid and lesser amounts of C_8, C_{10}, C_{14}, C_{16} and C_{18} acids. The unsaturated acid content is extremely low and hence the fat group has a longer shelf life. This fat group has low melting temperature. Commonly used fats of this group are obtained from coconut and palm seeds.

The oleic-linoleic acid group fats contain only fats and oils of vegetable origin. This group contains less than 20% of saturated fatty acids. Examples of this group are cotton seed oil, corn oil, sesame seed oil, peanut oil, sunflower oil, olive oil, and safflower seed oil.

Linolenic acid group fats contain linolenic acid predominantly along with oleic and linoleic acids. Examples include soybean oil, wheat germ oil, perilla oil and linseed oil.

The animal depot fat group consists of lard and tallow. These contain 30–40% of C_{16} and C_{18} saturated fatty acids and up to 60% of oleic and linoleic acids. The melting points of these fats are relatively high.

4.2.2 Role of Fats and Oils in Biological Systems

(i) Fats and oils are the most concentrated source of food energy. They provide energy at 9 k.cal/g. They perform the function of energy storage in seeds, fruits and in animals.

(ii) They function as carriers of fat-soluble vitamins.

(iii) They contribute to the flavour and palatability as well as to the feeling of satiety after eating.

(iv) Lipids in the form of triglycerides, phospholipids, cholesterol and cholesterol esters are important to the structure, and permeability of membranes and cell walls.

(v) Lipids are a major component of adipose tissue, which serves as thermal insulation for body, as protection against shock to internal organs and as a contributor to body shape.

4.2.3 Uses of Fats and Oils

Differences in triglyceride composition affects the physical properties of

fats and oils which in turn influences the use to which a given fat is put.

(i) Fats and oils are used for frying and cooking, providing a controlled heat exchange medium as well as changing the colour and flavour of cooked foods.

(ii) As shortenings, they impart a 'short' or tender quality to baked goods through a combination of lubrication and an ability to alter the interaction among other food constituents.

(iii) As salad oils, they contribute to mouth feel and also function as a carrier of flavours.

(iv) When emulsified with other ingredients they function as a carrier of flavours in the form of viscous pourable dressings or semi-solid fatty foods known as mayonnaise or salad dressings.

(v) Margarine (imitation butter) is used both for baking and cooking and also as a table spread.

(vi) Specially selected or manufactured fats are useful in confections, especially as enrobing or coating agents. These fats have a short melting range at body temperature.

(vii) Other fatty materials such as mono- and diglycerides and certain phospholipids such as lecithin have useful role as emulsifiers. Mono- and diglycerides contribute to shortening performance and act as staling inhibitors in bakery products. Lecithin is also used as mould-releasing agent in confections, to control fat bloom in chocolate candies and as an antispattering agent in cooking margarines.

4.2.4 Dietary Sources of Fats and Oils

The dietary fat intake for providing about 40% of the total energy requirement of an adult has been calculated to be in the range of 100 to 150 g/day.

Fats and oils are available in a variety of forms such as, cooking oils, salad oils, shortenings, margarines, etc. The various sources of fats include cereals, eggs, fish, meat, milk, legumes, fruits and vegetables.

Salad and cooking oils are prepared from the oils of cotton seed, soybean, corn, peanut and seeds of safflower, olive, sunflower and mustard. These oils are usually refined, bleached and deodorized. Some oils are lightly hydrogenated to provide stability of flavours.

Margarines are prepared from fats and oils of plant as well animal origin with other ingredients such as milk solids, salt, flavouring materials, and Vitamins A and D. The fat content in margarines is about 80%.

Commercial shortenings are semisolid plastic fats made from lard, tallow, cotton seed oil or soybean oil with or without emulsifying agents.

Butter obtained by churning cream is a water-in-oil emulsion containing 80% milk fats in a plastic form. Other minor constituents are casein, lactose, cholesterol, phospholipids, calcium salts, sodium chloride (1–3%) and Vitamins A, E and D. In addition, flavouring agents such as diacetyl, lactones and butyric acid are also present.

Cocoa butter from cocoa beans is the preferred fat in confectionery manufacture.

4.3 FATTY ACIDS IN FOODS

The fatty acids that occur in foods are straight chain monocarboxylic acids with an even number of carbon atoms. The acids are either saturated or unsaturated with one, two or even up to six double bonds. The fatty acids commonly found in foods are listed in Table 4.1. The cis-double bonds in polyunsaturated fatty acids, each separated by a single methylene group, bring about a kink or bend about 42° in the otherwise straight hydrocarbon chain of the fatty acid molecule. This change in the shape of the molecule seems to reflect in the physical properties such as melting points of fatty acids. For example, the saturated 18-C stearic acid melts at 69.6°C while the cis- unsaturated acids with 18-C atoms, e.g. oleic acid (one double bond), linoleic acid (two double bonds) and linolenic acid (three double bonds) melt at 10.5, − 5.0 and −11.0°C, respectively. However, insertion of tans- double bond seems to have no effect on the structure and melting point as indicated by elaidic acid, the trans-isomer of oleic acid, which melts at 43.7°C.

TABLE 4.1 Fatty Acids in Foods

Common name	*Systematic name*	**Structural representation*
Saturated fatty acids		
Butyric acid	n-butanoic acid	4:0
Caproic acid	n-hexanoic acid	6:0
Caprylic acid	n-octanoic acid	8:0
Capric acid	n-decanoic acid	10:0
Lauric acid	n-dodecanoic acid	12:0
Myristic acid	n-tetradecanoic acid	14:0
Palmitic acid	n-hexadecanoic acid	16:0
Stearic acid	n-octadecanoic acid	18:0
Arachidic acid	n-eicosanoic acid	20:0
Behenic acid	n-docosanoic acid	22:0

TABLE 4.1 Fatty Acids in Foods *(contd.)*

Common name	*Systematic name*	**Structural representation*
Unsaturated fatty acids:		
Palmitoleic acid	cis-9-hexadecanoic acid	16:1 *n* 9
Oleic acid	cis-9-octadecanoic acid	18:1 *n* 9
Linoleic acid	cis, cis-9,12-octadeca-dienoic acid	18:2 *n* 9,12
α-Linolenic acid	all cis-9,12,15-octa-decatrienoic acid	18:3 *n* 9,12,15
Arachidonic acid	all cis-5,8,11,14-eicosa-tetranoic acid	20:4 *n* 5,8,11,14
Clupanodonic acid	all cis-7,10,13,16,19-docosapentanoic acid	22:5 *n* 7,10,13,16,19

*A convenient structural representation of fatty acids gives the number of carbon atoms and the number of double bonds. The position of the double bonds is indicated by the symbol *n* followed by the carbon atom number(s).

The short chain fatty acids (with less than 14 carbon atoms) occur as triglyceride constituents in milk fat, coconuts and palm seeds. They also occur in free form or esters of low molecular weight alcohols in foods processed with microorganisms and contribute to the aroma of such foods. The odour threshold of free fatty acids vary in the range of 1–10 ppm. Fatty acids with more than 14 carbon atoms are flavourless and odourless. Fatty acids with odd numbers of carbon atoms such as valeric acid (5 carbon atoms), enanthic acid (7 carbon atoms) are present in foods only in traces and contribute to the aroma.

The unsaturated fatty acids contain one, two or three allyl ($-CH=CH-CH_2-$) groups. These acids are designated respectively as ω-3 (α-linolenic acid type), ω-6 (linoleic acid type) and ω-9 (oleic acid type). The ω-3 type includes hexadecatrienoic acid and the polyunsaturated fatty acids docosahexaenoic acid (DHA – 22:6) and eicosapentaenoic acid (EPA – 20:5). The ω-6 type includes γ-linolenic acid and arachidonic acid (found in meat, liver, lard and lipids of chicken eggs). Erucic acid found only in mustard family of seeds belongs to ω-9 group.

Substituted fatty acids include hydroxy fatty acids, oxo fatty acids and furanoid fatty acids. Ricinoleic acid (a hydroxy fatty acid) occurs in castor bean oil. D-2-hydroxy saturated 16:0 to 25:0 fatty acids are found in green leaves of a number of vegetables. ∂-Lacatones formed from hydroxy-carboxylic acids are found in milk fat, apricots and peaches. Oxo (or keto) acids are less common and found in milk fat at less than 1%. Furanoid fatty acids occur in fish liver oil and in some fresh water fish.

4.3.1 Essential Fatty Acids

Fatty acids of ω-3 and ω-6 types have been identified as essential fatty acids. Linoleic and arachidonic acids have been identified as precursors of a group of animal harmones known as the prostaglandins. The animal tissues cannot synthesize these fatty acids but convert them into C-20 acid moiety of prostaglandin. Prostaglandins play a regulatory role in many cellular processes. Different members of prostaglandins (about 14 are known) have different and often, opposite actions. These affect smooth muscle contraction, platelet aggregation, arterial wall tension and blood pressure. They inhibit gastric secretion and the release of fatty acids from adipose tissue. They are involved in regulation of immune function and thus resistance to infection.

The involvement of dietary fatty acids in the occurrence of atherosclerosis is associated with high levels of cholesterol in the blood due to consumption of fatty foods rich in saturated fatty acids. People consuming more of fish oils rich in polyunsaturated fatty acids and clupanodonic acid have a remarkably low incidence of arterial diseases. It is possible that these ω-3 fatty acids give rise to a different set of prostaglandins compared with those of ω-6 fatty acids such as linoleic or oleic acids. The prostaglandins from ω-6 fatty acids are known to promote formation of blood clots that cause thrombosis. The ω-3 family and ω-6 family suppress the metabolism of each other. The ω-6 family has been identified to suppress the formation of polyunsaturated fatty acids (PUFA) from oleic acid as is manifested in essential fatty acid deficiency. Hence it is suggested that consumption of ω-3 fatty acids should be enhanced and the ratio of ω-6 to ω-3 fatty acids be decreased.

4.4 TRIGLYCERIDE COMPOSITION AND STRUCTURE

Glycerol, a trihydroxy alcohol can form triesters with one, two or three different fatty acids. Thus tripalmitin (PPP) and tristearin (SSS) are triesters formed with a single fatty acid, namely, palmitic or stearic acid. Dipalmitoolein (P_2O) contains two different fatty acids, palmitic and oleic acids. Palmito-oleo-linolein (POL) contains three different fatty acids. The rule of this shorthand designation is that the acid with the shorter chain or in the case of an equal number of carbons in the chain, the chain with fewer double bonds is mentioned first. The Z number gives the possible different triacylglycerols which can occur in a fat with n different fatty acids.

$$Z = (n^3 + n^2)/2 \tag{4.1}$$

Thus for $n = 3$, the number of possible triglycerides is 18. Since Z value is based on the number of possible positional isomers within a molecule, the Z value in a triglyceride of the type POS, PSO and SOP becomes

smaller and designated as Z' (for $n = 3$, the Z' value would be only 10) as given by

$$Z' = (n^3 + 3n^2 + 2n)/6 \quad (4.2)$$

Properties of triglycerides depend both on the fatty acid composition and on the distribution of fatty acids in the triglycerides. The distribution of fatty acids in triglycerides is different for fats from plants and animals. Plants tend to distribute saturated and long chain acids in Positions 1, 3 and dienoic unsaturated fatty acids in the Position 2. In animal fats, saturated acids are in Position 1, short chain and unsaturated acids in Position 2 and long chain fatty acids in Position 3. Additional variables occur in fats of pigs, bird, mammals and fish. A simplified triglyceride composition of some animal and plant fats are given in Table 4.2.

TABLE 4.2 Simplified Triglyceride Composition of Some Fats

Triglyceride	*Mol per cent of components*			
	Beef fat	*Chicken fat*	*Palm oil*	*Groundnut oil*
UUU	3	24	10	47
USU	2	7	2	1
SUU	17	38	36	41
SUS	32	16	34	9
SSU	16	10	9	1
SSS	30	5	8	1

S = saturated and U = unsaturated fatty acids

4.5 PHYSICAL CHARACTERISTICS OF TRIGLYCERIDES

Physical characteristics of fats provide useful criteria for evaluating the utility of a fat for a specific product or for monitoring the stages of processing of foods. Among the physical properties that are useful in food processing are melting point, specific heat, heat of fusion, viscosity, density and refractive index.

Melting point. Sharp melting or narrow-range melting fats are used for confectionery while gradually melting or broad range melting fats are used for shortenings.

Specific heat. Knowledge of specific heat is useful in processing operations. Specific heat of fats increases with increasing unsaturation of fatty acids in both liquid and solid states of a fat. Of greater importance is the fact that liquid fats have specific heat values almost twice that of solid fats, and the solid α-form has a greater specific heat than the solid β-form. This property relates directly to the freedom of molecular motion in various states.

Viscosity. This is an important factor to be considered in designing systems for handling fats. Viscosity of fats increases with increasing chain length of fatty acids and decreases with increasing unsaturation. Viscosity is thus a function of molecular size and molecular packing or orientation.

Density. Knowledge of density is needed in designing equipment. This property provides a means of estimating solid-liquid ratio, called solid fat index (SFI) or solid content index (SCI) of fats. A dilatometer, which measures specific volume (reciprocal of density), is used for this purpose.

Titer. This is a measure of fat hardness and is a criterion for distinguishing between tallow and non-edible grease. Animal fats of uncertain origin, which have a titer of 40 or greater are 'tallow' while those with lower values are designated 'greases'. Titer is determined by saponifying a fat to obtain fatty acids. The melted fatty acids are allowed to cool slowly while the temperature is observed. At some point, crystallization sets in and the temperature rises sharply. The temperature during this initial crystallization is called the titer.

Cold test. This is a measure of time required to develop a cloudy appearance in an oil held in an ice bath. An acceptable minimum for salad oil is 5.5 hours. Oil to be used in mayonnaise must pass a cold test so that it cannot crystallize and break the emulsion during storage at refrigerated temperatures.

Refractive index. The refractive index of oils increases with increasing chain length of fatty acids in the triglycerides and also with increasing unsaturation. The iodine value of an oil is a measure of unsaturation and correlates with the refractive index. Hydrogenation of oils (vanaspati manufacture) is monitored by measuring the refractive index of the oil.

4.6 POLYMORPHISM OF TRIGLYCERIDES

Triglycerides have been found to exist in several crystalline forms, each of which has a characteristic melting point, X-ray diffraction pattern and IR spectrum. This phenomenon known as polymorphism, arises from different modes of packing in the crystal. For example, tristearin exists in three polymorphic forms, α, β and β', with melting points at 53°, 64.2° and 71.7°C respectively. The lowest melting α-form is obtained by rapid solidification of melted tristearin. Slow heating melts this form, which then resolidifies in the β' form. Repetition of this procedure results in the formation of the stable β-form. The β-form is also obtained by crystallization from a solvent.

Triglycerides containing different fatty acids with varying lengths and degrees of unsaturation exhibit exceedingly complex polymorphic behaviour. Some crystallize in units consisting essentially of double or triple chain lengths and are designated as β2 or β3 to indicate both the

packing arrangements and the number of acyl chains associated with it. Tri-unsaturated triglycerides (e.g. triolein) form double chain length structures. Mixed oleic acid containing triglycerides have a tendency to form triple chain length structures. Triglycerides composed of long and very short chain components have unusual properties, e.g. 1-stearodiacetin or 1-palmitodiacetin have unusually stable α-forms. Films of these can be stretched 200–300 times their lengths and are used as flexible coatings of food such as cheese. The β-form obtained by solvent crystallization cannot be stretched. The physical characteristics of the crystals are given in Table 4.3.

Table 4.3 Physical Characteristics of Fat Crystals

Polymorphic form	*Structure*	*Visual appearance*	*Size* (μm)
α	HCP	fragile, transparent platelets	5
β'	Orthorhombic	tiny delicate needles	1
β	Triclinic	large coarse crystals with clumping	25–50 (grow to 100 μm)

Fats and oils show a natural crystal forming tendency. Thus coconut oil, corn oil, olive oil, palmseed oil, peanut oil, safflower seed oil, sesameseed oil, sunflower seed oil, cocoa butter and lard exhibit a tendency to crystallize in β form while cottonseed oil, herring oil, milk fat of cow, palm oil, rapeseed oil, tallow and whale oil crystallize in β' form.

4.7 PLASTICIZING PROPERTIES OF FATS

The usefulness of a fat to a particular food application is dependent on the melting and crystallization character.

Shortenings are semisolid plastic fats and their quality depends on three important parameters, namely (i) incorporated air, (ii) plasticity and consistency and (iii) solid-liquid ratio. These parameters in turn depend on the polymorphic forms and the method of preparation. Plasticity, consistency and solid-liquid ratio depend on the melting range of the fat. Proper tempering is necessary to form the mixed crystals required for a broad melting range. Polymorphs differ in their abilities to incorporate air during plasticizing, i.e. whipping of air or nitrogen into a fat during crystallization stage to give whiter, creamier, smoother and more uniform shortenings. The β' crystals of the fat assist in the incorporation of an abundant quantity of very small air bubbles while mixing the fat with flour or sugar and hence give creamier shortenings compared to β crystals which incorporate air in small amount of large sized bubbles. Air incorporation in batter is similar to plasticizing . The volume, texture and tenderness of baked products depend on the size and number of air bubbles

in the batter. Thus fats with β'crystals are best for use in cakes and icings while fats with β crystals are totally unsatisfactory.

Good air incorporation properties can be obtained from random fats such as cottonseed oil and soybean oil by slight modification of the natural oils with regard to solids content. These have too small amounts of solids content and hence partial hydrogenation is carried out to increase the solids content. Tallow has too large an amount of solids content and hence it is destearinized and lightly hydrogenated to give improved physical properties and flavour stability. Adding flakes of hydrogenated fats aids in creaming performance. Cottonseed oil flakes or tallow flakes encourage formation of β' crystals in oils and are used for preparation of shortenings.

Margarines are produced from a mixture of fats blended to give desired properties. Oils that have been hydrogenated to two or three degrees of hardness are blended to provide SFI values in the range of 15–20 at 32°C, appropriate to the use. Cottonseed oil (which forms β' crystals) and soybean oil are blended for use in the manufacture of margarine.

In pastry, cakes, biscuits and other baked goods lard and butter are used to impart plasticity to the mix, improve cake volume and crumb structure, prevent fat blooming and give smoothness, soft and moist feel and palatability to the product.

4.8 ENROBING FATS

Cocoa butter has a short melting range at mouth temperature and is an important constituent in chocolate making as a coating or enrobing fat. The special characteristics of chocolate are its melting point (melts in the mouth but not in the hand), smooth texture and glossy surface. Cocoa butter can occur in six different polymorphic forms with melting points ranging from 17.3 to 36.4°C. Only one of these (the β-3 form) melts at 33.8°C and has the desired properties. The skill of the chocolate maker lies in ensuring that the fat is in this particular state in the product. This is achieved by tempering. The liquid is cooled to initiate crystallization and reheated to below the melting point of the desired polymorphic form so to melt out any of the undesirable types. The chocolate is stirred at this temperature for some time to obtain a high proportion of the fat as very small crystals.

Chocolate that has been incorrectly tempered or subjected to fluctuations in temperature as in shop windows develops bloom. This is a gray film, which resembles mould growth but is caused by the transition of some of the fat to a more stable polymorphic form which crystallizes out on the surface. Migration of triglycerides from the nut centres of chocolate or the crumb of chocolate coated biscuits can cause similar problems. Milk fat is an effective bloom inhibitor and is often included in amounts of about 25 per cent of the total fat in milk chocolates.

4.9 EMULSIFYING PROPERTIES OF FATS

Polar lipids such as glycerophospholipids, cholesterol and lipoproteins are important in food systems because of their ability to stabilize emulsions. Emulsions are colloidal systems of two immiscible liquids, one dispersed in the other continuous phase. Examples include 'oil-in-water' emulsions (milk and mayonnaise) and 'water-in-oil' emulsions (butter). In many cases the structure is complicated by the presence of suspended solid particles or air forming foam. Emulsions are prepared by vigorous mixing of the two immiscible liquids so that small droplets of the dispersed phase are formed. However, a simple emulsion breaks down rapidly as the dispersed phase droplets coalesce to form a layer which either floats to the surface or settles down to the bottom of the vessel. Emulsion stability is enhanced by the presence of substances, called emulsifiers or emulsifying agents, whose molecules have both polar and non-polar regions such as phospholipids. Such molecules orient at the interfaces between the two phases and thus form barrier to the coalescence of the droplets. In some foodstuffs, proteins also are used as emulsifying agents (e.g. sodium taurocholate which is a derivative of cholic acid secreted by liver).

The traditional source of emulsifying properties in food is the egg yolk. Approximately 33 per cent of egg yolk of hen is lipid and the protein content is about 16 per cent. Almost two-thirds of lipid is triglyceride and about 28 per cent is phospholipid and the rest of about 5 per cent is cholesterol. Lecithins are the predominant phospholipids along with small amounts of cephalins, cholines and spingomyelins. Yolk actually consists of a suspension of lipid/protein particles in a protein/water matrix. Egg yolk acts as an emulsifier in the preparation of sauce, mayonnaise, etc. Lecithin recovered from soybean oil is an alternative emulsifying agent for use in chocolate and other confectionery and for vegetarian food products.

Dairy products provide examples of both 'oil-in-water' and 'water-in-oil' type emulsions. The fat in milk contains fat globules of about 4–10 μm in diameter and there are about 1.5 to 3.0×10^{12} globules/litre of milk. Globules of such small size should according to Stoke's law take some 50 hours to float and stabilize as a layer of cream. But this happens in about a few minutes. This implies that larger particles of the size of about 800 μm are formed. This is due to the action of lipoproteins on the globules, which form a milk fat globule membrane, which in turn crosslinks proteins in water to form large size globules. On the other hand, evaporated milk which is sterilized after canning by holding temperatures above 100°C for several minutes shows no tendency to cream due to denaturation of lipoproteins. Creaming is also prevented in homogenized milk where the fat globules are reduced about 1 μm in size by passage through very small holes at high velocities (about 250 m/s). The increase in surface area results in adsorption of extra proteins from the aqueous phase of milk which prevent coalescence of globules.

Butter making is formation of a water-in-oil emulsion. Cream with a fat content of 30–35% is inoculated with a culture of bacteria and incubated for a few hours. The bacteria produce the flavour compounds such as diacetyl (butanedione). The cream is then mechanically agitated (churned) to disrupt the fat globule membrane and cause coalescence. During the process of mixing, a proportion of aqueous phase known as butter milk, is trapped as small droplets. These are prevented from coalescing by the rigidity of the fat and by the layer of proteins and polar lipids, which form at the fat/water interface. Butter contains about 20 per cent water. Salt is added as a flavouring agent as well as to deter the growth of microorganisms.

Margarine has butter milk and salt added to it during final blending along with emulsifying agents such as soya lecithin.

Many food products (ice cream, desserts, cheese, etc.) contain synthetic emulsifying agents (permitted emulsifier and stabilizer). These include glycerol derivatives and sorbitan esters. Of particular interest are the diacetyl tartaric acid of monoglycerides (DATEMS). They are added to bread recipes at 0.5% to improve both the loaf volume and crumb texture.

4.10 RANCIDITY AND REVERSION OF OILS

The chemical properties of relevance to the stability of fats and oils as food ingredients include rancidity and reversion. Rancidity indicates the deterioration of fats and oils. In dairy fats rancidity is due to hydrolysis of triglycerides (lipolysis) by microorganisms so that odorous short chain fatty acids are liberated. In other fats and oils and fatty parts of fish rancidity is due to auto-oxidation of the unsaturated fatty acids. The reaction is initiated by the highly reactive singlet oxygen generated by the reaction of oxygen with pigments such as chlorophyll, riboflavin or haem in the presence of light. The singlet oxygen then adds onto the olefinic carbon of an unsaturated fatty acid to give hydroperoxides which in turn produce hydroperoxy and alkoxy free radicals. These free radicals accumulate in the fat and initiate a chain reaction giving rise to ultimate breakdown products of aldehydes, ketones and alcohols which are characteristic of rancid fat. A method of assessing the deterioration of fats is based on the formation of malonaldehyde, a product of auto-oxidation, which gives a pink colour with thiobarbituric acid reagent. Other tests include measurement of the carbonyl content using 2,4-dinitrophenylhydrazine, the hydroperoxide content by reaction iodine and characteristic absorbance of conjugated dienes and trienes at 230 and 270 nm respectively.

Reversion of soybean oil is due to auto-oxidation of its labile linoleic acid giving off flavours. The compounds diacetyl (butanedione), cis-3-hexenal, 2,3-pentanedione and 2,4-pentadienal have been identified as responsible for the off flavour of reverted soybean oil. Fats and oils

exposed to atmosphere and heating over a long period show the final stage of oxidation sequence, namely, polymerization. This is particularly so with frying oils causing foaming and increased viscosity. The reaction proceeds via a free radical initiation followed by chain reaction of propagation and termination steps.

Antioxidants are used to stabilize fats and oils. These include butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), α-tocopherol (Vitamin E) and propyl gallate (PG). The antioxidants apparently block the propagation reaction by donating hydrogen atom to free radicals and terminating the auto-oxidation. Antioxidants do not reduce the ultimate degree of rancidity but lengthen the induction period in proportion to their concentration. Antioxidants PG, BHT and BHA are added in concentrations up to 200 ppm. Vitamin E occurs in plant tissue and is added up to 0.1%.

Questions

1. What are lipids? How are they classified?
2. Classify the fats and oils according to their group characteristics.
3. Explain the role of fats in biological system.
4. How are fats useful in foods?
5. List the fatty acids found in natural foods and classify them.
6. What are essential fatty acids?
7. Write a note on triacylglycerol composition.
8. Give an account of the various physical properties of fats and oils with reference to edible oil/fat processing.
9. What is polymorphism of fats? What are its implications?
10. What are enrobing fats?
11. Write notes on (a) plasticizing action of fats and (b) emulsifying properties of fats.
12. What is an emulsion? What types of emulsion are encountered in foods?
13. How does an emulsifying agent function?
14. What are the chemical properties of relevance to the storage stability of oils?
15. Give examples of antioxidants used in edible oils and fats.

CHAPTER

5

Proteins and Enzymes

Proteins are yet another important macro components of living systems. Proteins are polymers of amino acids with molecular weights ranging from around 10,000 to several hundred thousand. Proteins are of importance in human food with respect to two aspects namely, nutrition and desired textural quality of food. Protein imbalance in food leads to malnutrition. Enzymes are proteins with a special function, namely, biocatalysis.

5.1 CLASSIFICATION OF PROTEINS

Proteins may be classified on the basis of their solubility. Albumins are water soluble proteins while globulins are insoluble in pure water but soluble in salt solutions. Most of the enzymes and hormones belong to either of these two types. Glutelins are another group of proteins found in plant seeds soluble in acids or bases but not in pure water or salt solutions. Prolamines, also found in plant seeds, are soluble in about 70% alochol-water mixtures only.

Proteins may also be classified into five classes depending on their functional role in biological systems: (i) carrier proteins, such as haemoglobin and permeases which control the transport of substances across cell membranes, (ii) enzymes which are biocatalysts, (iii) immunoglobulins which form antibodies that provide an animal's defense against invading microorganisms, (iv) structural proteins, e.g. collagen and (v) proteins with nutritional function either in the transmission of nutrients from mother to offspring (e.g. casein of milk) or in the storage of nutrients to be utilized by an embryo (e.g. seed proteins of plants and egg proteins of birds).

In proteins with nutritional function, the physical characteristics of the proteins are of secondary importance compared to their overall chemical composition. For example, gluten, the protein of wheat, is rich

in glutamine and so, compared to other proteins it has higher nitrogen content. Among their distinctive characteristics, seed proteins and casein have in common a tendency to form more or less ill-defined aggregates. Biologically such behaviour ensures that large amounts of nutrients can be concentrated without the problems of osmotic pressure associated with high solute concentrations.

5.2 PROTEIN STRUCTURE

Proteins are polymers of amino acids and synthesized from 21 naturally occurring amino acids. They have a highly complex structure. The linear sequence of amino acids in a protein is referred to as the primary structure. The sequence varies from protein to protein. Proteins do not occur in nature in their primary structure. The secondary structure of protein arises due to a three-dimensional arrangement of the long protein chain with lowest feasible free energy. Examples of the secondary structure include the α-helix (three variations—right handed helix, left handed helix and 3_{10}-helix) and the β-pleated sheet (three variations—β-sheet parallel, β-sheet antiparallel and twisted sheet) configurations. The helical structure is stabilized by hydrogen bonding, almost parallel to the axis of the helix, between the nitrogen of peptide bond and oxygen of another separated by four amino acid residues along the chain. Most of the enzymes and respiratory proteins have the helical structure ultimately giving rise to a globular shape of the proteins. In the β-pleated sheet, found in many fibrous proteins, all the peptide bonds available for hydrogen bonding stabilize the configuration through maximum number of crosslinks between adjacent polypeptide chains. Collagen, the most abundant protein in vertebrates, has yet another type of secondary structure. Collagen contains about 33% glycine and about 25% proline or hydroxyproline. The rigid R (side chain) groups and the lack of hydrogen bonding by peptide linkages involving proline and hydroxyproline allow the formation of a kinked type helical structure rather than a α-helix. Peptide bonds composed of glycine form interchain hydrogen bonds with two other collagen polypeptide chains, giving rise to a triple helix, called tropocollagen. The tertiary structure of a protein arises due to folding of the regular units of the secondary structure as well as regions of peptide chain without a secondary structure. The tertiary structure is stabilized by hydrogen bonding between groups of the amino acid moieties, salt linkages, hydrophobic interactions and covalent disulphide (–S–S–) linkages. Proteins with single polypeptide chains thus exhibit primary, secondary and tertiary structures. Proteins with several polypeptide chains exhibit a quaternary structure due to interactions between individual polypeptide chains or subunits.

Most proteins in biological systems exist as conjugated proteins in combination with carbohydrates (glycoproteins), lipids (lipoproteins),

nucleic acids (nucleoproteins), metal ions (metalloproteins) and phosphate (phosphoproteins).

5.3 ROLE OF PROTEINS IN BIOLOGICAL SYSTEMS

Proteins are the source of dietary amino acids and are used for growth and maintenance of living systems. They are costlier sources of energy compared to carbohydrates and fats and hence the human body utilizes proteins mainly for biosynthesis rather than as an energy source, though the energy yield is 5 k.cal/g of protein.

The proteins in our diet provide the amino acids from which body synthesizes its own proteins, the major constituents of tissues. The action of hydrolytic enzymes in the stomach as well as the small intestine breaks down food proteins to their component amino acids. On absorption into the blood stream they become part of the amino acid pool of the body. Break down of the body's own tissue proteins (an essential part of the process of aging or redundant cells) also contributes to the pool. The amino acid pool is used for protein synthesis and also the synthesis of purines, pyrimidines and porphyrins. Mammals can synthesize non-essential amino acids if adequate amount of amino nitrogen and carbohydrates are available. Essential amino acids cannot be synthesized and hence must be supplied in the diet. These include histidine, phenylalanine, lysine, methionine, isoleucine, leucine, valine, threonine and tryptophan. Ideally, the protein in the diet should provide the amino acids in the same relative proportions as the body's requirement. Absence of even one particular amino acid will result in cessation of all protein synthesis since almost all proteins have at least one residue of all the amino acids. Adult requirement for essential amino acids in mg/kg of body weight include, isoleucine (10), leucine (11–14), lysine (9–12), methionine and cystine (11–14), phenylalanine and tyrosine (13–14), threonine (6–7), tryptophan (3) and valine (11–14).

5.4 PROTEIN REQUIREMENTS

The protein requirement unlike that of carbohydrates and fats, on the basis of human body weight varies only slightly with increasing age above 20 years. The recommended daily amount of protein for adults is about 1.8 g per kg of body weight. In infants up to the age of one year, the recommended amount is between 2.0 to 2.2 g per kg of body weight and between one year and twenty years, 1.0 to 1.8 g per kg of body weight. Protein requirement will be higher during pregnancy and lactation and during recovery from illness.

5.5 DIETARY SOURCES OF PROTEINS

The proportions of the different amino acids required by the human infant correspond closely to the amino acid composition of human milk and hence it is accepted as the standard against which nutritional value of other foodstuffs is judged. Animal foods such as egg, milk and meat do not differ significantly from human milk with respect to amino acid composition but plant proteins do not satisfy the requirements of humans. Wheat protein has a low lysine content and is only about half as effective as human milk as a source of protein. The legumes such as soy and peas provide ample proportions of lysine but are deficient in methionine. In addition to being of low proportions in proteins, the two amino acids, lysine and methionine cause problems during storage or processing of foodstuffs. Lysine participates readily in Maillard reaction and during processing at high temperatures, especially in alkaline condition, lysine side chains form crosslinks with other amino acids, thus reducing lysine availability.

5.6 FUNCTIONS OF PROTEINS IN FOODS

Apart from nutrition by way of contributing to the amino acid pool and energy, the functional role of proteins in foods are quite varied and include solubility, swelling, increased viscosity, gelation, coagulation, emulsification, contribution to foam volume and stability, elasticity, binding and extrusion. Proteins and their hydrolytic products, peptides and free amino acids contribute to the flavour and taste of foods.

Swelling and gelation of a protein food depends on the solubility of the protein, which in turn depends on its conformation and intermolecular interactions at the given pH and temperature. For example, during the cooking of meat, the insoluble collagen of the connective tissue is denatured and partially hydrolyzed into water soluble gelatin by heat and alkaline conditions. Gelatin contributes to the viscosity of the solution (gravy) and forms gels by entrapping water.

Water holding capacity of muscle proteins influence the taste, tenderness and colour of meat, and it plays an important role in the formation and stabilization of meat emulsions. Poultry and fish muscle proteins have similar properties. Soya protein improves water absorption by wheat flour to increase the viscosity of the mix.

Coagulation of milk gives casein the major component of cheese responsible for the texture. Casein is known to promote low-temperature stability of frozen foods.

Formation of stable foams in whipped creams and ice creams is aided by proteins. For instance, whey proteins have good whipping characteristics and hence stabilize foam structure in dessert toppings.

Structure and texture of baked products is strengthened by the native wheat proteins and also by the added soya protein isolates. They also

delay the staling process in baked products. Whey proteins are used in confectionery items to strengthen the structure and retention of moisture. Wheat proteins are responsible for the unique cohesive and elastic properties of dough used to produce bread, biscuits, cakes and other baked goods.

Emulsification and structure formation characteristics are exhibited by egg proteins and whole eggs. They find extensive use in a variety of food products for their coagulation, gelation, emulsification and structure formation properties. Soya protein finds use as emulsifier, emulsion stabilizer and for its fat absorption characteristics.

Flavour in dairy products is mainly due to milk proteins. Most of the peptides are either bitter or neutral in taste. However, dipeptide esters of aspartic acid are sweet in taste. Bitter tasting peptides may be formed by food processing as in the case of cheese due to faulty ripening.

Peptides also contribute to a variety of functional properties in foods. Glutathione (a tripeptide γ-L-glutamyl-L-cysteinyl-glycine) occurs in animals, plants and microorganisms. It is involved in active transport of amino acids and due to its ready oxidation, is also involved in red-ox reactions in foods. It influences the rheological properties of wheat flour dough through thiol-disulphide interchange with wheat gluten. High concentrations of reduced glutathione in flour bring about reduction in protein disulphide bonds and a corresponding decrease in molecular weight of protein constituents of dough gluten. Other peptides, such as carnosine, anserine and balenine found in the muscles of beef, chicken and whale respectively, (these peptides contain β-alanine as a constituent amino acid) are used analytically to identify the meat extract. Nisin, a peptide formed by several strains of *Streptococcus lactis* contains several unusual amino acids. Nisin is active against Gram-positive bacteria and anaerobic spore forming microorganisms and hence used as a preservative particularly in cheese and canned vegetables.

Free amino acids, generated by hydrolytic processes, contribute to the flavour of protein-rich foods such as meat, fish or cheese. Taste quality is influenced by the molecular configuration. D-amino acids are mostly sweet while the L-amino acids are bitter. The taste intensity of amino acids is dependent on the hydrophobicity of the side chain in the amino acid. Thus L-tryptophan and L-tyrosine are bitter with low threshold values (concentration required to recognize taste) in the range of 4–6 mmole/l while D-tryptophan is the sweetest amino acid with a threshold value of 0.2–0.4 mmole/l. L-glutamic acid has a meat broth flavour at higher concentrations and at low concentrations is a flavour enhancer. L-methionine has a sulphur like flavour.

5.6.1 Modification of Functional Properties of Proteins

The multipurpose functional properties of proteins in foods may be modified to suit the requirements of enhanced nutritional value and

flavour, taste and textural characteristics in processed foods. Modification of proteins by chemical and enzymatic methods is gaining importance as the modified proteins exhibit better features compared to native proteins. Modification of proteins provides a potential technological means to tap proteins from new raw materials or non-edible sources to alleviate the problem of protein malnutrition. Protein modification for food processing aims at (i) blocking undesirable reactions such as Maillard reaction, (ii) improving physical properties such as whippability, foam stability, solubility and texture and (iii) improving the nutritional value by increasing digestibility of the food, fortifying food with essential nutrients and inactivating toxic or undesirable constituents.

Chemical modification of proteins in foods include acylation of the free amino group, reductive alkylation of the amino group with formaldehyde to give dimethylamino moiety, hydrolysis of the amide to yield carboxylic acid, esterification of the carboxyl and alcoholic groups, oxidation of thiol to disulphide bridge and reduction of the disulphide bridge to thiol. Thus treatment of leaf proteins and yeast proteins with succinic anhydride (succinylation) improves the solubility, flavour and emulsifying properties. Modifcation of protein by reductive methylation of amino groups with formaldehyde/$NaBH_4$ retards Maillard reaction. Wheat gluten modified by reduction of the disulphide bonds to thiol groups and subsequent reoxidation of these groups under different conditions affect the rheological properties. Reoxidation of the thiol groups in dilute suspension in the presence of urea gives a weak, soluble, adhesive product (gluten A), while reoxidation of a concentrated suspension in the presence of a higher concentration of urea gives an insoluble, stiff, cohesive gluten C. Viscosity data have indicated that the disulphide bonds of gluten A are mostly intramolecular while those in gluten C are intermolecular.

Enzymatic modification of proteins are also possible. Plastein reaction enables the formation of larger polypeptides by joining smaller peptide fragments in protein hydrolysates. Plastein reaction using papain has been found to be useful for incorporation of selected amino acids so as to enrich proteins. Similarly soya proteins enriched with sulphur containing amino acids from partially hydrolyzed wool keratin have been prepared. Plastein reaction can be used to improve the solubility of proteins and to remove undesirable amino acids (e.g. to produce phenyl alanine free diet recommended for patients with certain metabolic defects).

Texturized proteins are produced to broaden the use of plant proteins (which are globular in structure) in processed foods with meat like texture in which proteins have essentially fibrous structure. Texturized proteins are used as meat extenders and meat analogues with desirable lumpy structure. Soya proteins, casein, wheat gluten, oilseed meals (of groundnut, sesame, sunflower, safflower, cottonseed or rapeseed), zein (corn protein), whey, blood plasma, offal and yeast are suitable for texturization. Soya flour is mainly used in the production of texturized vegetable proteins. The molecular weights of proteins should be in the range of 10–50 kDa.

Texturization involves unfolding the globular protein by breaking the intramolecular binding forces and stabilizing the protein chains through interchain interactions. In practice, texturization is carried out by spin process or extrusion process.

In the spin process the soya protein isolate is extracted from soya flour in alkali (20% solution, pH 11) to form a viscous solution which is then forced through tiny holes of a spinneret to give fine threads of spun material into a coagulating bath containing 10% sodium chloride and an acid (citric, acetic, phosphoric, lactic or hydrochloric acid) at pH 2–3. The protein fibres are stretched and twisted into bundles of fibres having a meat like texture. The molecular interactions are enhanced during stretching of the fibre increasing its mechanical strength. The bundles are pressed between rollers to remove the coagulating solvent and neutralized to pH 5–6 in a bath of sodium bicarbonate and sodium chloride. The fibre bundles may be combined to form larger aggregates. The aggregates are treated with additives such as egg protein, modified starch, aroma compounds and lipids. The aggregates are dried, sliced and marketed as spun vegetable protein (SVP).

In the extrusion process, the soya flour is moistened slightly with solution containing additives of sodium chloride, aroma compounds and buffers made into a dough. The dough is heated to 120–180°C under a pressure of 30–40 bar and extruded through a nozzle into atmospheric or reduced pressure. Hydration, partial unfolding, stretching and rearrangement of protein occurs under these conditions and the extruded mass expands to achieve the desired texture. It contains protein strands with fibre like particles rather than well defined fibres. The mass is cut into pieces and dried. It is marketed as texturized vegetable protein (TVP) in the form of chunks or granules for use as meat extender.

5.7 PHYSICAL AND CHEMICAL PROPERTIES OF PROTEINS

The most important physical property of proteins of relevance to food science is the ionic property which influences solubility of proteins. At pH values close to the isoelectric point of a protein, its solubility in water decreases due to hydrophobic interactions and agglomeration. Presence of salts at concentration up to about 0.2 M increases protein solubility due to salting-in but at higher concentrations of salt protein solubility decreases due to salting-out phenomenon.

The important chemical properties of proteins include denaturation, gelation, Maillard reaction and hydrolysis. Extremes of pH and temperature as well as addition of certain chemicals such as urea or guanidine at high concentrations or synthetic detergents denature proteins. Egg white (88% water and 12% protein) on heating becomes a solid gel due to denaturation of proteins. Similarly when liver is cooked

the proteins contained in the liver cells are denatured and if cooking is prolonged an unattractive hard texture results due to the rigid network of polypeptide chains. Denaturation of a protein makes it more amenable for hydrolysis by proteolytic enzymes, decreases its solubility while intrinsic viscosity increases. Denaturation of an enzyme results in loss of its enzymic activity.

Proteins with a high degree of asymmetric structure form gels by immobilizing water in a physically entrapped form. Two well-known examples of protenaceous gels in foods are those of casein and gelatin. Casein of milk is coagulated and precipitated as a gel by the action of the proteolytic enzyme rennin. Gelatin is formed during cooking of meat due to partial hydrolysis and dissolution of collagen.

Proteins undergo hydrolysis by the action of acids, alkali and proteolytic enzymes. The peptide bond is hydrolyzed releasing the amino acids.

Food processing operations also decrease the biological value of proteins by destroying essential amino acids, conversion of essential amino acids into derivatives which are not metabolizable and decreasing the digestibility of proteins as a result of intra- or inter-chain cross-linking. In addition, formation of toxic degradation products may also occur. Lysine content of the food is decreased due to Maillard reaction. Similarly loss of cystine, serine, threonine and arginine occurs at higher pH values.

5.8 PROTEINS OF SELECTED FOODSTUFFS

5.8.1 Milk

Milk of cow is an important protein source for humans and particularly for children. It is an aqueous solution of proteins (about 27 g/l), lactose, minerals, certain vitamins, emulsified fat globules and colloidal dispersion of casein micelles consisting of protein with phosphate, citrate and calcium. Skim milk or skimmed milk is milk from which fat has been removed. Whey or serum of milk is obtained by removing casein from skim milk by precipitation at a pH of 4.6. Whey obtained as a by-product in cheese making has a different composition as some of the casein is solubilized and much of the lactose is converted to lactic acid by the action of bacteria. The important proteins in whole milk are caseins (about 80% of total protein content) and whey proteins (20%).

Caseins are a heterogeneous group of associated phosphoproteins which are heat stable (well beyond 100°C) and include α_{s1}-casein (32% of total protein content), α_{s2}-casein (8%), β-casein (32%) and κ-casein (8%). Because of their high phosphate content they bind calcium ions strongly. Milk contains about 0.1% of calcium ions and the majority of caseins exists as large particles called casein micelles. The micelles are roughly spherical with an average diameter of 100 nm, containing about 5000

casein monomers and are highly hydrated. Together with fat globules these micelles scatter light considerably giving the white 'milky' appearance of milk. The micelles consist of sub-micelles held together by colloidal calcium phosphate, hydrophobic interactions and hydrogen bonds. The α- and β-caseins form hydrophobic core of the sub-micelle while κ-casein is located on the surface of the sub-micelle forming a protective layer which stabilizes the micelles against calcium induced flocculation and precipitation. The N-terminal segment (two-thirds of the κ-casein) is hydrophobic interacting with the other caseins while the C-terminal (one-third segment) is hydrophilic projecting out into the aqueous environment. These hydrophilic projections give the micelles a 'hairy' appearance promoting the solubility of the micelles.

Whey proteins include β-lactoglobulin (12%), β-lactalbumin (4%), immunoglobulins (3%) and serum albumin (1%).

Milk contains a variety of enzymes but only a few are of importance to food science. These include alkaline phosphatase, lipase, protease and xanthin oxidase. The presence of residual alkaline phosphatase activity is used to indicate the ineffectiveness of pasteurization of milk.

Milk and its products, which include cheese, yoghurt, whole milk and skimmed milk powders caseinates and whey powders, are used extensively in the confectionery and baking industry. Dried acid precipitated caseins are used for fortification of cereals, meat products and bread. Casein precipitated by rennin is used for the manufacture of cheese. Dissolving the acid precipitated casein in calcium or sodium hydroxide respectively at pH 6.7 and spray drying produces calcium or sodium caseinates. Caseinates find use because of their high solubility at neutral pH values. Calcium caseinate forms stable colloidal dispersions and sodium caseinate forms gels at concentrations above 17%. In addition, caseinates have useful functional properties such as absorption of water, fat binding, whipping and foam stability. Hence they find extensive use in margarine, coffee whiteners, whipped confectionery toppings and in semi-liquid cheeses. Casein micelles retain their structure and are resistant to heat and shear. They account for the water-holding capacity in gel and paste-like products such as yoghurt.

Milk whey powder has high nutritional value as the dried powder contains up to 70% of lactose. The denatured whole whey proteins obtained by heating to 90°C, known as lactalbumin, is used as a nutritional supplement in soups, cereals and snack foods. The native whey proteins are soluble over a wide range of pH and have desirable functional properties. They form foams and gels and can act as emulsifiers. In the manufacture of Fetta and Camembert cheeses, the whey proteins are retained in the cheese curd by ultrafiltration and hence these have enhanced nutritional value.

5.8.2 Meat

The edible muscle and flesh of cattle, sheep and pig are described as red meat because of the colour of beef, lamb or pork in contrast to the light or dark colour of poultry meat. A joint of lean meat from the butcher's shop devoid of all external fat contains about 18 to 20% protein. It consists of a number of muscles, each having its own independent attachment to skeleton, its own blood supply and nerves. A layer of connective tissues consisting of almost entirely of the protein collagen covers each muscle. Blood vessels, adipose (fatty) tissues and nerves also are embedded in the connective tissue. The composition of the muscle tissue is different from that of the organs, such as kidneys and liver, which are referred to collectively as offal. Structurally the muscle tissue is composed of fibres, each of which is made up of cells. Each fibre, about 100 μm in diameter and up to 30 cm length, consists of bundles of about 1000 myofibrils. The fundamental unit of meat is the myofibrillar sarcomere (Gk. 'sarco' = flesh) which contains thick filaments of protein myosin and thin filaments of actin, with smaller amounts of two other proteins—tropomyosin and troponin. The elastic fibrous protein, myosin can exist in stretched and unstretched forms and together with actin is involved in muscle contraction and relaxation. The energy required for muscle contraction and relaxation is provided by adenosine triphosphate (ATP) synthesized in mitochondria lying in the fibrils.

Immediately after the death of the animal there is still a demand for ATP in the muscle. However, as there is no further supply of oxygen from the blood stream, the supply of ATP is maintained for sometime by anaerobic glycolysis. Glycolysis comes to a stop due to lowering of the pH to 5.0–5.5 on the accumulation of lactic acid as the end product of glycolysis. When the ATP level in the muscle falls, myosin and actin combine irreversibly to form rigid chains of actomyosin and the muscle goes rigid, which is known as the state of rigor mortis. In the living muscle, ATP normally blocks this irreversible interaction between myosin and actin. It is impossible to get meat cooked before it goes into rigor and meat cooked during rigor is exceedingly tough.

The rate of postmortem glycolysis and the initiation of rigor mortis vary between different animals and even between muscles within the same carcass. Stress on the animal before its death, premature and/or incorrect cutting of the muscles and rapid cooling all influence the rates of postmortem changes in the meat. Decrease in pH due to glycolysis increases the stiffness of meat. Excessive shortening of the muscle fibres can be induced by cutting through the muscle shortly after death or cooling rapidly to temperatures less than 11°C. Cold shortening occurs in beef, sheep and turkey meat.

To avoid the problem due to rigor mortis, conditioning of carcass by hanging is adopted in the butcher's shop. Toughness can be avoided by holding the carcasses for a brief period at 35–40°C and ensuring that rigor

mortis has taken place before chilling the carcass below 15°C. The development of rigor can be brought forward by electrical stimulation. Aging by storage of meat increases tenderness. During aging, the rigor muscle relaxes from its rigid form (the resolution of rigor) and the fibrous structure starts to break down after about 10 days at 5°C. During conditioning or aging, the proteins of meat also undergo denaturation which is followed by breakdown of proteins into peptides and amino acids and an increase in tenderness.

Muscle tissue contains 55–80% of water. The tenderness and juiciness of meat is also influenced by the effect of intermolecular interactions between different meat proteins on the swelling of meat and its water holding capacity. Protein denaturation on cooking releases some of this water to give meat its desirable moist appearance. Water also comes out partly when the joint of meat is cut in the shop giving meat an attractive moist appearance.

The tenderness of meat is also finally determined by cooking. Meat cooked to an internal temperature of 60°C is described as rare, whereas at 80°C the meat is well cooked and changes to a more granular texture, although the softer fibrillar structure is a highly desirable characteristic.

The myofibrillar proteins also influence the binding properties of meat. A high binding strength is desirable as it enhances integrity and thus the quality of meat derived products.

The texture of meat is due to collagen which is present in the connective tissue surrounding the muscle fibres. The walls of muscle fibres contain mainly elastin, another fibrous protein. Collagen and elastin are insoluble and tough and hence difficult to digest. However, when meat is cooked in the presence of moisture, collagen is converted to soluble gelatin and digestion becomes easier. Meat of young animals is tender than that of older ones because of less crosslinking of the connective tissues in the young animals. Similarly the meat of inactive animals is tender and easy to digest than that of active animals.

The connective tissue also contains invisible fat which makes digestion of muscle more difficult as it coats the muscle fibres with a thin oily film. A much larger amount of fat is stored in fat depots of the animal mainly under the skin and around the internal organs and therefore, fat is not a part of lean meat. Beef, lamb and pork contain more of saturated fats while chicken contains more of polyunsaturated fatty acids (PUFA).

The red colour of meat is due to the interaction of myoglobin with oxygen to form oxymyoglobin. Higher concentrations of oxygen give meat a desirable bright red colour whereas meat at anaerobic conditions is purple in colour. Cut meat on aging at room temperature is brown in colour due to the oxidation of myoglobin to metmyoglobin. When meat is cooked the myoglobin is denatured along with most other proteins. The iron is oxidized to ferric iron by oxygen present giving the meat a brown colour. In the case of bacon (pig meat) low levels of nitrites are included with salt for curing. Due to enzyme action the nitrite gives NO which

combines with myoglobin and forms a red pigment. This gives uncooked bacon and ham their characteristic red colour.

Meat contains minerals and vitamins. It is a good source of iron, zinc and vitamins of B group, particularly niacin. The attractive flavour of meat is largely due to a variety of substances known as meat extractives that are soluble in water. These include compounds produced during muscular activity such as lactic acid, those derived from ATP, protein metabolism, amino acids such as glutamic acid and urea. Meat extractives aid digestion by stimulating the secretion of saliva and gastric juice.

Salting has always been an important method of meat preservation. Rubbing the carcass with dry salt, steeping in brine or modern injection methods all aim to raise the salt concentration of the tissues to inhibit microbial growth.

5.8.3 Fish

The flesh of fish is composed of bundles of short fibres called myomeres, held together by thin layers of connective tissue composed of collagen. Cooked fish is relatively easily digested as the tough elastin is absent in fish. Two classes of fish are consumed as food. The flesh of white fish such as cod, haddock, and whiting contain very little fat (less than 2%) and fat is concentrated mostly in liver which is removed for use as a source of vitamins, particularly fat soluble vitamins, i.e. A and D, e.g. cod liver oil. The second class of fish called fat fish, such as herring, trout and salmon, contains 10–25% fat. Fish is a rich source of proteins, the quality and amount of protein being similar to that in lean meat. Fish deteriorates rapidly because of the absence of bulky connective tissue and hence fish is frozen at sea immediately after being caught.

5.8.4 Eggs

Hen's egg is a complete food for a growing chick embryo though not complete for humans. The structure of egg plays an important role in preventing microbial attack on the egg. The egg consists of three main parts, the shell, the egg white and the yolk. The outer shell is a hard porous protective layer made of calcium carbonate. The viscous colourless liquid inside the shell is egg white. The egg white gel is a poor medium for the growth of microorganisms and is held firmly in position by its attachment to the inner shell membranes of the egg. It accounts for about 60% of the total weight of the egg. It is a dilute aqueous solution of proteins (12%), dissolved salts and riboflavin. The main protein is ovalbumin, and another protein called mucin is responsible for the viscosity of the liquid. The third component of egg, the egg yolk, is suspended in the egg white as a yellow or orange oil-in-emulsion stabilized

by lecithin and held centrally by the chalazae, which are anchored to the thick white gel. It is a rich medium for the growth of microorganisms. The egg yolk is a more concentrated source of nutrients containing about 33% fat, 15% protein and 50% water, in addition to minerals and vitamins. The fat of egg is concentrated in the egg yolk and it is a rich source of cholesterol.

The egg white proteins known as albumen (about ten of them have been identified) include ovalbumin, ovomucoids, lysozyme, globulins and avidin. Avidin, which is present in small proportion in the egg, has a strong affinity for the vitamin biotin in human diet and makes it unavailable to the body. However, avidin is inactivated during cooking. Egg yolk proteins are the phosphoproteins called lipovitellin and lipvitellenin. They also contain the lipid lecithin.

The nutritional value of egg is due to the good amount of proteins, iron, phosphorus, fat, Vitamin A and calcium. It is also a source of Vitamin D, riboflavin, thiamine and biotin. Ovalbumin, the major protein in egg white is easily denatured at 60°C to form a unique soft gel, acceptable as a solid food. However, the delicate gelatinous structure of heat-denatured egg is disrupted by freezing. Thus it is not possible to store boiled eggs in the frozen state for use in convenience foods.

Apart from their nutritional value, eggs also find many uses in food preparations mainly due to the versatility of egg proteins. Egg white is used in the formation of foams brought about by beating the egg white so that air is entrapped during cake making. Foam formation is promoted by acid as the pH is brought closer to the isoelectric point of ovalbumin. Ovomucins and lysozyme together with ovalbumin are effective in protecting foam volume and in the formation of stable foams. Whole egg, when beaten, also entraps air and is used to lighten the texture of baked goods such as sponge cakes.

Eggs are used as thickening agents. Coagulation of egg proteins is used to thicken sauces, custards and soups. Eggs are used as binding agents as the coagulation of egg proteins gives cohesion to a mixture of dry ingredients. Eggs along with bread crumbs are used as coatings for fish before it is fried. The coagulation of egg during frying forms a strong coating which holds the fish structure together. Beaten egg is used as protective covering for fried food thereby preventing penetration of food by oil. Eggs are good emulsifiers and assist in the formation of a stable emulsion from a creamed mixture of fat and sugar during cake making. Egg yolk is used to emulsify oil and vinegar in making mayonnaise. Whole egg is used to improve texture and flavour of baked goods such as cakes. The yolk contributes a rich yellow colour and the fat exerts a shortening effect in biscuits and cakes. When eggs and milk are beaten together and the mixture is heated, we get egg custard. The rate and duration of heating is adjusted to produce either a viscous stirred custard, produced by coagulation or a solid baked form, formed by complete coagulation.

Slow and even coagulation is essential in making the products if an acceptable texture is to be achieved. Rapid heating causes curdling.

5.8.5 Cereal Proteins

Of the different cereals such as wheat, rye, barley, oats, sorghum, maize and rice, only wheat and rye can be used for making bread with a leavened, open, crumb structure which is mainly due to the unique type of proteins in these. The grains (seeds) of cereals have similar structure consisting of three main components: (i) the embryo or germ of new plant, (ii) the endosperm which is the store of nutrients for the germinating plant and (iii) the protective layers of seed coat called the bran. The endosperm constitutes to about 80% of the bulk of grain and is the source of white flour used for making bread. It consists of cells packed with starch granules lying in a matrix of proteins, which constitute about 7–15% in the flour. Two types of proteins are available in the endosperm: (i) cytoplasmic proteins, mostly enzymes, which are soluble in water or dilute salt solutions and (ii) storage proteins of about 85% of the total protein content collectively called as gluten (or prolamins). The gluten proteins are insoluble in aqueous media and are responsible for dough formation on mixing the flour with water. Gluten can be extracted from wheat flour by washing off the soluble material from the dough. The remaining mass is tough, viscoelastic and sticky, consisting of about one-third gluten proteins and the rest, water.

Flours from different varieties of wheat vary in their protein content. Flours that are good for bread making (they give a good loaf volume) are obtained from spring-sown wheat varieties in North American continent. The protein content is quite high in this type of flours (12–14%). Wheat of this variety is described as 'hard' by the miller because the endosperm is brittle and disintegrates readily during milling, while the baker refers to good bread making flour as 'strong' indicating the greater elasticity and greater resistance to stretching characteristics of dough. Weak flours obtained from 'soft' winter wheats of Europe (protein content less than 10%) are used in biscuits and pastry.

The gluten proteins consist of soluble (in 70% ethanol) gliadins (about one-third of gluten) and insoluble glutenins. A single variety of wheat may have more than 40 different gliadin proteins in the molecular weight range of 30,000–40,000 containing a large proportion of glutamine (35–45 mol%) and proline (15 mol%). The glutenins have molecular weights in the range of 40,000 to 20 million, with most of them having molecular weight about 2 million.

The gluten proteins have a characteristic structure known as β-bend (the only other protein known to have an abundance of β-bends is the mammalian connective tissue protein, elastin). The presence of large number of β-bends arranged in sequence into a β-spiral seems to be

responsible to the elastic component of the dough's viscoelasticity. Addition of water to the flour leads to the formation of dough as the gluten proteins are hydrated. The damaged starch granules also absorb some water. The properties of the dough depend on the glutenin, which is capable of forming a three-dimensional network through different types of bonding between subunits of the protein. The high proportion of glutenins in wheat flour (hard flour) results in stronger doughs which require more mixing and give loaves of greater volume. The gliadin proteins seem to modify the viscoelastic properties of the dough. For biscuits, and pastries strong flours are unsuitable because they form hard rather than crisp biscuits and shrink erratically after moulding.

The properties of the wheat flour used for bread making can be improved by prolonged storage. During storage, auto-oxidation of polyunsaturated acids of flour lipids results in the formation of hydroperoxides which are powerful oxidizing agents. The hydroperoxides bleach the carotenoid pigments in the flour. The bleached flour gives an attractive whiter crumb. Aging also improves the bread loaf volume and the crumb becomes finer and softer. Aging of the flour beyond 12 months seems to be detrimental to retaining the properties. The aging process may be accelerated by the addition of oxidants such as chlorine dioxide or benzoyl peroxide (called bleaching agents) and flour improving agents such as ammonium or potassium persulphates, potassium bromate or iodate. In the Chorleywood Bread Process of UK, ascorbic acid is used as flour improving agent.

A satisfactory dough is capable of accommodating a large quantity of gas and retain its protein sets during baking. The process of dough development involves kneading in addition to mechanical mixing of the ingredients. The carbon dioxide bubbles liberated due to yeast fermentation also take part in the dough development. It is believed that the glutenin molecules are stretched out into linear chains which interact to form elastic sheets under the gas bubbles. The making and breaking of H-bonds between the gluten proteins and oxidation of free sulphydryl groups in the glutenin subunits permit the optimum expansion without the dough becoming excessively weak.

Rice contains alkali soluble glutelin to the extent of 80% of the total protein and a small percentage of about 5% of nutritionally poor quality prolamine. The lysine content of rice protein is relatively high (about 4%).

5.8.6 Seed Proteins

Most of the plant seeds contain more than 15% protein content as storage proteins which exist as aggregates. From the view point of human food, only the legume seeds of soya, peanut and dried peas and beans and oil seeds of sunflower, sesame and cotton are useful. Legume seeds contain between 18% and 42% protein, but are deficient in essential sulphur

containing amino acids. Because of the high protein content (about 40%) and absence of toxic compounds soybean seeds find extensive use in foods. The gelation property of soya proteins is useful in the manufacture of canned meat, soup and dairy products. A dispersion of about 2% w/v soya proteins on heating, forms a viscous progel which solidifies on cooling. Excessive heating, however, degrades the progel proteins and progel is converted into metasol which does not set on cooling. Gel formation is also induced by calcium ions as in 'tofu gel' or by alkali or alkali-alcohol solutions. The most important component for gelation has been identified as the 11S globulin.

Cottonseed contains globulins of which the 2S fraction is composed of low molecular weight proteins relatively rich in lysine and sulphur containing amino acids. The 7S and 12S fractions contain high molecular weight proteins and are present in protein bodies. Peanut proteins contain arachin, the major globulin to the extent of about 65% and another globulin is conarachin present in the cytoplasm.

5.9 ENZYMES

Enzymes are complex globular proteins and function as catalysts in biological systems. The catalytic efficiency of enzymes is very high with a turn over number (number of substrate molecules transformed into product molecules by one enzyme molecule within one minute) of several thousand. Unlike synthetic catalysts enzymes show a high substrate specificity as well as product specificity, function at lower temperatures (even at subzero temperature) and at atmospheric pressure. However, temperatures greater than 70–100°C, extremes of pH and the presence of denaturing agents destroy the catalytic activity of enzymes.

Enzymes have been classified into six main classes based on the type of reactions catalyzed by them. Each enzyme is assigned a number of four numerals, by Enzyme Commission. Each number is separated by a dot and arranged as follows. The first number is the main class to which the enzyme belongs and the six main classes are: (i) oxidoreductases (catalyze oxidation-reduction reactions), (ii) transferases, (iii) hydrolases (catalyze hydrolysis reactions), (iv) lyases, (v) isomerases (catalyze isomerization reactions) and (vi) ligases. The second number designates the subclass to which the enzyme belongs and the third number precisely defines the reaction catalyzed by the enzyme and the fourth number gives the serial number of the enzyme. Thus, α-amylase is given the code number EC 3.2.1.1 while glucose oxidase is given EC 1.1.3.4.

5.9.1 Distribution of Enzymes in Biological Systems

Most of the enzymes found in a typical cell exist bound to the subcellular organelles and only a few are found in cytoplasmic solution. Many of the

enzymes are isoenzymes or isozymes, i.e. they are found in the same species at different locations and catalyze the same reaction, but differ in their primary, secondary, tertiary and/or quaternary structure and hence in their catalytic efficiencies. The various types of enzymes present within animal and plant cells may be identified on the basis of their location and associated activities.

1. Enzymes found in nucleus are primarily associated with the biosynthesis of nucleic acids and nucleotides. In addition, enzymes involved in glycolysis, electron transport, hexose monophosphate shunt and citric acid cycle are also found in the nucleus.
2. Enzymes in mitochondria are involved in the production of ATP by oxidation of metabolites in tricarboxylic acid cycle, formation of acetyl coenzyme A from fatty acids and pyruvate and other anabolic processes. These enzymes are of importance in food processing/preservation. For example, enzymic activity persists during the ripening of post harvest fruits involving oxidative phosphorylation to supply energy for synthetic reactions. Similarly, in post slaughter meats, continued utilization of tissue oxygen for mitochondrial oxidation reactions inhibits anaerobic glycolysis. In addition, cytochromes and hydrolases present in the mitochondria may contribute to degradative reactions of lipids and tissue components in post slaughter meats. Coenzymes such as flavin adenine dinucleotide (contain riboflavin) may contribute the photodeterioration of tissues.
3. Over 100 enzymes are found in the endoplasmic reticulum and ribosomes of cells. Some of the enzymes are involved in post harvest ripening of fruits. Cytochromes and lipoproteins of this organalle possibly catalyze lipid oxidation giving rise to off-flavours in foods.
4. Enzymes in chloroplasts are involved in photosynthesis and also in the synthesis of pigments chlorophyll and carotenoids. The chloroplast is also a site for polyphenoloxidase and chlorophyllase activities.
5. Enzymes present in the plasma membrane include magnesium and calcium dependent ATPases, and also sodium and potassium dependent phosphatases which are concerned with maintenance of water and ion balance of cells. Operations such as freezing and thawing and osmotic shock due to concentration change can disrupt the cell membrane and release the contents of cell along with these enzymes.
6. Enzymes found in Golgi bodies include nucleoside diphosphatase, thiamine pyrophosphatase, acid and alkaline phosphatases, esterase and β-glucuronidase.

7. Enzymes in microbodies such as peroxisomes include catalase, D-amino acid oxidase, urate oxidase and L-α-hydroxy acid oxidase and glyoxysomes (plant microbodies) contain enzymes of glyoxylate cycle.
8. Lysosomes contain acid hydrolases which hydrolyze nucleic acids, lipids, proteins, polysaccharides, phosphates, glycoproteins and glycolipids. All the lysosome enzymes are of importance in food processing. For example, cathepsins are involved in the aging of meat and in autolysis of animal tissues. Lysosomal enzymes are resistant to lysosomal proteases. Freezing and thawing operations and osmotic shock disrupt the membrane structure and release the lysosomal enzymes.
9. Cytosol enzymes are involved in the synthesis of proteins, nucleic acids and fatty acids and in glycolysis. Glycolytic enzymes continue their activity in postmortem metabolism in meat.
10. Amyloplasts are starch granules which contain enzymes involved in the synthesis of starch from sugars.

5.10 ENDOGENOUS ENZYMES IN FOODS

Milk contains at least 20 enzymes distributed in the soluble phase, the micellar phase and the fat globules. The lysozyme content of egg is about 3.5% and in addition, about 12 enzymes have been detected in egg white and another 7 in egg yolk. About 55 enzymes have been detected in wheat kernel. Apart from these native enzymes, foods also contain other enzymes and chemicals of microbial origin. The endogenous enzymes of foods are of importance in processing and preservation because they cause changes in colour, texture, flavour and nutritive value of foods.

The endogenous enzymes and their activities are discussed in the following subsections.

5.10.1 Phenolase

Enzymic browning or phenolase browning is enzyme catalyzed browning in plant tissues responsible for changes in the colour and flavour of fruits and vegetables. Exposure of the cut surface of fruits such as apples and bananas and also of potatoes to air, particularly during unit operations of cutting, slicing, dicing or peeling results in rapid browning due to the enzymic oxidation of phenols to orthoquinones which undergo polymerization to yield brown pigments called melanins. The enzyme is commonly known as phenolase, catecholase, tyrosinase oxidase (EC 1.10.3.1). The prosthetic group of phenolase enzymes is copper. The

enzyme catalyzes two types of reactions: (i) hydroxylation (phenol hydroxylase or cresolase activity) in which monophenol or phenolic compounds such as caffeic acid, protocatechuic acid and chlorogenic acid present in the fruits is hydroxylated in the ortho position (a rate determining step) and (ii) oxidation (polyphenol oxidase or catecholase activity) in which the o-hydroxylated phenol is oxidized to orthoquinone. If tyrosine is the substrate as in the case of animal tissues, it is hydroxylated first to dopa which is then oxidized to dopaquinone. The orthoquinone undergoes non-enzymic oxidation and polymerization to yield melanins. The melanins further interact with proteins to form complexes. The enzyme is active between pH 5 and 7 and at lower pH of about 3 it is irreversibly inactivated.

Phenolase activity in fruits and vegetables is undesirable while it is desirable in coffee, tea, cocoa and dried fruits. In the manufacture of pasteurized apple puree, advantage is taken of enzymic browning to give an appealing pale yellow product. During fruit processing, enzymic browning is inhibited by a variety of methods. Phenolase activity is suppressed either by blanching or by addition of sulphur dioxide or sulphites, or citric, malic or phosphoric acids to lower the pH to about 3. Immersion of plant foods in brine or syrups and evacuation of atmospheric gases surrounding the food inhibits phenolase due to lack of oxygen. Addition of antiphenolase enzymes as o-methyl transferase which methylates orthophenols in the presence of methyl donors and protocatechuate oxygenase which modifies diphenols (initial product of phenolase activity) to dibasic acids incapable of undergoing oxidation and polymerization to melanins, inhibit enzymic browning. However, these antiphenolase enzymes require alkaline conditions and hence not useful in fruit processing.

5.10.2 Pectic Enzymes

Pectin is polygalactouronic acid with about two-thirds of carboxyl groups existing as methyl ester. Microbial spoilage of fruits and vegetables is primarily due to the action of pectic enzymes of pathogenic organisms which facilitate the invasion of tissues. Native or endogenous pectic enzymes present in foods are also involved in the ripening as well as in the spoilage of plant tissues. Pectic enzymes such as pectin methyl esterase (PE), polymethylgalactouranase (PMG), polygalactouronase (PG) and pectic acid transeliminase (PATE) are used for clarifying fuit juices, in controlling the extent of ripening of fruits, maintenance of viscosity in fruit products and in achieving textural changes. In jam manufacture the pectin esterase activity is inhibited by high concentration of sugar and thus the liquefaction of pectin gel is prevented.

5.10.3 Amylases

Two amylases, α- and β-amylases are well known for the hydrolysis of starch during ripening of fruits, processing of potatoes, production of corn syrup and corn sugar and in bread making. α-Amylase is a calcium dependent endoenzyme and hydrolyzes α-1,4-glycosidic linkages at random and hence used as 'liquefying enzyme' for reducing the viscosity of starch solutions. Starch hydrolysis by the use of this enzyme also increases reducing sugar content of processed foods. β-Amylase is an exoenzyme, occurring in vegetable tissues, which removes the maltose units from the non-reducing end of the starch chain by hydrolyzing alternate glycosidic linkages. Since maltose increases the sweetness of the starch solution, β-amylase is known as 'saccharifying enzyme'. β-Amylase activity stops at two or three glucose units from the branching point of amylopectins resulting in incomplete hydrolysis of starch to give 'limit dextrins'.

5.10.4 Cathepsins

These are intracellular acidic proteases located in the lysosomal fraction of animal tissues and are distinct from proteases such as trypsin and chymotrypsin secreted by cells. Five cathepsins designated as cathepsin A, B, C, D and E are known and may be involved in the tenderization of meat during aging.

5.10.5 Myosin ATPase

The enzyme plays an important role in muscular contraction in living systems and also in the rigidization of muscle due to extreme contraction and shortening during postmortem rigor mortis.

5.10.6 Flavour Enzymes

Enzymes participate either directly or indirectly in the formation of flavour compounds in fruits and vegetables. Four types of flavour forming mechanisms have been identified in foods and three of these mechanisms require enzymes directly or indirectly.

1. *Biosynthesis.* Flavour compounds such as terpenoids, esters, alcohols and acids are formed enzymically during the ripening of fruits such as apples, bananas, pears and citrus fruits and in pepper (these plant foods do not have any flavour during initial stages of growth up to harvest).
2. *Direct enzymic reactions.* A specific flavour compound is formed by the direct action of an enzyme on a specific precursor. For

example, the flavour of onion is due to the action of endogenous S-alkyl-L-cysteine sulphoxide lyase on the trans-(+)- S-propenyl-L-cysteinesulphoxide and other S-alkyl substituted L-cysteines naturally present in the onion to give volatile compounds with characteristic pungent aroma and taste of onions. A part of flavour precursors in onions is present in the form of γ-L-glutamyl peptides which can be released by the addition of exogenous γ-L-glutamyl transpeptidases so that full potential of onion flavour may be achieved. Another example of direct enzymic action on flavour precursors is that of cabbage flavour in cabbage, mustard and horseradish. Thioglycosidases act on thioglycoside precursors such as sinigrin or sinalbin to give alkylthiocyanates in cabbage flavour.

3. *Oxidative or indirect enzymic reaction.* The mechanism involves formation of the flavour components by oxidation of flavour precursors, the oxidizing agents being formed enzymically. In the black tea aroma, catechol oxidase oxidizes flavonols and the oxidized flavonols in turn oxidize amino acids, unsaturated fatty acids and carotenes present in tea to yield flavour constituents. Lipoxygenase present in plants, forms hydroperoxides which are capable of oxidizing susceptible food constituents to give flavour compounds.
4. *Pyrolysis.* Pyrolytic mechanism does not involve any enzyme. Flavours are produced by the action of heat on precursors during heat processing or baking. Examples include the flavours of coffee, chocolate and bread.

The use of enzymes to regenerate flavour in processed foods is known as flavourase concept. During heat processing native flavourase enzymes are inactivated and most of the flavours volatilize off. Exogenous enzymes from a homologous source when added to processed foods can act on flavour precursors which are non-volatile to regenerate flavours. Examples of processed foods in which flavour has been partially or fully regenerated include cabbage, potato, tomato, peas, carrots, bananas, grapefruit juice, orange juice, mustard, horseradish and onions.

5.10.7 Lipolytic Enzymes

Lipases hydrolyse ester linkages of emulsified glycerides at oil-water interface. These enzymes are present in cells, seeds, organs, tissues and fluids of plants and animals and in microorganisms. Generally lipases hydrolyze triglycerides in a stepwise manner to 1,2- or 2,3-diglycerides and then to 2-monoglycerides. Lipases show four types of specificities—glyceride, position, fatty acid and stereo specificities. Action of endogenous

lipases on triglycerides can lead to off flavours in foods. The bovine milk contains the lipase which is normally inactive but homogenization and cooling during milk processing activate the enzyme. This induced lipolysis results in the formation of short and medium chain length volatile fatty acids and spoilage of milk.

5.10.8 Lipoxygenase

Lipoxygenase or lipoxidase is present in legumes such as soybeans, mung beans and green beans, in cereals such as wheat, corn, oats and barley and also in fruits. The enzyme catalyzes stereospecific oxygenation of polyunsaturated fatty acids, esters and glycerides containing the cis,cis-1,4-pentadiene group to give optically active hydroperoxide containing conjugated cis,trans double bond system. Thus linoleic and linolenic acids present in most vegetables are subject to peroxidation by lipoxygenase resulting in the formation of off-flavours. The enzyme also bleaches pigments such as chlorophyll, carotene and xanthophyll. Because of the formation of free radicals during the action of this enzyme, vitamins are destroyed. Inactivating the enzyme by blanching vegetables before freezing or drying may prevent these undesirable effects. However, the enzyme is used beneficially in bleaching and improving the dough characteristics of wheat flour used for bread making by way of addition of soya flour. The addition of soya flour decreases the content of polyunsaturated fatty acids and aids in the release of bound lipids. The improvement of dough occurs due to lipid oxidation during mixing in the presence of oxygen and also possibly due to the oxidation of sulphydryl groups of proteins to give disulphide bridges resulting in crosslinked proteins.

5.10.9 Peroxidase

Milk and horseradish are best known sources of this enzyme. The enzyme catalyzes the reduction of peroxide group by electron donors such as ascorbate, phenols, amines or other organic compounds. The oxidation product is usually highly coloured and has been used for the colorimetric analysis of peroxidase activity. Enzymic analysis of foods uses these colour reactions. For example, in the estimation of glucose by glucose oxidase, the by-product is hydrogen peroxide which can be estimated with peroxidase reaction, and from the known stoichiometry of glucose oxidase reaction glucose can be estimated. Since peroxidase is very resistant to heat, it can serve as an indicator of the effect of heat treatment of a food such as pasteurization of milk. Peroxidases are involved in the destruction of Vitamin C, bleaching of carotenoids and anthocyanins and also in the peroxidative degradation of fatty acids to give volatile carbonyl compounds responsible for oxidized flavour.

5.10.10 Thiaminase

Thiaminase is present in most fresh water and marine fish and in unpolished rice, beans and mustard seed. The enzyme hydrolyzes thiamine and causes thiamine deficiency diseases such as beriberi. Thiaminase activity is destroyed by cooking or boiling fish. Salting the fish also reduces the enzyme activity.

5.10.11 Ascorbic Acid Oxidase

In the absence of the enzyme, Vitamin C is oxidized in the presence of air by copper or ferric ions to give dehydroascorbic acid and hydrogen peroxide. The hydrogen peroxide formed destroys ascorbic acid. Ascorbic acid oxidase is a copper containing enzyme which catalyzes the oxidation of ascorbic acid to dehydroascorbic acid and water. The enzyme is present naturally in oranges, citrus fruits and vegetables but not found in animals. The presence of reductases such as glutathione dehydrogenase in the whole fruit balances the activity of ascorbic acid oxidase by reducing dehydroascorbate. However, during extraction of fruit juices, reductases suffer maximal damage and ascorbic acid oxidase activity increases. Deaerating the juice to remove oxygen and also pasteurizing it subsequently inhibits the activity of the enzyme. In the case of marmalade manufacture, the orange peels are boiled in water to reduce the activity of the enzyme. Other oxidases such as cytochrome oxidase, peroxidase and phenolase also catalyze the destruction of ascorbic acid.

5.10.12 Other Enzymes

Foods also contain pigment degrading enzymes such as chlorophyllases and anthocyanases and phospholipases which hydrolyze phospho-glycerides.

5.11 ENZYMES AS FOOD PROCESSING AIDS

Enzymes catalyze a variety of reactions during processing and preservation of food products and form a large group of processing aids. Food industry utilizes enzymes in the production of bakery items, dairy products, fruit and vegetable juices and products, tenderization of meat, modification of oils and fats, flavour production and preservation of food products. Enzyme preparations may be added during processing or the process conditions may be manipulated to make use of native or endogenous enzymes.

5.11.1 Bread Making

Wheat flour, water, salt and yeast are basic ingredients of bread and other baked goods. A variety of additives are included to improve the flour quality, dough making and handling, baking characteristics and quality of the finished product. Endogenous proteases and protease preparations from *Aspergillus oryzae* have been used as processing aids in improving the quality of flour to achieve desired elastic characteristic of the dough. The high molecular weight gluten proteins initially break down due to the combined effects of hydration, mechanical mixing as well as the hydrolytic activity of proteases. Later, during fermentation the gluten proteins form a cohesive network through disulphide bridges and entrap carbon dioxide, thereby making the dough elastic. The formation of non-elastic 'bucky' dough has been attributed to the higher content of high molecular weight of gluten proteins in the flour which require longer mixing times. Proteases are added particularly to flours containing higher gluten content to hydrolyze the high molecular weight glutens partially and thereby enhance the elastic characteristics of the dough.

Endogenous redox enzymes such as peroxidases, ascorbate oxidase, dehydroascorbate reductase, glutathione oxidase and glutathione reductase along with chemicals such as bromate or ascorbic acid promote the disulphide bridge formation. Other redox enzymes such as glucose oxidase, polyphenol oxidase and peroxidase may be added to the flour though such addition has not been practised commercially.

Lipoxygenase present in the soya flour added to wheat flour for bread making promotes aggregation of gluten proteins and also catalyzes the oxidation of polyunsaturated fatty acids forming peroxides. The peroxides are strong oxidizing agents and bleach the carotenoid pigments thereby improving the colour of the wheat flour.

Wheat flour has relatively less endogenous α-amylase and the enzyme is supplemented by the addition of fungal α-amylase to enhance the rate of starch degradation in order to maintain yeast fermentation and obtain satisfactory loaf volume. Starch is hydrolyzed to produce limit dextrins which are hydrolyzed by endogenous β-amylase to fermentable glucose. Alternatively, fungal glucoamylase may be used to release glucose from starch. Thus the combination of protease, oxidases and glucoamylase improve the leavening of bread.

During baking, gelatinization of starch occurs during which the starch granules swell by absorbing water, lose their crystalline structure and exude soluble amylose resulting in a concentrated starch suspension which is transformed into a starch gel. The strength of this gel depends on the amylose concentration. Recrystallization of amylose during baking and cooling results in the formation of a rigid three-dimensional network and fixation of crumb structure. A fast crumb setting reduces the 'ovenspring' giving rise to a small loaf volume. The activity of α-amylase, which continues during baking till the internal dough temperature

reaches about 65°C, degrades the gel forming starch and maintains the dough's flexibility for a longer period. The quality of bread is affected by excessive starch degradation which reduces gel formation and results in soft, gummy and open crumb structure.

The insoluble hemicellulose fraction of wheat flour reduces the loaf volume and also gives a coarse crumb structure. Enzymes from fungi *Trichoderma sp.* or *Aspergillus sp.* may be used to break down the hemicellulose, but this can produce a sticky dough resulting in bread difficult to slice due to excessive water absorption. The problem of crumb firming or hardening of the bread during storage is due to retrogradation (recrystallization) of starch. This can be avoided by the use of fat to produce a softer crumb structure and also by limited degradation by α-amylase at low dosage.

The baking quality of rye flour and the shelf life of rye bread is improved by partial hydrolysis of rye pentosans using a mixture of glycosidases (cellusases and a mylases) and xylanase.

5.11.2 Cheese Making

Cheese is made by curdling or clotting of milk brought about by coagulation of casein. Rennet, a mixture of proteases chymosin (also called rennin) and pepsin is used for curdling. Chymosin specifically hydrolyzes the bond between phenylalanine and methionine at positions 105 and 106 on the κ-casein molecule producing para-κ-casein and glycomacropeptide fragments. Due to the release of the hydrophilic C-terminal glycomacropeptide, κ-casein protecting the casein micelles against calcium induced precipitation loses its protective function. The casein micelles coagulate, form a gel entrapping the available fat and precipitate as curd. Rennet is obtained by salt extraction from the fourth stomach of suckling calves. The chymosin content of rennet decreases as the calf grows and rennet containing more than 15% of pepsin is considered unsuitable for making typical Dutch cheese. Proteases derived from various microorganisms such as *Mucor miehei* and *Mucor pusillus* are used as alternative sources of rennet since they show nearly the same specificity of chymosin. The high thermal stability of microbial proteases, however, poses a problem in that their activity continues in the whey even after its pasteurization and interferes with the formulation of whey into dairy products and baby foods. Mild oxidation of microbial proteases reduces their thermal stability and these are commercially available for cheese manufacture.

Proteases and lipases are used for flavour production during ripening of cheese to give enzyme modified cheeses. Generally the cheese slurry is inoculated with specific proteases and lipases from different microorganisms and incubated at 37°C for about 2 to 15 days for flavour development. The enzymes are then heat inactivated and the products are formulated.

Milk used for cheese manufacture is pasteurized with hydrogen peroxide to reduce the microbial load as well as to give a whiter product. The excess hydrogen peroxide is then destroyed by catalase which decomposes it to water and oxygen.

5.11.3 Whey Processing

Whey obtained as a by-product during cheese manufacture is used as an ingredient of a variety of processed foods. The major constituent of whey is lactose. Lactose is less sweet and crystallizes as hard and sharp crystals in foods with high lactose concentration and thereby affects the texture of the food product. In addition, it causes digestion problems in individuals lacking lactase. The enzyme β-galactosidase or lactase (EC 3.2.1.23) derived from yeasts such as *Kluyveromyces lactis* or filamentous fungi such as *Aspergillus niger* hydrolyzes lactose into glucose and galactose. The optimum pH range of yeast enzymes is 6–7 and hence these are used for lactose hydrolysis in milk while fungal enzymes are used in whey processing because their optimum pH range is 4–5. The enzyme also finds use in milk based drinks to enhance their sweetness, in concentrated milk products to prevent sugar crystallization and in fermented milk products to enhance fermentation rate.

5.11.4 Meat Tenderization

Tenderness is a highly desirable quality of meat brought about by maturation of the slaughtered animal usually carried out by hanging the carcass for about ten days at 2°C. During maturation the highly crosslinked collagen of muscle tissue is hydrolyzed partially by endogenous neutral protease and collagenase. Proteases such as papain (from papaya) and bromelain (from pineapple) are capable of digesting the connective tissue and muscle proteins and hence are used for tenderizing meat. The enzyme preparations are either injected intravenously into the animal just before slaughtering or sprinkled on the surface of the meat.

5.11.5 Fruit Processing

Enzymes from plants and microorganisms which degrade cell wall are used as processing aids in fruit juice or fruit and vegetable puree manufacture. Enzymes are used for clarification of juices, to improve the pressing yield of the fruit, to increase liquefaction rate, to enable total liquefaction and to improve the colour and flavour extraction. Pectin esterases and pectin depolymerases are the two main groups of cell wall degrading enzymes.

Pectin esterase (EC 3.1.1.11), abbreviated as PE, hydrolyzes methylated pectin (methyl ester of polygalacturonic acid) which is the

main polysaccharide in the middle lamella of the cell wall. The enzyme from plants acts preferentially on methoxyl groups of esterified galacturonic acid at the non-reducing end of the polygalacturonan chain in a sequence while pectin esterases of fungal origin hydrolyze the methoxyl groups in a random manner and hence decrease the viscosity of the pectin solution.

Pectin depolymerases include polygalacturonase (PG), pectin lyase (PL) and pectate lyase (PAL). Two forms of PG, endo-PG (EC 3.2.1.15) and exo-PG (EC 3.2.1.67) act only on glycosidic linkages between non-esterified galacturonic acid molecules. PL (EC 4.2.2.10) is found only in microorganisms and depolymerizes pectins randomly breaking glycosidic linkages between esterified galacturonic acid units in the polygalacturonan chain producing unsaturated products. The enzyme is highly useful in fruit processing because of its pH optimum of 5 to 6, its enhanced activity on highly methylated pectin and its increased affinity for low methylated pectin at lower pH values, particularly in the presence of calcium and other cations. PAL (EC 4.2.2.2) is also found only in microorganisms and acts on glycosidic linkages beween non-methylated galacturonic acid moieties in low methoxyl pectin and pectic acid. However, PAL is of limited use in fruit processing because of its higher pH optimum in the range of 8.5 to 9.5 and requirement of calcium ions.

The enzymatic degradation of fruit cell wall during juice making include maceration and total liquefaction. Maceration with limited degradation of soluble pectin and middle lamella pectin by endo-PG and PE leading to tissue disintegration results in a cell suspension with a residual viscosity. Maceration with extensive degradation of pectin of soluble, middle lamella and cell wall by endo-PG and PE/PL results in a thin liquid with a small content of cell wall parts. Total liquefaction or solubilization of all cell wall polysaccharides including hemicellulose and cellulose components involves the use of a mixture of PL, endo-PG, PE, hemicellulase and cellulase enzymes.

Citrus fruit juices and purees, particularly of grapes, contain naringin (a dihydrochalcone) a bitter tasting compound. The bitterness is eliminated by converting the naringin to non-bitter naringenin by treating with the combination of α-L-rhamonosidase and β-D-glucosidase.

5.11.6 Modification of Oils and Fats

Natural oils and fats are used individually or in mixtures directly or in a variety of processed products. The triglyceride fractions crystallize in different polymorphic forms and triglyceride composition is responsible for a variety of properties, particularly their melting characteristics. In many cases it is necessary to modify the triglyceride composition with specific reference to the fatty acids to make oils and fats suitable for specific applications. This modification is achieved by lipase catalyzed interesterification. Many lipases are regiospecific in that they catalyze

reactions at 1- and 3-positions of glycerol. For example, lipase from *Mucor miehei* supported on diatomaceous earth has been used to catalyze regiospecific interesterification/modification of palm oil to give a product with fatty acid composition similar to that of cocoa butter, a valuable confectionery fat. The triglyceride composition of natural palm oil consisting of SSS (5%), POP (58%), POS (13%), SOS (2%), SLS (9%), SOO (4%) and others (2%) is modified similar to that of cocoa butter which consists of SSS (1%), POP (16%), POS (41%), SOS (27%), SLS (8%), SOO (6%) and others (1%) (S-stearic acid, P-palmitic acid, O-oleic acid, L-linoleic acid in 1, 2 and 3 positions) by lipase catalyzed interesterification. During interesterification, the steroyl groups are incorporated exclusively in 1- and 3-positions in exchange of palmitoyl groups. Cocoa butter equivalents may also be obtained by blending palm oil fractions rich in POP triglyceride with costly exotic tropical seed fats such as shea oil, illipe butter and sal fat which are rich in POS and SOS triglyceride fractions. The interesterification of oils and fats are used to produce triglyceride mixtures with better functional properties in margarine, low calorie spreads and bakery fats. Triglycerides containing two long chain fatty acids and one short or medium chain fatty acid are effective hardeners for margarines and other spreads.

5.11.7 Confectionery Items

Invertase (β-D-fructofuranase) produced by baker's yeast *Saccharomyces cerevisiae*, is used to convert cane or beet sugar to a 1:1 mixture of glucose and fructose (invert sugar) to enhance the solubility of sugar and decrease its tendency to crystallize in concentrated solutions. In addition the invert sugar has the advantages of higher sweetness and enhancement of fruit flavour due to fructose and higher stability at low pH. Invert sugar finds use in production of confectionery items.

Raffinose present in sugar beet prevents crystallization of sugar if its concentration is greater than 8%. The enzymatic preparations from *Mortiellla vinacea* containing α-D-galactosidase hydrolyzes raffinose and improve the yield of sugar during crystallization process. The enzyme α-D-galactosidase is also used to hydrolyze stachyose in legumes and thereby eliminate gas formation (flatulence) in the intestines due to consumption of legumes.

High fructose corn syrup (HFCS) is an important sweetener used in confectionery industry. It is manufactured from corn steep liquor by using three different enzymes in a sequence, amylase for liquefaction of starch followed by glucoamylase to produce glucose and finally glucoseiomerase to convert glucose to fructose. α-Amylase from *Bacillus subtilis, Aspergillus oryzae or Bacillus licheniformis* and β-amylase from the malting cereals hydrolyze corn starch into a mixture of glucose, maltose and dextrins to give starch syrup. The enzyme glucoamylase hydrolyzes the starch into D-glucose (saccharification process).

Pullulanase and isoamylase are used for the hydrolysis of amylopectin and starch to form starch syrup enriched with maltose for use in brewing industry.

Aspartame, a dipeptide is about 150–200 times more sweet compared to sucrose, is manufactured from enzymically produced optically pure L-amino acids to get the α-isomer in pure condition because the β-isomer is bitter. The sweet α-isomer is produced from L-aspartic acid and L-phenylalanine whereas β-aspartame produced from D-aspartic acid and D-phenylalanine is bitter. L-phenylalanine is prepared from trans-cinnamic acid using phenylalanine ammonia lyase from the yeast *Rhodotorula rubra*. Similarly fumaric acid can be converted to L-aspartic acid using aspartase.

5.11.8 Food Preservation

Enzymes can function as anti-microbials by (i) depriving spoilage microorganisms of an essential nutrient, (ii) producing substances toxic to spoilage organisms or (iii) by physically disrupting the cell (attacking the cell wall or membrane components). Enzymes also find use in catalyzing specific reactions in processed food products, and prevent deterioration of the quality with respect to flavour, colour or texture.

Glucose oxidase and catalase together deplete oxygen in foods stored in closed containers thereby inhibiting growth of aerobic microorganisms. Removal of oxygen from sealed containers or packages suppresses fat oxidation and oxidative degradation of natural pigments. Thus dipping crabs and shrimp in glucose oxidase/catalase solution prevents the change of colour from pink to yellow. The shelf life of citrus fruit juices, beer and wine is also enhanced by preventing oxidative reactions by the combination of these enzymes. Removal of glucose from eggs during egg powder manufacture by glucose oxidase prevents discolouration and deterioration of whippability of the product by preventing the Maillard reaction. Glucose oxidase is used for similar purposes in enhancing the golden yellow colour rather than the brown colour of potato chips, French fries and meat products.

Xanthine oxidase found naturally in milk can generate hydrogen peroxide in foods in the absence of catalase or peroxidase and hydrogen peroxide is toxic to many spoilage organisms. Similarly, lactoperoxidase coupled with a peroxide generating oxidase converts SCN^- to $SCNO^-$ which is lethal to microorganisms. Soft ice cream mix and pastry cream is preserved by this method. Myeloperoxidase-peroxide-halide system produces hypochlorous acid and chloramines toxic to microbes. Lysozyme can disrupt cell wall of gram positive bacteria and has been used in the preservation of certain varieties of cheese. It is effective against pathogens *Clostridium botulinum* and *Listeria monocytogenes*.

Aroma defects occur during soya processing due to oxidation of unsaturated fatty acids to aldehydes by native enzymes (e.g. lipoxygenase).

The aldehydes, particularly hexanal gives a bean like aroma in soya milk. The enzyme aldehyde dehydrogenase obtained from beef liver mitochondria oxidizes the aldehydes to carboxylic acids and thereby removes the aroma defect in soya milk. Similarly, diacetyl (a diketone, $CH_3.CO.CO.CH_3$) formed during beer fermentation causes a flavour defect. The enzyme, butanediol dehydrogenase from *Aerobacter aerogenes* reduces diacetyl to flavourless 2, 3-butanediol in the presence of NADH. Yeast cells also have this enzyme as well as the NADH required for the reaction. Moreover, the yeast cells also have a system to regenerate NADH. Hence yeast cells encapsulated in gelatin are used to correct flavour defect caused by diacetyl.

Tannases from *Aspergillus niger* hydrolyze polyphenolic compounds (tannins) and thereby prevent development of turbidity in cold tea extracts.

Staling of bakery products is retarded by lipase probably due to the formation of mono- and di-acylglycerols.

Questions

1. How are proteins classified? Give examples.
2. Write a note on the functional properties of proteins.
3. What are textured proteins? How are they obtained?
4. How do wheat proteins contribute to the texture of bread?
5. What is rigor mortis? How is it relevant in cooking meat?
6. How is meat tenderized?
7. What are the different types of proteins present in milk? How are they used in different food products?
8. What are the different functional properties and applications of egg proteins?
9. How are enzymes classified? Give examples.
10. How are enzymes useful as processing aids in confectionery and edible oil industry?
11. Write notes on the use of enzymes as processing aids in
    (a) fruit processing
    (b) meat tenderization
    (c) cheese preparation
    (d) bread making
    (e) food preservation.
12. What is enzymic browning in plant foods?
13. How is flavour produced by different enzymes?

CHAPTER

6

Vitamins and Minerals

6.1 DIETARY REQUIREMENTS OF VITAMINS

Vitamins are a diverse group of organic chemicals occurring in tiny amounts, but are considered to be the essential components of biochemical and physiological systems of animal and human life. These cannot be synthesized by animal or human body, unlike hormones, and hence must be supplied as part of diet. The recommended dietary allowance (RDA) for various vitamins and the diseases or symptoms caused in humans due to their acute deficiency are given in Table 6.1.

TABLE 6.1 Recommended Dietary Allowance of Vitamins and Diseases Caused due to Their Acute Deficiency

Vitamin	*RDA*	*Acute deficiency symptom*
A	5,000 IU	Night blindness
D	400 IU	Rickets
E	14 IU	Oxidation of fats and membrane lipids
K		Haemorrhage
C	45–80 mg	Scurvy, reduced defense against infections
B_1	1.5 mg	Beri-beri, neuritis
B_2	1.4 mg	Mouth and tongue lesions
Niacin	16 mg eqiv.	Pellagra, skin lesions
B_6 (pyridoxal)	2 mg	Atrophy of organs, lack of growth
Pantothenic acid		Disruption of metabolism
Biotin		Acidosis, skin rashes, immune deficiency and neurological problems
Folic acid		Megaloblastic anaemia
B_{12}		Pernicious anaemia

*For some of the vitamins dietary allowances are not recommended.

Vitamins are classified broadly into two groups based on their solubility as (i) fat soluble vitamins and (ii) water soluble vitamins. Fat soluble Vitamins A, D, E and K are generally stored in relatively large quantities in certain organs, e.g. Vitamin A, in liver. Water soluble vitamins of B group and Vitamin C are widespread in all metabolizing cells and are easily excreted and their deficiency occurs frequently and quickly.

6.2 FAT SOLUBLE VITAMINS

6.2.1 Vitamin A (Retinol)

Retinol is a primary alcohol derivative made of four isoprene units found in mammalian tissue. It is enzymatically oxidized to 11-cis retinal, required for night vision and retinoic acid which has been identified as mediator in cell differentiation and cell proliferation. Plant tissues provide the vitamin in the form of carotenoid pigments, e.g. β-carotene, found in carrots (2000 retinol equivalents or μg/100 g) and red palm oil (20,000 μg). Other sources include milk (40 μg), dried milk (350 μg), cheddar cheese (363 μg), eggs (190 μg), margarine (800 μg), cod liver oil (10^5 to 10^7 μg), polar bear liver (600,000 μg), pig liver (3,000 μg). The β-carotene is converted into retinol in the mucosa of small intestine. It is stored and transported as retinol ester of palmitic and stearic acids. Retinol esters and β-carotene are stable in foods under normal conditions. However, high temperatures in the absence of air, as in canning, isomerize them to neocarotenes which are biologically less active. In the presence of oxygen, they get oxidized and bleached due to hydroperoxides released by auto-oxidation of polyunsaturated fatty acids, particularly in dried fatty foods.

6.2.2 Vitamin D (Calciferol)

Vitamin D_3, cholecalciferol occurs naturally in diet. It is formed in mammalian tissues due to the action of ultra violet component of sun light on 7-dehydrocholesterol which is an intermediate in the biosynthesis of cholesterol. Ergocalciferol (Vitamin D_2) synthesized from ergosterol, is used in pharmaceutical applications and as food supplement. The liver oils of fatty fish are rich sources of cholecalciferol, e.g. mackarel (1.5 mg/100g). Other sources include the muscle tissues of fatty fish such as mackarel, salmon and herring (5 to 45 μg/100g), butter and cream (1–2 μg) and egg yolk (6 μg). In most developed countries legal act/ directives ensure sufficient Vitamin D content in foods such as milk, butter, margarine and breakfast cereals. Vitamin D is hydroxylated enzymatically into physiologically active 1,25-dihydroxycholecalciferol

(calcifetriol), one of the three hormones (the other two being calcitonin and parathormone) involved the control of calcium metabolism, particularly in the synthesis of proteins for calcium transport through cell membranes. Lack of sufficient quantities (RDA is 2.5–10 mg/day) prevents uptake of calcium from the intestine resulting in shortage of calcium for bone growth leading to the disabling disease, ricket (softening and weakening of bones). On the other hand, even modest overdose of the vitamin particularly in babies results in intestinal disorder and weight loss while massive overdose causes calcification of soft tissues such as lungs and kidneys (hypercalcemia).

6.2.3 Vitamin E (α-Tocopherol)

Eight naturally occurring compounds with Vitamin E activity include the α-, β-, γ- and δ-tocopherols and the corresponding tocotrienols. The primary sources of Vitamin E include the plant seed oils of olive, sunflower, soybean, safflower, sesame, cotton and wheat germ. The main physiological function of the vitamin in animals seems to be that of an antioxidant. It also functions as a scavenger of hydroxyl, superoxide and hydroperoxide radicals formed due to the action of chemical and enzymatic reactions involving oxygen, haemoglobin, xanthine oxidase and lipoxygenase.

6.2.4 Vitamin K (Phylloquinone, menaquinones)

The vitamins of K group are derivatives of menadione (2-methyl-1,4-naphthoquinone) or menadione itself. Phylloquinone (Vitamin K_1) has a phytyl side chain. These vitamins are widely distributed in biological systems in small amounts. Significant amounts are found in leafy green vegetables such as cabbage or (3 to 4 mg/100 g). Other important sources include peas, tomatoes and animal tissues and liver (0.1 to 0.4 mg) and cow and human milk (2 to 20 μg). Considerable quantities of Vitamin K are synthesized by bacteria in the large intestines. The physiological role of Vitamin K involves the conversion of the inactive prothrombin to the active proteolytic enzyme thrombin through carboxylation of a pair of glutamate residues in the polypeptide chain, necessary for clotting of blood.

6.3 WATER SOLUBLE VITAMINS

6.3.1 Vitamin B_1 (Thiamine)

Thiamine occurs in its free or pyrophosphate ester form, complexed with

protein in foods rich in carbohydrates such as legume seeds and the embryo of cereal grains and in pork to the extent of 0.1 to 1 mg/100 g. Other food sources include milk, raw peas and bovine muscle. Thiamine pyrophosphate is the prosthetic group of a number of enzymes catalyzing oxidative decarboxylation of α-keto acids such as pyruvic acid and α-ketoglutaracid which occur as intermediates in Krebs cycle. Thiamine is labile in that it is lost to the extent of 20 to 30% from food due to its water solubility and heat sensitivity above pH 6, particularly in bread making and canning. Addition of sulphites to fruits and vegetables to prevent browning also causes total destruction of thiamine.

6.3.2 Vitamin B_2 (Riboflavin)

Flavin mononucleotide (FMN) and flavin-adenine dinucleotide (FAD) are the coenzymes derived from riboflavin that occur in biological systems as prosthetic groups of respiratory enzymes known as flavoproteins. The flavoproteins are involved in oxidation/reduction reactions of respiratory processes that occur in mitochondria of living cells. Riboflavin is found in small amounts in almost all biological tissues and in abundance in meat (0.2 mg/100 g), liver (3 mg), milk (0.15 mg) and cheese (0.5 mg). Brewer's yeast and yeast extracts contain significant amounts of riboflavin and many other vitamins and riboflavin leaks into beer during fermentation. Riboflavin is unstable only in alkaline conditions and hence is not lost during washing or cooking. However, it is very sensitive to light. For instance, a glass of milk on two hours of exposure to sunlight can lose about 50% of its riboflavin. Irradiation of riboflavin results in the formation of lumichrome and lumiflavin which are powerful oxidizing agents and catalyze complete destruction of ascorbic acid (vitamin C). In addition, during such oxidations singlet oxygen (a very active form of oxygen) is produced which initiates auto-oxidation of unsaturated acids in the milk fat giving off flavours. Hence it is preferable to store milk in opaque containers.

6.3.3 Vitamin B_6 (Pyridoxine)

Pyridoxal phosphate is the naturally occurring form of the vitamin as a prosthetic group of a number of enzymes in tissues. Pyridoxine (pyridoxol), pyridoxal and pyridoxamine are the three forms of the vitamin found in foods, the first two predominantly in plant foods and the last two in animal foods. Pyridoxine, the active form of the vitamin is found in meat, egg yolk, wheat germ to the extent of 500 µg/100 g and in milk and cheese (about 50 µg). Pyridoxal phosphate is involved in transanimation reactions. Pyridoxal is stable to heat but the other two forms are a little less stable. Leaching during washing and cooking causes major losses of

the vitamin. Conversion of pyridoxal to pyridoxamine occurs during the boiling of ham and production of evaporated milk.

6.3.4 Niacin (Nicotinic Acid and Nicotinamide)

Niacin is the collective name to nicotinic acid and nicotinamide found in the living systems as components of nicotinamide adenine dinucleotide (NAD) and its phosphate derivative NADP. NAD and NADP are electron carriers in respiratory systems of living cells. NAD also occurs as a prosthetic group in a few enzymes such as glyceraldehyde-3-phosphate dehydrogenase. Niacin is present in small amounts in all biological materials. Rich sources of niacin include meat (5–15 mg/100 g), fruits and vegetables, particularly legume seeds (1–2 mg), whole wheat flour (5 mg) and brown rice (4.7 mg). Milling of the cereals reduces almost all vitamins drastically. In addition, niacin in cereals, particularly in maize, is tightly bound to hemicelluloses and is not liberated during digestion. However, cooking under alkaline conditions by the addition of baking powder or lime water to maize flour as done in the preparation of Mexican tortilla (dietary equivalent to bread) releases niacin for absorption during digestion. Human body has a limited ability to synthesize nicotinic acid from tryptophan. However, about 60 mg of dietary tryptophan is required in the synthesis of 1 mg of dietary nicotinic acid. Thus milk and eggs, though poor sources of the vitamin, have beneficial effects in the case of the disease pellagra that occurs due to deficiency of this vitamin. Niacin is stable under normal cooking conditions and is lost mainly due to leaching.

6.3.5 Vitamin B_{12} (Cyanocobalamin)

Cobalamin is present only in animal products and is synthesized only by specific microorganisms. Main sources of the vitamin include meat (1–2 μg/100 g), liver (40 μg), milk (0.3 μg), eggs (0.7 μg) and kidney (20 μg). Dietary requirements for humans is very low (about 1 mg) and deficiency can arise from vegetarian diet or mainly due to poor assimilation leading to pernicious anaemia. Patients with pernicious anaemia are unable to synthesize a carbohydrate rich protein called intrinsic factor which is normally secreted by gastric mucosa and facilitates the absorption of the vitamin by complex formation. Only two mammalian enzymes that require the vitamin B_{12} derivatives, are methyl malonyl CoA mutase and homocysteine methyl transferase which catalyze biomethylation. Cobalamin is stable to food processing and cooking conditions and losses occur mainly due to leaching.

6.3.6 Folic Acid (Folacin)

Folic acid includes a group of closely related compounds exhibiting vitamin activity similar to that of cobalamin. The active form is called tetrahydrofolate and consists of a pteridine nucleus and a p-aminobenzoic acid moiety containing glutamate residues. Animal liver (300 μg/100 g), leafy green vegetables (20–80 μg) and muscle tissues (3–8 μg) are the dietary sources of the vitamin. Tetrahydrofolate is involved in the metabolism of one-carbon units similar to cobalamin and hence the symptom of deficiency is similar to that of cobalamin. Leaching is the main cause of loss during cooking.

6.3.7 Biotin

Biotin is ubiquitous in the diet and also synthesized by intestinal microorganisms. However, eating large quantities of raw egg can lead to deficiency of biotin, because the egg white protein, avidin, has profound affinity for biotin. Avidin is denatured when egg is cooked. Biotin is stable to normal cooking and processing procedures. Deficiency symptoms include dermatitis, nervous irritability and anorexia with a marked fall in food intake. Biotin is required physiologically as a covalently bound cofactor of enzymes which catalyze carboxylation reactions, e.g. pyruvate carboxylase.

6.3.8 Pantothenic Acid

Pantothenic acid forms the central part of the coenzyme A (CoA) which is an acyl carrier protein and is also a prosthetic group of acyl carrier protein component of fatty acid synthetase complex. The vitamin is widely distributed in the biological tissues (0.2 to 0.5 mg/100 g of tissue). Liver is a rich source containing about 20 mg. Loss of the vitamin during food processing is mainly due to leaching and partly due to heating under alkaline or acid conditions.

6.3.9 Ascorbic acid (Vitamin C)

Vitamin C is synthesized by all mammals. L-ascorbic acid is easily oxidized into dehydro-L-ascorbic acid (DHAA) during enzymic browning of, particularly sliced, peeled or otherwise damaged, fruits and vegetables, catalyzed by copper dependent ascorbate oxidase. DHAA has the same vitamin activity but its enantiomer erythorbic acid has no vitamin activity. However, it behaves similar to ascorbate in food red-ox systems and hence used as a food additive. Fruits such as citrus fruits, pineapple,

strawberry, black currant, and vegetables such as tomato, pea, cabbage, cauliflower, horseradish, new potato and carrot are sources of the vitamin. Physiologically ascorbic acid takes part in a number of reactions, particularly in the formation of collagen, the connective tissue protein. Scurvy, failure of normal connective tissue formation, occurs due to the deficiency of the vitamin leading to internal bleeding, painful joints and failure of wound healing. Ascorbate is also required for the conversion of dopamine to noradrenaline and deficiency leads to disturbances in the nervous system which usually accompanies scurvy. Ascorbic acid seems to have important role in the absorption of iron and its transport in the bloodstream. Normal cooking, processing and storage can destroy ascorbic acid from vegetables and fruits. In the presence of air, DHAA is irreversibly oxidized to 2,3-diketo-L-gulonic acid (DKGA) which has no vitamin activity. DKGA is readily formed in alkaline conditions and hence it is advisable to cook green vegetables without adding sodium bicarbonate. The enzyme phenolase and metal ions such as ferrous iron and copper also destroy ascorbic acid in oxidation reactions in food products. Acidity in fruit juices ensures the stability of the vitamin which is further enhanced by the presence of metal chelating agents such as citric acid. Since ascorbic acid is manufactured at low cost and is acceptable as a nutrient, it finds use in vegetable and fruit products, particularly in dehydrated potato as an antioxidant to prevent browning reactions catalyzed by phenolase. In cured meat products it is used as a reducing agent to lower the concentration of nitrite and in bread making, as a flour improver.

6.4 MINERALS

Bulk minerals required in human diet include nitrogen, phosphorus, chlorine, sulphur, sodium, potassium, magnesium and calcium and trace minerals include iron, copper, zinc, selenium, iodine, fluorine, chromium, vanadium, manganese, cobalt and molybdenum.

6.4.1 Role of Minerals and Their Function

Sodium and potassium occur widely in plant and animal foods. Additional dietary sources of sodium include ham, bacon, cured meat and sodium glutamate. Sodium in the form of table salt is always included in the diet. The dietary sources of potassium include eggs, liver, milk, oranges, peas, potatoes, bacon, butter, chocolate and coffee. The human adult body contains about 1.4 g/kg (of body weight) of sodium and 2.0 g/kg of potassium. Daily requirements are in the range 1.7–6.9 g for sodium and about 0.8 g for potassium. Excessive intake of sodium results in hypertension (abnormally high blood pressure). Deficiency of potassium

leads to undernourishment. Potassium deficient foods include white bread, fat and oil. Sodium is the main cation of extracellular fluid required to maintain electrical neutrality and ionic strength. Generally, animal cells contain low concentrations of intracellular sodium and relatively high concentrations of potassium (about 100 mM) whereas in blood plasma concentrations of sodium and potassium are 140 mM and 5 mM respectively. The ionic gradients between extra- and intracellular fluids are maintained by sodium/potassium pump mechanism, which uses ATP as the energy source. The uptake of glucose and amino acids by the intestinal brush border cells is coupled to the simultaneous entry of sodium. The rate of entry of sodium depends on the sodium concentration gradient across the cell membrane and also on the efficiency of the ATPase activated sodium/potassium pump.

The intracellular potassium neutralizes acids and maintains cellular osmotic pressure as well as the ionic strength, which is necessary to maintain the active conformation of the enzymes. The importance of intracellular potassium compared to sodium is attributed to its larger ionic radius (1.33 vs. 0.95 A) and smaller hydration energy (55 vs. 72 k.cal/mole). Less energy is expended in the replacement of water molecules in the potassium shell by organic ligands. In addition, the smaller sized sodium is easily pumped out from the cells. The nutritional value of potassium over sodium arises from the lower polarization of water and ligands by potassium and its non-competing nature for active sites of enzymes which specifically require complexing of the divalent metals capable of participating in redox reactions.

Magnesium is found in plant and animal tissues and the human requirement is provided by cereal products, potatoes, vegetables and fruits. The human body contains about 250 mg/kg magnesium of which about 60% occurs in skeletal tissues. Daily requirement of magnesium is about 300–350 mg. The blood plasma level is constant at 20 mg/l. Magnesium dependent enzymes involve in the transfer of phosphate and carboxyl groups. The small size and high electrophilic character of magnesium ions allow ionic bonding with phosphate and carboxyl groups and facilitate the attack of substrates by nucleophiles.

Calcium and phosphorus are abundant in milk, cheese, leafy vegetables, legumes and sardines and other small fish. In mammals, Vitamin D is required for calcium absorption. Calcium deficiency leads to loss of bone and failure of muscle fibres to contract. The human body contains about 1250 g of calcium of which 99% occurs in bones and teeth. Daily requirement of calcium is about 0.8–1.0 g. Plasma levels of calcium are maintained constant at about 2.5 mM. The calcium ion concentration in cytoplasm is low (about 10^{-8} M) maintained by ATPase calcium pumping system and thereby preventing it from competing with magnesium ions for the active sites of enzymes. The larger ionic radius,

lower hydration energy and less geometrical demands in complex formation are useful properties in the rapid exchange of calcium between hydrated and bound forms. Thus calcium induces structural changes in calmodulin, troponin and sarcoplasmic reticulum proteins of animal muscle; triggers enzymatic reactions of calcium activated kinases for lipolysis and glycolysis and in post mortem activity of tissues. Calcium forms gels with alginates and pectic acid, whereas in egg white, magnesium stabilizes the glucoprotein gel.

Phosphorus occurs in foods as inorganic phosphate and organic esters. The human body contains approximately 700 g of the element with about 80% of it in the bones. Daily requirement is about 0.8–1.2 g and the Ca/P ratio should be about 1. The bones act as a metabolic reserve for both calcium and phosphorus as well as a structural component. Phosphorus in its highest oxidation state (+5) is used in metabolic reactions. The stability of the P–O bond and lower electronegativity of P permits the formation of stable organic esters and anhydrides like ADP and ATP. These esters function as energy reservoirs. The oxyanions of phosphorus, sulphur, vanadium and molybdenum along with the cations of sodium, potassium, calcium, and magnesium contribute to the electrochemical properties of living systems and also maintain osmotic pressure and volume of fluids in the cells.

Organic nitrogen occurs in the reduced state with three covalent bonds and a lone pair of electrons. Its electronegativity and relatively 'hard base' character facilitate complex formation with metal ions in compounds having heterocyclic rings such as imidazole of histidine and pyrrole of porphyrin. It is also possible that nitrogen with its five valence electrons introduces an essential distortion into the symmetry of carbon compounds which provides additional properties of coordination, basicity, charge and red-ox activity.

Sulphur exists in oxidation states from +2 to +6. It is found in animals and plants mainly in coenzyme A, glutathione and the amino acids cystine and methionine. Sulphur may be regarded as an alternative substituent for the more electronegative oxygen in organic compounds. As a 'soft base' it forms complexes with soft Lewis acids such as ferrous iron, zinc and copper. Sulphur is involved as a ligand in a number of enzymes and metalloproteins of which the iron-sulphur proteins and copper proteins are the most important. The sulphydryl groups participate in redox reactions and the sulphur atoms form disulphide bonds to give structural stability in proteins. Organic sulphonates enhance the water solubility of organic compounds.

Chloride content in the body is 1.1 g/kg of body weight. It serves as a counter ion for sodium. It is excreted in urine.

A summary of biological functions of some of the trace elements is given in Table 6.2.

TABLE 6.2 Biological Functions of Some Trace Elements

Element	*Content in human body*	*Daily requirement*	*Biological function*
Chromium	6–12 mg	5–200 μg	part of a glucose tolerance factor
Manganese	10–40 mg	2–18 mg	cofactor for arginase, superoxide dis-mutase, glycosyl transferases, pyruvate carboxylase and amino acid peptidases
Iron	4–5 g	1–3 mg/kg	oxygen transport, enzyme catalyzed red-ox reactions, activation of diatomic oxygen
Cobalt	1–2 mg		DNA synthesis
Copper	100–150 mg	1–1.5 mg	required for synthesis of haemoglobin, cofactor of cytochrome oxidase, superoxide dismutase, lysyl oxidase, amino oxidase, galactose oxidase, phenolase, ascorbic acid oxidase and plastocyanin.
Zinc	2–4g	6–20 mg	cofactor for carbonic anhydrase, carboxypeptidase, several dehydrogenases, superoxide dismutase, RNA and DNA polymerases, alkaline phosphatases and phospholipases.
Selenium	10–15 mg	0.05–0.1 mg	cofactor for glutathione peroxidase
Molybdenum	8–10 mg	0.3 mg	cofactor for xanthine oxidase, nitrogenase and sulphite oxidase
Iodine	10 mg	100–200 μg	part of thyroxine
Fluorine	2–6 g	0.5–1.5 ppm	increases stability of bones and teeth by substituting for hydroxyl groups on hydroxyapatite
Boron and vanadium	17–45 mg	1–2 mg	required for growth of plants and rats
Nickel			activator of a number of enzymes, e.g. alkaline phosphatase, oxaloacetate decarboxylase
Silicon	1g		promotes growth
Tin			promotes growth in rats
Aluminium	50–150 mg		

Questions

1. How are vitamins classified? List the vitamins under each class.
2. What do you understand by the term recommended dietary allowance?
3. What are deficiency diseases? What are the causes?
4. Discuss the role of minerals in human metabolism.
5. What are trace elements? How are they important?

CHAPTER

7

Food Colours and Flavours

Colour is very important for appreciation of food. There is a natural desire to maintain colours of processed and preserved foods as close as possible to the natural colours of the original raw materials. Some foods acquire their recognized colours during processing, e.g. brown of the bread crust, pink of cured meat, etc.

The principal groups of colourants are plant pigments such as anthocyanins, chlorophylls and carotenoids, brown colours of polymerized, oxidized polyphenols, purple red betanins and chemically synthesized dyes.

7.1 CHLOROPHYLL

Chlorophyll, the green pigment of plants gives colour to vegetables and skin of fruits, particularly unripe fruits. It is lost naturally from leaves. This breakdown accompanies a general breakdown of chloroplast membranes. When green vegetables are heated, either during cooking or when they are blanched prior to freezing, or during canning, the phytol side of the chlorophyll is lost to give chlorophyllide. Magnesium is also lost particularly in acidic conditions to give pheophytins of dirty brown colour seen in overcooked green vegetables, e.g. cabbage. The loss can be minimized by the addition of a small quantity of sodium bicarbonate to render the medium alkaline. However, too high an alkalinity affects the texture and flavour of the food and also enhances the loss of Vitamin C. Thus in canned peas, the acidity of the plant cells leads to loss of green colour. Artificial colourants such as tartrazine and green S (dyes) are added to restore the colour. A derivative of chlorophyll, sodium copper chlorophyllin (i.e. chlorophyll from which the phytol side chain has been removed and magnesium is replaced by copper) gives an acceptable blue-

green colour to canned peas. Though copper is toxic, the concentration level is too low to be a toxic hazard.

7.2 CAROTENOIDS

Carotenoids are pigments responsible for yellow and orange colour of fruits and vegetables. Chemically they are classified as terpenoids. Terpenoids having one, two, three or four isoprene units are well known; but steroids which are triterpenoids and carotenoids, the only known tetraterpenoids, are the most important from the point of view of food chemistry. The carotenoids are of two groups, carotenes, which are hydrocarbons and xanthophylls which contain oxygen. Due to their instability and insolubility in water they are not used as food colourants. However, there are two exceptions—crocetin and bixin. Crocetin is the major pigment of the spice, saffron. Bixin is the major constituent of a colouring matter known as annatto, an extract of the fruit of the plant *Bixa orellana*, a large shrub of tropical origin. Annatto is the colourant of red cheeses (Leicester cheese) and is also used to colour margarine.

Palm oil contains high levels of β-carotene (13–120 mg/100 g) and is also used to colour margarine. Three commercially synthesized carotenes are β-carotene, β-apo-8'-carotenal and canthaxanthin used in colouring ice creams, baked goods such as cakes and biscuits, margarine and cheese. The carotenes get extracted into oils or organic solvents during cooking/processing. Egg yolk owes its colour to two xanthophylls lutein and zeaxanthin, with only a small proportion of β-carotene. The carotenoids which give the depot fat of animals its yellow shade, are derived from the vegetable material in the diet. The dark greenish purple pigment of lobster is a complex of protein with astaxanthin. When lobster is boiled, the protein is denatured and colour reverts to the more typical reddish shade of a carotenoid. Astaxanthin is also the source of the pink colour of salmon flesh.

7.3 ANTHOCYANINS

Anthocyanins cause pink, red, mauve, violet and blue colour of flowers, fruits and vegetables. Cyanidin is the most important anthocyanin pigment. Grapes contain anthocyanin pigments. Anthocyanins complex metal ions giving rise to unusual colours in canned fruits. Sulphites or meta-bi-sulphites usually added as microbial preservatives in wines bleach the anthocyanin pigments, particularly in red wines. Anthocyanins also undergo a number of reactions slowly with other flavanoids present in the grapes resulting in extensive polymerization products giving a brown colour to old wines. In most food processing operations the anthocyanins are quite stable, particularly in low pH conditions.

7.4 BETALAINES

Betalaines give the characteristic red-purple colour of beet-root, *Beta vulgaris*. These are fairly stable under food processing conditions. However in the presence of air at neutral pH, heating causes breakdown of these compounds. Betacyanins obtained from beet-root juice are used in confectionery. The beet-root juice is fermented with yeast to eliminate its high sugar content and then dried to give a powder having 6–7% of betacyanin.

7.5 MELANINS

Melanins are formed in vegetables and fruits such as apples and bananas, as a brown colour when the plant tissues are damaged or cut portions are exposed to air. The browning is due to polyphenolic substances of melanin type pigments produced by action of the enzyme phenolase (also known as polyphenol oxidase) present naturally. During fruit and vegetable processing, the activity of phenolase is minimized by immersion in water to reduce contact with air. Phenolase activity is inhibited by agents such as malic acid and citric acid which complex copper that is a component of active sites of the enzyme. Sliced fruits and vegetables (peaches, potatoes) are often immersed in solutions of these acids supplemented with ascorbic acid or sulphite. Ascorbic acid is not only valued as a vitamin but as a good chelating agent and antioxidant. Dehydrated potato powders include ascorbic acid to ensure whiteness by preventing browning due to traces of oxygen present in sealed cans.

The enzymatic oxidation of polyphenolic substances is a highly desirable feature of tea. The tea leaves contain about 30% by weight of polyphenolic materials. Caffeine, the stimulant in coffee and cola-based drinks, is also present in tea leaves (dried) to the extent of 3–4% by weight. After the freshly plucked leaves have been allowed to wither, i.e. allowed to lose 20% of their water content, the leaves are macerated so that the phenolase of the leaves can come into contact with polyphenolic substances of the cell vacuoles. The macerated leaves are then left at ambient temperature for a few hours to 'ferment' during which time a high proportion of the polyphenolic substances is oxidized and polymerized. The leaves are then 'fired', i.e. dried out at temperatures up to 75°C, to bring the reactions to a halt and give the final product, tea.

7.6 NATURAL FOOD COLOURANTS

Natural pigments such as β-carotene and the beet pigments are extracted from natural sources. Curcumin is the yellow pigment in the spice turmeric, the dried powdered roots of the turmeric, *Curcuma longa*. It is

used in colouring baked goods such as cakes and also in ice-cream. Cochineal is a group of red pigments from various dead female insects used in the form of dried powder. Carminic acid obtained from cochineal by hot water extraction, gives bright coloured complexes with aluminium salts. These can be precipitated by ethanol to give water soluble powder.

7.7 ARTIFICIAL FOOD COLOURANTS

Food industry makes use of these colourants in confectionery, soft drinks, sweets, ice cream, jam and canned fruit. Most of these are azo dyes such as amaranth, ponceau 4R, sunset yellow FCF, tartrazine and triarylmethane dyes (patent blue V, food green S), xanthenes (erythrosine), quinolines (quinoline yellow) and indigoid dyes (indigo carmine). The amount of food colourants varies between 20–100 mg/kg.

7.8 FLAVOUR

Flavour and appearance of food rather than the vitamin or nutrient content are more appealing to the consumer. Flavour is a collection of sensory phenomena, which includes taste, aroma, texture or mouth feel as detected by different sensory organs. Food flavour rarely depends on a single substance, e.g. scotch whisky has 313 different volatile compounds including 32 alcohols and 22 esters as analyzed by gas chromatography. History provides the evidence that flavour development in foods was achieved by trial and error with a primary aim to preserve foods such as cheese, yoghurt, beer, wine and vinegar rather than to improve flavour. The use of microorganisms in preservation of these foods was found to improve the flavour due to the formation of metabolites such as ethanol, acetic acid, propionic acid, esters, amino acids and methylketones. Food processing involving heating facilitates Maillard and Strecker degradation reactions which lead to characteristic flavours as in baked bread, boiled or baked meat and roasted coffee and cocoa.

Visual characteristics of foods also play an important role in consumer acceptance and are considered to be indicative of certain flavour traits of a food. The consumer recognizes, discriminates and selects foods by visual examination and expects that an item of a certain shape and colour to have a specific odour, taste and texture. Fading colour, discolouration and shrinking on the surface are considered to be less desirable in some foods. In contrast appearance has no relevance in foods such as black olives, dried figs, raw oysters or squid.

7.9 SENSORY PERCEPTION OF FLAVOUR

Sensory evaluation of flavours is mainly based on taste and smell. The sense of taste refers to the ability of taste buds in mouth and tongue to

recognize the four basic tastes, namely, sweet, salty, sour and bitter. The components of food responsible for tastes are usually nonvolatile and soluble in aqueous media including saliva. The taste buds exist in grooves around little projections on the upper surface of the tongue. The anterior of the tongue is more sensitive to sweet stimuli, the posterior portion, to bitterness and the lateral portions, to sour and salty tastes.

Aroma substances are volatile compounds which are sensed by the odour receptor sites in the olfactory tissue in the nasal cavity. Human olfactory system is very sensitive and some odourants can be detected even at 10^{-18} M concentrations. The critical duration for olfaction is about 0.18 seconds. Experienced judges can distinguish as many as 10,000 different odour qualities and for each quality, about 20 different levels of intensity.

The number of volatile aroma substances in food is very large and these occur in very low concentrations. The aroma of a food may be dependent on a single compound, or on the so called character impact compound/compounds. For example, the butter and cheese aroma are decisively formed by 2.3-butanedione and supported by acetaldehyde and dimethylsulphide in the case of butter, and 2-heptanone and 3-nonanone in cheese. Food aroma may be obtained by fermentation in combination with or without thermal processing as in the case of processed meat, roasted coffee, tea or bread. The loss of character impact compounds or a change in the composition of the aroma compounds results in an aroma defect or off-flavour which often occurs during food processing or storage.

The lowest concentration of a compound that can be directly recognized by its odour or taste is designated as the threshold value of the flavour compound. Odour threshold values of different aroma compounds depend on their vapour pressure which is affected by both temperature and medium. A comparison of the threshold values is helpful in evaluating the intensity or potency of such aroma substances. For example, the odour threshold value of ethanol is 100 mg/l while that of pyrazine is 300 mg/l which are very high compared to threshold value of 2-methylbutyric acid ethyl ester (0.1 μg/l) or that of methylmercaptan (0.02 μg/l).

Once the food is placed in the mouth a multiple impression is formed as the tissues of the mouth, throat and nasal cavity are interrelated and all the sensory systems operate simultaneously. The cutaneous sensibilities of the mouth react to texture, tingle and astringency as well as to chemical burn, biting or cooling. When the intensity of the stimulus, whether it is visual, auditory, tactile, olfactory, thermal or gustatory, becomes too strong, the receptors register pain. The physiological functions of the oral cavity include biting, chewing and regulation of the volume of food and drink while sensory functions include registering mainly tactile and thermal stimuli.

7.10 SPECIFIC FLAVOURS

The various flavour components and specific flavours of foods are summarized in Table 7.1.

TABLE 7.1 Flavour Components

Taste	*Aroma*	*Texture, temperature and physiological factors*
Mainly due to	Mainly due to	Soft/hard characteristics
Non-volatile components	volatile components	Crisp/juicy
		Hot/cold temperatures
Sweetness	**Fruity flavour**	
sugars, polyhydroxy alcohols,	alcohols, esters	
glycosides, carboxylic acids,	terpenes, sesqui-	
amino acids, proteins,	terpenes,	
isocoumarin derivatives,	γ-decalactone	
indole alkaloids,		
synthetic sweeteners	**Bready aroma**	
	2,5-dimethylfuranone,	
Saltiness	pyrrolidine	
alkali metal halides		
	Caramel flavour	
Sourness	hydroxyfuranones,	
organic acids	and pyrones	
	maltol (3-hydroxy-2-methyl-	
Bitterness	4H-pyran-4-one)	
alkaloids, terpenes,		
flavanoids,	**Roasted aroma** (of coffee,	
amino acids, peptides,	nuts and cocoa)	
β-D-mannose	thiophenes, pyrroles, pyrazines	
	and methyl pyrazines	
Other taste sensations		
pungency, astringency	**Buttery aroma**	
burning, cooling	4-butyrolactone	
	diacetyl	
Umami (at certain concentrations	**Tomato aroma**	
these substances do not have	thiazoles	
identifiable sweet, sour, bitter	**Aroma defects** sunlight flavour in	
or salty tastes)	milk and beer,	
flavour enhancers and potentiators	cat urine taint in canned beef	
such as MSG, IMP and GMP		
Taste modifiers		
Sweet suppressors-gymnemic acid		
Salty–MSG		

7.10.1 Sweetness

Sweetness is detected by taste buds at the tip of the tongue. However, buds at the base of the tongue detect a few sweet substances (dihydrochalcones). Sweetness has been attributed to be a special characteristic of sugars. However, sugars are much less sweet than sucrose, and some varieties of sugars are not at all sweet. The molecular structure responsible for sweetness of substances has been identified as the saporous unit consisting of a proton donor/acceptor system designated as A,H-B system. The A,H-B system of the sweet molecule has to match steric requirements and can interact with a complementary receptor system in the taste buds through hydrogen bonding, as shown in Chapter 3 (page 30). The concept has been extended to non-sugar sweet molecules also. The A,H-B system and the γ-site of a few sweet molecules are shown in Fig. 7.1.

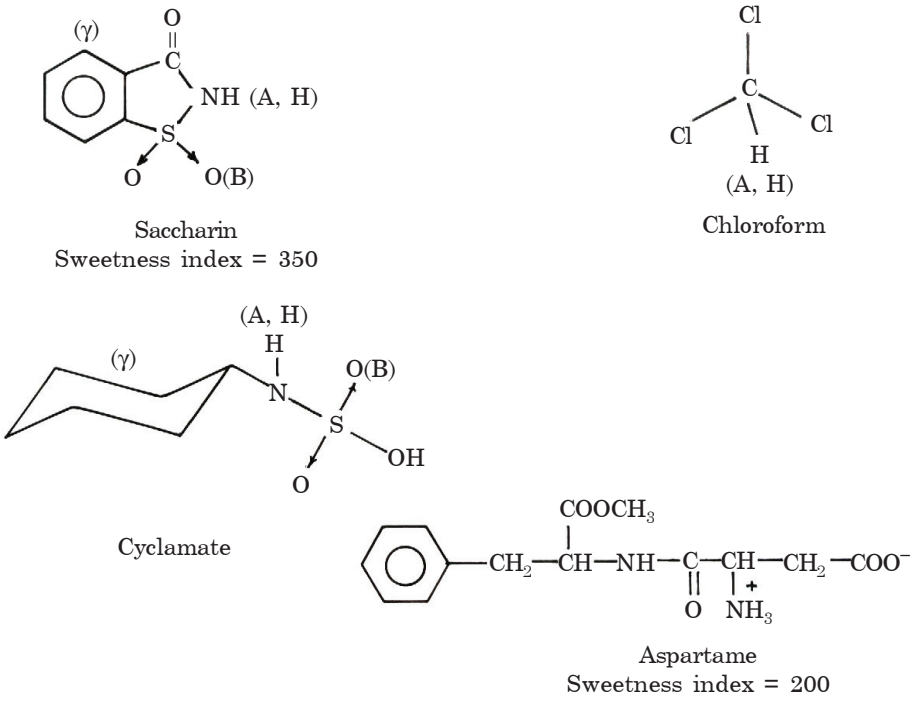


Figure 7.1 The saporous unit and lipophilic site in a few sweet molecules.

Non-sugar sweet substances. A variety of substances show sweet characteristics. The intensity of sweetness of the various substances are indicated by sweetness index (S.I.) values based on sucrose as the standard (S.I. = 1.0).

Aspartame (S.I. = 200) is a dipeptide ester.

Monellin (S.I. = 3000) is a sweet protein of 11.5 kDa present in the pulp of the fruit *Dioscoreophyllum cumminsii*. The presence of two peptide chains A and B which are not covalently bound have been identified. The separated individual chains are not sweet and when the chains are recombined, sweet taste is restored slowly but the intensity of sweetness of native protein is not reached.

Thaumatins (S.I. = 2000) are two sweet proteins (thaumatin I and II) isolated from the fruit of *Thaumatococcus daniellii*. The fruit also contains three other thaumatin proteins (a, b and c) in smaller amounts. Thaumatin I with 207 amino acid residues and monellin A and B peptides have been shown to have five identical tripeptide units of Thr-Arg-Gly, Ala-Asp-Ile, Ile-Asp-Ile, Asp-Ile-Ser and Gln-Tyr-Gly positively identified with sweetness of the proteins.*

Miraculin is a glycoprotein present in a tropical fruit known as miracle berry. It is a taste modifier because the protein has no taste but changes the taste of acidic solutions into a sweet taste. When the mouth is first rinsed with miraculin solution, lemon juice tastes sweet.

Other non-sugar sweet molecules with S.I. values given in parentheses include sodium cyclamate (30), saccharin (350), neohesperidin (1000), a dihydrocahlcone derivative, perillaldehyde antioxime (2000), stevioside (300) extracted from the leaves of *Stevia rebaudiana*, (+)-hernandulcin (1250) a sesqui-terpene from *Lippia dulcis Trev. (Verbenaceae)* and 1-n-propoxy-2-amino-4-nitrobenzene (4000). Sugar alcohols sorbitol and xylitol are sweet. Xylitol imparts cooling flavour in chewing gums.

7.10.2 Saltiness

Saltiness is detected on the sides and tip of the tongue and elicited by many inorganic salts besides common salt. Sodium chloride, particularly the sodium ion has a major role in imparting flavour to foods. For example, when sodium ions were eliminated from a mixture of amino acids, nucleotides, sugars, organic and other compounds known to mimic the flavour of crab meat, the mixture lost the flavour. Besides this tendency to stimulate meaty flavours, salt tends to decrease the sweetness of sugars, to give a richer, and more rounded flavour that is required in many confectionery products.

7.10.3 Sourness

Sourness is always assumed to be a property of acidic solutions. In most fruits and fruit juices, citric acid and malic acid are responsible for all the

*Thr–Threonine, Arg–Arginine, Gly–Glycine, Ala–Alanine, Asp–Asparagine, Ile–Isoleucine, Ser–Serine, Gln–Glutamine, Tyr–Tyrosine.

acidity and sourness. Tartaric acid is an essential characteristic of grapes while isocitric acid is that of black berries. Oxalic acid is present in rhubarb. The individual content of organic acids varies in fruit juices. For example, orange juice contains citric acid (51 mM) and malic acid (13 mM), while lemon juice contains higher concentration of citric acid (220 mM) and malic acid (17 mM). Grape juice contains tartaric acid (80 mM) together with smaller amounts of malic (7 mM) and citric acid (16 mM).

Acetic acid is diluted to give vinegar (10–15% acetic acid) and is responsible for the sourness of many foods. However, in many pickled products such as pickled cabbage (sauerkraut in Germany) and cucumbers, it is lactic acid, derived from the sugar in the vegetable by bacterial fermentation, which is responsible for low pH and sourness. Lactic acid derived from lactose fermentation in cheese (about 2% in concentration) contributes some of the sharpness to cheese.

7.10.4 Bitterness

Bitterness is associated with several classes of chemicals and it is detected by taste buds at the back of the tongue. Structural criterion (size) distinguishes saltiness from bitterness in the case of alkali metal halides. When the sum of the ionic diameters is below that of KBr (0.658 nm) which tastes both salty and bitter, then the salty taste predominates as in NaCl (0.556 nm). KI (0.706 nm) and $MgCl_2$ (0.85 nm) are bitter. Phenolic substances in the form of flavanoids are important sources of bitterness in fruit juices, particularly citrus fruits. Naringin, a glycoside of the falvanone naringenin, occurs in grape fruit and Seville oranges. Its bitterness can be detected even at a dilution of 1 in 50,000. Limonin is another bitter substance in citrus juices.

Bitterness is preferred in beer. Bitterness is achieved by adding hops to the wort (unfermented sugary extract of the malt). The mixture is boiled, cooled and allowed to undergo fermentation. Hops are dried flowers of the hop plant *Humulus lupulus*, rich in volatile compounds, and these give beer its characteristic aroma together with resins and α-acids which give bitterness to beer.

Nicotine, atropine and emetine are alkaloids which are extremely bitter but these have undesirable pharmacological effects. Quinine is used as a bittering agent in soft drinks (bitter lemon and tonic water).

Amino acids and oligopeptides also taste bitter. Proteins are used as nutritious food additives and also to modify the texture of foods. In peptides, the hydrophobic amino acids—valine, leucine, isoleucine, phenylalanine, tyrosine and tryptophan—are responsible for bitterness. The structural requirement for bitterness is similar to that for the sweetness in molecules. Correctly spaced pair of hydrophilic groups, one basic and another acidic, together with a third hydrophobic group in the spatial relationship to the hydrophilic groups is required by a molecule to

exhibit bitterness. Proteins and protein hydrolysates are used as food additives. A knowledge of amino acid content of proteins and peptides is helpful in predicting their bitterness. For example, casein and soya protein are rich in hydrophobic acids and hence yield bitter peptides on hydrolysis, which in turn impart bitterness to the food product. Bitterness of the food product can be avoided if protein hydrolysates with molecular weights above 6000 (obtained by limited hydrolysis of proteins) are used as additives.

7.10.5 Astringency

Astringency is a sensation related to bitterness but is registered within the oral cavity as well as on the tongue. It is a desirable characteristic of fruits and cider but it is only in red wine and tea that astringency is most important. It is associated with the high content of polyphenolic compounds which are also involved in the production of colour. In tea and wine the polyphenols are called tannins.

7.10.6 Pungency

Pungency is another sensation experienced by the entire oral cavity and an essential characteristic of spices such as green and red chillies, pepper, ginger, mustard, onions, garlic and to a lesser extent, in leafy vegetables (cabbage).

7.10.7 Meatiness

The flavour of meat is an amalgamation of our responses to both volatile and non-volatile compounds. Meaty taste is associated with two substances which occur in many other foods besides meat. They are ionosine monophosphate (IMP) and monosodium-L-glutamate (MSG). They are known as flavour potentiators or modifiers because they have no flavour of their own, but can enhance desirable flavours. A concentration of 300 mg/dm^3 is necessary to make an impact. When present together they show synergism. A mixture of equal proportions of IMP and MSG is about 20 times stronger than the same total amount of one alone. MSG is used as flavour enhancer in dried soups, at 1 g/dm^3 concentrations. Food manufacturers often use IMP together with guanosine monophosphate (GMP). The differences in flavour between different meats depend on the proportions of IMP and MSG. For instance, IMP content of beef and pork are roughly similar (100–150 mg/100 g) while beef has more free amino acids and twice as much MSG as pork. Lamb is especially rich in MSG but has low levels of dipeptides such as carnosine which also contribute to the meaty flavour.

Fatty acid breakdown is also an important source of characteristic odours of meat products. Arachidonic acid that occurs in polar lipids of muscle tissue breaksdown on heating to give a mixture of aldehydes, which together have a distinctive cooked flavour of chicken. The aldehydes are 3-cis-nonenal, 4-cis-decenal, 2-trans-5-undecadienal and 2-trans-4-cis-7-cis tridecatrienal.

7.10.8 Fruity Flavours

The tastes of most fruits are a blend of sweetness due to sugars (mostly mixtures of glucose, fructose and sucrose) and the sourness of organic acids (malic and citric acids). However, to recognize the fruity flavour we rely heavily on the distinctive characteristics of a fruit's volatile constituents. A typical fruit's flavour may have different volatile constituents at a total concentration at levels of ppm. These include acids, alcohols, esters, aldehydes and ketones, e.g. the typical apple flavour has at least 20 aliphatic acids ranging from formic to n-decanoic acids, 27 alcohols, over 20 esters, 26 aldehydes and ketones and other compounds such as ethers, acetals, terpenoids, etc. (a total of 131 compounds). Many of the compounds are common to many different fruits. These are produced enzymatically by the action of the enzyme lipoxygenase on unsaturated fatty acids of lipids during ripening. If tender vegetables such as peas and beans are rapidly blanched to destroy the accumulation of aldehydes, off-flavours develop.

A particular fruit flavour is dependent on one or two fairly 'character impact' substances, e.g. isopentylacetate is the crucial substance in banana flavour although eugenol contributes to the mellow, full bodied aroma of ripe banana. Benzaldehyde is another character impact substance of cherries and almonds. In apples it is ethyl-2-methylbutyrate. Raspberry aroma is due to 1-(p-hydroxyphenyl)-3-butanone but the fresh grassy aroma (notably lacking in raspberry flavoured products) is contributed by cis-3-hexenol supplemented by α-and β-ionones.

Terpenoids are character impact substances of many herbs, spices and citrus fruits. They are isolated from plant materials by steam distillation. The oily fraction of the distillate is as essential oil. The essential oil of citrus fruits including peels is mainly terpenes of which the monoterpene (+)- limonene is almost 80%. Grapefruit juice contains only 16 ppm of limonene. But the characteristic aromatic flavour of grapefruit is due to nootkatone. The character impact substance of lemon is citral which is actually a mixture of isomers geranial and neral.

7.11 SYNTHETIC FLAVOURS

Compositions for a few synthetic flavours are listed as:

Raspberry flavour

Vanillin	20 parts
Ethyl vanillin	8
α-ionone	1
Maltol	30
1-(p-hydroxyphenyl) butanone	100
Dimethyl sulfide	1
2, 5-dimethyl-N-(2-pyrazinyl)pyrrole	1

Pineapple flavour

Ethyl butyrate	60 parts
Isoamyl butyrate	20
Allyl caprate	5
Glycerol	5
Lemon oil	1
Ethyl acetate	1

Chocolate flavour

Dimethyl trisulphide	1 part
2, 6-dimethyl pyrazine	3324
Ethyl vanillin	143
Isovaleraldehyde	100

7.12 FLAVOUR RETENTION DURING FOOD PROCESSING

Food processing aims at (i) prevention of loss of desirable flavours during handling, heat treatment, transport and storage of food products, (ii) avoidance of development of undesirable flavours, such as oxidative rancidity in fats and oils, light induced off-flavours in milk and beer, metallic flavours in canned food products, (iii) development of desirable flavours as in the roasting of coffee beans, baking of bread, aging and cooking of meat, (iv) fortification of foods with substances which enhance natural and/or mask off-flavours. In some cases the volatile substances can be collected from the product during processing and added back in a subsequent stage of processing while in some others, specific enzymes can act upon precursors to regenerate lost flavours.

7.13 NATURAL FLAVOUR DEVELOPMENT IN FOODS

The interest in developing natural food flavours has paved way for biotechnological processes. Two broad strategies are adopted for flavour development in foods. Microorganisms or enzymes are used to generate

flavouring complexes or multi-component flavour systems as practised in the production of dairy products. Alternatively, biosynthesis may be facilitated to develop single-flavour compounds.

Multi-component flavour systems are developed by the use of specific strains of lactic acid bacteria and fungi to arrive at specific cheese flavours. For example, the specific flavour of blue cheese is developed by *Penicillium sp.*, mainly *P. roqueforti* which converts the free fatty acids formed by lipolytic reactions in cheese to methylketones, the key flavour components of blue cheese. A variety of methylketones are formed in a two-step process of β-oxidation of the fatty acids to β-keto acids which undergo decarboxylation to give methylketones of which the main products include 2-pentanone, 2-heptanone and 2-nonanone. Another example of the development of multi-component flavour systems involves the use of short-chain fatty acid specific lipases and proteases for the production of enzyme modified cheeses. Other examples are included in Section 10.8 (page 139).

Biosynthesis and isolation of individual flavour compounds is the second strategy involving biotransformation through hydration/dehydration or oxidation/reduction reactions. For example, the buttery flavour of fermented dairy products such as sour cream and buttermilk is primarily due to diacetyl, biosynthesized from citric acid naturally present in milk. The pleasant coconut and peach like fruity flavours natural to fruits, vegetables, nuts and milk products are due to lactones such as γ-decalactone formed by yeast species (*Candida sp*) and fungi (*Sporobolonyces odorus, Trichoderma viride*) using ricinoleic acid (12-hydroxyoctadecyl-9-enoic acid) as the substrate.

Biosynthesis of esters such as ethyl acetate, ethyl butyrate, ethyl isovalerate and ethyl hexenoate responsible for the fruity aroma and flavour is facilitated by the use of species of *Pseudomonas*, *Lactococcus*, *Lactobacillus* and *Candida utilis*. Pyrazines and their derivatives are associated with roasted and nutty flavours (found naturally in green beans and potatoes) and are biosynthesized by species of *Lactococcus lactis, Pseudomonas perotens, Bacillus subtilis* and *Corynebacterium glutamicum*. The pungent taste of mustard, horseradish and cress is due to the formation of isothiocyanates from glucosinolates by the action of the enzyme myrosinase. The flavour and aroma of cucumber, apple and tomato are identified with green flavour components such as nonenal, nonadienal and hexanal. These compounds are formed by degradation of unsaturated fatty acids via a lipoxygenase catalyzed formation of hydroperoxides followed by cleavage by a hydroperoxide lyase.

Flavour enhancers such as inosine-5'-monophosphate (IMP) and guanosine-5'-monophosphate (GMP) can be produced by degradation of RNA with 5'-phosphodiesterase. RNA is a by-product of plant processing and single cell protein (SCP) production.

Questions

1. Write notes on the following food colours: (a) chlorophyll (b) carotenoids (c) anthocyanins (d) betalaines and (e) melanins.
2. What are synthetic food colourants? Give examples.
3. Write a note on sensory perception of flavour.
4. What is aroma due to in foods? List a few examples.
5. Why are sugars sweet?
6. Explain the molecular structure in relation to sweetness of a molecule.
7. What are non-sugar sweeteners? Name a few.
8. What is sourness due to in fruits?
9. Write a brief note on the taste characteristics of alkali metal halides.
10. Write a note on bitterness in foods.
11. What are character impact compounds? Give examples.
12. What are flavour enhancers?
13. Discuss in detail the development of flavours in foods naturally.

CHAPTER

8

Food Additives

8.1 DEFINITION

A variety of non-nutritive chemicals are added intentionally to food, generally in small quantities, to improve its appearance, flavour, texture, or storage properties. Other chemicals such as vitamins and minerals are also added primarily for their nutritive value. In addition, certain chemicals find their way into food unintentionally such as pesticide and fertilizer residues, chemicals that may leach out from the packaging or trace amounts of lubricants from food processing equipment. Thus intentional additives are those substances purposely added to perform specific functions, whereas unintentional or incidental additives are those substances which have no intended function in the finished food but become part of the food product through a certain phase in the production process or subsequent handling.

The Food Protection Committee has defined a food additive as a substance or a mixture of substances, other than the basic foodstuff, which is present in a food as a result of any aspect of production such as processing, storage or packaging. The term does not include chance contaminants. Such definition encompasses intentional as well as incidental or accidental additives.

8.2 LEGITIMATE USES AND UNDESIRABLE ASPECTS OF ADDITIVES IN FOOD PROCESSING

The use of food additives is justified when it serves at least one of the following purposes: (i) maintenance of nutritional quality, (ii) enhancement of keeping quality or storage stability with reduction in food losses, (iii) making the food attractive and acceptable to the consumer and at the same time not leading to deception and (iv) as essential aids in food processing.

The use of food additives is not in the best interest of consumer if it is practised to deceive the consumer by disguising the use of faulty processing and handling techniques resulting in reduced nutritive value of the food, or when the desired effect can be achieved by good manufacturing practice.

8.3 FUNCTIONAL CHARACTERISTICS OF CHEMICAL ADDITIVES

The functional characteristics of chemical additives may be varied.

1. *Preservatives.* These are compulsorily used for prevention of microbiological spoilage, chemical deterioration and control of insects and rodents.
2. *Nutritional supplements.* These include vitamins, amino acids, minerals and calories.
3. *Colour modifiers.* These are natural colouring matter, certified food dyes and derived colours.
4. *Flavouring agents.* These may be synthetic or natural and are mainly flavour enhancers or extenders.
5. *Firming agents and maturing agents.* These are mainly chemicals which affect functional properties of foods—control of colloidal properties of gels, emulsions, foams and suspensions
6. *Antifoaming agents, chelating agents and yeast nutrients.* These are chemicals used to process foods—for sanitation, public health and or aesthetic purposes, to facilitate the removal of unwanted coverings (skins, hides, feathers, hair, etc.),
7. *Anticaking agents and waxes.* These are chemicals used to control moisture.
8. *Acids, bases and salts.* These commonly known chemicals are used to control pH.
9. *Ripening agents.* These chemicals are used to control physiological functions in order to hasten the ripening process.
10. *Miscellaneous chemicals.* These are other chemicals such as gases which are used for pressure dispensing

8.4 INTENTIONAL FOOD ADDITIVES

The intentional food additives therefore include preservatives, anti-oxidants, sequestrants, surface active agents, stabilizers and thickeners, bleaching and maturing agents and starch modifiers, buffers, acids and

alkalies, food colours, non-nutritive special dietary sweeteners, nutrient supplements, flavouring agents and miscellaneous additives.

8.4.1 Acids, Bases and Their Salts

Some of the organic and inorganic acids occur naturally in plants, vegetables and fruits and are deliberately added during food processing. With bases, they participate in buffering action. Natural buffering systems based on citric acid (lemon, tomato), malic acid (apple, tomato and lettuce), oxalic acid (lettuce, rhubarb) and tartaric acid (grape, pineapple), function in conjunction with phosphate salts. Milk may be considered as a complex buffer containing proteins, phosphate, citrate and calcium.

Acids used in food have many other functions. Monocarboxylic acids and hydroxy acids such as lactic acid (acetic acid) are used to induce coagulation of milk in the manufacture of cheese, sour cream and cultured products. Lactic acid is used for improving the whippability of egg white, flavour improvement in beverages and vinegar pickled vegetables, and for prevention of discolouration of fruits and vegetables. Lactic, butyric and acetic acids are added as flavouring agents to give sour or tart taste and also to modify and intensify the taste perception of flavouring agents. Malic acid is used in marmalades, jellies and beverages and in canned fruits and vegetables. Tartaric acid with a 'rough' and 'hard' taste is added to fruit juice and sour candies.

Dicarboxylic acid additives include adipic, succinic and fumaric acids. Adipic acid is used in baking powder and in powdered fruit juice drinks, in marmalades and jellies to improve the gelling properties and in cheese for improving its texture. Succinic acid anhydride is used for binding water in dehydrated food products and in baking powder for slow acidification. Fumaric acid increases the shelf life of some dehydrated food products such as pudding and jelly powders. It is also used together with benzoic acid as a food preservative and as an additive for promoting gel setting.

Benzoic acid and sorbic acid function as antimicrobial agents. Ethylene diamine tetra acetic (EDTA) and citric acid are used as chelating agents. Citric acid is used as a preservative and to suppress browning of fruits and vegetables and as an aroma improver in processed cheese and buttermilk. Phosphoric acid is used as an acidulant in carbonated beverages such as cola and root beer.

Bases are added during food processing mainly to maintain pH, carbon dioxide evolution, enhancement of colour and flavour and also for chemical peeling of skins of vegetables. Thus, ripe olives are treated with 0.25–2.0% solution of sodium hydroxide to remove bitterness and produce dark colour. Hot sodium hydroxide solution (3% at 60–80°C) is used in the peeling of fruits and vegetables. Sodium bicarbonate is used in the manufacture of peanut brittle candy to enhance sugar-amino acid browning and release of carbon dioxide to produce a porous texture. In

cocoa, sodium bicarbonate enhances sugar-amino acid browning to give a darker colour, and also gives a smoother, bitter chocolate flavour and improved solubility. Alkaline salts such as di-sodium phosphate, sodium citrate and tri-sodium phosphate are used in processed cheese to increase the pH value from 5.7 to about 6.3 and to aid in protein dispersion, improve water binding and emulsification capacity of proteins. Sodium carbonate, magnesium carbonate, calcium hydroxide and sodium hydroxide are used to neutralize acids in foods as in cultured butter.

Salts such as di- and tri-sodium phosphates, sodium hexametaphosphate, tetrasodium pyrophosphate promote a uniform, smooth texture and also function as emulsifying agents. Phosphates increase water binding capacity of raw and cooked meat and sausage. They are also used in curing ham and to decrease drip losses in poultry and sea foods.

8.4.2 Antioxidants

These are compounds used to prevent oxidation of fats in foods such as potato chips, breakfast cereals, salted nuts and fat-containing dehydrated foods. Examples of antioxidants include butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT) propyl gallate (PG). Ascorbic acid, stannous chloride and tocopherol (vitamin E) are also used.

8.4.3 Sequestrants

Metals such as iron and copper occur in traces in foods and are active catalysts of undesirable oxidation reactions. They also participate in reactions giving rise to off-colour in foods. Sequestrants or chelating compounds are also referred to as metal scavengers, since they combine with trace metals and remove them from solution. Examples of sequestrants include EDTA, polyphosphates and citric acid. Polyphosphates and EDTA are added to canned sea foods to prevent the formation of glassy crystals of struvite (magnesium ammonium phosphate). Addition of chelating agents to vegetables prior to blanching inhibits metal induced discolouration and also removes calcium from pectin in cell walls, thereby promoting tenderness. Citric acid and phosphoric acid are added in drinks to prevent metal induced oxidation of flavour compounds such as terpenes. In fermented malt beverages added sequestrants complex copper and thus prevent polyphenol oxidation and formation of haze.

8.4.4 Stabilizers, Thickeners and Gel Builders

Hydrocolloid materials such as gum arabic, guar gum, carboxymethyl cellulose (CMC), carageenan, agar, starch, dextrins, hydrolyzed vegetable

proteins, gelatin and pectin are used in foods for a variety of purposes. They are used to increase viscosity (thickening) of gravies and pie fillings, to form gels and to inhibit crystallization in food products. They are also used to stabilize texture of foam, emulsion or suspension, as in jellies, puddings, salad dressings and chocolate milk drinks. They improve icings on baked goods by reducing stickiness. They are also used for encapsulation of flavours.

8.4.5 Surface Active Agents

These are emulsifiers used to stabilize oil-in-water and water-in-oil mixtures, gas-in-liquid mixtures and gas-in-solid mixtures. These include emulsifiers of natural origin such as lecithin, synthetic mono- and di-glycerides and their derivatives.

8.4.6 Bleaching and Maturing Agents and Starch Modifiers

A variety of chemicals are used as bleaching and maturing agents and starch modifiers. Freshly milled flour has an unattractive yellowish tint due to carotenoid pigments and also gives a sticky dough with sub-optimum baking qualities. Both colour and baking properties improve slowly in normal storage and can improve rapidly with the use of certain oxidants such as benzoyl peroxide, chlorine dioxide and oxides of nitrogen. Bromate and iodate oxidizing agents are also used to condition bread dough for optimum baking performance. Inorganic salts such as ammonium chloride, ammonium sulphate, calcium sulphate, ammonium phosphate and calcium phosphate are used as dough conditioners as they facilitate growth of yeast and aid in the control of pH. Emulsifying agents such as calcium stearyl lactylate improve the mixing quality of the dough and improve loaf volume of bread. Carageenan and carboxymethyl cellulose have baking applications. They retard retrogradation and staling of bread. They are also used in the preparation of doughnuts to decrease the absorption of fat during frying. Sodium hypochlorite oxidizes starch to a higher degree of water solubility and functions as a starch modifier. Hydrogen peroxide is used to whiten the colour of milk (by oxidation) used for cheese manufacture.

8.4.7 Leavening Agents

Bicarbonate salts of sodium or potassium are used as leavening agents to release carbon dioxide in dough and batter during baking and thus give the characteristic porous, cellular structure to finished products such as bread, cake and biscuits.

8.4.8 Water Binding and Retaining Agents (Humectants)

Alditols such as glycerol are formed as fermentation products in wine and beer, while sorbitol occurs naturally in fruits such as apples and pears. These and other polyhydric alcohols such as propylene glycol, mannitol and polyethylene glycol (PEG 6000), (ethylene glycol is toxic) exhibit water binding properties and hence are used as additives in a variety of foods to control viscosity, texture, bulking, moisture retention, reduction of water activity, control of crystallization of sugars and to improve softness and rehydration of dehydrated foods. In intermediate moisture foods (IM foods), glycerol (25%) and in dietary sugar-free candies sorbitol (40%) are used to retain moisture. IM foods contain moisture to the extent of 15 to 30%, with a corresponding water activity of 0.7 to 0.85, yet are shelf-stable and are not susceptible to microbial attack even without refrigeration. This is due to the presence of polyhydric alcohols or sucrose, glucose or sodium chloride which are also effective humectants and preservatives. The additives also stabilize their plastic texture. Examples of IM foods include dried fruits, jams, jellies and fruit cakes.

8.4.9 Anticaking Agents

Anticaking agents are added to maintain the free-flowing nature of granular and powdered food materials which are normally hygroscopic. These additives function by absorbing excess moisture present in the food and coating the food particles to provide a water repelling characteristic. They also function as insoluble dilutents in the mixture. Examples include calcium silicate used in baking powder (5%) and in table salt (2%); calcium stearate used in dehydrated vegetable products, salt, onion and garlic salt. Calcium stearate also functions as lubricant in the manufacture of tablet form of candies.

8.4.10 Firming Agents

Heat processing or freezing of fruits and vegetables such as apples, berries and tomatoes softens the cell walls and ultimately destroys the integrity and texture of the products. Additives such as calcium salts of chloride, citrate, sulphate, lactate and phosphate stabilize the tissues and retain the texture by stabilizing the cell walls. However, most calcium salts are less soluble and also contribute bitter flavour to the finished product. Other firming agents used in food industry include sodium, or ammonium alum, generally added to stabilize and retain crispness of cucumber pickles.

The firmness and texture of some vegetable and fruit products of potatoes, cauliflower and sour cherries can be retained without the addition of firming agents by proper control of processing conditions. For

example, the enzyme pectin methyl esterase naturally present can be activated by low temperature blanching at 70–80°C for about 3 to 15 minutes to yield pectinic and pectic acids from pectin. The enzyme is deactivated at usual blanching temperatures of 88 to 100°C. These acids are relatively less soluble compared to pectin, particularly in the presence of calcium salts. Hence they remain in the cell wall during processing and stabilize the texture while pectin migrates out.

8.4.11 Clarifying Agents

These are added to prevent haze formation, sedimentation and deterioration in products such as wine, beer and fruit juices. Naturally present phenolic substances such as anthocyanins, flavanoids, tannins and leucoanthocyanins along with proteins and pectins form haze forming colloids which may be removed by filter aids such as diatomaceous earth. Bentonite is used as a clarifying agent in wine to prevent protein precipitation. Polyvinyl pyrrolidone (PVP) is used in the stabilization of wine to prevent browning of white wine and also to prevent haze formation in beer. Gelatin finds use in apple juice to precipitate out tannin.

8.4.12 Gases and Propellants

Reactive gases are used during food processing for specific purposes. Examples include the use of hydrogen for hydrogenation of unsaturated fats, chlorine to bleach flour and sanitize equipment, sulphur dioxide to inhibit enzymic browning in dried fruits, ethylene to promote ripening of fruits, ethylene oxide to sterilize spices and air to oxidize ripe olives for colour development.

The inert gas mostly used in food industry is carbon dioxide for carbonation of soft drinks, beer and some wines and fruit juices so that the products become effervescent, tangy and tart.

Gases such as nitrogen, nitrous oxide and carbon dioxide are used as propellants to dispense liquid, foam or spray products. Liquified propellants such as freon C-318 and freon 115 are used to dispense whipped creams and foamed toppings.

8.4.13 Preservatives

These are food additives which are specially added to prevent the deterioration or decomposition of a food. Microorganisms, food enzymes or chemical reactions may cause the deterioration in the quality of food. The inhibition of the growth and activity of microorganisms is one of the main purposes of the use of chemical preservatives. Preservatives may inhibit

microorganisms by interfering with their cell membranes, their enzyme activity or their genetic mechanisms. Other preservatives may be used as antioxidants to hinder the oxidation of unsaturated fats, as neutralizers of acidity, as stabilizers to prevent physical changes, as firming agents, and as coatings or wrappers to keep out microorganisms, prevent loss of water, or hinder undesirable microbial and chemical reactions.

Ideally, a chemical preservative should have a wide range of activity and should not be inactivated by the food or any substance present in the food. It should not aid or encourage the development of resistant strains and should kill rather than inhibit the microorganisms. At the same time it should be non-toxic to human beings or animals, economical, with no effect on the flavour, taste or aroma of the original food. So far, such an ideal preservative has not been found. Most of the preservatives are inhibitory at acceptable concentrations, and only ethylene oxide, propylene oxide and diethyl pyrocarbonate are lethal to microorganisms at normal concentrations of use. Many food products need to remain stable for some time after the package has been opened, e.g. jam and ketchup, and therefore they need the preservative to be present in the food itself.

The antimicrobial preservatives may be classified into two broad categories as (i) naturally occurring preservatives such as organic acids (lactic, malic and citric acids and their salts), vinegar (acetic acid), sodium chloride, sugars, spices and their oils, wood smoke, carbon dioxide and nitrogen and (ii) substances generally recognized as safe (GRAS) for addition to foods such as propionic acid and its sodium and calcium salts, caprylic acid, sorbic acid and potassium, sodium and calcium sorbates, benzoic acid and benzoates and derivatives of benzoic acid such as methylparaben and propylparaben, sodium diacetate, sulphur dioxide and sulphites, potassium and sodium metabisulphite and sodium nitrite.

Organic acids and their salts. Lactic, acetic, propionic and citric acids and their salts may be added to or developed in foods during fermentation. Citric acid is used in syrups, drinks and jellies as a substitute for fruit flavours and for preservation. Lactic and acetic acids are added to brines of various kinds and green olives. Sodium propionate and calcium propionate are used most extensively for the prevention of mold growth and 'rope' inhibition in bread and other baked goods and for mold inhibition in many cheese products and spreads. They are effective against molds but have no inhibitory effect on most yeasts and bacteria. Their effectiveness decreases with an increase in pH with an optimal upper limit being pH 5 to 6, depending on the food item. They are ideal preservatives for bread and baked goods because even after the destruction of molds by heat during baking, recontamination of bread can occur during slicing and wrapping. They can be added to the dough of yeast-raised baked goods, as they do not inhibit yeast activity and leavening of bread.

Sodium benzoate has been incorporated in jams, jellies, margarine, carbonated beverages, fruit salads, pickles and fruit juices as an antimicrobial agent at concentrations of 0.05 to 0.1%. It is ineffective at pH values near neutrality and the effectiveness increases with increase in acidity. It is most effective in the pH range of 2.5 to 4.0. Benzoate is excreted by humans as hippuric acid (N-benzoyl glycine) in urine and hence is not accumulated in the body. Two esters of p-hydroxybenzoic acid, methyl paraben and propyl paraben, are used in many foods, and effective at higher pH values. They are primarily antifungal agents and also active against yeasts.

Sorbic acid salts of calcium, sodium and potassium are used directly (fungistatic agents) in foods and as spray or dip coating on packaging materials. They are used in cheeses, cheese products, baked goods, beverages, syrups, fruit juices, jellies, jams, fruit cocktails, dried fruits, pickles and margarine up to 0.3%. They inhibit yeasts and molds but are less effective against bacteria. They are most effective at low pH values with a maximum effectiveness at about pH 6.5. They are more effective compared to sodium benzoate at pH values above 4.0.

Acetic acid in the form of vinegar is used in mayonnaise, pickles, and pickled sausages. It is more effective against yeasts and bacteria than against molds. At lower pH values it is more effective as undissociated acid. Sodium diacetate ($CH_3COOH.CH_3COONa.½H_2O$) has been used in cheese spreads, ketchup, mayonnaise, acid-pickled vegetables, bread and malt syrups and for treatment of wrappers used on butter.

Other organic chemicals. *Sugars.* Sugars, such as glucose or sucrose function as preservatives because of their ability to lower the water activity, by tying up water molecules and providing osmotic effect. Foods preserved by high sugar concentrations include sweetened condensed milk, fruits in syrups, jellies and candies.

Alcohols. Ethanol, a coagulant and denaturizer of cell proteins, is most germicidal in concentrations between 70 and 95%. Flavouring extracts, e.g. vanilla and lemon extracts are preserved by their high content of alcohol. The alcoholic content of beer, ale and unfortified wine is not high enough to prevent their spoilage by microorganisms, but limits the growth of these organisms. Liqueurs and distilled liquors usually contain enough alcohol to ensure freedom from microbial attack. Methanol is poisonous and should not be added to foods. The trace quantities ingested during smoking of foods are not enough to be harmful. Glycerol is antiseptic in high concentrations because of its dehydrating effect but is not important in food preservation. Propylene glycol has been used as a mold inhibitor and as a spray to kill airborne microorganisms.

Ethylene oxide. Ethylene oxide kills all microorganisms but propylene oxide is not as effective. These have been primarily used as sterilants for

packaging materials, fumigation of warehouses and cold sterilization of numerous plastics, chemicals, pharmaceuticals, syringes and hospital supplies. They have been tried successfully in dried fruits, dried eggs, gelatin, cereals, dried yeast and spices. The use of ethylene oxide has however been restricted to walnut meats, copra and whole spices. Propylene oxide is permitted only as a packaging fumigant for dried fruits, cocoa, gums, spices, starch and processed nutmeats (but not peanuts).

Formaldehyde. Formaldehyde is not permitted as an additive to food but is one of main constituents in wood smoke used for preservation of food. Since it is effective against molds, bacteria and viruses, it is used in the sanitation of walls, shelves and floors.

Inorganic salts. *Sulphur dioxide and sulphites.* Sulphur dioxide and sulphites are used in the wine industry to sanitize equipment and to reduce the normal flora of the grape must (unfermented grape juice) at concentrations up to 100 ppm to achieve a differential effect. The desirable wine yeast *Saccharomyces cerevisiae* is able to grow and ferment sucrose to ethanol while some undesirable yeast species such as *Kloeckera apiculata* as well as lactic acid producing bacteria are suppressed. Many strains of wine yeasts produce sulphur dioxide themselves by the reduction of sulphate in the grape juice. When wine is bottled, further quantity of sulphur dioxide up to 500 ppm is added to prevent secondary fermentation of the residual sugar in the bottle. In aqueous solutions, sulphur dioxide and various sulphites, bisulphites and metabisulphites of sodium and potassium form sulphurous acid which is the active antimicrobial agent. Most light coloured dehydrated fruits are treated by exposure to the fumes of burning sulphur. Vegetables are exposed to a spray of neutral bisulphites and sulphites before drying. Sulphites also prevent enzymatic and nonenzymatic changes or discolouration of foods.

The disadvantages of the sulphites as preservatives include (i) the destructive reaction of bisulphite with the vitamin, thiamine, e.g. potatoes which are often stored in sulphite solutions lose a large proportion of their thiamine content and (ii) an unagreeable flavour and easily detectable disagreeable odour at concentrations higher than 500 ppm.

Nitrates and nitrites. Nitrates and nitrites (sodium and potassium salts) have been used in curing solutions and curing mixtures for meat. Nitrites decompose to nitric oxide which forms nitrosomyoglobin by reacting with the heme pigments in meat, giving it a stable red colour. Nitrites can react with secondary and tertiary amines to form nitrosamines which are known to be carcinogenic. Hence the permitted level of nitrites in foodstuff has been gradually reduced to about 50 μg/g. Moreover, ascorbic acid is added to curing salt mixtures to prevent the formation of nitrosamines. Nitrites have been shown to prevent the growth of *C. botulinum* in meat products, particularly in bacon and canned or processed hams. The effect of residual nitrite in ham

in inhibiting toxin formation is known as *Perigo effect*, named after its discoverer. During cooking, the residual nitrite breaks down to nitrogen oxide which is known to be a specific and potent inhibitor of at least one enzyme, pyruvate: ferredoxin oxidoreductase that has an essential role in energy metabolism and therefore, inhibits the growth and toxin production of anaerobes such as *C. botulinum.*

Salt. Salt tends to tie up moisture and thus has an adverse effect on microorganisms. Salt is used in brines and curing solutions, or is applied directly to the food. A large quantity may be added to slow down or prevent the growth of microorganisms. Sometimes only small quantities may be added to permit an acid fermentation to take place. The effectiveness of salt as a preservative varies directly with its concentration and the temperature. Salt functions as a preservative through several mechanisms. The various mechanisms include (1) creation of high osmotic pressure and hence plasmolysis of microbial cells, (2) dehydration of the food by drawing out and tying up water molecules, and dehydration of microbial cells, (3) harmful effect of chloride ion on organisms, (4) reduction of the solubility of oxygen in the moisture and sensitization of the cells against carbon dioxide and (5) interference with the action of proteolytic enzymes.

Woodsmoke. The smoking of foods is practised mainly for two purposes: adding desired flavour and preserving food. Other desirable effects may also be obtained, e.g. improvement in the colour of inside of meat and in the finish or gloss of the outer surface and a tenderizing action on meats. The smoking process helps preservation by impregnating the food near its surface with chemical preservatives from the smoke. The combined action of heat, preservatives and drying at the surface is responsible for the preservation action.

Smoke is obtained from the burning wood, preferably wood like hickory, but it may be generated from burning corn cobs or other materials. Oak, maple, beech, birch, walnut and mahogany wood are also used particularly for apples. Sawdust is added to the fire to give a heavy smudge. Temperature and humidity are controlled to remain at certain levels favourable to the product being smoked, and the duration of smoking depends upon the kind of food. Smoking temperatures for meats vary from 43 to 71°C and the period lasts from a few hours to several days.

Wood smoke consists of two phases, a disperse phase of liquid droplets and a continuous gas phase. The gas phase of smoke contains over 200 different volatile compounds that have bacteriostatic and bactericidal effect. Formaldehyde is the most effective of these compounds. Other compounds include phenols, cresols, aliphatic acids from formic to caproic acids, primary and secondary alcohols, ketones, acetaldehyde and other aldehydes, waxes, resins, guaiacol and its methyl and propyl isomers, catechol, methylcatechol and pyrogallol and its methyl ester. Smoke also

contains carcinogens such as benzpyrenes and other polynuclear hydrocarbons. These various compounds of wood smoke are grouped under the name *pyroligneous acid*. Wood smoke is more effective against vegetative cells than against bacterial spores and the rate of germicidal action of the smoke increases with its concentration and the temperature and varies with the kind of wood employed. The application of liquid smoke (solution of chemicals similar to those in wood smoke) to the outer surface of foods has little or no preservative effect although it contributes to flavour.

Spices and condiments. Spices and condiments do not have any bacteriostatic effect in the concentrations usually used but may help other agents in preventing the growth of microorganisms in food. Mustard flour and volatile oil of mustard are very effective against *Saccharomyces cerevisiae* but not against most bacteria. Cinnamon and cloves containing cinnamic aldehyde and eugenol, respectively, are more bacteriostatic than other spices. Extracts of horseradish, garlic, onion, cabbage and turnip are inhibitory to *Bacillus subtilis* and *E. coli*.

Antibiotics. Only a few antibiotics are permitted to be used as food additives. Nisin, a polypeptide produced by certain strains of *Streptococcus lactis* and occurring naturally in milk and Cheddar cheese, is a nontoxic and heat stable antibiotic. It has good storage stability, does not cause any off-flavour or off-odours and is destroyed by digestive enzymes in humans. It has been used to suppress anaerobes in cheese and cheese products since it makes the cheese immune to spoilage from gas forming *Clostridia* which causes cracks and blow holes in the chese. Nisin is most effective against Gram-positive organisms and ineffective against Gram-negative organisms. Hence it is used as a mopping-up preservative for heat processed foods such as canned food. Aureomycin (chlortetracycline) and terramycin (oxytetracycline) are used up to 5 ppm for preserving fresh fish and unpeeled shrimp by dipping. Natamycin, also known as pimaricin, tennecetin and myprozine, is a polyene isolated from *Streptomyces natalensis* and is effective against yeasts and molds but not bacteria.

Other chemical preservatives. Chlorine and hypochlorites of calcium and sodium are used for treatment of drinking and process water and for cleaning the processing equipment. Iodine impregnated wrappers are used to wrap fruits. Phosphoric acid is used in some soft drinks, e.g. colas. Hydrogen peroxide is used as preservative in conjunction with heat in pasteurizing milk used for cheese manufacture. Borax is used to wash vegetables and whole fruits such as oranges.

8.4.14 Flavour Enhancers

These compounds enhance the aroma of a food apparent to the senses as 'feeling', 'volume', 'body' or 'freshness' particularly in thermally

processed foods. Monosodium glutamate (MSG) has a pleasant, salty-sweet taste and a property often described as 'mouth satisfaction'. It promotes sensory perception of meat-like aroma and is used as additive in frozen, dehydrated or canned fish or meat products. Inosine monophosphate (IMP) and guanosine monophosphate (GMP) have properties similar to that of MSG and find use in soups, sauces, canned meat or tomato juice at 75–500 ppm levels. Maltol (3-hydroxy-2-methyl-4-pyrone) has a caramel like odour and enhances the perception of sweetness of carbohydrate rich foods such as fruit juices, marmalades and jelly. Addition of maltol at 5–75 ppm allows a decrease of sugar content by about 15% without affecting the sweetness intensity.

8.4.15 Other Additives

These include colourants, nutrient supplements, e.g. vitamins, minerals such as iron, calcium and iodine and lysine. Non-nutritive dietary sweeteners are also used as food additives.

Questions

1. What are food additives? What are their legitimate uses?
2. Classify the different additives used in food industry, giving one example for each class.
3. How do bleaching and maturing agents function in bread making?
4. What are stabilizers and thickeners? Give examples.
5. What are IM foods? How is their moisture content maintained?
6. Write notes on (a) anticaking agents and (b) firming agents.
7. How are gases and propellants used in food industry?
8. What is a food preservative? What are its requisites?
9. Describe how organic acids and their salts function as preservatives.
10. How do sulphur compounds function as preservatives? What are their disadvantages?
11. What is Perigo effect?
12. How does wood smoke function as a preservative?
13. What is the use of ethylene oxide in food industry?
14. Write a note on antibiotics which find use in food preservation.

CHAPTER

9

Microorganisms Associated with Food

9.1 RELEVANCE OF MICROBIOLOGY

Food serves as an interacting medium between various living species because it is a source of nutrients for humans, animals as well as microorganisms. This is a natural consequence of cohabitation. Human and animal food is basically derived from plant and animal sources. Food fit for human consumption is also a medium for the growth and activity of microorganisms. Hence human food is always associated with a variety of microorganisms. Since the primary function of microorganisms is self perpetuation, they use the human or animal food as a source of nutrients for their own growth and activity. Microbial activity in a food can be beneficial in certain cases but in most cases it leads to deterioration of the food and renders it unfit for human consumption. Four aspects of microbial activity are of relevance to processing and preservation of food.

Fermented foods. Microorganisms can be used as processing aids in the production of fermented foods. New and modified foods with better shelf stability, palatability, flavour and organoleptic properties are produced by fermentation using specific microorganisms under controlled conditions.

Food chemicals from microorganisms. A variety of food chemicals and additives may be produced by fermentation involving select species of microorganisms. In addition, microorganisms themselves may be used as food. The biomass produced by fermentation can be harvested and used as a protein rich raw material for the formulation of foods.

Food poisoning and food borne diseases. Pathogenic microorganisms grow in the food utilizing the nutrients in the food and produce toxins which are detrimental to the health of the consumer when such

food is consumed. Food also serves as a vector or medium for certain pathogens that cause food infections and diseases.

Food spoilage. The metabolic activity of various microorganisms not only utilizes the nutrients in food but also causes the spoilage of food through undesirable enzymatic changes affecting the quality of the food. The enzymatic changes include the formation of products which contribute off-flavours and affect the organoleptic, textural and keeping qualities of the food.

9.2 BACTERIA, YEASTS AND MOLDS

Thousands of genera and species of microorganisms have been identified and classified. Several hundreds of these are associated in one way or other with food products. Many of them are of industrial importance as they find use in the production of new foods and food chemicals by fermentation and also in the preservation of food products. Microorganisms are capable of spoiling food and causing diseases. The microorganisms which are of importance in food microbiology include bacteria, yeasts and molds.

Bacteria are unicellular organisms of aerobic or anaerobic nature and exhibit many morphological forms. Three principal shapes have been well recognized, namely, spherical shape of cocci, rod shape of bacilli and spiral form of spirilla and vibrios. All bacteria associated with food are small in size, typically a few micrometers long and smaller in diameter. Bacteria form spores, which are seed like and far more resistant to heat, presence of inhibitory chemicals and other adverse conditions during food processing. Most bacteria multiply best at temperatures between 16 and 38°C and are termed mesophilic. Psychrotropic or psychrophilic bacteria can grow at low temperatures while thermophilic ones can grow at higher temperatures. Generally, under favourable conditions, bacteria can double their numbers every 30 minutes.

Yeasts are unicellular fungi of larger size compared to bacteria of oval, elongate, elliptical or spherical cell shape having length in the range of 20 μm and diameter of about 6 to 7 μm. These organisms produce pigments of many colours, with red and black pigment producers being common. Most common yeasts grow best in the presence of plenty of moisture. However, they also grow in the presence of relatively high concentrations of solutes such as salt, sugar or alcohol. Yeasts require more moisture compared to molds. The optimum temperature of growth for most yeasts is about 25 to 30°C and the maximum temperature range is about 40 to 47°C. Some are psychrotropic and can grow at 0°C or less. Yeasts grow better in acidic pH of 4 to 4.5 and do not grow well in alkaline conditions. They grow best under aerobic conditions but fermentative types can grow under anaerobic conditions, though slowly.

Yeasts are grouped into oxidative or fermentative types according to their function or activity. *Oxidative yeasts* are also called film yeasts and they oxidize organic acids and alcohol. *Fermentative yeasts* produce alcohol and carbon dioxide. Top yeasts carry out the fermentation of sugars to alcohol at the top of a vessel while bottom yeasts do the same function at the bottom of the vessel. Yeasts are used in the manufacture of foods such as bread, beer, wine and vinegar and for surface ripening of cheese. They are grown for producing enzymes and for use as food. Yeasts also cause spoilage of sauerkraut, fruit juices, syrups, honey, jellies, meat, wine, beer and molasses.

Molds are still larger and more complex in structure. They are multicellular filamentous fungi growing as a network of hair like fibres called mycelia with a fuzzy or cottony appearance. They form fruiting bodies and mold spores referred to as conidia.

Most molds are mesophilic with optimal temperatures in the range of 25 to 30°C while some such others as *Aspergillus sp.* grow well at 35 to 37°C or above. A few are thermophilic. A number of molds are psychrotropic and grow well at refrigerator temperatures while some molds can grow slowly at temperatures below freezing. Molds require less of available water for their growth and activity compared to bacteria and yeasts. Hence they tolerate low water activity. Molds are aerobic. They grow over a wide range of pH between 2 to 8.5 of the substrate but prefer acidic pH. Molds can utilize many kinds of foods including complex foods as they possess a variety of hydrolytic enzymes. Some molds produce compounds which inhibit the growth of other organisms, e.g. penicillin is produced by *Penicillium chrysogenum* and clavacin, by *Aspergillus clavatus*. Sorbic acid, propionates and acetates inhibit the growth of molds in foods and are called mycostatic.

Molds find use in the manufacture of certain foods such as Oriental food that includes soya sauce, sonti, miso, etc. Food ingredients such as amylase for bread making or citric acid for use in soft drinks are produced using molds. Some kinds of cheese such as Roquefort, Camembert, Gammelost, Brie, etc. are ripened by using molds. The black colour of spoilt bread and blue coloured veins of blue cheese are due to the formation of conidia. Some molds produce toxic metabolites called mycotoxins.

Bacteria, yeasts and molds attack virtually all the constituents of foods. Depending on their nature and availability of enzymes some ferment sugars and hydrolyze starches and cellulose while others hydrolyze fats and produce rancidity. Some organisms digest proteins and produce putrid and ammonia like odours. Some participate in the formation of acid and make the food sour, while others produce gas and make the food foamy, still others produce pigments that contribute to the discolouration of food. Some organisms produce toxins giving rise to food poisoning while some others cause food borne diseases through infection.

9.3 PRIMARY SOURCE OF MICROORGANISMS COMMONLY ASSOCIATED WITH FOOD

The primary source of microorganisms associated with food is the environment with which the species are associated. The sources include: (i) soil, water and air, (ii) plants and their products, (iii) animals and human beings and (iv) food handling equipment.

Soil, water and air. Food borne bacteria generally found in soil and water include species of the genera *Acinetobacter, Alcaligenes, Bacillus, Citrobacter, Clostridium, Corynebacterium, Enterobacter, Micrococcus, Pseudomonas,* and *Streptomyces*. Molds such as *Aspergillus, Rhizopus, Penicillium, Trichothecium, Botrytis,* and *Fusarium* are commonly found in soil and in water. A large number of yeast genera are associated with plants and hence may be found in soil, though their numbers in water is low. In addition, bacteria of genera *Bacillus* and *Micrococcus* and the yeast genus of *Torulopsis (Torula)* and many mold genera are also found in air and dust.

Plants and their products. Plant borne bacteria include the species of the genera *Acetobacter, Flavobacterium, Lactobacillus, Leuconostoc, Listeria, Pediococcus* and *Streptococcus*. Molds associated with plants cause spoilage of vegetables and fruits (causing diseases known as market diseases). These include *Botrytis cinerea, Geotrichum candidum, R. stolonifer* and *Phytophora sp*. Notable yeast species associated with plant products include genera *Saccharomyces, Rhodotorula* and *Torula*.

Animals and human beings. The intestinal tracts of humans and animals are primary sources of a variety of microorganisms, principally bacteria. These include *Bacteroides, Escherichia, Lactobacillus, Proteus, Salmonella, Shigella, Staphylococcus, Streptococcus, Clostridium, Citrobacter, Enterobacter* and *Pseudomonas*. The yeast, genus *Candida* is often found in the intestinal tract of man. In addition, there are several genera of bacteria that are specifically associated with the hands, nasal cavities and mouth of man. A food handler in a food processing plant can cause contamination of food through *Micrococcus and Staphylococcus sp*. Animal hides, animal feeds and fecal matter are also sources of microorganisms which can contaminate the food.

Food handling equipment. Food handling equipment and utensils can be a source of contamination of food if they have not been properly sterilized prior to use.

9.4 FACTORS INFLUENCING MICROBIAL ACTIVITY

The growth and activity of the microorganisms depend on the nature of the food and its composition. The factors that govern the microbial

activity in a food include both intrinsic and extrinsic factors. The *intrinsic factors* include pH, water activity, oxidation-reduction potential, nutrient content and presence of inhibitors in the food. The *extrinsic factors* include temperature, relative humidity and the atmosphere surrounding the food. These factors operate individually as well as in combination and affect the growth and activity of microorganisms in foods.

9.4.1 Intrinsic Factors

pH value of the food. Plant and animal foods have a natural pH in the range of distinctly acidic pH of about 2 to a neutral pH of about 7 as shown in Table 9.1.

TABLE 9.1 Approximate pH Values of Plant and Animal Foods

Plant food	pH	*Animal foods*	pH
Fruits		**Meat and poultry**	
Apples	3.0–3.3	Beef	5.1–-6.2
Bananas	4.5–4.7	Ham	5.9–6.2
Lime	1.8–2.0	Chicken	6.2–6.4
Orange	3.6–4.3	**Fish**	
Melon	6.3–6.7	Most species	6.6–6.8
Grapes	3.4–4.5	Tuna	5.3–6.1
Vegetables		Salmon	6.1–6.5
Cabbage	5.4–6.0	Shrimp	5.2–6.0
Carrots	5.0–6.0	Crabs	7.0
Cauliflower	5.6	Oysters	4.8–6.3
Onions	5.3–5.8	**Dairy products**	
Potatoes	5.3–5.6	Milk	6.3–6.5
Tomatoes	4.2–4.4	Butter	6.1–6.4
Beans	4.6–6.5	Cheese (Cheddar)	5.9–6.0

Every microorganism has an optimum pH at which its growth is maximum and has minimum and maximum pH values it can tolerate. In general, yeasts and molds are more tolerant to lower pH values compared to bacteria. Foods with relatively acidic pH of less than 4.5 are not usually affected or spoiled by bacteria. Thus fruits have pH values below those required by many spoilage organisms. This is attributed to the fact that the biological function of the fruit is to protect the plant's reproductive body, the seed. The pH of a living animal favours the growth and activity of most of the spoilage organisms. However, other intrinsic factors prevent microbial attack of a living animal. The keeping quality of meat from fatigued animals is low compared to that of well-rested animals

because of pH attained upon completion of rigor mortis. The final pH of the meat of a well-rested animal is lowered from 7.4 to about 5.6 because of the glycolytic activity and production of lactic acid. This is due to sufficient reserve of glycogen available in a well-rested animal, which is converted to lactic acid and such meat has better keeping quality.

Some foods have buffering capacity in that they resist changes in pH. In general, the buffering capacity of meat is greater because of the higher protein content compared to vegetables which lack buffering capacity. When microorganisms grow in acid media their metabolic activity results in the substrate becoming less acidic. Similarly, microorganisms growing in alkaline environments tend to bring down the pH of the substrate. However, microorganisms cannot survive in too acidic or too alkaline substrates.

Moisture content and water activity. The water requirements of microorganisms are generally expressed in terms of water activity of the substrate in which they grow. As already mentioned in the chapter on water, the water activity of most fresh foods is in the range of 0.9 to 0.99 and such an environment is highly conducive for the growth and activity of both spoilage as well as pathogenic microorganisms. Many preservation methods are in fact oriented towards decreasing the water activity in foods.

Oxidation–Reduction potential (O–R potential, Eh). Microorganisms are sensitive to the oxidation–reduction potential of the substrate. The O–R potential of a substrate is defined as the ease with which the substrate loses or gains electrons and is expressed by the symbol Eh. A substance that loses electrons readily is a good reducing agent while a substance which gains electrons is a good oxidant. When electrons are transferred from one compound to another, a potential difference exists between the two compounds which is expressed in millivolts (mV). A highly oxidized substance has more positive potential while the more reduced substance has a negative potential. When the concentrations of oxidant and reductant are equal the potential difference is zero. Fruit juices are highly oxidized substrates with Eh values in the range of +400 mV. Solid meat is a reducing medium with Eh values in the range of –200 mV while minced meat is an oxidizing medium with Eh values in the range of +200 mV. Aerobic microorganisms such as aerobic bacteria require positive Eh values (oxidized) for their growth while anaerobes such as *Clostridium* species require negative Eh values (reduced). Some aerobic bacteria grow better in slightly reduced conditions and these organisms are referred to as *microaerophiles*. Examples of microaerophilic bacteria include lactobacilli and streptococci. Facultative organisms are capable of growing under aerobic as well as in anaerobic conditions.

The O–R potential of a food is determined by (i) the characteristic O–R potential of the original food, (ii) the poising capacity, i.e. the

resistance to change the potential of the food, (iii) the oxygen tension of the atmosphere surrounding the food and (iv) the access of atmosphere to the food. Most fresh plant and animal foods have a low and well-poised O–R potential in their interior. The presence of reducing sugars and ascorbic acid in plants and fruits and the presence of –SH groups in meat are responsible for the low O–R potential. In living plant and animal cells these reducing substances tend to poise the O–R potential at a low level resisting the effect of oxygen due to respiration and diffusion from the outside. Thus a cut of fresh meat or a whole fruit would have aerobic conditions only close to the surface that supports the growth of aerobic microorganisms while the interior of the meat or fruit supports the growth of anaerobic bacteria. Processing of foods alters the poising power of food by destroying or altering the reducing or oxidizing substances and also may allow more diffusion of oxygen. Thus clear fruit juices do not contain the natural reducing substances which are lost during extraction and filtration and become substrates with more positive Eh values. Microorganisms affect the Eh values of foods during their growth. Thus aerobes consume oxygen and reduce the Eh values while anaerobes cannot affect the Eh values. In the presence of limited amounts of oxygen the same aerobic and facultative organisms may produce incompletely oxidized products such as organic acids from carbohydrates while in the presence of large amounts of oxygen, complete oxidation of carbohydrates to carbon dioxide and water occur. Similarly, protein decomposition under anaerobic conditions results in putrefaction while under aerobic conditions the products may be less obnoxious.

Nutrient content. Microorganisms require water, source of energy, source of nitrogen, vitamins, minerals and other growth factors for their growth and activity. Hence the nutrient content of food is an important factor in determining the type of organism that is likely to grow in a given food. Each microorganism has a definite range of food requirements. For some species the range is wide and growth occurs in a variety of food substrates, while for most of the pathogenic organisms the range of food requirements is narrow and hence they can grow only in a limited number of foods. Most microorganisms use carbohydrates, especially fermentable sugars and other carbon compounds such as alcohols, esters, organic acids, amino acids and peptides as a source of energy. Only a few microorganisms are capable of utilizing complex carbohydrates such as pectin, cellulose and starch. Similarly, only a few organisms are capable of using fats and proteins as sources of energy. Microorganisms differ in their ability to use even some of the soluble sugars. Most organisms are capable of using only glucose. Organisms, which cannot use lactose, do not grow well in milk. Bacteria are identified and classified on the basis of their inability to use various sugars and alcohols.

Microorganisms utilize nitrogenous compounds such as ammonia, urea, amino acids, peptides and proteins for their growth. They differ in

their ability to use various nitrogenous compounds as a source of nitrogen for their growth. Microorganisms require B group vitamins and minerals in low quantities. Most natural foods have these constituents, but foods which lack a particular vitamin or mineral will not support the growth of microorganisms which depend on such nutrients for their growth. Processing and long-term storage of foods usually reduces the vitamin content.

Presence of inhibitory substances. The presence of inhibitory substances (antimicrobial constituents) in the food also affect the growth of microorganisms. The antimicrobial substances may be present naturally as in the case of spices, which have essential oils with antimicrobial characteristics. Milk contains antimicrobial constituents lactoferrin, conglutinin and lactoperoxidase. Eggs contain lysozyme and conalbumin. Similarly, vegetables and fruits also have antibacterial and antifungal constituents. The antimicrobial agents may also be added deliberately during food processing.

9.4.2 Extrinsic Factors

For the growth of the microorganisms certain external factors should be favourable.

Temperature. Microorganisms grow over a wide range of temperatures. The lowest temperature at which a microorganism grows is about –34°C and the highest temperature is about 90°C. The various microorganisms are usually grouped on the basis of their temperature requirements for growth. Thus organisms that grow well below 20°C and have optimum temperature in the range of 20 to 30°C are referred to as psychrophiles or psychrotrophs. Organisms that grow well between 20 and 45°C with optimum temperature in the range of 30 and 40°C are called mesophiles. Thermophiles are those organisms that grow well at temperatures above 45°C with optimum temperature in the range of 55 to 65°C.

The type and quality of the food must be taken into account for selecting a storage temperature. For example, bananas keep better at 13 to 17°C while a large number of vegetables store better at about 10°C. In addition, the relative humidity of the storage chamber has to be maintained at desired levels together with temperature control for effective preservation of foods against microbial spoilage.

Relative humidity of environment. The relative humidity of the storage environment must be such that excessive drying of the food or absorption of moisture by the food does not occur.

Gaseous atmosphere surrounding the food. It is possible to inhibit surface spoilage of foods by a controlled atmosphere surrounding the food. Increased amounts of carbon dioxide in the atmosphere by about 10%

retards fungal rotting of fruits while higher percentage of carbon dioxide is effective with meat, fish and other animal products. Addition of ozone has been found to be effective against several spoilage microorganisms. Ethylene has been found to be effective in maintaining the natural resistance of fruits against fungal attack.

9.5 IMPORTANCE OF BACTERIA IN FOODS

Bacteria that play significant roles in foods are often grouped on the basis of their activity in foods without regard to their systematic classification.

Lactic acid bacteria ferment sugars to lactic acid and include species belonging to genera of *Leuconostoc, Lactobacillus, Streptococcus* and *Pediococcus*. Their activity is desirable in a variety of foods such as sauerkraut and other pickled vegetables and dairy products for the production of flavour. They cause spoilage of wines.

Acetic acid bacteria oxidize ethanol to acetic acid. Species of genera *Acetobacter* and *Gluconobacter* are the most common. *Acetobacter* species are capable of oxidizing acetic acid to carbon dioxide. They are useful in vinegar manufacture but are undesirable in alcoholic beverages.

Butyric acid bacteria are mostly the spore forming anaerobes of the genus *Clostridium*. They produce butyric acid by fermenting sugars. Propionic acid bacteria produce propionic acid and belong the genus *Propionibacterium*.

Proteolytic bacteria include a heterogeneous group of bacteria which produce extracellular proteases. The species include aerobic, facultative and spore forming organisms. Most species belong to the genera of *Clostridium, Bacillus, Pseudomonas* and *Proteus*. *Bacillus cereus* is an aerobic, spore forming organism while *Pseudomonas fluorescens* is non-spore forming, aerobic as well as facultative organism.

Lipolytic bacteria are also a heterogeneous group of bacteria which produce lipases. Many of the aerobic proteolytic bacteria are also lipolytic. Organisms of the genera *Pseudomonas, Alcaligenes, Staphylococcus, Serratia* and *Micrococcus* are lipolytic.

Saccharolytic bacteria hydrolyze disaccharides and polysaccharides to simpler sugars. *Bacillus subtilis* and *Clostridium butyricum* are examples of saccharolytic bacteria which are also amylolytic, i.e. produce amylase to hydrolyze starch outside the cell.

Pectinolytic bacteria produce pectinases responsible for softening of plant tissues or loss of gelling power in various plant foods. Species belonging to genera of *Bacillus, Achromobacter, Aeromonas, Arthrobacter* and *Flavobacterium* are pectinolytic.

Thermophilic and thermoduric bacteria are resistant to high temperatures. Thermophilic bacteria cause spoilage of low acid canned foods. Important species include *Bacillus stearothermophilus* and *C. thermosaccharolyticum*. Thermoduric organisms survive heat treatment

such as pasteurization. Bacteria of genera *Bacillus, Clostridium, Micrococcus, Streptococcus, Lactobacillus* and *Microbacterium* are found in foods.

Psychrotrophic bacteria are able to survive and grow at refrigeration temperatures though their optimum temperature of growth is around 20 to 30°C. In contrast, psychrophilic bacteria have optimum temperature of growth at refrigeration temperatures. Psychrotrophic bacteria spoil non-sterile refrigerated foods such as meat, fish, poultry, eggs and other foods. Species belonging to genera of *Pseudomonas, Achromobacter, Alcaligenes* and *Flavobacterium* are psychrotrophic.

Halophilic bacteria include species of the genera *Bacillus, Micrococcus, Vibrio, Moraxella, Halobacterium, Corynebacterium, Streptococcus and Clostridium.* These organisms require certain minimal concentrations of dissolved sodium chloride for their growth and survive at higher concentrations of salt to different extents depending on whether they are slightly, moderately or extremely halophilic or halotolerant species.

Osmophilic or saccharophilic bacteria grow in high concentrations of sugar, e.g. *Leuconostoc sp.*

Pigmented bacteria produce colours during their growth in foods. Examples include species belonging to genera *Flavobacterium* (yellow to orange), *Serratia* (red), *Halococcus* (red to orange) and *Halobacterium* (pink, red and orange). *Lactobacillus plantarum* produces rust colour pigment discolouring cheese. *Flavobacterium* species cause discolouration on the surface of meat and spoilage of shellfish, poultry, eggs, butter and milk.

Slime or rope forming bacteria include *Alcaligenes viscolactis, Enterobacter aerogenes,* and *Klebsiella oxytoca* and some species of *Streptococcus* and *Lactobacillus* which cause slime or ropiness in milk. *Lactobacillus plantarum* causes ropiness in various fruit, vegetable and grain products such as cider, sauerkraut and beer while some species of the genus *Bacillus* cause ropiness in bread. Meat curing solutions become slimy due to the activity of *Micrococcus. Gluconobacter* species includes *G. oxydans*, which causes ropiness in beer.

Gas forming bacteria include species of the genera of *Leuconostoc, Lactobacillus* and *Propionibacterium* which produce carbon dioxide. Species of *Escherichia, Enterobacter, Proteus, Bacillus* and *Clostridium* produce both carbon dioxide and hydrogen. *Clostridium* species ferment carbohydrates with the production of butyric and other acids and gases (carbon dioxide and hydrogen). The species are found commonly in soils and involve in the gaseous spoilage of canned vegetables and putrefaction of foods. *C. perfringens* causes stormy fermentation of milk.

Off-flavour forming bacteria include those of genus *Streptomyces* which produce undesirable flavours and musty or earthy odour and taste. Many species of *Pseudomonas* produce a variety of metabolites that affect the flavour of foods deleteriously.

Coliform bacteria are aerobic and facultatively anaerobic non-spore forming bacteria which ferment lactose with gas formation. They cause spoilage of a variety of foods producing off-flavours and sliminess. The important species include *Escherichia coli* and *Enterobacter aerogenes*. *E. coli* counts in foods are used as indicators of fecal pollution.

Several strains of the genus *Bacillus* are important as test organisms in sterility testing of food products. *B. subtilis* and *B. stearothermophilus* are recommended for testing steam sterilization procedures. *B. subtilis* is also the test organism for detecting penicillin in milk. *B. pumilus* is recommended for determining the suitability of gamma radiation sterilization. *B. subtilis* var. *niger* is recommended for ethylene oxide sterilization testing.

Some of the important disease causing bacteria include the following.

C. botulinum produces a neurotoxin in canned meat products and causes the fatal disease botulism.

Corynebacterium species includes the diphtheria organism *C. diptheriae*.

Erwinia species are plant pathogens and damage plants and plant products causing bacterial soft rot.

E. coli species includes some serotypes which are pathogenic to humans.

Mycobacterium species includes the tubercle bacilli *M. tuberculosis* that causes tuberculosis especially through raw milk from infected cows.

Salmonella species are enteric pathogens that grow in foods and cause food infection.

Shigella species is transported by foods and causes bacillary dysentery.

Staphylococcus species includes the important *S. aureus* which produces an enterotoxin causing food poisoning.

Streptococcus species includes pathogenic *S. agalactiae* which causes mastitis in cows and *S. pyogenes* which causes septic sore throat, scarlet fever and other diseases in humans.

Vibrio species is pathogenic to humans.

9.6 IMPORTANCE OF YEASTS IN FOODS

Saccharomyces species are the most widely used yeasts. The leading species *S. cerevisiae* is used in the manufacture of many foods, with special strains used for the leavening of bread and for the production of ale, wine, alcohol, glycerol and invertase. Top yeasts are active fermenters and grow rapidly at 20°C. The clumping of the cells and the rapid evolution of carbon dioxide sweep the cells to the surface and hence the name *top yeasts*. *Bottom yeasts* do not clump and settle to the bottom, hence the name bottom yeasts. They grow more slowly and ferment at lower temperatures of 10–15°C.

S. cerevisiae var ellipsoideus is a high alcohol yielding variety used in the production of industrial alcohol, wine and distilled liquors. *S. uvarum* is a bottom yeast used in beer manufacture. *S. Fragilis* and *S. lactis* have the ability to ferment lactose and hence are important in milk products.

Zygosaccharomyces species are osmophilic and are involved in the spoilage of honey, syrups and molasses. They find use in the fermentation of soya sauce and some wines.

Candida species spoil foods high in acid and salt content. *C.utilis* is grown for food and *C. krusei* is used with dairy starter cultures to maintain the activity and longevity of lactic acid bacteria. *C. lipolytica* spoils butter and oleomargarine.

Brettanomyces species are involved in the fermentation of Belgian and English beers and in French wines, while *Kloeckera* species are found commonly on fruits and flowers and in soil. *Trichosporon* species (*T. pullulans)* grow at low temperatures and are found in breweries and on chilled beef.

Rhodotorula species cause discolouration of foods by forming coloured spots on meat and sauerkraut.

9.7 IMPORTANCE OF MOLDS IN FOODS

Even though *Mucor* species are involved in the spoilage of some foods, they find use in the manufacture of certain foods such as Oriental foods and ripening of cheese (e.g. Gammelost). *M. rouxii* is used for the saccharification of starch. *Rhizopus stolonifer*, the so-called bread mold is a common food spoilage mold involved in the spoilage of fruits, vegetables, berries and bread. *Thamnidium elegans* spoils refrigerated meat by forming whiskers.

Aspergillus is a widespread species. Some of the *Aspergilli* are involved in spoilage of foods, e.g. *A. glaucus* and *A. repens*. The widespread *A. niger* species is used for the manufacture of citric and gluconic acids and also a variety of enzyme preparations. *A. flavus-oryzae* group includes molds useful in the making of Oriental foods and production of enzymes. Some molds of this group are also spoilage organisms.

Penicillium genus is widespread in occurrence and some of the species cause spoilage of foods while certain others are useful. *P. expansum* causes soft rots of fruits. *P. digitatum* and *P. italicum* cause rotting of citrus fruits. *P. camemberti* and *P. roqueforti* are used in ripening of cheeses. Other molds belonging to genera such as *Trichothecium, Botrytis* (causes gray mold on many plants and plant foods), *Trichoderma, Cladosporium* (produces black spots on beef), *Endomyces, Fusarium* (causes neck rot of banana and spoilage of other fruits and vegetables) and *Sclerotinia* cause spoilage of foods.

Questions

1. What is the relevance of microbiology in food processing and preservation?
2. Write a note on the different types of microorganisms commonly found in food.
3. What are the sources of microorganisms associated with food?
4. Write in detail about the various intrinsic factors which influence microbial activity.
5. What is the influence of O–R potential of a food on the activity of microbes?
6. What do you understand by the term poising capacity of a food with respect to O–R potential?
7. What are extrinsic factors of importance in influencing microbial acitivity?
8. Name a few bacteria which are useful in food processing. Give details.
9. What are pathogenic bacteria? How do they cause food borne illness?
10. How do microorganisms affect the flavour of foods?
11. Write notes on the use of the following in food processing: (a) yeasts and (b) molds.

CHAPTER

10

Fermented Foods and Food Chemicals

10.1 FOOD FERMENTATION

Fermentation involves the use of microbial metabolic process in which carbohydrates and other nutrients are oxidized partially to a variety of breakdown products such as alcohols, acids, amino acids, other metabolites and antibiotics and a small amount of energy. Fermentation is carried out for (i) the modificiation/production of new or desired form of foods such as cheese, bread, idli, dosa, tofu, coffee, alcoholic beverages, soya sauce, pickled vegetables such as sauerkraut and meat sausages such as salami, pepperoni, bologna, etc. (ii) production of flavours as in the case of cheese, butter milk and other dairy products, (iii) preservation of foods as in the case of sauerkraut, cucumber, carrot, olive, mixed vegetables, green tomatoes and hot cherry pepper pickles and (iv) production of food chemicals and additives such as proteins, enzymes, vinegar, amino acids, citric acid, etc.

Fermented foods have several advantages such as appetizing flavour and aroma as well as textural characteristics. They also have longer shelf life compared to the stability of starting materials from which they have been derived. The digestibility of such foods is also greater. In certain cases vitamin content is enriched due to microbial fermentation and hence nutritional content is enhanced.

The process involves mostly modification and preservation of vegetable and animal products by natural acidification mainly due to the formation of lactic acid from fermentable sugars.

Microorganisms used in food fermentation may be added as pure or mixed cultures as in the manufacture of dairy products such as fermented milk, butter and cheese, bread, malt beverages, wine, distilled liquor and vinegar. In some cases, the desired microorganism may be present in

sufficient numbers in the original raw material as in the manufacture of sauerkraut, fermented pickles, green olives and processing of cocoa, coffee, poi and citron.

The steps involved in food fermentation include: (i) selection of cultures, (ii) maintenance of activity and purity, (iii) preparation of starter cultures and (iv) fermentation.

Cultures are selected for food fermentation primarily on the basis of their ability to produce the desired products or changes in the raw food material, their efficiency and their stability. The selection process involves screening a number of naturally existing or mutant strains or genetically engineered species.

Once a strain with the desired traits has been identified and selected, it is maintained in pure and active condition by periodic transfer of the culture into proper culture medium, whereby incubation till maximal stationary phase of growth is attained and this is followed by storage at low temperatures to prevent further growth. Stock cultures are usually prepared by lyophilization (freeze drying) or freezing in liquid nitrogen (–196°C). Bacterial cultures may be preserved for months, even extending to years at room temperature on slants of agar in which 1% of NaCl had been added. The purity of the stored cultures is checked periodically by microscopic examination as well as specific tests.

The starter culture called the mother culture is usually prepared daily from a previous mother culture or from the stock culture. The mother culture is used to inoculate a large culture medium to prepare the bulk culture for use in fermentation. Cultures used for fermentation should contain only the desired microorganism in sufficient and uniform numbers, which is highly active as indicated by the rate of its growth and production of desired products and should have adequate resistance to unfavourable conditions so that the uniformity and consistency of the end product is ensured. Mixed cultures of microorganisms may be grown together or grown separately and mixed at the time of use.

Fermentation is carried out by a combination of selective and elective conditions to achieve the desired results. Selective conditions are those in which the growth of the unwanted organisms is selectively inhibited by the presence of an inhibitory agent such as salt or acid. Elective conditions are those in which the growth of the desired organism is selectively promoted by proper temperature, anaerobic conditions and nutrients that can be used only by the starter cultures but not by the unwanted organisms.

10.2 MICROBIAL CULTURES USED IN FOOD INDUSTRY

10.2.1 Bacterial Cultures

Bacterial cultures are used as starters in the manufacture of dairy

products, bread and sausage. Most of them are mixed cultures of lactic acid bacteria. Propionic acid bacteria is used in the production of Swiss cheese and acetic acid bacteria in vinegar manufacture.

Lactic acid bacteria belonging to the four genera of *Lactobacillus, Leuconostoc, Pediococcus* and *Streptococcus* are commonly used. Other species related to lactic acid bacteria include *Aerococcus, Microbacterium* and *Propionibacterium*. The lactic acid bacteria have a common trait of fermenting hexoses to lactic acid. These bacteria lack amylases and hence cannot utilize starch or other polysaccharides. The lactic acid bacteria are subclassified into two types: (i) homofermentative or homolactic and (ii) heterofermentative or heterolactic. Homolactics produce lactic acid as the sole or major product of glucose fermentation. All species of *Pediococcus* and *Streptococcus* are homolactic. Others include species belonging to the genus *Lactobacillus* such as *Lb.acidophilus, Lb. bulgaricus, Lb. casei, Lb. coryniformis, Lb. delbrueckii, Lb. helveticus, Lb. lactis* and *Lb. plantarum*. Homolactics possess aldolase and hexoseisomerase enzymes but lack phosphoketolase. They use Embden-Meyerhof-Parnas (EMP) pathway to produce two lactate molecules per glucose molecule and four ATP molecules. Heterolactics include certain species belonging to the genus *Lactobacillus* such as *Lb.brevis, Lb. bruchneri, Lb. fermentum* etc. and *Leuconostoc* species such as *L.cremoris L.lactis,* and *L.mesenteroides*. They possess phosphoketolase but lack aldolase and hexoseisomerase and hence use hexose monophosphate or pentose pathway-producing equimolar amounts of lactate, CO_2 and ethanol and two molecules of ATP as end products.

The lactic acid bacteria require amino acids, B vitamins and purine and pyrimidine bases for their growth and activity. They are mesophilic but some of them are capable of growing at temperatures as low as 5°C and some as high as 45°C. Most of the species can grow in pH range of 4.0–4.5 while some can grow up to pH 3.2, and a few, even at pH 9.6. The bacteria are weakly proteolytic and lypolytic.

Propionic acid bacterial cultures consist of lyophilized or spray dried cultures of *Propionibacterium freudenreichii* used in the manufacture of Swiss cheese to improve flavour and assist in 'eye' formation.

Acetic acid bacterial cultures consist of pure cultures of *Gluconobacter* or *Acetobacter*. Since they are not efficient, impure mixed cultures are allowed to develop naturally during vinegar manufacture.

10.2.2 Yeast Cultures

The most widely used yeast is of genus *Saccharomyces* and the species is *S. cerevisiae*. Baker's yeast is manufactured from single cell isolates of *S. cerevisiae* and grown in a medium of cane or beet molasses containing mineral salts, ammonium salts, urea, malt sprouts, inorganic phosphates,

accessory growth substances and vitamins. The pH is adjusted to 4.3 to 4.5 and incubated at 30°C under aeration.

Yeasts for malt beverages are derived from specific strains as bottom yeast for beer and as top yeast for ale. The top yeasts used are strains of *S.cerevisiae* and for bottom yeast, *S.uvarum* is used. For wine making, a special strain of *S.cerevisia* var. *ellipsoideus*, adapted to the making of the specific type of wine is selected. Distiller's yeast is a high alcohol yielding strain of *S.cerevisiae* var. *ellipsoideus*, adapted to growing in the chosen medium or mash to be used for fermentation. The fermentation medium is malted grain of corn or rye for whisky, molasses for rum, or juices/ mashes of fruits for brandy. The liquors are obtained by distillation of the fermented mashes.

10.2.3 Mold Cultures

Spores of *Penicillium roqueforti* are used for manufacture of blue, Roquefort and Stilton cheeses while *P. camemberti* spores are used for manufacturing Camembert and Brie cheeses. In the manufacture of soya sauce, mixed cultures of the mold *Aspergillus oryzae* together with a yeast and a bacterial culture of *Lactobacillus delbrueckii* are used.

10.3 FERMENTED DAIRY PRODUCTS

Milk is a highly unstable food as it is a very good medium for the growth of many microorganisms. Fermentation of milk is primarily aimed at preservation by conversion into more stable, nutritious and desirable products such as cheese, yoghurt, butter milk and butter. Milk from cows, sheep, goats and other mammals may be used directly for fermentation or after preliminary treatment such as pasteurization, ultrafiltration or lactose hydrolysis. The fermentation may be initiated spontaneously by native microflora or by addition of specific starter cultures or material from previous fermentation. The fermented milk may undergo further processing to yield a more concentrated product or may be blended with other ingredients such as salts, fruits, herbs, spices, sugars and natural colourants to change flavour, appearance and texture of such products.

The microorganisms used in the industrial scale of fermentation of cow's milk include mainly lactic acid producing bacteria such as the mesophilic *Lactococcus* and *Leuconostoc* species with optimum growth temperatures in the range of 20 to 30°C and thermophilic *Lactobacillus* and *Streptococcus* species used at temperatures up to 45°C. Other bacteria such as *Corynebacterium, Propionibacterium*, yeasts such as *Saccharomyces* and molds such as *Penicillium camemberti* and *Penicillium candidum* are also used.

The nutrients in cow's milk include lactose (~4.5%), protein (~3.5%), fat (~4%) and citrate which are utilized by the microorganisms. Lactose is converted into lactic acid homolactically or into lactic acid, carbon dioxide and ethanol heterolactically. The milk proteins are hydrolyzed to individual amino acids by the proteolytic enzymes of the starter culture. The amino acids contribute to the flavour either by themselves or by yielding aldehydes, amines, alcohols and various sulphur compounds via dicarboxylation, tansamination, deamination and desulphurization reactions. For example, *Propionibacterium* produces proline containing peptides and proline is considered to be an important contributor to the sweetness in Swiss type of cheese. Another protein metabolic product is ammonia, responsible for the flavour of Camembert cheese. *Penicillium sp.* produces hydrogen sulphide, dimethyl sulphide and methane thiol from methionine. The flavour of yoghurt is due to acetaldehyde derived from the degradation of threonine catalyzed by threonine aldolase of *Lb. bulgaricus*. Citrate is metabolized by *Lb. lactis, L. cremoris* and others to an intermediate product α-acetolactate, which is decarboxylated to diacetyl, the characteristic flavour compound of butter, buttermilk and cottage cheese. In addition, carbon dioxide is also an end product of citrate metabolism contributing to the formation of 'eyes' in certain types of cheese such as Gouda cheese. The milk fat is not utilized directly by starter cultures of lactic acid bacteria as they are weakly lipolytic. But other organisms constituting the secondary flora hydrolyze milk fat to liberate free fatty acids.

10.4 FERMENTED MEAT PRODUCTS

The sterile tissues of animals on slaughtering become contaminated by spoilage and pathogenic Gram-negative bacteria such as *E. coli, Salmonella* species, *Pseudomonas sp.* and *Clostridium perfringens* and also Gram-positive species. As a measure of preservation of meat, its water activity is reduced by salting or curing by addition of nitrite or drying. Acid fermentation of meat is carried out to enhance the stability, texture, colour and flavour of the product by choosing appropriate starter cultures. In the manufacture of fermented meat products, starter cultures mainly of *Pediococcus cervisiae*, *Lactobacillus plantarum* and *Micrococcus* are used. The majority of fermented products may be classified into dry sausages with a moisture content of 25 to 45% (a_w ~0.91) and semi-dry sausages with a moisture content of ~50% (a_w ~0.95). Meat curing compounds and starter cultures are mixed and stuffed in casings at low temperatures of about –5°C and incubated at a temperature between 20 and 40°C for fermentation. The fermented product is dried at 10–20°C. Acid formation during fermentation also quickens the process of drying. Examples of dry fermented meat sausages include pepperoni and salami while bologna and summer sausages are semi-dry fermented meat

sausages. Fermented poultry sausages include dry as well as semi-dry turkey sausages.

10.5 FERMENTED VEGETABLE PRODUCTS

The major fermented vegetable foods include sauerkraut from cabbage and pickles from cucumber, carrots, mixed vegetables, green tomatoes and olives. Gram-negative bacteria are commonly found on vegetables in large numbers while lactic acid bacteria are less in numbers. Fermentation basically involves the repression of Gram-negative bacteria by salting and stimulation of the growth of lactic acid bacteria by anaerobiosis and low water activity. Heterolactic *Leuconostoc mesenteroides* and *Lactobacillus brevis* as well as homolactic *Lactobacillus plantarum* and *Pediococcus cerevisiae* are the naturally present lactic acid bacteria in cabbage. Temperature and salt concentration control the activity of the type of lactic acid bacteria in the primary and secondary stages of fermentation. At a salt concentration of 2.25% and low temperatures (~7.5°C), heterolactics predominate yielding sauerkraut of superior flavour and colour. Higher temperatures favour the more acid resistant homolactics.

Cucumber and olive pickles are made by fermentation at higher salt concentrations in the range of 5–8% and the fermentation is mainly brought about by the homolactic *Lactobacillus plantarum*.

10.6 FERMENTED ORIENTAL FOOD PRODUCTS

Oriental food fermentations are primarily based on solid state or solid substrate fermentation. Solid substrates of wheat, rice or soybeans are fermented aerobically in the absence of free water. However, water content varies between 20 and 80% depending on the water binding properties of the substrate. Since the nutrients in the substrate are in complex forms of starch and proteins, only those organisms capable of degrading these substances are capable of growth in these substrates. Fungi with their enzymes of amylases and proteases are capable of utilizing the nutrients in such substrates. The different types of solid state fermentation may be classified into five types depending on the steps involved: (i) fungal fermentation followed by brine processing as in production of soya sauce, shoyu, (ii) one-step bacterial fermentation to produce pastes with meat flavour such as natto (from soya), (iii) acid soaking followed by fungal fermentation as in tempeh production from coconut, peanut or soya, (iv) lactic acid bacterial fermentation of doughs as in the production of idli or dosa from rice and minchin from wheat and (v) fermentation with yeast or other fungi to yield alcoholic foods such as tape ketella from cassava or tape ketan from rice.

10.7 OTHER FERMENTED FOODS

These include bread and beverages. Bread making uses baker's yeast for achieving leavening and aroma due to the formation of carbon dioxide and alcohol. The manufacture of stimulating beverages such as coffee, tea and cocoa and alcoholic beverages such as wine, beer and ale involves fermentation spontaneously or by the use of specific starter cultures as an important step.

10.8 FERMENTATION FOR FLAVOUR PRODUCTION

A variety of organisms are used exclusively for flavour production in foods. The organisms and the flavour compounds produced in foods are listed in Table 10.1.

TABLE 10.1 Microorganisms Used in Food Processing and Flavour Compounds Produced

Food item	*Microorganism*	*Flavour compounds produced*
Buttermilk	*Streptococcus lactis* *Streptococcus cremoris*	Lactic acid, diacetyl, small amounts of acetaldehyde
	Lactobacillus bulgaricus	Acetaldehyde
Yoghurt	*Streptococcus thermophilus* *Lactobacillus bulgaricus*	Acetaldehyde and diacetyl Acetoin
Alcoholic fermented milk	*Saccharomyces sp.* *Lactobacillus sp.*	Ethanol Acetoin and diacetyl
Sauerkraut	Mixed cultures of *Lactobacillus brevis,* *Leuconostoc mesenteroides* *Lactobacillus plantarum*	Acetate and small amounts of short chain fatty acids
Soybean milk	*Lactobacillus sp.* *Streptococcus thermophilus*	Aldehydes including pentanal
Soya sauce (Japan)	*Aspergillus oryzae* *Lactobacillus sp.* *Saccharomyces rouxii*	Organic acids, alkyl phenols and pyrazines
Tempeh (Indonesia) (Wheat based)	*Rhizopus sp.*	Fatty acids
Cocoa	*Saccharomyces sp.* *Lactobacillus sp.* *Acetobacter sp.*	Fatty acids and aromatic acids
Cheese	*Mucor meihei*	
Swiss cheese	*Propionibacterium shermanii*	Propionic acid
Bread	*Saccharomyces cerevisiae*	Ethanol

10.9 MICROORGANISMS AS FOOD—SINGLE CELL PROTEIN

Microorganisms themselves may serve as food fit for human consumption or animal feed. In addition, they find use in the production of chemicals, proteins and enzymes required in food processing industries.

The term *single cell protein* (SCP) refers to microbial cells grown and harvested primarily for use as human or animal food. The development of a microorganism as a direct primary food for human consumption is based on the conversion of the abundant carbohydrates (starch and sugar) of plants into protein. SCP offers a possible solution to meet a shortage of protein under unusual conditions when agriculture and fishing may not be able to produce sufficient quantities of protein so as to meet the demand. Any new protein food must satisfy the conditions of safety and organoleptic characteristics fit for human consumption. A number of microorganisms have been identified as potential food for humans and animals. Bacteria, yeasts, molds and algae may be used for production of SCP, but till today, only yeasts have been produced commercially and used as food. Recently *Spirulina maxima*, an algae is being grown commercially as a nutrient rich food. Organisms may be grown directly for the purpose of food (primary organisms) or they may be recovered as by-product of a fermentation (secondary organisms). Advantages of SCP include the inherent high protein content (about 60 to 70% of the cell), rapid increase in cells in a short time and independence from climatic conditions for growth.

Yeasts are widely used for SCP production because of certain advantages such as better public acceptance, lower content of nuclei acid, easier harvesting due to size and concentration and better growth characteristics in substrates of low pH. Strains of *Candida utilis* (torula yeast) grow rapidly, utilizing pentoses and hexoses and are capable of synthesizing accessory nutrients required for growth even from comparatively poor culture media. *Candida lipolytica* strains are used for SCP production using alkanes and gas oil as substrates. Other yeasts such as *Rhodotorula* and *Saccharomyces* and fungi such as *Aspergillus, Penicillium* and *Fusarium* have also been used for SCP production. A typical SCP called mycoprotein, capable of being formulated into a variety of processed foods, has been produced by using *Fusarium graminearum* .

10.9.1 Mycoprotein Production

The organism *Fusarium graminearum,* a filamentous fungus capable of growing on starch was first isolated from soil samples (in Marlow, Buckinghamshire, UK) by a process of elective culture. The soil sample was dried and exposed to p-dichlorobenzene vapours to remove mites. The sample was suspended in sterile distilled water in a flask for about 30

minutes. A small quantity of the suspension was inoculated into petri dishes containing Malt Extract Agar (MEA) with antiobiotic oxytetracycline added to suppress bacterial growth. A number of decimal dilutions of the suspension were also inoculated into petri dishes containing MEA and all the plates were incubated at 30°C until the colonies were well developed. The fungal strains (about 30 in number) were isolated from the soil sample by this procedure. The colonies were purified by removing a number of spores with a sterile loop and streaking the spores on fresh MEA plates. This procedure was repeated until only one mold of one morphological type was present on these isolation plates which were grown subsequently in shake flask cultures.

The flow chart for the production of mycoprotein is shown in Figure 10.1. The steps involved include: (i) biomass production using a

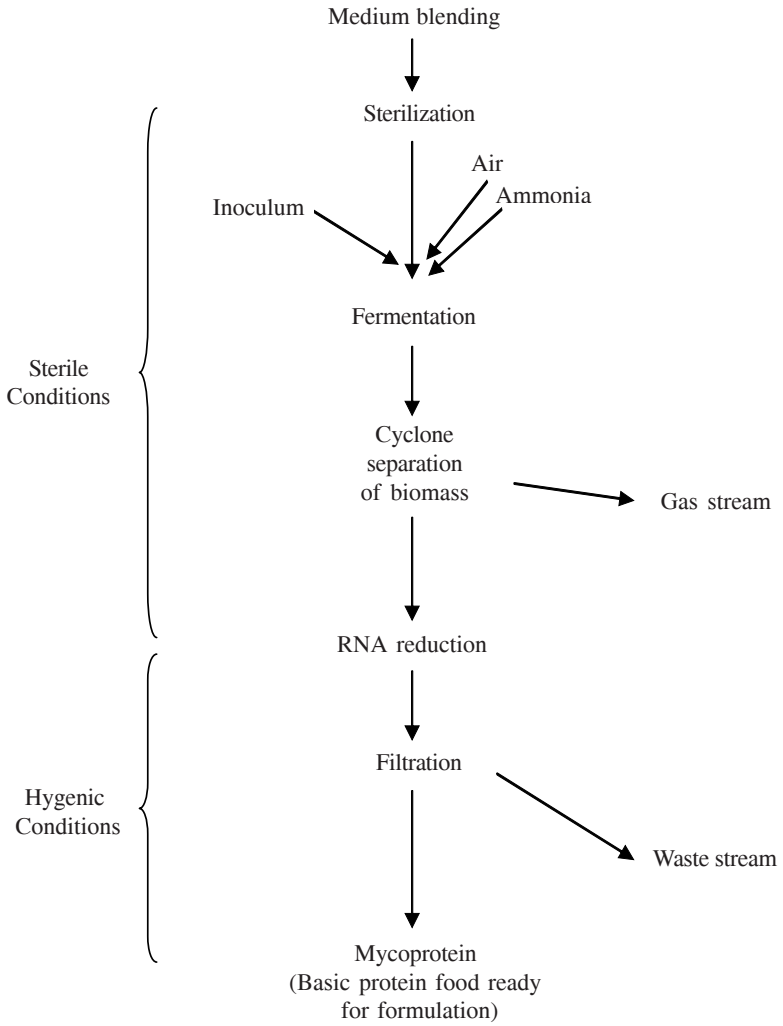


Figure 10.1 Flow chart for the production of mycoprotein.

continuous culture of the microorganism, (ii) reduction of the RNA content of the biomass and (iii) recovery of biomass by separation of the mycelium and formulation of the mycoprotein into a food product.

Biomass production. The organism is grown under steady state conditions for periods of several weeks in a CSTR type fermenter. The culture medium consists of carbohydrate nutrient (hydrolyzed starch derived from potato, wheat or maize or molasses after removal of inorganic salts and gums) and other constituents such as ammonium phosphate, sulphates of potassium, magnesium, iron, zinc, manganese and copper, biotin, choline chloride, sulphuric acid (10%) and ammonia gas (fed directly to the fermenter via air stream). The blended culture medium is sterilized and continuously pumped into the fermenter together with metered amounts of air and ammonia gas. The medium is then inoculated and fermented, the biomass formed leaving the fermenter at a rate equal to the inflow of culture medium. The biomass is separated in a continuous cyclone separator into a slurry stream (allowing the gas stream to escape), which is passed on to an aseptic RNA reduction stage.

RNA reduction. The RNA reduction of the biomass increases the protein percentage of the final product. The reduction is achieved by using the endogenous Rnase of the cell to degrade its own RNA. A heat shock given to the cells initiates a series of complex responses which include the inactivation of protease and Rnase inhibitor, disruption of ribrosomes to release of RNA and degradation of RNA into nucleotides and nucleosides by Rnase. The cell loses its viability. The degradation products of RNA diffuse out through the cell wall. The RNA content is thereby reduced to about 1% from its initial value of about 10% on a dry weight basis. In addition, compounds with low molecular weight, such as amino acids and TCA cycle intermediates also diffuse out from the cell resulting in a total loss of about 35% of the initial cell dry weight. Because of this elimination of other intermediates the protein percentage in the final product is effectively higher compared to the protein content in the intact cells.

The RNA reduction is carried out aseptically in a stirred vessel provided with a steam jacket. Sterile air is supplied to the vessel head space to maintain a positive pressure. The optimum temperature for RNA reduction process is about 64°C. The pH of the culture medium usually remains unaltered at 6 during this process.

Recovery of biomass. The RNA reduced slurry is concentrated from about 1% to about 30% on dry weight basis by mechanical filter presses or centrifuges. The filter cake, called the mycoprotein, is similar to any moist perishable food and can be stored at 25°C in closed unsealed container for about 16 hours or at 4°C for about 48 hours. The product may be spray dried for storage at room temperature for months or even years. The mycoprotein is amenable for formulation into a variety of foods.

10.9.2 Brewers' and Bakers' Yeast Production

Secondary yeasts recovered as a by-joint product of alcohol fermentation include strains of brewers' yeasts, *Saccharomyces cerevisiae and S. carlsbergensis* and distillers' yeast, *S. cerevisiae* var. *ellipsoideus.* Brewers' yeast is debittered by washing with caustic soda solution and water, adjusted to pH of about 5.5 with phsophoric acid, salted, fortified with thiamine, riboflavin and niacin and finally dried in a drum dryer.

Bakers' yeast is usually single cell isolates of the strain *S. cerevisiae.* The culture is built up from the original mother culture through several intermediate cultures of increasing size to the final seed culture. The cells from the seed culture are concentrated into a cream by centrifugation and this heavy suspension is added to a build-up culture of molasses containing minerals, salts, and accessory growth substances. The pH is adjusted to about 4.5 and incubation temperature is about 30°C. The medium is aerated at a rapid rate and molasses is added gradually during the growth of the yeast. After four or five budding cycles, the yeast is centrifuged out in the form of a cream. The cream is filter press dried, mixed with small amount of vegetable oils and made into cakes. The cells in the cakes survive for some months in their ability to leaven the dough.

10.10 FOOD CHEMICALS

Amino acids. Amino acids are essential nutrients in foods, and are produced through various processes such as fermentation, enzymatic synthesis and bioconversion. Strains of *Corynebacterium, Arthrobacter, or Brevibacte-rium* lack the enzyme α-ketoglutarate dehydrogenase and hence glutamate accumulates in the cells due to transamination of α-ketoglutarate. Hence these organisms are used in the production of glutamic acid. Most of the amino acids are produced by enzymatic or bioconversion processes. Glutamic acid is used as a flavour enhancer (as monosodium glutamate, MSG) in food products. Alanine and glycine are used as flavours. Aspartic acid and phenylalanine find use in the production of the sweetener aspartame. Lysine, threonine and tryptophan are used as dietary supplements and cysteine as a reducing agent (antioxidant) in food industry.

Enzymes. The changes brought about in foods during microbial fermentation are primarily due to the various enzymes produced by the organisms. It is possible to isolate, purify and even immobilize the specific enzymes required for a desired function. Microbial enzymes produced by bacteria, yeasts and molds and their application in food industry is summarized in Table. 10.2

Lactic acid. Lactic acid is produced industrially by homofermentative lactic acid bacteria. The bacteria used depends on the raw material to be fermented. *Lactobacillus delbrueckii* produces lactic acid from glucose,

sucrose or maltose from molasses or a starch hydrolysate mash. Nitrogenous minerals and growth factors required by the bacteria are added in the form of malt sprouts, corn-steep liquor or milk. *Bacillus coagulans* and *Lactobacillus delbrueckii* subsp. *bulgaricus* are used for making lactic acid from whey. Anaerobic conditions, an optimum temperature of 45–50°C and a slightly acidic pH are maintained during fermentation which occurs over several days. Lactic acid formed is neutralized by the addition of calcium hydroxide or calcium carbonate and edible grade lactic acid is obtained from recrystallized calcium lactate. Lactic acid is used to acidify jams, jellies, confectionery, soft drinks and other extracts. It is also used as a preservative in brined pickles of olives and horseradish and fish. Its addition makes milk more digestible for infants. Calcium lactate is used as an ingredient in baking powder.

TABLE 10.2 Sources and Uses of Microbial Enzymes

Enzyme	*Source*	*Use*
Amylase	*Aspergillus oryzae*	Mashing of malt
	A. niger	Flour additive in bread making
	Bacillus subtilis, Rhizopus sp., Mucor rouxii	Syrup manufacture, processing of foods
Cellulase	*A. niger, Trichoderma viride*	Preparation of coffee concentrates
Dextransucrase	*Leuconostoc mesenteroides*	Stabilizing sugar syrups, ice creams, confections
Invertase	*Saccharomyces cerevisiae*	Artificial honey, prevention of crystallization of sugars in confections, ice creams
Lactase	*S. fragilis*	Hydrolysis of lactose in dairy products
Lipase	*A. niger, Mucor sp., Rhizopus sp.*	Flavour production in cheese
Pectinase	*A. niger, Penicillium sp., Rhizopus sp.*	Liquefaction, maceration and clarification of fruit juices and clarification of wine
Protease	*B. subtilis*	Bread making, meat tenderization
	A. oryzae	Prevents chill hazing in beer
Rennets	*Mucor miehei, M. pusillus*	Curdling of milk for cheese
Glucose oxidase	*A. niger*	Glucose removal from egg solids
Glucose isomerase	*Streptomyces sp.*	High fructose corn syrup

Citric acid. Citric acid is made by fermentation of beet molasses by the mold *Aspergillus niger*. Minerals such as iron, zinc and manganese are required by the organism in addition to ammonium salts or urea. Fermentation period lasts up to ten days under acidic conditions at temperatures in the range of 25 to 30°C. The yield of citric acid is maximum under conditions of a slight deficiency of nitrogen and phosphate foods. Citric acid is added to flavouring extracts, soft drinks and candies to decrease the pH, as a synergist with antioxidants for preservation of oils, as a dip coating material for sliced peaches to delay browning and to prevent discolouration of crab meat.

Vinegar. Vinegar is defined as a condiment made from sugary or starchy materials by an alcoholic fermentation followed by an acetous one. It must contain at least 4 g of acetic acid per 100 ml. Vinegar may be classified on the basis of the materials from which it has been made. Vinegar may be made from the juices of fruits such as apples (cider vinegar), grapes, oranges, from starch vegetables such as potatoes after hydrolyzing their starch into sugars, from malted cereals of barley, rye, wheat or corn (malt vinegar), from molasses, honey, or from waste alcoholic liquor.

Manufacture of vinegar involves two steps: (i) fermentation of sugar to ethanol followed by (ii) oxidation of alcohol to acetic acid. The first step is an anaerobic process carried out by *Saccharomyces cerevisiae* var. *ellipsoideus*. The second step involves aerobic oxidation by acetic acid bacteria of genera *Acetobacter* and *Gluconobacter*. The oxidation of the alcohol is carried out in a cylindrical tank of wood called generator. The interior of the tank is divided into three parts, an upper section, a middle section containing a slimy growth of acetic acid bacteria supported on beechwood shavings, corncobs, charcoal, coke or pomace and a bottom section. The alcoholic liquid from the first step of yeast fermentation is fed into the upper section and is allowed to trickle down over the supported acetic acid bacteria to undergo oxidation. Air enters through the middle section and is vented out at the top. The oxidation process is exothermic and hence control of temperature at about 30°C is necessary. The vinegar collects in the bottom section of the generator and undergoes finishing steps of (i) aging to improve in body, taste and bouquet, (ii) filtration or clarification to remove suspended materials and (iii) final pasteurization in bulk or in bottles at about 60 to 66°C for a few seconds before marketing.

Dextran and xanthan. Dextran is used as a stabilizer for sugar syrups, ice cream and confections and as a plasma extender. The gummy polysaccharide is made by *Leuconostoc mesenteroides* fermentation of molasses. Xanthan is a polysaccharide produced by *Xanthomonas campestris* grown in a glucose based medium. It is used as a stabilizer in food industry. It is not degraded by human or animal digestive system and is excreted as such.

Beta carotene. The microalgae *Dunaliella salina* is cultivated (similar to spirulina) as it is a rich source of beta carotene which is a natural antioxidant capable of neutralizing free radicals generated during oxidation reactions within the body.

Questions

1. What is food fermentation? What is its use?
2. Describe in detail the steps involved in food fermentation.
3. Write a note on bacterial cultures used in food fermentation.
4. How are yeasts and molds useful in food fermentation?
5. How are fermented dairy products obtained?
6. Write a note on the fermented meat products.
7. What are Oriental fermented foods? How are they produced?
8. Give examples of organisms used for flavour production in foods.
9. What is SCP? Why is it required?
10. Describe in detail the production of mycoprotein for use as food.
11. What are food chemicals? How are they produced?
12. Write a brief account of enzymes used in food processing and their microbial sources.
13. How is vinegar manufactured?

CHAPTER

11

Food Borne Diseases

The cause of food borne diseases is mainly the ingestion of infected food. Apart from gastrointestinal problems and related illness due to improper diet, nutritional deficiencies and overeating, the causes of food borne illness may be due to either (i) food infection or (ii) food intoxication. The entry of pathogenic organisms and parasites into the body and the reaction of the body to the presence of organisms or their metabolites cause food infection. Pathogenic organisms make their entry through the food chain i.e. by consuming foods contaminated with such organisms. Bacterial food infection refers to food borne illness caused by the entry of bacteria into the body. Food intoxication or poisoning is caused by consuming food contaminated with toxic substances.

11.1 FOOD INFECTIONS

Food infections may be broadly sub-classified into two types.

(i) Those in which the contaminated food does not ordinarily support the growth of the pathogenic organisms but merely carries them. Such food infections include diseases such as diphtheria, dysentery, typhoid, cholera, tuberculosis, infectious hepatitis, etc.

(ii) Food infections in which the food serves as a culture medium for the growth of pathogens to increase in number and cause infection when such food is consumed leading to diseases such as salmonellosis, shigellosis, gastroenteritis, etc.

Food infections can lead to outbreak of diseases in epidemic proportions. Some of the common food infections are detailed below.

11.1.1 Salmonellosis

The disease results due to ingestion of viable cells of *Salmonella sp*. The

organisms belonging to this genus are Gram-negative non-spore forming rod shaped bacteria. They ferment glucose with gas formation but do not ferment lactose or sucrose. They grow over a wide temperature range of about 10 to 45°C in good culture media in the pH range of 4 to 9. The minimum water activity level required is about 0.93. The organisms differ in their heat resistance depending on the type and the nature of the food. Holding the food at 66°C for about 12 minutes kills the organisms. The primary sources of the organism include human beings and animals (rats, mice) and infected animal food products. Most common sources include unrefrigerated meat and poultry products.

The disease requires an incubation period of about 12 to 36 hours to manifest the principal symptoms of gastrointestinal infection such as nausea, vomiting, abdominal pain, diarrhoea, greenish foul smelling stools, muscular weakness, moderate fever, restlessness, twitching and drowsiness. Mortality due to the disease is low. The duration of the illness varies from two to six days. The outbreak of this infection may be prevented by avoiding contact with infected sources such as dead animals, eliminating contaminated foods, cooking or heating the foods for sufficient time and providing adequate refrigeration for storing.

11.1.2 *Clostridium Perfringens* Gastroenteritis

The disease occurs due to the enterotoxin released in the gut during sporulation of the cells. The organism is Gram-positive anaerobic spore forming rod shaped bacteria. The organism grows in the temperature range of 20 to 55°C and in a pH range of 5 to 9. Growth of the organism is inhibited by using 5% NaCl. The spores of the food poisoning strains have considerable heat resistance. Raw foods and cooked meats which have been allowed to cool slowly and then held for some time before consumption, are the most common sources of the organism. Cooking the food destroys the cells but the spores of some strains have high heat resistance. Germination and growth of surviving spores occur in cooked foods which have been inadequately refrigerated. The symptoms of the disease appear in about 10 to 24 hours following ingestion of the contaminated food and include abdominal pain, diarrhoea and gas formation. The occurrence of the disease may be prevented by adequate heating (above 60°C) and rapid cooling of cooked meat and other foods, reheating leftover foods and good personal hygiene.

11.1.3 *Bacillus cereus* Gastroenteritis

Bacillus cereus is a Gram-positive aerobic spore forming rod shaped bacteria capable of growing in the temperature range of 10 to 49°C with a pH range of 4.9 to 9.3. Ingestion of contaminated food containing

relatively a large number of viable cells results in gastroenteritis of diarrhoeal or emetic syndrome. The incubation period for the appearance of the symptoms is about 2 to 8 hours, with nausea, abdominal cramps, vomiting and diarrhoea lasting a day or less. Foods involved include cereal products, mashed potatoes, vegetable sprouts, meat loaf, puddings and custards. The illness can be prevented by holding hot foods above 65°C, reheating leftover foods to at least 72°C and personal hygiene.

11.1.4 *E. coli* Infection

Hundreds of strains of *E.coli* are known and most of which are harmless and live in the intestines of healthy humans and animals. However, enteropathogenic *E. coli* (EEC) is implicated in human diarrhoeal diseases and food poisoning. Two types of human disease syndrome occur due to the ingestion of EEC. The enterotoxigenic illness is due to the enterotoxin produced by certain strains. The symptoms include diarrhoea, vomiting, dehydration and shock similar to cholera appear after an incubation period of 8 to 24 hours and the illness lasts a day or two. Certain other strains of EEC cause invasive illness resulting in fever, headache, abdominal cramps and watery diarrhoea similar to shigellosis. Preventive measures include cooking foods thoroughly, rapid chilling of foods, good sanitation, use of protected water and ensuring personal hygiene.

11.1.5 Shigellosis

The illness is bacillary dysentery caused by ingestion of moist foods such as milk, potato, shrimp, tuna, turkey, apple cider, etc. contaminated by *Shigella sp.* The symptoms appear after about four days and include fever, abdominal cramps, chills, watery stools containing blood, mucus or pus, headache, nausea and dehydration. Preventive measures include maintaining personal hygiene, control of flies, use of covered and purified water, maintaining sanitary conditions during preparation of foods and quick chilling of foods.

11.1.6 Other Food Borne Infections

These include: (i) yersiniosis (organism involved is *Yersinia enterocolitica* and other *Yersinia sp.*) caused by eating contaminated pork (pig is the main carrier), meat products, raw milk and leftover foods, (ii) Vibrio (organism involved is *V. parahaemolyticus*) caused by eating raw marine foods, shell fish, salt water fish, crustacea and other fish products, (iii) beta hemolytic streptococcal infection resulting in scarlet fever and

septic sore throat (organism involved is *Streptococcus pyogenes*) caused by consuming contaminated milk, ice cream, eggs, steamed lobster, potato salad, custard and milk or egg containing products.

11.2 VIRAL INFECTIONS

Food borne infections in which viruses have been implicated include poliomyelitis and infectious hepatitis.

11.3 INFECTIONS DUE TO FOOD BORNE PARASITES

Parasitic infections transmitted by foods include: (i) amebiasis (causative organism is *Endamoeba histolytica*) due to sewage contaminated water resulting in diarrhoea (may be fatal sometimes), (ii) trichinosis (causative organism is a nematode) due to raw or insufficiently cooked contaminated pork and pork products and meat products of whale, seal, or walrus resulting in nausea, vomiting, diarrhoea, fever, swelling of eyelids and laboured breathing (occasionally fatal) and (iii) illness with abdominal pain, vague discomfort and digestive disorders due to tapeworm infected beef, fish or pork. Most of the parasitic infections can be prevented by proper cooking of foods, quick freezing, using purified water and keeping personal hygiene.

11.4 FOOD INTOXICATION

Food poisoning or intoxication also results in food borne illness. Ingestion of contaminated food due to the presence of poisons or toxicants causes food poisoning. Food poisonings may be classified into four types depending on the type of poison contaminating the food: (i) bacterial, (ii) fungal, (iii) biological and (iv) chemical.

11.4.1 Bacterial Food Intoxication

Bacterial food intoxication refers to food borne illness caused by the presence of a bacterial toxin formed in the food. The intoxications include botulism and staphylococcal intoxication. Sometimes gastroenteritis caused by *Clostridium perfringens* and *Bacillus cereus* are also included as illness due to food intoxication because of the toxin release in the intestinal tract due to cell lysis.

Botulism. The disease is caused by the ingestion of food containing the neurotoxin produced by *Clostridium botulinum*. The organism is found in soil. It is a spore forming, gas forming anaerobic organism. Seven types

(Types A, B, C, D, E, F and G) have been identified on the basis of the serological specificity of the toxins produced by the organism. Toxins produced by Types A, B, E (commonly found in fish and fish products) and Type F have been identified to cause human botulism. Toxin of Type C causes botulism of fowls, cattle and other animals. Type D is associated with forage poisoning of cattle. Toxin production by the organism depends on the ability of the cells to grow and autolyze in a food. Meat, fish and low or medium acid canned foods support the growth and toxin production. The organism requires glucose or maltose for toxin production. The optimal conditions for the growth and toxin production include a temperature range of 26 to 35°C, moisture content greater than 30% and a pH above 4.5. Sodium chloride at concentrations greater than 8% inhibits growth and toxin production. The spores of the organism have a high heat resistance. The heat treatment required to destroy all the spores in a food varies with the type of the organism, kind of food, the numbers and age of spores, the nature of the medium and temperature in which the spores were originally formed. Studies have indicated that heat treatment conditions to destroy all the spores of the organism in a food are 360, 36 and 3.6 (~4) minutes at temperatures 100, 110 and 120°C, respectively.

The toxin, which is a protein of high potency, is absorbed in the small intestine and paralyzes the involuntary muscles of the body. The toxin is thermostable and can be destroyed by heat treatment at temperatures above 90°C. The toxin can be destroyed by gamma radiation of 7 to 8 Mrad. The toxin is unstable at pH values above 6.8.

Foods involved in causing botulism include inadequately heat processed canned foods of low acidity such as canned vegetables, fruits, fish and fish products, meat and meat products. The growth of the organism in some foods such as meat and proteinaceous low acid vegetables produces an obnoxious odour. Acidic foods and foods low in protein content may not show putrefaction or spoilage but may still contain the toxin. Proteolytic strains of *C. botulinum* cause food spoilage. As a preventive measure, all foods—raw as well as canned foods—which show indication of spoilage should be rejected. Canned foods that exhibit any pressure in the container should also be rejected.

Ingestion of food contaminated with the toxin, even in very small amounts, causes botulism. The incubation period for the appearance of typical symptoms of botulism is about 12 to 36 hours. The early symptoms include acute digestive disturbance followed by nausea and vomiting, diarrhoea, fatigue, dizziness and headache. Other symptoms include double vision and dryness of mouth and constriction of throat, swollen tongue and difficulty in swallowing and speaking. The toxin paralyzes the involuntary muscles and paralysis spreads to heart and respiratory system and death results due to respiratory failure. Administration of an antitoxin, particularly before the advent of symptoms, is the only known treatment for the disease.

Staphylococcal gastroenteritis. Certain strains of *Staphylococcus aureus* produce an enterotoxin which causes gastroenteritis or inflammation of the lining of the intestine. Most of the strains of the organism are facultative but grow better aerobically than anaerobically and induce coagulation of blood plasma. Some of the strains tolerate sodium chloride to about 10–20% (w/v) and nitrites. Such organisms can grow in meat curing solutions and even on cured meat. They are also tolerant to sugars (about 50% sucrose). Six serologically distinct toxins (A, B, C_1, C_2, D and E) differing in toxicity are known. Most of the food poisoning is due to Type A toxin.

The source of the organism is mostly human or animal. The nasal passage of humans, infected wounds and boils are laden with the organism. The optimal conditions for the growth of the organism and production of the toxin vary with the type of food and include a temperature range of 20–45°C with rapid growth occurring at close to 45°C, a pH range of 4.8 to 8.0 and a minimum water activity of 0.86. The organism, if present in large numbers in foods, may be killed by heating at temperatures above 60–65°C for about 15 minutes or by irradiating the food with about 0.4 to 0.5 Mrad of gamma radiation.

The enterotoxin is a protein with molecular weight of about 30,000 daltons. The toxin is produced at appreciable rate in the temperature range of 15 to 45°C with best rate at about 40°C. The presence of starch and protein in appreciable amounts in the food enhance toxin production. The toxin is quite stable towards heat. Normal cooking of foods, or pasteurization at 72°C for 15 seconds or ultra high temperature heating at 143°C for 9 seconds are not sufficient to render the toxin inactive and hence, deactivation requires higher temperature and longer duration. Further, toxins inactivated by heating at low-temperature can undergo reactivation in some foods.

Foods involved in causing staphylococcal poisoning include custard and cream-filled bakery goods, ham, poultry, meat and meat products, fish and fish products, milk and milk products, salads, puddings, pies, salad dressings and cream sauces. Low acid foods, leftover meat products and dressings kept out of refrigerator and heating foods for extended periods with improper control of temperature facilitate the growth of the organism and production of the toxin. Good sanitation and avoiding infected or spoiled raw foods, would save us from getting infected by the organism. Adequate refrigeration and adequate heat treatment can kill the organism and deactivate the toxin and thus prevent food poisoning.

Ingestion of contaminated food leads to gastroenteritis with symptoms such as salivation, nausea, vomiting, retching, abdominal cramps and diarrhoea. Blood and mucus may be found in stools and vomits in severe cases. The incubation period for the appearance of symptoms is about 2 to 4 hours, the illness lasting about a day or two with complete recovery. No treatment is necessary ordinarily, but in extreme cases saline solution may be given.

E. coli poisoning. The strain 0157:H7 causes infection resulting in bloody diarrhoea and kidney failure. Pre-school children and the elderly are susceptible to this type of poisoning due to toxins produced by the organism. Consuming contaminated food and water can be fatal. Improperly processed meat and unpasteurized fruit juices have been implicated in the occurrence of the poisoning.

11.4.2 Mycotoxins (Fungal Toxins)

Mycotoxins are fungal metabolites, some of which are toxic to many animals and potentially toxic to humans. Fungi include molds, yeasts, mildews, rusts, mushrooms and blights. Some are edible (e.g. mushrooms) while others are used in food processing. Species of the genera *Penicillium* and *Aspergillus* are known to produce mycotoxins. Mycotoxicosis is the syndrome resulting from the ingestion of mold contaminated food. Ergotism was quite common during Middle Ages particularly due to the fungus *Claviceps purpurea* infecting rye and other food grains. Other mycotoxins of importance include aflatoxins, patulin, ochratoxin, luteoskyrin, penicillic acid, etc.

Aflatoxins. Aflatoxins are produced by certain strains of the fungi *Aspergillus flavus* and *A. parasiticus* and other organisms. Optimal conditions for the production of aflatoxin include water activity of about 0.85 and temperature of 25–40°C. The two major aflatoxins have been designated as B_1 and G_1 because they fluoresce blue and green, respectively when exposed to long wavelength ultraviolet light. Other closely related aflatoxins designated as B_{2a}, G_{2a} and GM_1 and hydroxylated derivates of B_1 and B_2 such as M_1, M_2 and P_1 are also known. Aflatoxin B_1 is the most toxic one and the other aflatoxins are toxic or carcinogenic to fish, mammals and poultry. Many foods such as dairy products, bakery products, fruit juices, cereals, forage crops, peanuts and peanut meals, cottonseed meal, cowpeas, sorghum, sweet potatoes, etc. can support the growth of the molds if contaminated or inoculated. Peanuts, cottonseeds and corn are susceptible to fungal invasion, growth and production of toxins before the harvest itself and contamination of formulated food products can occur in large scale resulting in outbreaks of food poisoning.

Patulin. Patulin produced by several molds such as *Penicillium expansum, P. clavifome, P. patulum, Aspergillus clavatus, A. terreus,* etc. was first isolated as an antibiotic as it is effective against many bacterial species. It exhibits strong fungistatic activity and is also toxic to seeds and seedlings of sugar beets, corn, wheat, peas, tomatoes, cucumber and flax. At lower concentrations, it is toxic to mice and rats resulting in pathological changes in organs, while at higher concentrations it is fatally toxic to mice and rats. It is considered as a potential human carcinogen. Patulin has a relatively high heat resistance in that it is stable at 100°C

for about 15 minutes. Patulin, even if present in foods, is inhibited by several food ingredients and compounds such as peptone, glycine, methionine, asparagine, p-aminobenzoic acid, sodium sulphate, sodium thiosulphate and casein hydrolysates and hence patulin poisoning is rare.

Ochratoxin. Molds such as *Aspergillus ostianus, A. sclerotiorum, A. melleus, Penicillium cyclopium* and *P. commune,* produce ochratoxin which is toxic to some animals. Ochratoxin and another mycotoxin named citrinin produced by *P. viridicatum* and *P. palitans* act probably synergistically to result in nephrotoxicity in pigs, rabbits and rats by causing lesions in the kidneys of the animals. Foods such as wheat, rice flour and oats are involved in supporting the growth of toxin producing organisms.

Other mycotoxins. Other less-known mycotoxins with potential human intoxication include luteoskyrin, sterigmatocystin, penicillic acid, alimentary toxic aleukia, Roquefortine, etc. are mostly fungal metabolites which have been investigated with respect to their toxicity and carcinogenic activities in experimental animals.

11.4.3 Biological Toxins

A significant number of foods in their natural state contain toxic substances in small amounts. Normally the body is able to deal with regular but small amounts of such type of foods tuff. Biological food poisoning occurs when such foods of plant or animal origin is consumed in excess. Examples of such toxins and their effect on health are shown in the Table 11.1.

TABLE 11.1 Biological Toxins and Their Effects

S. No.	*Food involved*	*Toxin present*	*Effect on health*
1.	Almonds, lima beans	Cyanogens (produce cyanide)	Inhibition of respiratory system (possibly fatal)
2.	Alcoholic drinks	Ethanol	Vomiting, hangover, unconsciousness
3.	Black pepper, celery seed, nutmeg	Myristicin	Headache, cramps, nausea
4.	Green or sprouting potatoes	Solaine and chaconine	Stomach upsets, nervous effect
5.	Rhubarb leaves	Oxalic acid	Interference with calcium absorption
6.	Cabbage	Goitrogens	Interference with iodine absorption
7.	Raw beans such as soya	Protease inhibitors	Interference with protein digestion and absorption

TABLE 11.1 Biological Toxins and Their Effects *(Contd.)*

S. No.	*Food involved*	*Toxin present*	*Effect on health*
8.	Bread and cereal products	Phytic acid	Interference with iron and calcium absorption
9.	Raw red beans	Haemaglutenin	Clumping of red blood cells
10.	Mushroom	Amanitin	Inactivation of metabolic enzymes
11.	Shellfish	Alkaloid	Fatal
12.	Fish (tuna, mackerel)	A variety of toxins	Digestive problems
13.	Mustard	Sanguinarine	Dropsy (fluid retention)
14.	Tea, coffee, cola drinks	Caffeine	Diuretic and stimulant
15.	Cheese, red wine, yeast extract	Tyramine	Migraine, increased blood pressure
16.	Celery, parsley and parsnips	Psoralens	Genetic mutation

11.4.4 Chemical Poisoning

Food poisoning due to chemicals in the food is usually characterized by appearance of the symptoms within a short time after eating the poisoned food. Arsenic, antimony, cadmium, chlorinated hydrocarbons, copper, cyanide, fluoride, lead, zinc and nicotinic acid may enter foods from utensils, processing equipment, pesticide or insecticide spray residues on vegetables and fruits, from improper packaging materials or added accidentally in place of normal additives.

Questions

1. Distinguish between food infections and food intoxications.
2. What are food borne diseases? How are they caused?
3. What causes Salmonellosis? How can it be prevented?
4. Write a note on gastroenteritis and the causative agents.
5. What are food intoxications? Give examples.
6. How is botulism caused? How can it be prevented?
7. Name the fatal food borne diseases and the organisms responsible.
8. Give an account on fungal toxins in food.
9. What are biological toxins found in food? Give examples.

CHAPTER

12

Food Spoilage

12.1 FOOD FIT FOR CONSUMPTION

The consumability of a food is mostly judged on the basis of individual consumer's acceptance. Thus a food considered fit for consumption by a consumer may not be considered fit by another. A hungry or starving consumer might eat food which he may not eat under normal conditions. A tentative definition of the fitness of food may be stated as follows. A food product is fit for consumption, if a discriminating consumer with knowledge of the details of its production eats it on seeing the material itself. Conversely, the food is considered spoiled and unfit for consumption when such a consumer refuses it as food. Thus the definition of fitness of a food seems to be based more on the visual characteristics of the food and on the personal judgement of consumers. However, certain criteria are universally accepted for assurance of fitness of a food. The criteria include:

(i) The desired stage of development or maturity—fruits and vegetables should be at certain but differing stages of maturity or ripeness and poultry meat preferably from young birds.

(ii) Freedom from contamination—food must be free from contamination by flies, rodents, or insects. Handling of the food products in dirty equipment or by diseased workers is not acceptable. Foods such as vegetables grown with water contaminated with sewage or oysters grown in water mixed with sewage should be rejected.

(iii) Freedom from unacceptable changes in foods—microorganisms and endogenous food enzymes cause unacceptable changes as in the case of putrefied meat or cheese and over-ripened fruits and vegetables. In addition, the food should be free from undesirable chemical changes that occur due to environmental factors such as prolonged exposure to air or oxygen or fluctuations in temperature and humidity.

12.2 DETERIORATION OF FOOD QUALITY

Foods undergo deterioration to varying degrees in their organoleptic properties, nutritional values, safety and aesthetic appeal. The progressive deterioration of food ultimately leading to spoilage is a natural process and cannot be prevented but only delayed. From the moment food is harvested or slaughtered, it undergoes progressive deterioration in its quality and depending on the type of food, deterioration may be slow as in the case of seeds and nuts, or may be rapid rendering the food spoiled within a few hours as in the case of milk and meat.

Foods may be classified broadly into three groups based on their ease of spoilage.

(i) Stable or nonperishable foods which do not spoil normally, such as sugar, salt, flour and dry beans. Improper handling or storing will, however, spoil them.

(ii) Semiperishable foods which remain unspoiled for relatively long period under proper handling or storing conditions. These include potatoes and certain varieties of apples.

(iii) Perishable foods which spoil readily unless special preservative methods are adopted. Most of our daily foods belong to this group which includes meat, fish, poultry, milk, vegetables and fruits.

12.3 CAUSES OF FOOD SPOILAGE

Food is mostly subjected to physical, chemical and biological changes and these cause the deterioration in the quality and ultimately the spoilage of food. The major causes of food spoilage include: (i) microorganisms, their growth and activity, (ii) action of native enzymes, (iii) insects, rodents and parasites, (iv) chemical reactions of the constituents of food, (v) environmental factors such as temperature, moisture, air and light and (vi) time. These factors operate simultaneously affecting the quality of food. It is not possible to completely eliminate these factors and secure total prevention of food spoilage. Preservation methods attempt to minimize the effect of these factors and enhance the storage stability and maintain the quality of foods for prolonged periods.

12.3.1 Activity of Microorganisms

Microorganisms capable of spoiling food are available commonly in soil, water and air, on the skins of cattle, fruits and vegetables, on the feathers of poultry, on the hulls of grains, and shells of nuts, on the clothing and skin of handling personnel, on processing equipment and within the intestines and body cavities of animal and human body. However, they are not found within the healthy living tissues of plants and animals.

Microorganisms invade the plant and animal tissues through a break in the skin or penetrate through the skin that has been weakened due to disease or death. Most raw foods also contain a variety of bacteria, yeasts and molds and contamination increases their numbers and also introduces new kinds. Food spoilage involves *association* of microorganisms with each other. Microorganisms may be *symbiotic,* i.e. mutually helpful or grow together without hindering or aiding each other. They may be *antibiotic,* i.e. mutually antagonistic. Some are *synergistic,* which means, by growing together they bring about changes that neither could produce when growing alone. For example, the growth of *Pseudomonas syncyanea* alone in milk causes a light brownish tinge and the growth of *Streptococcus lactis* alone causes no change in the colour. However, the simultaneous growth of these two organisms results in a bright blue colour. Competition among the different kinds of microorganisms in a food supported by the type of food and environmental factors will usually result in the enhanced growth of one type at the cost of others ultimately resulting in the characteristic type of spoilage of food. In most cases, the microorganisms produce a *metabiotic* effect, i.e. one organism grows and makes the conditions favourable for the growth of a second organism. Most natural fermentations in raw foods are primarily due to metabiosis involving the succession of microorganisms. For example, raw milk at ambient temperatures supports acid fermentation by the growth of *Streptococcus lactis* and coliform bacteria until the acid formed inhibits their growth. In the next stage of fermentation, acid tolerant lactobacilli grow producing more acid. Then a film of yeasts and molds grow over the top and reduce the acidity facilitating the growth of proteolytic bacteria at a subsequent stage.

Molds are aerobic organisms and grow as fine threads or filaments forming a complex network or mat called mycelium which is visible by its cottony or fuzzy appearance on the surface of the food. Molds also form spores or seeds and these can be transported in air and thus can infect other foods. Meat, cheese and sweet foods are most susceptible foods for mold growth. They grow best at pH 4 to 6 and at temperatures close to 30°C. At lower temperatures their growth rate is slow but does not stop. However, psychrophilic molds grow at refrigerated temperatures while a few molds are thermophilic. Some of the food spoilage molds include the following. *Mucor racemosus; Rhizopus stolonifer* (bread mold) spoils bread, berries, fruits and vegetables. *Aspergillus sp.*, *Penicillium expansum* and *Penicillium digitatum* cause soft rot in fruits. *Sporotrichum* causes white spot in chilled meat. *Botrytis cinerea* causes diseases in grapes. *Cladosporium herbarum* causes black spots in many foods. *Sclerotinia* causes rots of vegetables and fruits and *Helminthosporium sp.* are plant pathogens. Certain molds produce poisons called mycotoxins, e.g. *Aspergillus flavus* grows on peanuts and cereals and produces aflatoxin.

Most of the preservation methods aim at controlling the growth and activity of spoilage and pathogenic microorganisms. This may be achieved

by eliminating contaminated raw food materials, by the use of high temperatures, chemicals, radiation to kill the organisms or by storing the food at low temperatures. In addition, a rigorous check on the hygienic conditions at different stages of food processing is also necessary.

12.3.2 Action of Native Enzymes

The activity of the endogenous enzymes in plant and animal foods is often intensified after harvest or slaughter due to lack of control mechanisms in the harvested plant food or slaughtered animal. For example, pepsin in a living animal helps in the digestion of protein but does not affect the intestine itself but when the animal is dead, pepsin does contribute to proteolysis of the organs containing it. Similarly, uncontrolled ripening of vegetables and fruits results in their spoilage. The enzymatic changes in foods are desirable to some extent as in the case of natural tenderization of meat during aging and ripening of fruits but continued enzymic activity beyond an optimum point becomes food deterioration due to rotting. The native enzymes may be inactivated by heat, radiation or by the use of specific chemicals.

12.3.3 Insects, Parasites and Rodents

Insects destroy cereal grains, fruits and vegetables by not only consuming the food but by contaminating the food. They also facilitate microbial attack on foods. Fumigation of stored plant foods with chemicals such as methyl bromide and ethylene oxide is carried out to control the insects. However, insect eggs persist in the food. In the case of cereal flours, the insect eggs are destroyed by throwing the flour with high impact against a hard surface in a centrifuge type machine called entolater. Parasites enter the human body mostly through poultry and hogs which have been improperly cooked. Rodents apart from consuming considerable quantity of food also contaminate the food through their droppings, urine and filth. Rodents are also carriers of pathogenic bacteria.

12.3.4 Chemical Reactions

The quality of foods deteriorate due to chemical reactions of the constituents of food. The unsaturated fatty acid components undergo oxidation due to exposure to atmospheric air giving rise to oxidative rancidity in fat rich foods. Free fatty acids may also be released due to hydrolytic reactions causing odour as well as undesirable changes in the texture of food. Losses of vitamins due to oxidation or light induced reactions also occur.

12.3.5 Environmental Factors

Environmental factors which affect the quality of food include temperature, moisture and humidity, air and light.

The temperature range over which much of the food is handled is about 10–40°C. Excessive heat denatures proteins, breaks emulsions, destroys vitamins, enhances the rates of chemical as well as enzymatic reactions and dries out food by removing moisture. On the other hand, excessive cold also spoils foods. The texture of fruits and vegetables allowed to freeze and thaw on the trees is disrupted. Skins crack rendering the food susceptible to microbial attack. Uncontrolled freezing also spoils liquid foods, e.g. milk tends to curdle and loses protein due to denaturation and the emulsion breaks separating the fat.

Moisture content in the food and humidity of the surrounding atmosphere also have an important role in the deterioration of the quality of food. Surface moisture causes lumping and caking of granular foods. Surface defects such as mottling, crystallization and stickiness are caused. Condensation of moisture on the surface of the food facilitates the growth and activity of microorganisms. Even in moisture proof packages, fruits and vegetables give off moisture from respiration and transpiration sufficient to support the growth of microorganisms.

Air and oxygen can have detrimental effects on Vitamins A and C, food colours, flavours and other food constituents. Oxidation reactions are facilitated due to the presence of air. The growth of aerobic organisms particularly molds is also supported.

Light destroys riboflavin, Vitamins A and C and also promotes light induced oxidation reactions affecting flavour and colour of foods. Light also causes destruction of proteins as in the case of milk.

12.3.6 Time

The quality of food remains at its peak for some time soon after its harvest or slaughter and thereafter as time progresses, the deterioration in the quality of the food also progresses. There are two simple options for maintaining the food quality, though temporarily.

(i) We should keep the food alive as long as possible, e.g. keeping fish, poultry, etc. alive without slaughtering till required. Similarly, fruits and vegetables may not be plucked till required. This option is, however, limited in its application.

(ii) The harvested or slaughtered food must be cleaned and cooled immediately. This delays the onset of deterioration of food quality but does not prevent it.

12.4 SPOILAGE OF VARIOUS FOODS AND FOOD PRODUCTS

The various types of foodstuff ranges from cereals, vegetables and fruits, dairy and poultry products, seafood and products made from raw food materials.

12.4.1 Vegetables and Fruits

Market diseases of vegetables and fruits are commonly caused by microbial spoilage. Bacterial soft rot of fruits and vegetables is caused by *Erwinia carotovora* and pseudomonads such as *P. marginalis*. The organisms break down pectins, giving rise to a soft, mushy consistency, sometimes a bad odour and a water soaked appearance. Vegetables such as onion, garlic, bean, carrot, celery, beet, lettuce, rhubarb, potato, cabbage, spinach, cauliflower, radish, turnip, tomato, cucumber, pepper and watermelon are affected by this disease. Sour rot or watery soft rot of vegetables is caused by *Geotrichum candidum* and other organisms, and vegetables thus affected, include bean, carrot, lettuce, cabbage, onion, garlic, radish, cauliflower, tomato and turnip. Rhizopus soft rot in carrot, potato, cabbage, cucumber, pumpkin, watermelon, radish and tomato is caused by *Rhizopus stolonifer* making the vegetables soft and mushy. Other market diseases include bacterial blight of celery (*P. apii*), black rot of cabbage and cauliflower (*Xanthomonas campestris*), ring rot of potatoes (*C. sepedonicum*), bacterial speck of tomatoes, green mold rot of citrus fruits (*Penicillium digitatum*), sour rot of tomatoes and citrus fruits (*Geotrichum candidum*), slimy brown rot of vegetables (*Rhizoctonia sp.*), brown rot of citrus fruits (*Phytophora sp.*), etc.

12.4.2 Cereals and Cereal Products

The low water activity of wheat, rice, rye, corn and other cereals protects them from microbial attack if stored properly. The cereal flours are also relatively protected by the action of bleaching agents used during milling operation. Under favourable conditions of water activity, bacteria of the genus *Bacillus* and molds grow. Many of the aerobic sporeformers produce amylase and hence are capable of utilizing starch in flours.

12.4.3 Bakery Products

The moisture content of commercially made bread is insufficient for the growth of most organisms except molds. *Rhizopus stolonifer* is the common bread mold causing the spoilage known as ropiness in bread,

particularly in bread stored in high humidity or in bread wrapped while still warm. The composition of cakes with high concentration of sugar does not allow bacterial spoilage. However, molds cause spoilage of cakes. The baking process is sufficient to destroy many of these organisms but icings, toppings, nuts and fruits added after baking can be the source of molds. The growth of molds in bread and cakes causes hardening of these items.

12.4.4 Dairy Products

Milk, cream, butter and cheese are easily susceptible to microbial spoilage. Milk is an excellent growth medium for a variety of spoilage organisms. Fresh raw milk and refrigerated raw milk contain many bacteria belonging to the genera of *Streptococcus, Leuconostoc, Lactobacillus, Pseudomonas, Bacillus, Microbacterium, Propionibacterium, Micrococcus, Proteus* and coliforms. The growth of *Alcaligenes viscolactis* in raw milk favoured by low temperatures causes ropiness and a slimy layer of bacterial products. Pasteurization eliminates many of the species. Heat resistant streptococci cause spoilage of pasteurized milk. They produce lactic acid by fermenting lactose and decrease the pH to about 4.5 causing curdling of the milk. Lactobacilli, if present, can grow and continue their activity below pH 4.

Butter is not highly perishable but still it undergoes microbial spoilage. Two important types of spoilage of butter include surface taint or putridity and rancidity. Putridity is caused by *Pseudomonas putrefaciens* on the surface of the butter producing organic acids, particularly isovaleric acid, causing bad odours. Rancidity occurs due to hydrolysis of butter fat resulting in the formation of free fatty acids. Lipase producing organisms *Pseudomonas fragi* and *P. fluorescens* are involved in bringing about rancidity. Other bacterial spoilage conditions in butter include malty flavour due to the growth of *S.lactis var. maltigenes*, skunk-like odour due to the growth of *Pseudomonas mephitica* and black discolouration due to *P. nigrifaciens*. Fungal spoilage of butter is also common due to the activity of *Cladosporium, Alternaria, Aspergillus, Mucor, Rhizopus, Penicillium* and other species. The yeast *Torula* also causes discolouration of butter.

Cheese, particularly cottage cheese, gets spoiled to a condition called slimy curd. Gassiness of cheese is caused due to the formation of carbon dioxide formed by lactose utilizing organisms of *Clostridium sp*.

12.4.5 Nutmeats

The relatively dry products such as peanuts, walnuts, cashew nuts and almonds are safe from spoilage bacteria. However, molds grow if sufficient moisture is available during storage.

12.4.6 Meat, Poultry and Seafoods

Fresh meat is spoiled by microflora coming from the animal's lymph nodes, intestinal tract, hide and from the handling equipment and personnel engaged in this business. Microbial spoilage of meat occurs due to bacteria from internal sources such as *Clostridium perfringens* and other enterobacterial species. In the case of refrigerator stored meat the spoilage is only on the surface caused by bacteria from external sources. Fungal spoilage of fresh meat is caused by species such as *Mucor, Rhizopus* and *Thamnidium* which produce whiskers on beef, *Penicillium* which produces green patches and *Cladosporium* which causes black spot. Ground beef and hamburger meat are spoiled by bacteria belonging to the genera *Pseudomonas, Alcaligenes, Aeromonas, Moraxella* and *Acinetobacter*. Beef cuts such as steaks or roasts undergo surface spoilage mostly by bacteria.

Fresh livers of beef, pork and lamb, because of their high content of carbohydrates and near neutral pH value (about 6.4), undergo fermentative spoilage with pH decreasing below 6.0. Lactic acid bacteria are the predominant bacteria causing spoilage of the organ.

Bacon and cured hams are relatively not susceptible to bacterial spoilage because of processing procedures such as smoking and brining employed in preparing them. The high fat content and low water activity of bacon makes it to susceptible to moldiness, the most common form of bacon spoilage attributed to the growth of *Aspergillus, Fusarium, Mucor, Rhizopus, Penicillium* and other molds. Cured hams differ in their spoilage compared to fresh or smoked hams. Cured ham contains fermentable sugars and a variety of lactobacilli cause sourness in ham.

Poultry is mostly spoiled by bacteria of the genus *Pseudomonas*. Poultry spoilage occurs mostly on the surface as the inner portions of poultry tissue are generally sterile. Fresh poultry stored in humid conditions is susceptible to aerobic bacterial growth. As spoilage occurs, off-odours and slimy appearance develop.

Fish of both fresh-water and salt-water contain high levels of proteins and nitrogenous constituents with low fat content and practically no carbohydrates. Fresh fish and iced fish are spoiled by bacteria while salted and dried fish are spoiled by fungi. *Pseudomonas, Acinetobacter* and *Moraxella* species are involved in the bacterial spoilage. In addition intestinal bacteria of the fish and native proteolytic enzymes also contribute to the spoilage and develop off-odours and sliminess. Hence fish need to be eviscerated (removal of viscera) immediately.

Shell fish such as shrimp, lobster, crab and crayfish undergo bacterial spoilage by *Pseudomonas, Acinetobacter* and *Moraxella* species, similar to that of fresh fish. Mollusks such as oyster, clam, squid and scallops contain higher carbohydrate content compared to seafoods and undergo fermentative type of spoilage.

12.4.7 Eggs

Hen's egg is well protected by three structures—an outer waxy shell membrane, the shell and the inner shell membrane. Internally the egg white is not susceptible to microbial spoilage because of its high pH of about 9.3 and because of the presence of ingredients which function as antimicrobial agents. These include lysozyme which is effective against Gram-positive bacteria, avidin which forms a complex with biotin rendering the vitamin unavailable to microbes and conalbumin which forms a complex with iron preventing its use by microorganisms. In contrast the egg yolk is an excellent medium for the growth of most microorganisms because of its neutral pH (6.8) and nutrient content. Bacteria grow in the yolk producing hydrogen sulphide and other foul smelling compounds and also cause the yolk to become runny and discoloured. Freshly laid eggs are sterile but become contaminated quickly. Bacteria of genera *Pseudomonas, Acinetobacter, Proteus, Aeromonas, Alcaligenes, Escherichia, Micrococcus, Salmonella, Enterobacter, Flavobacterium* and *Staphylococcus* are commonly found. Molds of the genera *Mucor, Penicillium, Cladosporium* and the yeast species of *Torula* are also found. Storing eggs in highly humid conditions favours the growth of microorganisms on the surface and facilitates microbial penetration. On the other hand, at very low humidity eggs lose water and become undesirable.

The most common bacterial spoilage of eggs is rotting. Green rot caused by *Pseudomonas sp.*, black rot by *Aeromonas, Pseudomonas, Proteus sp.*, colourless rot by *Acinetobacter* and *Pseudomonas*, red rot by *Serratia* and custard rot by *Proteus sp.* are common. Bacteria also cause a spoilage condition known as mustiness. Molds cause pinspots due to mycelial growth which can be seen by candling. *Penicillium* and *Cladosporium sp.* cause the pinspots and fungal spoilage of eggs.

12.4.8 Spices

Molds and a few bacteria grow on spices if sufficient moisture is available. Propylene oxide treatment of spices reduces the bulk of microorganisms, and those that remain, do not cause spoilage if moisture level is low.

12.4.9 Mayonnaise and Salad Dressings

The yeast species *Saccharomyces* and *Zygosaccharomyces* cause spoilage of mayonnaise and salad dressings by producing gas and destroying the emulsion.

12.4.10 Sugar and Confectionery Items

Sugar gets spoiled only under improper storage conditions such as high

humidity. *Torula* and osmophilic *Saccharomyces* can cause inversion of sugar. *Leuconostoc mesenteroides* converts sucrose to dextran, a gummy and slimy polymer of glucose. Chocolate creams undergo spoilage causing explosion due to the activity of *Clostridium sp*.

12.4.11 Alcoholic Beverages

The spoilage of beer and ale is commonly referred to as beer infections caused by bacteria and yeasts. Four different spoilage patterns are common. These are known as ropiness, sarcinase sickness, sourness and turbidity. Ropiness is caused by *Acetobacter, Lactobacillus, Pediococcus cerevisiae* and *Gluconobacter oxydans*, the beer becoming viscous and oily. Sarcinase sickness is caused by *P. cerevisiae* which produces diacetyl giving rise to a honey like odour to the beer. Sourness is caused by *Acetobacter sp*. and is due to the acetic acid formed by the oxidation of ethanol by the organism. Turbidity and off-odours in beer is caused by *Zymononas anaerobia* and yeasts such as *Saccharomyces sp*.

Table wines undergo spoilage by bacteria and yeasts. *Acetobacter* oxidizes ethanol to acetic acid causing spoilage of wine. The yeast *Candida mycoderma* grows as a film on the surface of wine and creates an appearance that is referred to as wine flower. Tourne disease in wine is caused by a facultative organism that utilizes sugars under low alcohol condition resulting in a silky type cloudiness and a mousey odour and taste. Malo-lactic fermentation is yet another spoilage condition in wine in which malic and tartaric acids in grape must and wine are converted to lactic acid and carbon dioxide thereby reducing the acid content and affecting the flavour of wine.

Distilled liquors do not undergo microbial spoilage due to their high alcohol content.

Questions

1. What are the criteria used for defining the fitness of a food for human consumption?
2. Classify the various foods on the basis of their ease of spoilage. Give examples.
3. Discuss in detail the role of various factors in the spoilage of foods.
4. How do native enzymes cause spoilage of food? How can the spoilage be prevented or minimized?
5. What are market diseases? How are they caused?
6. What are the organisms responsible for the spoilage of (a) bread (b) dairy products and (c) meat products?

CHAPTER

13

Food Engineering Operations

The various processing operations in food industry may conveniently be grouped under four broad headings as (i) preparative operations, (ii) conversion operations, (iii) preservation operations and (iv) ancillary and allied operations. These are discussed in chapters 13–18.

The seasonal nature of food raw material availability makes food processing itself a seasonal operation. However, the food processor prefers to have a continuous supply of the raw material in good quality and quantity throughout the year, with uniform characteristics and stability towards storage with a predictable price. Hence it is necessary to evaluate the food raw material characteristics to arrive at a balanced assessment of the suitability of the raw material for a particular processing operation.

13.1 CHARACTERISTICS OF FOOD RAW MATERIALS

The important characteristics of food raw materials include geometric, physical, functional, growth, mechanical, thermal and electrical properties in addition to its ready availability in sufficient quantities.

13.1.1 Geometric Properties of Food Raw Materials

Foods with regular geometry are best suited for high speed mechanical processes such as cleaning, peeling of skins, size reduction, sorting, grading, filling into containers, packaging and heat processing. The important parameters of geometry include shape and uniformity of shape, freedom from surface irregularities and blemishes, size and weight. The dimensional shape of a food is important in packaging, filling by weight and heat processing. Uniformity of shape is important in filling into

containers, conveying, heat treatment, dehydration, freezing, sorting and grading operations. Surface projections and depressions on a food raw material impairs cleaning as well as processing operations. The surface of a food unit influences the rate of processes such as extraction, smoking, brining and heat treatment. Uniformity in size and weight is helpful in sorting and other processing operations.

13.1.2 Physical Properties of Food Raw Materials

Colour, texture, mechanical strength, aerodynamic and hydrodynamic properties are the important physical properties of foods to be considered. The colour of the food raw material is taken as an index of its suitability for processing operations. In low temperature processes such as freezing and freeze drying, changes in colour are minimal. However, in heat treatment processes such as canning and dehydration, colour changes do not provide any indication. For example, apples and pears may develop a pink tinge on canning while cherries are bleached due to migration of colour into canning syrups. Similarly, chlorophyll of green vegetables is converted into pheophytin with a brownish green colour during heat processing. Some degree of browning is essential in potato chip processing while browning is not required for puree. Colour control of the product is best achieved by (i) selecting varieties of known process performance, (ii) by using pretreatment such as blanching and (iii) by using optimum process conditions designed by preliminary trials to retain the natural food colour.

Textural and mechanical properties of raw materials must be such that the raw material is robust enough to withstand the stresses of mechanized cleaning, sorting and packing operations. The raw material must also withstand processing conditions to yield a product with desired texture. Differences in aerodynamic and hydrodynamic properties between the desired material and unwanted contaminants are used in cleaning, sorting and grading operations.

13.1.3 Functional Properties of Raw Materials

A raw material of ideal functionality is one, which may be processed to give a first quality product while permitting maximum process effectiveness. The functional properties required in a raw material depend on the process to which it is to be subjected. For example, sheep are bred exclusively for wool alone or meat alone. Wheat varieties of soft low protein content is used for biscuits and cakes while hard wheat with high protein content is preferred for use in bread manufacture.

13.1.4 Other Properties of Raw Materials

Other important properties of raw materials include flavour, resistance to processing stresses, freedom from defects and maturation.

13.2 PREPARATIVE OPERATIONS IN FOOD INDUSTRY

The preliminary preparative operations in food processing include cleaning, sorting and grading of food raw materials. These may be considered as separation operations. Cleaning involves the separation of contaminants from the desired raw materials. Sorting involves the separation of the raw materials into different categories based on their physical characteristics such as size, shape and colour. Grading involves the separation of the raw materials into categories based on the differences in their overall quality. This classification is not rigid because cleaning and sorting result in upgrading the quality, and grading always involves sorting.

13.3 CLEANING OF FOOD RAW MATERIALS

Cleaning is an essential preliminary operation in any food industry. The ultimate quality of the finished product, storage stability, safety from health hazards, organoleptic properties and consumer acceptance depend on cleaning process. The methods adopted depend on the type of raw material, type and extent of contamination, the degree of cleaning to be achieved and the type of finished product.

13.3.1 Contaminants in Food Raw Materials

Different food raw materials are associated with different types of contaminants. These include: (i) mineral contaminants—soil, sand, stones, metallic particles, grease and oil, (ii) plant parts—twigs, foliage, stalks, pits, husks and rope, (iii) animal parts and contaminants—excreta, hair, insect eggs and body parts, (iv) chemical contaminants—sprayed residues of pesticides, insecticides and fertilizers and (v) microbial contaminants—microorganisms and their metabolites.

13.3.2 Cleaning Objectives

The two main objectives of cleaning food raw materials are: (i) removal of contaminants which constitute a health hazard or which are aesthetically unacceptable and (ii) control of microbiological loads and biochemical reactions which impair subsequent process effectiveness and product quality.

The chosen cleaning process must satisfy the following requirements in order to achieve the aforesaid objectives.

- The separation efficiency of the process must be high and consistent and should produce minimum wastage of good material.
- Damage of cleaned raw material must be avoided.
- Recontamination of the cleaned food should be avoided by complete removal of the contaminants.
- The design of the process equipment should be such that recontamination of the cleaned food due to flying dust or wash water is prevented.
- The cleaning process must leave the cleaned surface in acceptable condition.
- The volume and concentration of liquid effluents must be kept to minimum and the effluents should be disposed off effectively.

Complete cleaning of a raw material is not possible and in practice, a balanced approach, considering the economic aspects of cleaning and the need to produce good quality food, is usually adopted.

13.3.3 Cleaning Methods

The cleaning methods can be classified into two groups, namely (a) dry cleaning methods which include screening, brushing, aspiration or winnowing, abrasion and magnetic separation and (b) wet cleaning methods which include soaking, spraying, fluming, flotation, ultrasonic cleaning, filtration and settling. Cleaning food raw materials invariably involves the use of more than one cleaning method in combination, to achieve the desired results in efficient and cost effective manner.

Dry cleaning methods. These methods are relatively cheap and convenient as the cleaned surface is dry. However, a major drawback is the spread of dust which can recontaminate cleaned raw material and also lead to fire and explosion. A careful control of dust is important in milling, high-speed conveying and other similar operations but especially so in dry cleaning. Remedial measures include dust-proofing of equipment, dust extraction and rigorous house-keeping.

Screening. Screens are primarily size separators or sorting machines but may be used as cleaning equipment for removing contaminants of different size from that of the raw material. Flat bed screens consist of one or more flat screen decks fixed together in a dust-proof casing, the assembly being shaken by a variety of devices. These machines are useful in cleaning fine materials such as flour and ground spices but must be frequently cleaned to remove oversized contaminants which may otherwise get pulverized due to abrasion and spread contamination of the raw material. Rotary drum screens (called as trommels, reels or centrifugal

screens) find numerous applications in the food industry in removing weed-seeds, grit and small stones from cereals and in cleaning of flour, salt or sugar from which strings and bag-hairs are removed. These machines have high capacity and inexpensive to install, maintain and operate but difficult to clean unless designed properly.

Abrasion cleaning. Abrasion between food particles or between the food and moving parts of cleaning machinery is used to loosen and remove adhering contaminants. Trommels, tumblers, vibrators, abrasive discs and rotating brushes are used for this purpose. Dust removal to prevent recontamination and explosion is essential in this type of cleaning.

Aspiration cleaning. Aspiration (or winnowing) is based on the differences in the aerodynamic properties of materials, i.e. materials differing in buoyancy. The raw material to be cleaned is fed into a stream of air flowing at controlled velocity to separate the raw materials into two or more streams (e.g. light, and heavy streams). The cleaned products are usually discharged as the middle stream leaving the heavy debris (stones, pieces of metal or wood) behind while floating off the light debris such as stalks, husks and hairs. This method is used in cleaning cereals, nuts, beans, onions, melon, eggs and other foods which are not amenable to wetting. Aspirators are capable of very precise adjustment and may be used to remove bran particles from flour and even discriminate between protein and starch fragments in the production of enriched flour. Aspirators use large quantities of low-pressure air and hence energy intensive. The method cannot be used with oxidation-sensitive materials.

Magnetic cleaning. This involves cascading the contaminated stream of food raw material over one or more magnets which are located underneath a conveyor. Magnetic separators used for this type of cleaning include rotating or stationary magnetic drums, magnetized belts, magnets located over belts carrying the food or staggered magnetized grids through which the food is passed.

Miscellaneous dry-cleaning methods. Theoretically any procedure which can detect the properties of the food raw material and its contaminant can be used to effect separation and hence clean, sort and possibly grade materials. Such cleaning methods include: (i) electrostatic cleaning, (ii) radio isotope separation and (iii) X-ray separation.

Electrostatic cleaning. This method takes advantage of differences in electrostatic charging of materials under controlled humidity conditions, charged particles being removed by oppositely charged or earthed rollers, grids, etc. This procedure is used in dust extraction and in other processes such as cleaning of tea. Tea fannings, initially adjusted to a moisture content of 3–4%, are cleaned to remove dust, stalks and other contaminants by flowing the material over a roller charged to a potential of 5–20 kV and rotating at 70–350 rpm. The separated matter is removed from the roller continuously by rotating brushes.

Radio isotope separation. Clods of earth and stones may be separated from potatoes using low energy gamma radiation emitted by ^{241}Am isotope to which potatoes are opaque.

X-ray separation. Stones, glass and metal fragments in foods such as confectionery can be separated by this method. The food is passed through an X-ray scanner and the image is viewed on a fluorescent screen by an operator who stops the conveyor when a contaminant is seen so that it can be manually removed.

Wet cleaning methods. Wet cleaning has the advantage of removing firmly adherent soils and allowing the use of detergents and sanitizers. However, wet methods have a number of disadvantages such as the use of large amounts of high quality water and generation of large volume of effluent (about 15,000 litres per ton of canned food) which necessitates costly effluent treatment process. Further, wet surfaces of foods spoil more rapidly and hence wet cleaning often requires dewatering step to provide material suitable for processing or storage. Wet cleaning methods include soaking, spray washing, flotation washing and ultrasonic cleaning methods.

Soaking. This is the simplest method and is often used as a preliminary stage in the cleaning of heavily contaminated root vegetables and other foods. Soaking softens adhering soil and also facilitates the removal of sand, stones and other abrasive materials. The efficiency of soaking is improved by agitating the raw material to be cleaned by means of propeller-stirrers built into the tank. Alternatively the raw material is fed into a rotating horizontal perforated drum partially submerged in a water tank. The use of warm water and detergents increase the efficiency but the use of chemicals may affect the texture of the food, e.g. sodium hexametaphosphate softens peas while some metal ions toughen peas and peaches destined for canning. Chlorination is used to decrease bacterial load of water in the soak tank. High levels of chlorine affects the food, e.g. potatoes develop blackened flesh.

Spray washing. This is the most widely used method for wet cleaning of fruits and vegetables. The surface of the food is subjected to water sprays. The efficiency of spray washing depends on several parameters such as water pressure, volume of water, temperature, the distance of the food from jets, the time of spraying and number of spray jets used. A small volume of water at high pressure is the most effective combination. High pressure sprays may be used to cut out parts of peaches and tomatoes and to remove adherent soil and black moulds on citrus fruits. However, it may damage ripe fruits and vegetables such as straw berries and tomatoes and delicate vegetables such as asparagus.

A *spray drum washer* consists of a reel constructed of metal slats or rods, spaced so as to retain the food whilst allowing to be washed through. The drum is inclined to the horizontal and rotates slowly. The

speed of rotation and the angle of inclination control both the movement of the food within the drum and the duration of washing cycle. The washer is equipped with a central spray rod which is fitted with jets for spraying water. Any adhering dirt is loosened by abrasion, but abrasion may also cause damage to some foods.

A *spray belt washer* consists of a conveyor, a vibratory conveyor or a perforated belt which carries the food beneath banks of water sprays. With spherical foods such as apples, contact is improved by roller conveyors which cause the fruit to spin beneath the sprays.

A *rubber disc cleaner* requires less amount of water for cleaning. It uses soft rubber discs spinning axially at about 500 rpm which brush the pre-soaked fruits and vegetables conveyed mechanically in a shallow channel. The soil is flung off into the base of the channel. The residual soil adhering to the food is spray washed. The disc cleaner uses only about 20 litres of water per ton of fruit while other washers use 1500–5000 litres. Average removal of bacteria, soil and spores range between 94 and 97%.

Flotation washing. The method depends on the differences in buoyancy of the desired and undesired parts of the food raw material to be cleaned. For example, bruised or rotten apples sink in water and are removed by fluming the fruits in a tank and collecting the good fruit as overflow. Heavy debris may be removed by fluming the dirty raw material over a series of adjustable weirs. The less buoyant contaminants are trapped and remain behind the weirs. The partly cleaned material with contaminants of the same or higher buoyancy is further washed by passage over a vibratory where water sprays remove the fine contaminants. The flotation washer effectively removes stones, dirt and plant debris from peas, beans, dried fruits and similar materials. Water requirement is about 4,000–10,000 litres per ton of raw material to be cleaned. Froth flotation has been used to separate peas from weed seeds by immersing the peas in dilute mineral oil-detergent emulsion through which air is blown. The contaminants float on foam and are removed. The cleaned peas are given a final wash to remove the emulsion.

Ultrasonic cleaning. Sound waves of frequencies above 16 kHz are used. Insonation of a fluid with ultrasonic waves at frequencies in the range of 20–100 kHz produces a rapidly alternating pressure in the path of the waves and this leads to the rapid formation and collapse of bubbles in the fluid. These effects, called the cavitation and decavitation, result in the release of energy causing a violent agitation of particles immersed in fluid. This phenomenon is utilized to effect the loosening of adherent dirt, e.g. grit on vegetables, grease and wax on fruits and dirt on eggs. The contaminants after loosening are removed by other cleaning methods.

Dewatering. Wet cleaning results in a cleaned product that may have some excess water adhering to it. Dewatering may be effected by passing the food over vibratory screens or specially designed rotary screens. In the case of cleaned peas for freezing, or washed wheat for milling, centrifuges

may be used. Occasionally it may be necessary to resort to drying procedures, as in the case of cereals or fruits, which are to be stored or sold as fresh.

13.4 SORTING OF FOODS

Sorting may be regarded as a separation operation based on the differences in physical properties of the food raw materials or products. Separation based on single properties such as colour, size, shape or weight of the food raw material are generally considered as sorting operations. Sorting is an important operation in controlling the effectiveness of many processes in food industry. For example, sorted vegetables and fruits are betted suited for mechanized operations of peeling, pitting and coring or blanching. Similarly, food materials of uniform size or shape are better suited for efficient heat transfer during sterilization, pasteurization, dehydration or freezing. Mechanized filling or packing of sorted food products into standard containers is more efficient and less time consuming. Catering packs containing sorted foods of uniform size, shape or weight are easier to serve. Sorted foods are more attractive and have greater consumer acceptance.

Sorting and grading can both damage the food raw material or product because of improper handling by human operators (operator damage), dumping (dumping damage) or dropping of material (drop damage). Such damages have to be eliminated or minimized as otherwise the quality of the finished product, the effectiveness of food process and economic aspects on the whole will be severely affected.

13.4.1 Sorting Methods

Sorting methods include weight sorting, size sorting, shape sorting and photometric or colour sorting.

Weight sorting. The weight of a food unit is proportional to the cube of its characteristic dimension and hence weight sorting is more precise compared to dimensional sorting. Meat cuts, fish fillets, fruits such as apples, pears and citrus fruits, vegetables such as potatoes, carrots and onions and eggs are sorted by weight. Typical weight sorting machines carry fruits in tared canvas pockets attached to the pivoted beams fitted with counterbalance weights. As the beams pass along the sorter, the beam fulcrum is moved towards the counterbalance weight until such time as the weight of the fruit causes the beam to tip, discharging the fruit into padded chutes in weight categories. Weight sorting is also practised using computer controlled weighing heads on which the food units are weighed. The best combination of weighed food items to make up a required weight is selected by the computer which are then grouped together on a

conveyor for processing or packaging. High speed weighing uses a machine that measures electronically, the cross-section dimensions of potatoes and converts the dimensions into weight by a computer. The machine sorts the potatoes into different weight categories at rates up to 4 or 5 tons per hour.

Size sorting. Different types of screens are used for size separation of foods. The screen designs commonly used in food industry may be grouped into two types: (i) variable aperture screens using cable, belt, roller or screw sorters and (ii) fixed aperture screens using stationary, vibratory, rotary, gyratory or reciprocating screens. Fixed aperture screens of flat-bed type are used in preliminary sorting of potatoes, carrots and turnips. Multi-deck screens are used in size sorting of cereals, nuts and also partly processed and finished foods such as flour, sugar, salt, ground spices and herbs. Drum screens are used for sorting peas, beans and other similar foods capable of withstanding tumbling action in a rotating drum screen. Variable aperture screens with continuously variable apertures of roller, belt or screw type find use in size sorting of fruits and vegetables.

Shape sorting. Shape sorting is adopted when food raw materials contain undesirable material even after size or weight sorting and cleaning. For example, cleaned and size or weight sorted wheat may still contain weed seeds of similar size and weight compared to wheat. Shape sorting on the basis of a combination of length and diameter is useful under such circumstances. A disc sorter is used for shape sorting wheat, rice, oats and barley. The disc sorter specific for each of the above food material has indentations on both sides of the vertically rotating, discs mounted on a motor driven horizontal shaft. The vertically rotating discs pass through unsorted food held in a trough at the lower end of their rotation, collect the food of correct shape (wheat or rice) in the indentations and discharge the collected material into bins at the top of their rotation. Undesirable materials such as weed seeds will not fit into indentations on the discs and hence will not be collected by the rotating discs. Spherical weed seeds may be removed from wheat and other cereals by shape sorting using their tendency to travel down an inclined surface. The raw material to be sorted is fed onto an upward-moving belt held at an inclination. The weed seeds will easily roll down while the desired material moves upward along with the belt and is discharged at the upstream end.

Photometric sorting. Photometric sorting uses optical properties of foods to effect separation of desired material from contaminants. Optical properties such as reflectance and transmittance of foods are useful indicators in a variety of ways. Reflectance properties are used to indicate: (i) raw material maturity (e.g. colour of fruit, vegetables and meat indicates ripeness and freshness characteristics), (ii) the presence of surface defects (e.g. worm holed cereals or nuts and bruised fruits) and

(iii) the extent of heat processing (e.g. in the manufacture of biscuits, bread and potato chips or crisps). Transmittance properties of foods are useful in indicating the internal properties of foods such as ripeness and presence of any core defects in fruits and presence of blood spots or any extraneous matter in eggs.

Mechanical reflectance colour sorters are used to sort, clean and grade foods such as nuts, cereals and coffee beans containing worm-holes or other defects. Other foods such as diced vegetables, whole potatoes, onions and fruits which are incompletely peeled, citrus fruits and tomatoes exhibiting maturity differences and finished food products such as jams, peanut butter, butter and crisps exhibiting differences in processing are also sorted by colour sorters. The colour sorter functions on the basis of photometric scanning of each food unit as it moves past or rotated before a photocell. The signal generated by the photocell is compared to an adjustable pre-set standard signal automatically, and any imbalance between the two signals is amplified and used to activate the sorting gear. Rejection systems of different types are used. A pneumatic rejection system is used for small particulate foods, the signal from the discriminator being used to operate an air jet which deflects substandard material into the reject chute. With larger food units such as diced vegetables and whole fruits, scanning is done individually on a vacuumed pick-up wheel combined with a mechanical displacement rejection system. Electrostatic separation may also be used, the rejected material being charged by an electrode and then withdrawn from the bulk by an oppositely charged deflector.

Transmittance sorting is a non-destructive internal examination of liquid and solid foods. The method requires a very sensitive detection system as only about 0.1% of the incident light energy is transmitted through the food. Apples are sorted by measuring the transmittance at 692 ± 3 nm which is related to the chlorophyll content of the flesh of the fruit and hence to its maturity.

Other sorting methods. Sorting on the basis of surface roughness or stickiness may be used for separating seeds. Dodder mill separates seeds with rough skin-coats which are preferentially retained when passed through an inclined rotating cylinder lined with muslin, velvet or flannel. Sticky weed-seeds (e.g. buck thorn when damp) may be removed from cereals as agglomerates by screening. Worm-holed and cracked peas are removed in a slowly rotating inclined drum lined with fine needles projecting inwards. Damaged peas are picked up by the needles and discharged into separate bins located at the centre of the drum. Whole peas are swept aside by the needles and discharged at the lower end of the drum.

13.5 GRADING OF FOODS

Grading is quality separation on the basis of an overall assessment of those properties, which affect the acceptance of the food raw material for

processing, and finished food product for consumer acceptance and safety. Thus grading involves simultaneous evaluation of multiple properties and is mostly done manually. The grading factors, which determine the quality of the food, include: (i) process suitability, (ii) consumer safety, (iii) conformity with legal requirements and (iv) consumer acceptance.

The grading parameters commonly used in food industry include the following: (i) size and shape as functional and acceptability factors, (ii) maturity to describe the freshness of eggs, ripeness of fruits and aging of meat, (iii) texture to grade the crumb structure in bread and cakes, crispness in apples and viscosity of creams, (iv) flavour and aroma as indicators of ripeness of fruits as well as effectiveness of processing conditions, (v) colour as indicator for consumer acceptability and effectiveness of process, (vi) blemishes such as cloudy yolk, blood spot and shell cracks in eggs, bruises in fruits and insect holes in coffee beans and cereals to indicate their defect and impurity. Contaminants and undesired parts such as rodent hair and insect parts in flour, soil and spray residues on fruits and vegetables, microorganisms and their metabolites on meat, toxic metals in shell fish, bone fragments in meat products, pod residues in peas and beans and stalks and stones in fruits all these are the adverse qualities of the raw food materials.

13.5.1 Grading Methods

Grading methods may be classified into two types:

- Quality control procedures in which the quality of the food is determined by laboratory tests on samples drawn statistically from a batch of food.
- Procedures in which the total quantity of food is subjected to physical separation in quality categories. This grading may be carried out manually or by specialized machines.

For proper grading, the food unit must be presented singly before the human grader or machine for assessment. Presentation devices may be roller or vibratory tables or rotating wheels equipped peripherally with pneumatic devices which pick up food pieces, rotate them for viewing and then release them at a given signal.

Manual grading is done by trained operators who are able to assess a number of grading parameters simultaneously. For example, eggs are graded manually by candling. This is a non-destructive examination of the egg contents through the translucent shell when the egg is spun in front of a powerful light in a darkened booth. The presentation of each egg is automatic and a candling operator can separate several thousand eggs an hour into 3 or 4 grades taking into account up to 20 different grading parameters. In certain cases, light of specific wavelengths may be used to grade food units. Thus blood spots in eggs are detected when light of

wavelengths 577 and 597 nm is used while green rot is revealed by fluorescence at 490 and 510 nm. Manual grading is costly and boredom and fatigue reduce grading efficiency.

Machine grading is carried out by combining a group of sorting operations so as to separate the food units on quality basis. Thus wheat of a particular variety may be graded by a combination of cleaning and sorting operations. Sometimes a single property may be helpful in grading the food. Thus peas of small size are recognized to be most tender and of highest quality so that size sorting of cleaned peas results in quality grading. Peas may also be graded on the basis of their density using flotation in brines of varying densities. Similarly, potatoes of high density, desirable for manufacturing french fries, potato crisps and dehydrated mashed potato, may be graded using flotation in brines. Mechanical grading is cost effective and efficient.

Questions

1. What are the characteristics of food raw materials?
2. What are preparative operations in food industry?
3. List the common contaminants found in food raw materials.
4. Give an account of the dry cleaning methods.
5. Write notes on (a) spray washing, (b) flotation washing and (c) soaking.
6. What are the advantages and disadvantages of wet cleaning methods?
7. Describe in detail the different sorting methods used in food industry.
8. Write a note on grading parameters and grading methods.

CHAPTER

14

Food Conversion Operations

The various conversion operations commonly carried out in food industry include size reduction and screening of solids, mixing, emulsification, filtration, membrane separation, centrifugation, extraction and crystallization.

14.1 SIZE REDUCTION AND SCREENING OF SOLIDS

Size reduction or comminution is an important unit operation in food industry carried out for a variety of purposes. Size reduction aids the extraction of a desired constituent from food raw material, e.g. flour from wheat grains and juice from sugar cane. Size reduction increases the surface area of the solid and thereby increases the rates of many processes. The rates of drying of moist solids, solvent extraction as in the case of edible oils and also the rates of unit operations of cooling, blanching and mixing or blending of dry powders such as infant foods, cake mixes and soup mixes are enhanced.

The nature of forces used in size reduction may broadly be grouped under three types: (i) *compressive forces* employed for crushing hard materials such as nuts in roller crushers, (ii) *impact forces* used in hammer mills for coarse, medium and fine grinding operations and (iii) *shear* or *attrition forces* in disc attrition mills and ball mills, used for fine grinding of softer, non-abrasive materials. Size reduction may be operated in open circuit or closed circuit modes as required. The term *reduction ratio* expresses the ratio of the average size of the feed to that of the product.

The characteristics of the feed material decide the nature of forces and the type of equipment to be used for comminution. The characteristics of the feed considered, include hardness, toughness, abrasiveness, stickiness

and softening or melting temperature. Hard and abrasive materials are difficult to comminute and require more energy. The equipment is subject to increased wear and tear. The mechanical structure of the feed also plays a role in determining the ease of size reduction. Crystalline materials are easy to comminute. However, many food materials have a fibrous structure and are not easily disintegrated by compressive or impact forces. In such cases, shredding or cutting is required. The presence of moisture can both aid and hinder size reduction. If the moisture content is more than 2–3% it can lead to clogging of the mill. Small quantities of water are useful in suppressing dust and also preventing fire hazard. In some applications as in corn milling, a large amount of water is used to transport the solid particles as a slurry. The feed may become sticky or clog the mill if it is temperature-sensitive as the temperature increases during size reduction.

14.1.1 Disintegration of Fibrous Materials

Many food raw materials have a fibrous structure as well as appreciable amounts of liquid (e.g. fruits, vegetables and meat). Size reduction of such materials is achieved by impact and shear forces but not compressive forces. Modified equipment such as percussion mills with a series of knives replacing hammers are used to apply the impact force along the thin edges of knives. Similarly, disc attrition mills with discs provided with studs or serrations on the faces of the discs, impart a tearing action. In addition to these modified size reduction equipment, specially designed equipment for slicing, dicing, shredding and pulping operations also are used. *Slicing* involves the use of rotary knives to cut fruits into slices for dessert dishes. *Dicing* involves cutting the material into cubes in two steps—slicing to give slices of desired thickness followed by cutting into cubes by rotary knives. *Shredding* involves tearing the fibrous food into small fragments. Squirrel cage disintegrators consisting of twin concentric cylindrical cages with cutting edges on their surfaces and rotating in opposite directions are used for shredding fibrous food as a preliminary step prior to dehydration of such food. *Pulping* is used in crushing fruits for jam making. Fruits which are edible but unacceptable for whole fruit processing are used in jam making. A combination of pulping and sieving is used. A pulper consists of a cylindrical perforated screen containing high speed rotating brushes. Material is fed into the inside of the cylinder, the brushes forcing the pulped material through the perforations. Stalks, skins and stones pass over the screening surface as rejects. Some fruits are softened by heating before pulping the same to yield more pulp.

14.1.2 Screening

Screening is a separation operation used in combination with size reduction to separate the different sized solids. Bar screens or grizzlies are used for separating large sized particles while vibrating screens, trommels and cylindrical screens are also used for sorting and grading operations.

14.2 MIXING

Mixing aims at a uniform distribution of two or more components interspersed in space with one another. Uniform distribution is achieved easily by mechanical mixing in the case of miscible liquids and solids that are readily soluble in liquids. In contrast, mixing immiscible liquids, pastes or dry solids is difficult and uniformity in distribution of components is much less.

14.2.1 Mixing Equipment

Mixing equipment commonly used in food industry may be grouped under three broad types based on the nature of food items to be mixed.

1. Mixers generally consisting of a stationary vessel containing a moving stirrer, agitator, paddle or impeller mounted on rotating shaft are used for mixing liquids of low or moderate viscosity. Paddle agitators and impeller mixers are the most commonly used ones in this type. They are used for blending low viscosity oils in the manufacture of margarine and cooking fats, for diluting concentrated solutions and for manufacturing fruit squashes. They also find use in preparing liquid sugar mixtures for sweet manufacture, for preparing brines and syrups, for dissolving soluble dyes and colouring matter and for reconstituting dried products such as milk powder.

2. Mixers consisting of a stationary vessel containing moving vanes, knives, paddles, ploughs or screws are used for achieving relatively higher consistency of mixing as in the case of mixing viscous liquids, dough, pastes and fats. Pan mixers, kneaders, dispersers, horizontal trough mixers, z-blade mixers and masticators belong to this type. They find use in dough mixing for bread manufacture, batter mixing for cakes and biscuit manufacture, for preparing meat and fish pastes, blending of margarine, cooking fats and butter, whipping of dairy and artificial creams, for preparing cheese spreads and blending of cheese and in the manufacture of chocolate products.

3. Mixers consisting of a moving vessel and moving or stationary paddles, vanes, knives, ploughs or screws are used for achieving very high consistency of mixing as required in pastes, dry solids and plastic materials. Tumbler mixer, ribbon blender (horizontal screw mixer) and

vertical screw mixer are commonly used for mixing dry solids such as blending of grains prior to milling, blending of flours and incorporation of additives to flours, preparation of custard powders, cake mixes, dry soup mixes and for incorporation of additives in dry products such as baby foods.

14.3 EMULSIFICATION

Emulsification involves the intimate mixing of two immiscible liquids so that one liquid (usually called the dispersed or discontinuous or internal phase) is dispersed in the form of fine droplets or globules in the second liquid called the dispersing, continuous or external phase. In food industry the two most common immiscible liquids encountered are water and oil. The water phase usually consists of aqueous solutions of salt, sugar or other organic and colloidal hydrophilic materials while the oil phase consists of oils, fats, hydrocarbons, waxes, resins and other hydrophobic materials. Two types of emulsions are commonly found in food products, namely, oil-in-water (O/W) emulsions, e.g. milk, and water-in-oil (W/O) emulsions, e.g. butter.

14.3.1 Interfacial Tension and Emulsion Formation

The emulsion formed depends on the relative proportions of the phases, the type of emulsifying agents used to stabilize the emulsion and the method of preparation of emulsion. In a heterogeneous liquid system or emulsion the interface between the two immiscible liquids is associated with free energy (called interfacial tension) arising from an imbalance in the cohesive forces of the two liquids. The interfacial tension causes the interface to contract to form smallest possible interfacial area. Spherical droplets of a liquid, dispersed in another liquid, represent the smallest possible interfacial area per unit volume of the liquid. The droplets tend to coalesce to form larger drops thereby reducing the interfacial area. This coalescence of drops continues leading ultimately to the formation and separation of layers of the two immiscible liquids and destruction of the emulsion. Crude or improperly prepared emulsions are less stable and layer separation occurs in a short span of time. The formation of an emulsion involves the creation of a large number new interfaces in the form of finely dispersed droplets of the internal phase and reducing the interfacial tension usually by mechanical mixing or agitation.

14.3.2 Emulsifying Agents

The emulsion is stabilized by use of emulsifying agents. Emulsifying agents or emulsifiers have two important functions, namely (i) to reduce the interfacial tension between the liquids of the emulsion and (ii) to

protect or stabilize the emulsion by forming a film or protective coating around the droplets of the internal phase and preventing their coalescence. A food grade emulsifier should be specific to the type of emulsion it promotes, non-toxic, odourless, tasteless, colourless and should be physically and chemically stable under processing, handling and storage conditions. The emulsifiers used commonly in food products include esters of glycerol, propylene glycol, sorbitan esters of fatty acids, cellulose ethers, carboxymethyl cellulose. Proteins, phospholipids, sterols

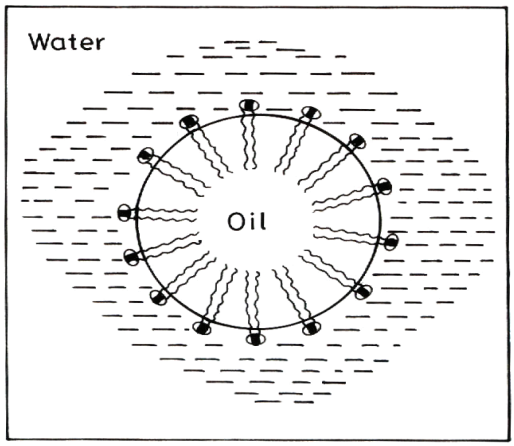


Figure 14.1 Orientation of emulsifying agent at the interface.

and finely divided solids such as bentonite and carbon black. Organic emulsifiers have a structure which may be considered as consisting of a polar head and a non-polar tail. The emulsifier molecule orients itself at the interface such that the polar head faces the water phase and the non-polar tail faces the non-polar oil phase (Figure 14.1).

If the polar and non-polar groups of emulsifying agents are balanced perfectly, then the emulsifier can promote both the types of emulsions. Emulsifying agents with a small imbalance of these groups can promote specifically O/W or W/O emulsions. Usually the phase in which the emulsifying agent is most soluble tends to become the external phase. Finely divided solids adsorb at the interface and act as emulsifying agents though less efficiently. Solids easily wetted by water promote oil-in-water emulsion and vice versa.

14.3.3 Emulsion Properties

The properties of the emulsion formed are mostly the properties of the external phase liquid. Thus an O/W emulsion may be diluted with water, coloured with water soluble dyes and exhibits electrical conductivity. In

contrast, the W/O emulsion can be diluted with oil and coloured with oil soluble dyes. It has a low electrical conductivity characteristic of oils.

14.3.4 Emulsification Methods

Theoretically, the work required to be done on an emulsion system to create new interfaces is equivalent to the product of the newly formed surface and the interfacial tension. In addition, energy must be provided to keep the liquids in motion and to overcome frictional resistance. In practice, the work is done on the liquids by subjecting them to violent agitation so that drops of the internal phase are subjected to shear forces. The large drops are broken down to smaller droplets. Under such favourable conditions, the emulsifying agent is adsorbed at the interface and a stable emulsion is formed. The time and speed of agitation need to be optimized. Shorter times of agitation and insufficient agitation lead to unstable emulsions while longer time of agitation or intense agitation can lead to the breaking of the emulsion.

The factors, which influence the formation of emulsions, include: (i) use of appropriate emulsifying agent, (ii) the phase volume ratio and (iii) temperature of emulsification. Use of appropriate emulsifying agent is necessary to favour the type of emulsion to be formed, i.e. O/W or W/O type. The phase volume ratio (i.e. the percentage by volume of the internal phase) decides the external phase; usually the phase that is present in larger proportion tends to become the external phase and hence emulsion with phase volume ratio of more than 50% is difficult to produce and handle. The temperature of emulsification must be controlled because interfacial tension and viscosity decrease with increase in temperature and the upper limit of temperature depends on the heat sensitiveness of the ingredients. Milk products can be emulsified at temperatures up to 70°C while products containing egg yolk solids as emulsifying agent (e.g. mayonnaise) need to be emulsified at much lower temperatures.

The two phases are usually prepared separately with emulsifying agent being added to the external phase. However, hydrophilic gums and colloids used as emulsifiers are dispersed in oil phase to prevent their swelling and lump formation. The internal phase is usually added gradually to the external phase with simultaneous stirring.

14.3.5 Emulsification Equipment

Mixers, homogenizers, colloid mills and ultrasonic devices are used for emulsification.

Mixers such as slow speed paddle agitators, z-blade mixers and high speed turbine or propeller mixers used for mixing doughs are used in manufacture of salad creams, creams for cake filling and margarine.

Homogenization brings about the reduction of the size of droplets of the internal phase by forcing the crude emulsion through a narrow opening at high velocity. A pressure homogenizer consists of a homogenizing valve providing an adjustable narrow opening of a few thousandths of an inch and a high pressure pump capable of operating at pressures up to 10,000 psi. The liquids of the emulsion are forced through the opening and accelerated at velocities up to 300 m/s. The droplets of the internal phase are subjected to high shear forces as they come out of the valve to atmospheric pressure and simultaneously they impinge on a hard surface set normal to the direction of flow of the liquid. The impact breaks down the droplets further. The final size of the droplets of the internal phase are in the range of 0.02 μm. Pressure homogenization is used in processing of milk and in the production of low fat cream, evaporated milk and sterilized milk, ice cream mixes, salad creams and cream soups.

Colloid mill consists of a stationary surface (stator) and a rotating surface (rotor) separated by a small adjustable opening of 50 to 150 μm. The crude emulsion is fed under gravity at the centre of the rotation, passes through the opening and is discharged at the periphery. The liquid is subjected to high shear and turbulence. Colloid mills are more effective than pressure homogenizers for handling high viscosity products such as salad creams, mayonnaise and artificial creams.

Ultrasonic emulsification involves the use of ultrasonic waves for producing cavitation and turbulence. The device is used in the manufacture of fruit and vegetable purees, tomato juice, peanut butter, baby foods and essential oil emulsions.

14.4 FILTRATION

The application of filtration in food industry may be considered under three categories, viz. cake filtration, clarification and microfiltration.

Cake filtration is used in the case of slurries containing appreciable amount (1–2% by weight) of insoluble solids to separate the solid and liquid components, one or both the components being valuable. The solid cake is formed on the upstream side of the filter medium. Cake filtration is used in edible oil industry to remove higher melting triglyceride fractions, waxes from salad oils and in refining edible oils and in breweries for filtering mash and recovering yeast after fermentation. Vertical plate pressure filters, horizontal plate filters, shell and leaf filters, vacuum filters and continuous rotary drum filters are used for cake filtration.

Clarification aims at removing small quantities of unwanted insoluble solids to produce a clear valuable liquid. Beer, wine, vinegar, fruit juices, yeast and meat extracts, salad oil, brine and sugar syrup are clarified. Vertical and horizontal plate filters and tube filters are used for clarification in food industry. The solids are usually fine and of colloidal or slimy nature and hence require the use of filter aids or precoating of the filter medium.

Microfiltration involves the removal of very fine particles of the order of 1 μm or less, and is generally used for removing microorganisms from liquid foods such as milk and fruit juices and for removing yeast and microorganisms from beer, wine and yeast extract. Both horizontal and vertical plate filters and tubular filters are used for microfiltration.

14.5 MEMBRANE SEPARATION

Membrane separation methods include ultrafiltration (UF) and reverse osmosis (RO). Both are pressure activated techniques in which solutes of different molecular weights are separated from solution. Ultrafiltration involves the separation of high molecular weight solutes from low molecular weight solutes. The mechanism of separation may be considered basically as that of filtration using a filter medium with micropores. Reverse osmosis involves the separation of solute molecules or ions of comparable size to those of solvent molecules. The separation mechanism is not simple filtration but more complex involving diffusion and capillary flow. Reverse osmosis requires higher pressure for its operation (about 50–75 atm. or 5 to 7.5 MN/m^2) compared to ultrafiltration (2–10 atm. or 0.1 to 1.0 MN/m^2) because of the higher resistance offered by the membrane to the flow of solvent.

The equipment and operating features of both ultrafiltration (UF) and reverse osmosis (RO) are similar. Dilute feed is pumped under pressure by high pressure positive displacement pumps, to the membrane modules via a manifold. The number of modules used depends on the required rate of separation. Different types of membrane module configurations are available. These include flat sheet modules, spiral wound modules and hollow fibre membrane modules. The feed flows over the membrane in cross-flow, i.e. parallel to the membrane surface in contrast to conventional filtration, to avoid cake formation and consequent fouling of the membrane. The retentate is recycled between the membrane module and the feed storage vessel. The permeate passes through the membrane and is collected separately. In food processing applications, membrane separation is commonly used to effect concentration and hence the desired stream is the retentate. A heat exchanger is often included in the system to maintain desired temperatures so as to inhibit microbial growth in the food.

Ultrafiltration is used to recover whey proteins from the whey, a by-product obtained during cheese manufacture. The whey proteins are concentrated to more than ten fold so that the concentrate is amenable for spray drying. Ultrafiltration is also used in the production of protein rich concentrate from skim milk for use in the manufacture of soft cheese. Reverse osmosis is used primarily in water treatment for recovering potable water from ground, surface or sea water by desalination. In food industry, RO is used for recovering lactose from whey after the removal of proteins by the ultrafiltration process. Potential applications of RO

include the recovery of low molecular weight organic substances of economic value from waste streams of food processing industries, thereby reducing the BOD load of waste water, e.g. sugar recovery from waste water of confectionery industries.

14.6 CENTRIFUGATION

Centrifugation as a separation operation finds use in food industry for different types of applications. These include: (i) separation of immiscible liquids, (ii) clarification of juices, (iii) desludging and (iv) centrifugal filtration.

Centrifugal separators for liquid-liquid separations and clarification include tubular bowl centrifuge and disc bowl centrifuge. The tubular bowl centrifuge consists of a long narrow cylindrical bowl suspended from the top rotating at high speed of about 10,000 rpm, in an outer stationary casing. Bowl dimensions range from 3 to 6 inches in diameter and up to 5 feet in height. The feed is introduced at the bottom of the bowl and the discharge occurs through an annular opening at the top. In food industry, tubular bowl centrifuges find applications in dewatering animal fats, vegetable and fish oils and in the clarification of fruit juices, cider and sugar syrup.

The disc bowl centrifuge consists of a shallow wide cylindrical bottom driven bowl rotating at moderate speed (about 6000 rpm) in a stationary casing. The bowl, about 12 to 40 inches in diameter contains a number of closely spaced metal discs, located one above the other with a fixed clearance of 0.02 to 0.05 inches between them. The discs have one or more sets of matching holes which form channel through which the feed material flows as the discs rotate with the bowl. The feed is introduced at the bottom through a centrally located feed pipe from above. Under the influence of centrifugal force the dense phase of the feed travel towards the bowl wall streaming down the undersides of the discs while the lighter phase, displaced towards the centre, flows over the upper faces of the discs. The liquids are thus divided into thin layers and the distance a drop of one liquid has to travel to get into the appropriate outgoing stream is very small. The shear forces at the liquid-liquid interface break down emulsions and improve separation of the phases. Disc bowl separators are used in dairy industry for separating cream from milk, in the refining of animal fats, vegetable and fish oils.

Clarification of liquids, particularly when the solids content is low in the range of about 1% by weight, is carried out by both tubular bowl and disc bowl centrifuges. Other types of clarifying centrifuges include nozzle discharge and valve discharge centrifuges.

Clarifying and desludging centrifuges are used for clarification of beer, wort, fruit juices, dewatering of corn, wheat, rice starches and recovery of yeast in brewery.

Centrifugal filtration involves the use of filtering centrifuges (basket centrifuge), screen conveying or screw conveying centrifuges. These

machines are used in recovering animal and vegetable protein, separating cocoa, coffee and tea slurries and in preparing fish meal. Filtering centrifuge also finds use in sugar refining for recovering washings and drying sugar crystals and in the recovery of vegetable proteins for freeze concentration.

14.7 EXTRACTION

Extraction and expression involve the recovery of valuable liquid from coarse solid particulates, which have cellular structure. Many food materials such as fruits, vegetables, and seeds contain valuable liquid constituents within the cell structure. Ground nuts, coconuts, soyabeans, sunflower seeds, sesame seeds, olives, cottonseeds yield edible oils and fats. Fruits yield juices used in the manufacture of soft drinks and wine. The cell walls of these plant materials normally require disruption such as pulping or heating, before the liquid can be separated out. Extraction, also called liquid-liquid or solvent extraction, involves the use of a solvent to recover the valuable liquid constituent of such plant foods. Expression uses compressive forces to crush the solid particles to release the liquid constituents within the cell structure.

14.7.1 Applications

1. *Sugar from sugar beet.* Sugar is extracted from sliced beets using water as solvent. The beet is sliced to provide an enhanced surface area for efficient extraction and also for limiting cell damage. Excessive cell damage results in the extraction of undesirable non-sugar components from the beet. Temperature control during the extraction is critical because high temperatures lead to peptization of the beet and also extraction of non-sugar components. The water extract from beet contains about 15% dissolved solids. The solution is purified by settling and filtration and concentrated by vacuum evaporation. Sugar is obtained by crystallization from the concentrated solution.

2. *Extraction of edible oil.* Edible oil is extracted from nuts, oil seeds and from fish by using organic solvents such as hexane, heptane and cyclohexane. Acetone and ethyl ether are used for extraction of fish liver oil and carbon disulphide for extraction of olive oil.

3. *Manufacture of instant coffee.* The roasted coffee beans are ground and extracted with hot water to give a final extract containing about 25–30% solids. The concentrated extract is spray dried. Extraction is carried out in counter current multistage static bed system containing 5–8 units. The initial temperature of water used for extraction is about 100°C and increases to about 180°C as extraction progresses. Higher

temperatures produce off-flavours. The extraction cycle time is about 60 minutes and coffee to water ratio varies between 1:5 and 2:7.

4. *Manufacture of instant tea.* Extraction of dried blended leaves with hot water is an essential stage in the manufacture of instant tea. Extraction is usually carried out in a 3–5-stage static bed system in the temperature range of 70–90°C. The final extract contains about 3–5% of solids. The extract is stripped of its volatile aroma constituents and then concentrated by vacuum evaporation to a solid content of about 30–50%. The aroma constituents are added back to the concentrated extract and the solution is spray dried or freeze dried.

5. *Other applications.* Extraction of essential oils from flowers, oils from wheat germ and rice bran are some of the applications of extraction in food industry.

14.8 EXPRESSION

Three methods of expressing the liquid by use of compressive forces are commonly used in food industry. These include hydraulic pressing, roller pressing and screw pressing. The various factors which affect the efficiency of expression are: (i) the yield stress of the solid, i.e. the resistance of the solid to undergo deformation, (ii) porosity of the cake formed, (iii) viscosity of the liquid expressed and (iv) compressive force applied.

14.9 CRYSTALLIZATION

Crystallization is a process where solid particles are formed from a homogeneous phase. It may also be considered as a solid-liquid separation process in which mass transfer of a solute occurs from the liquid to the solid phase. Crystallization may be initiated either by cooling or by evaporation. The solution is concentrated either by evaporation or cooling until a supersaturated solution is obtained. The solute then forms crystals of pure composition of the solute. The method of concentration may be decided based on the phase diagram. From the phase diagram of sucrose-water system (Figure 14.2) it will be clear that the quantity of sucrose obtained by cooling is limited because below the eutectic temperature T_c, the equilibrium state is a single solid phase, while above the eutectic temperature, the system consists of crystals of one component in equilibrium with a solution of eutectic composition. Since only water is volatile in the sucrose-water system, evaporation of water can raise the concentration to induce sucrose crystallization.

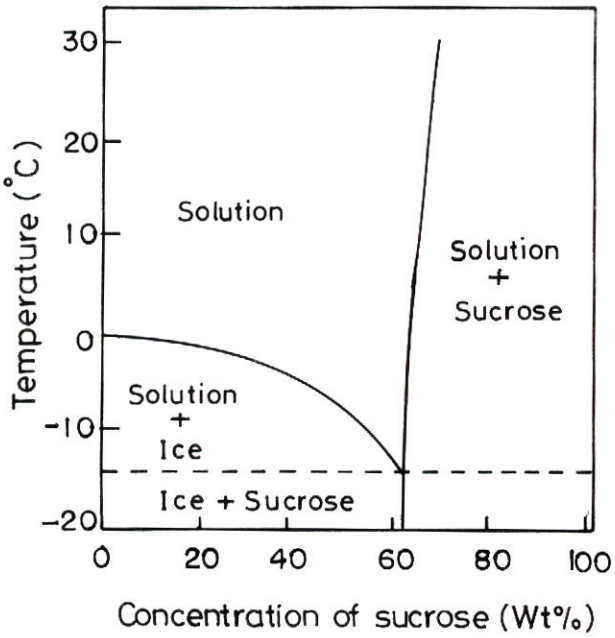


Figure 14.2. Phase diagram of sucrose-water system.

In commercial crystallization the yield and purity of crystals as well as the sizes and shapes of the crystals are important. Size uniformity of crystals is desirable to minimize caking in the package, for ease of pouring, washing and filtering. Sometimes crystals of a certain shape are required, such as needles rather than cubes. There are seven classes of crystals, depending on the arrangement of the constituent atoms or ions. These include the cubic, tetragonal, orthorhombic, hexagonal, monoclinic, triclinic and trigonal systems.

14.9.1 Crystallization Theory

During crystallization from a homogeneous solution, a new solid phase is created. Understanding the mechanism of crystal formation and its growth is helpful in designing and operating crystallizers. The process of crystallization is considered to consist of the basic steps of nucleation and crystal growth. In the absence of any solid particle, nucleation must first occur before crystal growth. The driving force for both these steps is supersaturation. The degree of supersaturation of a solution is measured in terms of the supersaturation coefficient S, given by the ratio of the concentration of solute in solvent at a given temperature (C_t) to the

concentration of solute in solvent in a saturated solution at the same temperature (C_{t0}).

$$S = C_t/C_{t0} \tag{14.1}$$

The solution is supersaturated if $S > 1$ and $S = 1$ defines the solubility curve. Crystallization in supersaturated solution may be initiated by primary nucleation or homogeneous nucleation due to rapid local fluctuations on a molecular scale in the homogeneous phase. Solute molecules come together to form clusters. Alternatively, crystallization may be initiated by the addition of solute crystals which give rise to formation of additional crystal nuclei (secondary nucleation). Dust, gas bubbles, mechanical shock or ultrasonic shock may also bring about nucleation (heterogeneous nucleation).

In industrial crystallization process either of the following two approaches are practised. Finely divided solute crystals are added to a solution maintained in the metastable zone of supersaturation and the seed crystals grow to the finished size without further nucleation. Alternatively, nuclei may be formed either spontaneously or by secondary nucleation from solution in the labile zone. Agitation of the solution allows the formation of new crystals in the upper range of metastable zone.

Crystal growth will occur subsequent to nucleation or the addition of seed material. The solubility of small crystals is greater than that of a large crystal. In a supersaturated solution the small crystals will dissolve and large crystals will grow. The rate of growth of a crystal depends both on the transport of material to the surface of the crystal and on the mechanism of surface deposition. Stirring the solution during crystallization helps in the transport of material to the surface. However, the rate of growth of crystals is diffusion controlled only above 50°C in the case of sodium chloride and above 45°C in the case of sucrose crystals. The presence of impurities usually reduces the rate of crystal growth. The external form or habit of a crystal depends on the conditions of growth. The different faces of a growing crystal grow at different rates resulting in the overlapping of the faster growing face by a slower growing face, the faster growing face ultimately being absent in the finished crystal. Crystals formed rapidly from highly supersaturated solution grow into long needles and have a dendrite (tree-like) structure because such a structure has high specific surface area and can easily dissipate the heat released during crystallization. The shape of the crystals may be selectively modified by chemicals called habit modifiers. Habit modifiers may be naturally present or deliberately added during crystallization. For example, raffinose occurs naturally in sugar beet, and modifies the sucrose crystals to a cubic appearance at concentrations of about 1%, while at

concentrations of about 2%, it modifies sucrose crystals into thin, narrow plates. Addition of a trace of potassium ferrocyanide to brine modifies sodium chloride crystals to dendrite type, an extreme form of habit modification is crystal inhibition where growth is reduced to negligible proportions on all faces. Habit modification occurs during crystal growth when crystals come together and adhere to give twinned crystals. Crystal aggregation may also occur giving rise to large irregular shaped crystals.

14.9.2 Equipment for Crystallization

Crystallizers may be batch or continuous type, the latter being generally preferred. Crystallization cannot occur without supersaturation. The main function of any crystallizer is to cause a supersaturated solution to form. Crystallizers bring about supersaturation either by (i) cooling the solution with negligible evaporation, (ii) evaporation of the solvent with little or no cooling as in evaporator crystallizer or (iii) combined cooling and evaporation as in adiabatic or vacuum crystallizers.

Crystallizers producing supersaturation by cooling are used for solutes having solubility curves that decrease markedly with temperature. However, in the case of common salt, the solubility curve changes little with temperature and hence evaporation of the solvent water is necessary for producing supersaturation. In adiabatic crystallizers, a hot solution is introduced into a vacuum, where the solvent evaporates and the solution is cooled adiabatically. This method is widely practised in large scale operations.

14.9.3 Applications of Crystallization

In food industry crystallization process is used for two specific purposes. Firstly, it is used to separate out a solid phase of different composition from a liquid phase and one or both the fractions may be valuable. Alternatively, crystallization is used without effecting separation of fractions in order to control or bring about desirable changes to the texture of the solid product.

Crystallization process involving separation. In order to achieve efficient separation, crystallization should yield maximum deposition of the desired component in the solid form with minimum amount of liquid trapped within the crystals or wetting the surface of the crystals. It is desirable to have crystals of uniform size and shape.

Crystallization processes involving separation of solid phase may be classified into two types depending on whether the liquid or the solid phase is more valuable. Examples of crystallization processes in which the

liquid phase is more valuable include winterization of oils and freeze concentration of fruit juices.

Winterization of oils. Vegetable oils particularly cottonseed oil and lightly hydrogenated soybean oil (iodine value 105) are used as salad oils in the manufacture of mayonnaise. These oils contain some triglyceride fractions with high melting points which deposit as solid crystals at refrigerator temperature spoiling the appearance of salad oils (due to turbidity) and also impairing their pouring properties. Deposition of solids would also destroy the emulsion in mayonnaise. The high melting triglyceride fractions are removed by fractional crystallization and the process is known as winterization of oils. The process involves super cooling of the oils with a low rate of crystal nucleation so as to allow the formation of large sized crystals of high melting triglyceride fractions. The process of cooling is spread over two or three days and once nucleation has started, stirring is stopped to prevent the breaking of crystals. The mixture is filtered to get the clear salad oil. Alternatively, crystallization from a volatile solvent such as acetone or hexane allows high melting triglycerides to crystallize out easily and quickly due to the lower viscosity of the liquid phase. The winterized salad oil is further stabilized by the addition of a fat crystallization inhibitor such as lecithin.

Freeze concentration of fruit juices. This process involves the fractional crystallization of ice from fruit juices. Apple juice is freeze concentrated to enhance the alcoholic content of cider. Citrus fruit juices are freeze concentrated so as to be preserved by freezing. The process also finds use in concentrating alcoholic beverages, vinegar and to adjust the alcohol content of wines. The advantage of this method is that heat degradation of labile food products is avoided and high quality products are obtained. However, the process is expensive and the degree of concentration achieved is limited.

Crystallization processes involving separation in which the solid phase is more valuable include salt manufacture and sugar manufacture.

Salt manufacture. In coastal areas with high insolation (i.e. exposure to considerable solar radiation) sea water is evaporated in shallow lagoon like salt pans by natural agencies. In large scale industries, multiple effect vacuum evaporators are used for salt manufacture. In a triple effect evaporator a temperature difference of about 13°C across each calandria is sufficient to maintain adequate heat transfer. The boiling point of water is increased by about 9°C due to the presence of salt. The vapour from the first calandria is superheated by about 9°C and hence a temperature difference of about 22°C exists in the boiling temperatures of the solutions in adjacent effects. Thus in a triple effect evaporator, temperature difference of about 44°C between the boiling temperatures in the first and

third effects exists. In the third effect, an adequate crystallization rate is obtained with a boiling temperature of 46°C and the vapour produced can be condensed with cooling water at ambient temperature. The boiling temperature in the first effect will be about 90°C which is quite acceptable since salt is completely stable at this temperature.

The salt crystals formed in such multiple effect evaporators are kept suspended in circulating brine and allowed to grow until they are large enough to settle to the base of the evaporator. The system can be operated continuously but due to the increase in concentration of impurity in the evaporating brine, the latter is completely replaced periodically. The normal crystal from vacuum evaporated salt is cubical. Potassium ferrocyanide is added as a crystal habit modifier to yield dendritic form of salt crystals required for specific purposes. These coarse absorbent crystals are required in curing hides and for making sausage casings.

Sugar manufacture. Sucrose solutions cannot be boiled at temperatures above 85°C as coloured impurities are formed. The crystallization rate is slow and uneconomical below 55°C. Because of the temperature constraint, multiple effect evaporator cannot be used for crystallizing sucrose as the temperature difference to be maintained between adjacent calandria will be much greater than 20°C because of the high thermal resistance of the solution. Hence single effect short-tube evaporators are used. The high fluid viscosity and the requirement of uniform crystal size lead to batch operation.

The syrup is drawn into the evaporator in amounts sufficient to cover the calandria and concentrated to a metastable supersaturated solution. Finely ground sugar is added as seed material to initiate crystallization by growth of the seed material. Syrup is added as the process proceeds and gradually the syrup/crystal mixture (called massecuite) level rises in the evaporator. At the end of the crystallization process the syrup/crystal ratio is allowed to fall (called tightening of the massecuite) to a degree limited by the need to maintain the fluidity of massecuite sufficiently to facilitate easy discharge. The discharged massecuite is centrifuged to separate the crystals from the syrup and the latter is fed to another evaporator pan. About two to four evaporation stages may be used, the syrup becomes increasingly impure and crystal growth becomes increasingly slower. The massecuite is discharged into crystallizers which are tanks where the massecuite is cooled in a controlled manner to maintain the syrup in the metastable zone in order to complete the crystallization process. The tanks are equipped with stirring apparatus designed to minimize crystal damage.

Crystallization processes not involving separation. A number of food manufacturing processes involve crystallization operation in which

no separation of the crystals is desired. Such crystallization occurs in the manufacture of frozen foods such as ice cream, sweetened condensed milk, butter, margarine, chocolate and certain sugar confectionery. The common requirement in such processes is that the crystals produced should be smaller than a certain critical size so as to give a desirable texture to the finished product. In candy manufacture, the sugar crystals are quite small and the crystal size is controlled to give a desired texture depending on the type of confectionery item.

Crystallization of ice in foods. The texture of thawed frozen foods is influenced to a large extent by the size of the ice crystals formed during freezing. A higher rate of nucleation and formation of small sized ice crystals are achieved by rapid cooling of foods during freezing. On the other hand, slow cooling of fruits, vegetables, meat and fish results in sparce nucleation and the formation of large sized ice crystals which disrupt the cellular structure of the food. The desirable smooth texture of ice cream is obtained by controlling the size of ice crystals during freezing. The ice cream mix is rapidly cooled by scraped-surface heat exchangers to a temperature where nucleation occurs freely. Mechanical agitation during cooling favours secondary nucleation resulting in the formation of small sized ice crystals. The frozen product is pumped out of the heat exchanger in a plastic condition and subsequently hardened by prolonged cooling. Repeated thawing and freezing due to fluctuations in the temperature of the freezing chamber results in the melting of ice crystals and recrystallization into larger size crystals which gives an undesirable sandy texture to the finished product. The addition of gel forming agents such as carboxymethyl cellulose (CMC) reduces the mobility of water and acts as a habit modifier preventing the growth of large sized crystals in ice cream and stabilize the foam.

Lactose crystallization. It is necessary to control the size of lactose crystals in a food product to less than 10 μm in order to obtain a smooth texture, because lactose crystals dissolve slowly in the mouth and if they are over 30 μm in size they impart an undesirable sandy texture to the product. Ice cream mix contains milk-solids-not-fat (MSNF) and prolonged storage of ice cream, particularly at fluctuating temperatures, promotes the growth of lactose crystals and hence MSNF content has to be decreased to prevent high lactose concentration. However, this reduces the overrun characteristics of ice cream.

Cow's milk contains about 5% of lactose contributing about 40% to the total solids content. A saturated solution of lactose at 20°C contains 16% lactose by weight and it is difficult to crystallize lactose in the presence of sucrose. In the manufacture of sweetened condensed milk, the excess lactose is crystallized out as small crystals by cooling to a temperature of about 30°C, where the product is supersaturated with

respect to lactose but not too viscous, then seeded with fine lactose crystals and held at a constant temperature with agitation to promote secondary nucleation.

Fat crystallization. Triglycerides of edible oils exhibit polymorphism, i.e. different crystalline forms due to differing molecular arrangements within the crystal lattice. *Enantiotropism* is reversible polymorphism as it involves a reversible transition between two different crystalline forms at a specific transition temperature. In contrast, monotropism is irrevesible polymorphism. Almost all triglyceride polymorphs exhibit monotropism, i.e. a lower melting crystal can only transform into a crystalline form of higher melting point but the transition is not reversible. Hence it is easy to supercool fats down to the temperature of fusion of lowest melting form (i.e. least stable form). Crystallization of low melting form of fats occurs rapidly. These crystals can transform to crystals of higher melting forms, the rate of transformation being a function of temperature. If the crystals are melted and the liquid fat cooled, low melting crystalline form of fat can again be produced.

Monotropic crystallization behaviour of fat crystals poses a problem in chocolate manufacture. Once the chocolate is solidified in the last step of manufacture, the fat crystals produced must be small enough to give a smooth texture to the finished product and must be in stable crystalline form so that subsequent recrystallization is avoided. Recrystallization of fat crystals in a finished chocolate results in the loss of surface appearance and is called fat bloom. Hence during chocolate manufacture, production of fat crystals of suitable type and stable form is an important step known as tempering. During tempering, the triglycerides in chocolate are melted at about 49°C. On cooling to about 27°C, nuclei of unstable form of fat crystals are formed. On re-heating to about 32°C, the unstable form of fat crystals are converted to the more stable β polymorphic form. Subsequent crystal growth, as the chocolate is cast or enrobed, will be in this form. Chocolate bloom can occur due to improper tempering during manufacture and also when the finished chocolate is subjected to repeated temperature fluctuations as in shop windows. This can be inhibited by addition of milk fat (bloom inhibitor) during manufacturing.

Margarine is a water-in-oil emulsion, the oil phase containing a mixture of oils and hence a whole range of different triglycerides. At room temperature, a good finished table margarine contains about 20–25% of the total triglycerides in solid form with the size of the fat crystals being in the range of 3–10 μm. Bigger fat crystals will give a grainy texture while very small sized crystals give a product lacking in plasticity. Hence it is necessary to control the size of the fat crystals. The emulsion is passed through a scraped surface heat exchanger so that it is supercooled to a hard mass. Slight warming and mechanical working of the

supercooled mass causes recrystallization of the fat into the required final composition.

Peanut butter needs to be stabilized because peanut oil is a liquid at room temperature and separates out. Hydrogenated peanut oil and other fats are used for stabilizing the peanut butter. The stabilized peanut butter has a matrix of fine crystals of extreme habits which complex or adsorb much of the liquid oil and retain the emulsion characteristics.

14.10 HEAT PROCESSING

Heat processing is chemical conversion operation in which heat is applied to the food raw materials primarily to effect chemical changes ultimately yielding the processed food. The term *heat processing* is applied to unit processes such as cooking, boiling, frying, baking, grilling, blanching and other heat activated chemical conversion operations.

Heat processing is effected by heat transfer to food by all the three modes—conduction, convection and radiation, sometimes in combination of two or all the three modes. Heat processing also changes the physical properties of the food such as viscosity and density and these in turn affect the thermal behaviour of the food. Food is a complex system because of its compositional heterogeneity. In addition, the geometry of food units vary widely and hence theoretical treatment of heat transfer in food is difficult. Heat processing improves the quality of food by enhancing natural flavour of food, softenes hard material and increases the digestion. Heat processing destroys microorganisms and toxins in the food and renders it safe for consumption. Heat processing if carried out improperly can also destroy the thermolabile nutrients, vitamins, flavour and texture of the food.

14.10.1 Methods of Applying Heat to Foods

The main sources of heat for use in food processing include solid fuels (coal, coke or wood), liquid fuels (kerosene, fuel oil or paraffin), gaseous fuels (natural gas, petroleum gases or coal gas) and electrical energy.

The heat generated from the fuels is applied to food either directly or indirectly. Direct heating is effected by the use of solid, liquid or gaseous fuels, infrared sources, dielectric or microwave ovens.

Infrared heating is carried out by radiant heaters located in a tunnel through which the food is conveyed. The radiant heaters consist of electrically heated filaments which emit infrared radiation in the wavelength region of 0.75 to 350 μm. Only wavelengths up to about 50 μm are of practical importance in heating of foods. Water in the food absorbs

best at wavelength of about 1 μm. Infrared heating is characterized by low penetration and rapid surface cooking and surface browning of the food. Water and volatile flavour are retained and hence the food is juicy and full of flavour. Heat penetration to the interior of the food piece is by conduction, a slow process. Hence it is necessary to achieve a predetermined centre temperature. Radiant heating is used in grilling, toasting, baking, cooking, drying of sugar cubes and nuts, melting of fats and in freeze drying of food items.

Dielectric heating of foods occurs when an electrically insulated material (food item) is subjected to alternating electric field (27 MHz is common) by placing it sandwiched between capacitor plates connected to a high frequency alternating generator. Heat is produced due to molecular friction caused by the rapid orientation of the electric dipoles under the influence of the high frequency alternating applied field. The dielectric heating equipment consists of a belt, which conveys the food at a controlled rate between the plates of the capacitor. Control of the heat generated in the product is achieved by power input or by adjusting the distance between the plates. Heating occurs at some depth below the surface of the food and hence heat losses are minimal. There will be no surface browning of the food. Dielectric heating is used to thaw frozen eggs, meat, fruit juices and fish. It is also used to melt butter, fats and chocolate, in baking biscuits, drying of sugar cubes and crisp bread, heating of peanuts used in confectionery.

Microwave heating of foods involves the use of microwave radiation of 897 or 2450 MHz frequency. Heat is produced by molecular vibration by absorption of radiation (in contrast to electrostatic interaction in dielectric heating). No heating occurs if the food is transparent. Microwave heating differs from dielectric heating in that higher frequencies (higher energy) are used but at a lower voltage. Hence dielectric stress within the food is reduced and 'flash-over' is mostly eliminated. Microwave heaters consist of a high frequency generator from which energy is transported by a hollow rectangular waveguide or by a coaxial cable to a heating chamber usually a metal oven. Microwave heating finds use in heating precooked, frozen foods and in conjunction with browning effect of infrared heaters for high speed cooking of food in large canteens and hospitals. It is also used in potato crisp finishing, apple juice evaporation, in pasteurization of fruit juices and reducing mold growth in bread, cakes and jam.

Indirect heating is effected by the use of hot gases (steam or air), liquids (water or organic heat exchanging liquids) or electrical resistance heating. Electrical resistance heating is carried out by attaching resistors to the process vessel walls or by immersing sheathed resistors in the food material to be heated (immersion heating). Nickel-chromium heating elements working at about 800°C are used.

14.10.2 Food Conversions

Blanching, baking, cooking, roasting, broiling and frying are the most important food conversion operations, the first two being carried out in industrial scale.

Blanching. Blanching is an essential step of heat processing used for agricultural products. Fruits and vegetables are blanched prior to canning, freezing or dehydration. The aims of blanching are: (i) to inactivate the native enzymes and to destroy enzyme substrates such as peroxides, (ii) to reduce the microbial load in the food raw material, (iii) to soften and shrink the food facilitating the filling into containers (iv) expel cellular gases thereby reducing can-corrosion and improve the texture particularly of dehydrated foods. However, loss of heat sensitive vitamins and leaching out of water soluble nutrients may occur. Overblanching damages the texture of the product.

Canning of unblanched food requires more time to reach the desired temperature to inactivate the enzymes. In the case of low temperature processes such as spray drying, freeze drying or vacuum evaporation the temperature may not be sufficient to inactivate the enzymes and enzyme activity will continue in unblanched food resulting in the loss of quality of the processed food due to the formation of unacceptable odours, flavours, colours or texture.

Blanching is effected by heating the food rapidly to a selected temperature and holding the food at this temperature for a specified time followed by rapid cooling or passing on to subsequent processing operation. Two methods are commonly used for blanching. These include: (i) immersion blanching using hot water and (ii) steam blanching. Immersion blanching involves passing the food at a controlled rate through a perforated drum revolving in tank of water maintained at prescribed temperature in the range of 75–100°C. Immersion blanching leads to loss of solubles in food. A rigorous practice of plant hygiene is necessary to avoid contamination from thermophilic organisms. Steam blanching involves conveying the food on a mesh belt into a steam chamber at atmospheric or lower pressure, the residence time being controlled by the speed of the conveyor. The blanched product is washed and cooled as steam blanching does not clean the product as efficiently as immersion blanching. A costlier method is microwave blanching applicable for fruits and vegetables packaged in film bags.

Baking. Baking is yet another important heat conversion operation in the manufacture of bread, biscuits and cakes. A series of complex chemical reactions occur at different temperature ranges during the baking of fermented food such as bread. These include dough softening (40–80°C), starch gelatinization (55–120°C), dextrinization (40–60°C),

enzyme activity (30–80°C), gas expansion (40–120°C), protein coagulation (80–130°C), yeast activity (30–75°C) and volatilization, oxidation, esterification and changes in flavour (40–250°C). It is necessary to control the heating carefully to achieve desired colour, flavour, aroma and texture of the finished bread.

Cooking, roasting, grilling and broiling. Cooking is carried out in different media such as air, water, steam, fat or oil. Grilling, roasting and baking involve cooking in air. In grilling, the food is heated by radiation only leading to surface browning and the heat is conducted through the food. Roasting and baking are similar in that they are carried out in an air oven at temperatures in the range of 120–260°C. Heat is transferred by radiation, conduction and convection. The food is cooked partially by dry heat and partially by moist heat. Broiling consists of cooking food (particularly meat) by direct radiant heat such as open fire of a gas flame, live coal or electric oven. Heat is conducted through the food by direct contact with the hot broiler. Heat transfer occurs by radiation to a smaller extent. The method of cooking food kept immersed in water is generally known as boiling, simmering or stewing. Heat transfer occurs through the water medium via conduction and convection. Simmering and poaching involve cooking food immersed in water at temperatures just below the boiling point. Cooking in a pressure cooker using steam involves moist heat. The food wrapped in aluminum foil or plastic sheet may be cooked with no loss of flavour. Cooking in fat is frying which may be shallow frying or pan frying (sauteing) or deep frying. Sliced or thin pieces of food are pan fried with the food turned from one side to other to complete the cooking. Heat transfer occurs partly by conduction due to contact with the heated pan and partly by convection currents. Deep fat frying is similar to boiling but higher temperatures is achieved during frying. Rapid drying of the surface of the food occurs resulting in a hard, crisp, brown surface. Appreciable quantity of fat is absorbed by the food enhancing its calorific value.

Questions

1. How are fibrous food materials disintegrated?
2. Write a note on the types of equipment used in food industry for size reduction.
3. What are emulsifying agents? How do they function?
4. Write a note on mixing and emulsifying equipment.
5. Discuss briefly the principles and applications of membrane separation methods in food industry.

6. Give an account of the applications of extraction in food processing.
7. What are the different types of centrifuges used in food industry? List their applications.
8. Write notes on the unit operation of crystallization in the manufacture of the following food products: (a) salt, (b) sugar and (c) salad oil.
9. Give an account of the applications of crystallization operation without involving separation of products.
10. What are heat processing operations? How is heat applied to food?
11. What is blanching? Why is it carried out?
12. Write a note on the different types of heat processing of foods.

CHAPTER

15

Food Preservation and Use of High Temperatures

15.1 PRINCIPLES OF FOOD PRESERVATION

The basic principles of food preservation primarily involves the process of inhibiting

(i) the growth and activity of microorganisms

(ii) activity of endogenous enzymes

(iii) chemical reactions which may deteriorate the quality of food and

(iv) invasion and spoilage by insects and rodents.

In addition, spoilage of food may be caused during mechanical handling, processing, packaging, storing and transportation. Appropriate care has to be exercised to prevent deterioration of quality of food.

Several methods are available for preservation of food based on the above principles. The methods include: (i) preventing the accessibility of food to microorganisms by asepsis and packaging, (ii) physical removal of microorganisms from food by filtration or centrifugation, (iii) hindering the growth and activity of microorganisms by use of preservatives, use of low temperatures, atmospheric control in packaging and storing of foods and decreasing water activity in foods by drying or evaporation, (iv) killing the microorganisms by use of high temperatures and ionizing radiation, (v) inactivation of endogenous enzymes by moderate heating, (vi) inhibition of chemical reactions through the use of chemical additives and (vii) fermentation of foods to yield more stable or less perishable food products. Food preservation as it is practised in the industry always involves the use of combination of methods for achieving maximum effectiveness.

Asepsis or preventing the accessibility of food to microorganisms is well exemplified in nature. The protective covering on natural foods such as skins on fruits and vegetables, shells on eggs and nuts and skins and membranes on livestock and fish prevent the attack by microorganisms and maintain the living tissues in healthy condition. Microbial attack is facilitated only after the death of the animal or when the skin is physically damaged. Packaging of foods and food products in a variety of materials such as metal cans, plastic films, pouches, bags or boxes, paper bags or cartons and glass bottles provide effective protection against microbial attack.

Filtration or *centrifugation* is adopted to physically remove microorganisms particularly in liquid foods such as milk, soft drinks, fruits juices and alcoholic beverages such as wine and beer.

The use of preservatives and additives in food processing and preservation and fermentation to enhance the shelf life of the foods have already been discussed in earlier chapters.

15.2 USE OF HIGH TEMPERATURES FOR FOOD PRESERVATION

Preservation of food by the use of heat finds very wide applications compared to other methods. Heat may be used either for processing or conversion of foods or simply as a means of preserving the food. In heat processing or conversion the application of heat is used primarily to effect chemical changes in food. Cooking, frying, and baking involve both processing and preservation operations. Cooking makes food palatable and tender and also destroys a large proportion of microorganisms and natural enzymes. Cooked foods can be stored for several days provided they are protected from recontamination. Refrigeration of cooked food is a normal household practice to prolong the storage time. However, cooking will not sterilize a product. Cooking also destroys the toxin formed by *Clostridium botulinum* during a ten minute exposure of the food to moist heat at 100°C. Thus cooking provides a final measure of protection for the consumer from food borne diseases.

The killing of microorganisms by heat is due to thermal denaturation of proteins and enzymes of the microorganism required for its metabolic activity and growth. The heat treatment necessary to kill the organisms or spores varies with the kind of organism, its state and the environment during heating. The type of heat treatment will depend on the kind of organisms to be killed, other preservative methods to be employed and the effect of heat on the food. The use of heat also affects the food adversely and hence it is necessary to use only mild heat treatment that ensures freedom from pathogens and enzyme activity and enhance the shelf life of the food.

15.3 FACTORS AFFECTING HEAT RESISTANCE OF MICROORGANISMS

Cells and spores differ widely in their ability to resist high temperatures. Even within a population of cells and spores the heat resistance varies as indicated by the thermal death time or frequency distribution. In general, a small number of cells have low resistance, most of the cells have a medium resistance and a small number have a high resistance. The various factors influencing the thermal death time (heat resistance) of microorganisms include the following.

(i) *Temperature-time relationship.* The time required for killing cells or spores under a given set of conditions decreases as temperature is increased. For example, the time required to kill the spores of *C.botulinum* at an initial population of 6×10^{10} in buffered medium at pH 7 has been estimated to be 260 minutes at 100°C, 120 minutes at 105°C, 36 minutes at 110°C and 5 minutes at 120°C.

(ii) *Initial concentration of spores or cells.* The more the number of spores or cells present, greater is the heat treatment required to kill them. For instance, the time required to kill spores of a thermophilic organism in spoiled canned corn juice with a pH of 6.0 at a temperature of 120°C has been found to be 14 minutes for an initial concentration of 50,000/ml, 10 minutes for 5,000/ml and 9 minutes for 500/ml. Similarly, the thermal death time of the spores of *C.botulinum* (in a buffer of pH 7) at 100°C was found to be 110 minutes for an initial population of 3.2×10^7 and 50 minutes for an initial population of 1.64×10^4.

(iii) *Previous history of the vegetative cells or spores.* The initial conditions under which the cells have grown and spores produced and the subsequent treatment to which they have been exposed will influence their resistance to heat. The parameters of importance include: (a) culture medium, (b) temperature of incubation, (c) phase of growth or age and (d) desiccation. The better the medium for growth, the more the resistance of cells or spores. The temperature of growth of cells and the temperature of sporulation influence their heat resistance. In general, heat resistance increases as the incubation is raised toward the optimum value for the organism. For example, *E.coli* is more heat resistant when grown at 38°C which is near its optimum temperature for growth than at 25°C. The heat resistance of vegetative cells varies with the stage of growth and spores with their age. Cells have greatest resistance during the lag phase but almost as great resistance during their maximum stationary phase. They are least resistant to heat during their logarithmic phase of growth. Young and immature cells are less resistant to heat than mature ones. Dried spores of some bacteria are harder to kill by heat than those kept moist.

(iv) *Composition of the substrate.* The composition of the substrate or food in which the cells or spores are heated has a profound influence on the heat resistance of the organism. The following factors are of relevance with respect to the composition of the substrate.

(a) *Moisture content*. Moist heat is more effective as a killing agent than dry heat.

(b) *pH of the medium*. Cells or spores have greater heat resistance at or near neutral pH values. Hence foods may be classified on the basis of their pH into four broad categories.

- High acid foods: pH < 3.7; e.g. berries, sauerkraut.
- Acid foods: pH 3.7–4.5; e.g. tomato, pear, pine apple.
- Medium acid foods: pH 4.5–5.3; e.g. beet, pumpkin, spinach.
- Low acid foods: pH >5.3; e.g. corn, peas, beans, meat, fish, poultry and milk.

The heat process required in the canning of food increases with their pH. The effect of pH of the substrate is complicated because of the fact that heating at high temperatures causes a decrease in the pH of low or medium acid foods; the higher the original pH, the greater the drop in pH caused by heating. Foods with pH less than 5.5–5.8 change little in acidity on heating. Foods adjusted to more alkaline pH give increasing protection to spores against heat as the pH increases towards 9.0.

(c) Other constituents in food, particularly, sodium chloride in low concentration has a protective effect on some spores. Sugars may protect some spores; glucose protects *E.coli* and *Pseudomonas flourescens* against heat better than sodium chloride at water activity levels near the minimum water activity required for growth. Glucose is harmful to *Staphylococcus aureus*, whereas sodium chloride is very protective.

(d) The concentration of solutes may affect the heat process necessary for sterilization. Hence canners classify foods further as high-soluble-solids foods (e.g. syrups and concentrates) and low soluble-solids foods such as fruits, vegetables and meat.

(e) Antiseptic or germicidal substances in the substrate aid heat in the destruction of microorganisms. Thus hydrogen peroxide with heat is used to reduce bacterial content of sugar and milk for cheese making. The combination of antibiotic or plant extract or irradiation with ionizing rays and heat has been suggested for the treatment of canned foods.

15.4 THERMAL DEATH TIME OF BACTERIAL CELLS AND THEIR SPORES

The heat resistance of microorganisms is expressed in terms of their thermal death time which is defined as the time required at a certain temperature to kill a stated number of organisms (or spores) under

specified conditions. This is referred to as 'absolute thermal death time' to distinguish it from the 'majority thermal death time' for killing most of the cells or spores present. The 'thermal death rate' expresses the rate of killing. Thermal death point, now used little, is defined as the temperature necessary to kill all the organisms in a fixed time, usually 10 minutes. The thermal death time varies with the type of organism and temperature as shown in Table 15.1. Similarly, the thermal death times of different bacterial spores in minutes at a heat treatment temperature of 100°C vary with the type of spore. Thus the time required to kill bacterial spores of *B.anthracis* is 2 minutes while the spores of *B.subtilis* require 15–20 minutes. Spores of *C.botulinum* require 100–300 minutes and the thermophilic spores of flat sour bacteria require more than 1030 minutes.

TABLE 15.1 Thermal Death Times of Bacterial Cells

Bacterium	*Time* (min.)	*Temperature* (°C)
Gonococcus	2–3	50
Salmonella typhosa	4	60
Staphylococcus aureus	19	60
E. coli	30	57
Streptococcus thermophilus	15	75
Lactobacillus bulgaricus	30	71

15.5 PRINCIPLES OF THERMAL DESTRUCTION OF SPOILAGE MICROORGANISMS

The quality of heat processed foods are influenced both by temperature and duration of the thermal process. Underprocessed food will be liable to bacterial spoilage and overprocessed food will be nutritionally and organoleptically inferior and hence unacceptable. The parameters of a suitable thermal process may be estimated on the basis of the knowledge of the resistance of spoilage microorganisms and the temperature history of the food during processing.

The rate of killing of bacteria by heat can be described in terms of first order kinetics. The death rate is almost proportional to the number of surviving cells present in the system being heated. The rate of destruction may be expressed as

$$(dN/dt) = -kN \tag{15.1}$$

where N is the number of viable organisms at any time, t is time in minutes and k is rate constant in min^{-1}. The rate constant is a function of the type of microorganism and temperature. Rearranging and integrating, the above equation gives

$$\ln(N_i/N) = kt \quad \text{or} \quad N = N_i\, e^{-kt} \tag{15.2}$$

where N_i is the initial spore or vegetative cell concentration in one ml at time $t = 0$ (also called the contamination level).

Microbiologists usually refer to decimal reduction time, D, which is the time in minutes during which the initial number of microorganisms is reduced to $1/10^{th}$, i.e. 90% reduction. Substituting for the ratio of (N/N_i) in the above equation, we get

$$(N/N_i) = 1/10 = e^{-kD} \tag{15.3}$$

and taking $\log_{10}$ on both sides and solving for D as

$$D = 2.303/k. \tag{15.4}$$

Combining Equations 15.2 and 15.4 gives Equation 15.5

$$t = D \log (N_i/N) \tag{15.5}$$

A graphical plot of log (N/N_i) versus t gives a straight line. This is referred to as a logarithmic order of death, i.e. under constant conditions the same percentage of the bacterial population will be destroyed in a given time interval, regardless of the size of the surviving population, as given in Figure 15.1. (If a given temperature kills 90% of the population in one minute, 90% of the remaining population will be killed in the second minute and so on). The decimal reduction time or D-value decreases the surviving population by one log cycle. The logarithmic plot is applicable for most vegetative cells and approximately for spores. Bacterial spore plots sometimes deviate from the logarithmic death rate curve, particularly during a short period immediately following exposure

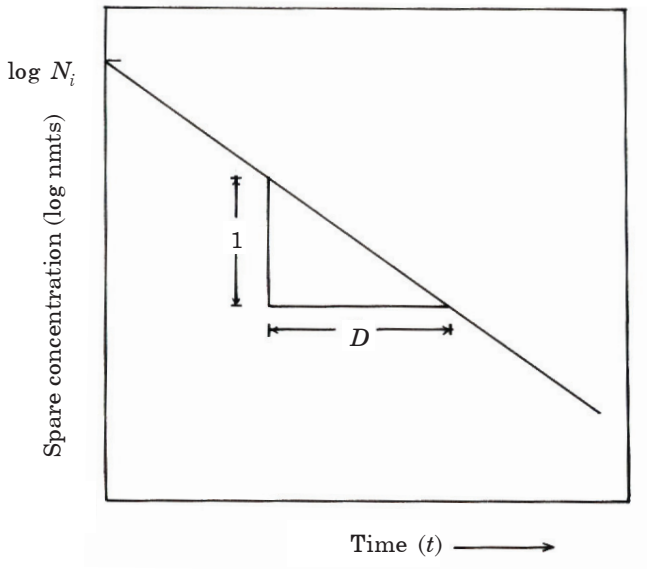


Figure 15.1. Logarithmic order of death of microorganisms.

(Source: Food Engineering Operations, 2nd ed., Brennan, Butters, Cowell, and Lilly, Applied Science Pub. Ltd., London, 1979, p. 252.)

to heat. However, for thermal processing purposes a logarithmic curve is used for cells as well as for spores.

The microbial death rate is experimentally determined for a given cell or spore suspension in a capillary or test tube. A number of these tubes containing a known count (number) of cells or spores are suddenly immersed into a hot bath at constant temperature for a specified time, taken out and chilled. The number of cells or spores remaining alive after the heat treatment is determined by plate count.

15.5.1 Thermal Process Time for Sterilization

The thermal process time required for sterilization is usually based on the necessity to kill all the spores of *Clostridium botulinum* which produces the deadly poisonous neurotoxin, particularly in canned meat products. Complete sterility with respect to this spore is the purpose of thermal processing. As a measure of safety a large initial population of the organism is assumed to be present and any heat treatment process should reduce the number of spores by a factor of 10^{12}. Such a large initial population would make the food unfit for consumption. Since D-value represents the time in minutes required to reduce the initial spore population by 10^1 (i.e. one log cycle), substituting into equation 15.2 and solving for t gives

$$t = 12\,(2.303/k\,) = 12\,D \tag{15.6}$$

The time t is equal to 12 D, often called as 12 D concept, a mathematical concept which has been found empirically to give effective sterilization. The 12 D concept, though somewhat arbitrary, is universally used as a measure of safety.

15.5.2 Effect of Temperature on Thermal Death Time

The decimal reduction time is one type of thermal death rate providing information on the rate of destruction of a specific organism in a specific medium or food at a specific temperature. It is necessary to have information of thermal death at different temperatures in order to arrive at time-temperature relationship and optimize process conditions. The effect of temperature on the death rate constant k may be expressed by an Arrhenius type equation

$$k = A\,e^{-E/RT} \tag{15.7}$$

where A is an empirical constant, R is the gas constant in kJ/g mol.K (or cal/g mol.K), T is absolute temperature in K and E is the activation energy in kJ/g mol (or cal/g mol). The value of E is in the range of 210

to 418 kJ/g mol (50–100 k.cal/g mol) for vegetative cells and spores and much less for enzymes and vitamins. Substituting equation 15.7 into 15.2, we have

$$\ln\ (N_i/N) = A \int_{t=0}^{t} e^{-E/RT}\, dt \tag{15.8}$$

At constant temperature T, equation 15.8 becomes equation 15.2. Since k is a function of temperature, the decimal reduction time D is also a function of temperature. Hence D is often written as D_T to show that it is temperature dependent and also to specify the temperature.

The approximate heat resistance of the spores of thermophilic and mesophilic spoilage organisms may be compared by using the D_T values. Thus the most heat resistant spores of *C.botulinum*, Types A and B have D_{250} values of 0.1–0.2, while the D_{250} values for other spores are 4.0–5.0 for *B. stearothermophilus*, 3.0–4.0 for *C. thermosaccharolyticum* and 0.1–0.15 for *C.sporogenes*.

A plot of the thermal death rates (D_T) versus T in degree Fahrenheit (°F) (or °C) on a semilog scale gives a straight line (thermal death time curve or TDT curve) over the range of temperature used in food sterilization, as shown in Figure 15.2.

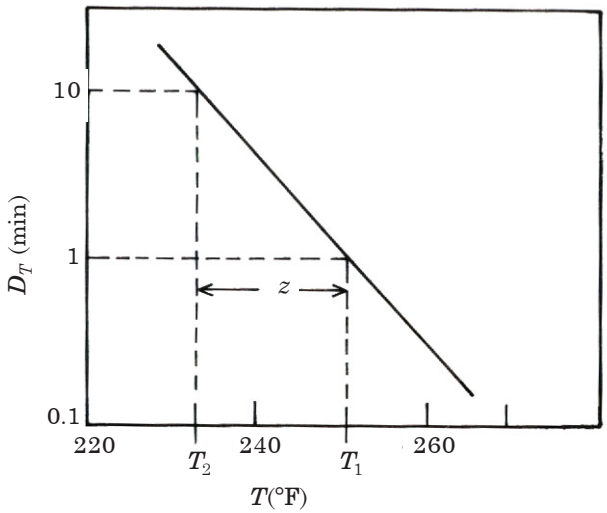


Figure 15.2 Effect of temperature on decimal reduction time.

(Source: Transport Process and Unit Operations, 3rd ed., by C.J. Geankoplis, Prentice-Hall of India, New Delhi, 1997, p. 572.)

Actually, by combining Equations 15.4 and 15.7, it can be shown that the plot of log D_T versus $1/T$ (in K) is a straight line. A straight line is also obtained by plotting log D_T is against T in °F or °C. The term z represents

the number of degrees required for a specific TDT curve to pass through one log cycle (temperature range in °F for a 10:1 change in D_T). Since the plot is a straight line,

$$\log D_{T2} - \log D_{T1} = 1/z\ (T_1 - T_2) \tag{15.9}$$

Taking T_1 as 250°F (121.1°C) as the standard temperature against which thermal processes are compared and setting $T_2 = T$ the above equation may be rewritten as

$$D_T = D_{250} \cdot 10^{(250 - T)/z} \tag{15.10}$$

For *Clostridium botulinum*, the value of z = 18°F (or 10°C). It means that each increase in temperature by 18°F the death rate will increase by a factor of 10 (one log unit) which is quite high compared to a factor of 2 for chemical reactions in general. While D reflects the resistance of an organism to a specific temperature, z provides information on the relative resistance of an organism to different destructive temperatures. Different organisms in a given food will have different z values which characterize the resistance of the microbial population to change in temperature. Similarly, a given organism will have different z values in different foods. It allows for the calculation of equivalent thermal processes at different temperatures. For example, if heat treatment for 3.5 min. at 140°F is adequate for an organism with a z value of 8 in a given food, heat treatment for 0.35 min. at 148°F or 35 min. at 132°F can be considered as equivalent for destroying the specified organism in the specified food.

The lethal capacity of a thermal process is usually referred to in terms of F values. The F value represents the number of minutes to destroy a specified number of an initial population of an organism with a specific z value at a specific temperature. Taking the temperature of 250°F as standard temperature the integrated lethal value of heat received by all points in a container during heat processing is designated as F_0 value which is derived as

$$F_0 = D_{250}\ (\log N_i - \log N) \tag{15.11}$$

The D_{250} value for *Clostridium botulinum* is 0.21 and considering the 12 D concept to achieve sterilization of a food item with respect to this organism, the F_0 may be arrived at as

$$F_0 = 0.21\ (\log 1 - \log 10^{-12})$$
$$= 0.21 \times 12 = 2.52$$

Thus a heat processing for 2.52 min. at 250°F will reduce the population of the organism by a factor of 10^{12}. A minimum safe public health sterilization value for low acid canned foods is considered to be F_0 = 3.0 minutes (F_0= 2.52 rounded off). The conditions during heating and the composition of the food have profound effect on the heat resistance of organisms. The equivalent minimum public health sterilizing values at 250°F for *C. botulinum* would be 3.0 min. at pH 6.0–7.0, 2.3 min. at pH

5.5, 1.6 min. at pH 5.0 and 1.2 min. at pH 4.6. Heat treatment with different combinations of time and temperature that give the same lethality as the reference F_0 value are also given the same F_0 values. Most of the commercial processes exceed this requirement as F_0 = 6.0 to 8.0 are quite common to eliminate the more resistant thermophilic spore formers.

15.5.3 Factors Affecting Thermal Process Time

The thermal process time necessary for preservation of a chosen food depends on the rate of heat penetration into a food. Since every part of the food in a container or can must receive the same adequate heat treatment to prevent its spoilage, the process time would depend on the part which receives heat most slowly. The food part near the centre of the container usually receives heat most slowly. Heat penetration in food is mainly due to conduction as well as convection, which may operate simultaneously or successively. Conduction occurs in solid particles of foods and is slow while heat transfer by convection occurs in liquids. Some foods such as sugar syrup, thick soup and tomato juice undergo changes in their consistency during heating. A variety of factors influence the time required for thermal processing for a food. These include: (i) the material of the container—glass having a slower rate compared to that of a metal, (ii) the size and shape of the container—smaller and slim cylindrical cans attain the desired temperature faster, (iii) initial temperature of the food—a higher initial temperature provides the lethal range for the microorganisms for a longer time and its average temperature during heating is higher than the food in the can with a lower initial temperature. A higher initial temperature is necessary for foods which heat slowly, e.g. meat, pumpkin etc., (iv) retort temperature—a higher retort temperature heats the food rapidly, (v) consistency of the food—nature, sizes and shapes of pieces decide the rate of heat penetration. The pieces of food may retain their identity as in peas, plums, beets, asparagus and whole grain corn with smaller pieces requiring less time. Pieces may become mushy or change into a viscous mass as in the case of pumpkin, sweet potatoes, squash and cream-style corn, which get heated slowly as conduction is the main mechanism of heat transfer. Pieces may form layers as in the case of spinach and asparagus. In addition, the consistency of the food depends on the different types of sauce, starch, salt or sugar that may be present. For example, tomato sauce on baked beans slows down heat penetration more compared to plain sauce. Starch concentrations up to 6% decreases heat penetration, but at higher concentrations, has no additional effect. Sodium chloride does not influence heat penetration, as its concentration is usually low. On the other hand, increasing concentration of sugar decreases heat penetration. However, this effect is counteracted by the decrease in viscosity of sugar solutions with increasing temperature and (vi) rotation

and agitation of the container during heat processing hasten heat penetration, particularly if the food is a fluid, but may cause undesirable physical damage to some solid foods.

Cooling of the canned foods also is influenced by the same factors as the heating process. Rapid, artificial cooling is usually adopted to effect a greater control on the cooling rate, because slow cooling may cause overcooking of the food and also allow the growth of thermophilic organisms.

15.5.4 Calculation of Thermal Process Time and Adequacy of Heat Treatment

For calculating the thermal process time for a chosen food the information about the following are important.

(i) TDT curve for the most heat resistant organism likely to be present in the food—in low acid foods a thermophile such as flat sour organism is common.

(ii) The heat penetration and cooling curve for the food in the chosen container of specific size and shape should be known or determined experimentally.

The calculation of time in minutes necessary for the heating process in order to destroy a certain number of heat resistant organisms or spores with known z and F values, in a given container of food by heating at a temperature T is done by using Equation 15.12.

Since

$$(\log t - \log F)/\log 10 = (250 - T)/z$$

$$\log t/F = (250 - T)/z \quad \text{and} \quad t/F = \text{antilog } (250 - T)/z$$

or

$$t = F \text{ antilog } (250 - T)/z \tag{15.12}$$

The ratio t/F = time to destroy the organism at temperature T if $F = 1$ and F/t = lethal rate at T. The adequacy of a heat treatment process for a given food in a chosen container may be calculated by using Equation 15.13 arrived at by combining 15.5, 15.10 and 15.11.

Thus

$$F_0 = t \cdot 10^{(T - 250)/z} \tag{15.13}$$

where T is degree Fahrenheit and $z = 18\ F$, or T is in degree Celsius and $z = 10\ C$.

The effects of different but successive sterilization processes in a given material are additive. Hence for several different temperature stages T_1, T_2 and so on each having different times t_1, t_2 ... the F_0 values for each stage are added to arrive at the total F_0 value.

$$F_0 = t_1 \cdot 10^{(T_1 - 250)/z} + t_2 \cdot 10^{(T_2 - 250)/z} + \ldots. \tag{15.14}$$

For example, if a given food is heated in a retort for sterilization, the temperature varying for different periods from 0–20 min. at 160°F (t_1=20 min. and T_1 = 160°F), 20–40 min. at 210°F (t_2 = 20 min. and T_2 = 210°F) and 40–73 min. at 230°F (t_3 = 33 min. and T_3 = 230°F), the total F_0 value of the heat treatment may be calculated using Equation 15.13 as follows. Taking the F_0 and z values for *C.botulinum* as 2.50 min. and 18 F in the given food and substituting the temperatures and corresponding times in the above equation, we have

$$F_0 = 20.10^{(160-250)18} + 20.10^{(210-250)/18} + 33.10^{(230-250)/18}$$

$$= 0.0020 + 0.1199 + 2.555 = 2.68 \text{ min.}$$

Since the required F_0 value is 2.50 min. for the chosen food, the heat treatment provided by the different time-temperature combinations is adequate.

15.6 COMMERCIAL HEAT PRESERVATION METHODS

Commercial heat preservation methods may be classified into three types: sterilization, pasteurization and blanching. When heat treatment is done above 100°C it is known as sterilization, and when it is done below this temperature it is called pasteurization. Blanching is also a kind of pasteurization in which the natural enzymes in fruits and vegetables are inactivated.

15.6.1 Sterilization

The aim is complete destruction of microorganisms. Because of the resistance of certain bacterial spores to heat, a treatment at 121°C (250°F) of wet heat for 15 minutes or its equivalent is necessary for sterilization. Every particle of the food must receive this heat treatment. If a can of food is to be sterilized, immersing it into a pressure cooker or retort at 121°C for 15 minutes will not be sufficient because of the low rate of heat transfer through the food. Depending upon the size of the can and the type of food, sterilization may require several hours. During this time, many changes can occur to depreciate the quality of the food. Fortunately, many of the foods need not be sterilized completely. Most of the canned and bottled products are commercially sterile. It means a degree of sterilization at which all pathogenic and toxin forming organisms as well as all other types of food spoilage organisms have been destroyed. Commercially sterilized food may still contain a very small number of resistant bacterial spores, but these will not normally multiply in food. However, if they are isolated from the food and given suitable environmental conditions they will multiply. Commercially sterile canned foods have a shelf life of 2 years or more. Even after longer periods, deterioration in the quality of such

foods is generally due to texture or flavour changes rather than to growth of microorganisms.

Canning. A variety of foods are canned. These include whole fruits and vegetables, sliced or diced fruits and vegetables, meat and meat products, fish and fish products and soups. Two general methods of sterilization are commonly used in industry: (i) in-pack sterilization (sterilization inside containers) and (ii) sterilization of the food before placing in the containers.

The in-pack sterilization involves several steps. These include: (i) cleaning and grading of food raw materials, (ii) blanching to inactivate native enzymes (particularly in vegetables and fruits), (iii) filling or placing the cleaned raw food in a sealable container, with added brine in the case of vegetables, meat and fish, or sugar syrup in the case of fruits, (iv) deaeration (to prevent bulging or bursting of can during heating) followed by closing and sealing the container, (v) heating the container in a retort at specified temperature for a specific holding time, (vi) cooling partially under pressure in the retort followed by cooling in a cooling tank and (vii) labeling, storing and/or marketing. The sterilization process may be operated in batch or continuous mode involving vertical or horizontal retorts, pressure cooker, rotary or reel cooker or hydrostat cooker or non-pressurized methods. All these procedures use temperatures above 100°C and hence the pressure developed is greater than atmospheric pressure. The containers may be cans, necked-in flange type (to control post-canning infection) or tin-free steel type cans, glass jars, film pouches or sterilizable plastic containers of rigid, semi-rigid or flexible forms. Heat penetration in these packs is controlled by the geometry of the pack and by the thermal properties of the food, the container and the heating medium. Hence sterilization of food placed within the containers is a slow process.

Asceptic canning involves: (i) sterilizing the food separately outside the containers, (ii) placing the food in previously sterilized cans under asceptic conditions and (iii) sealing the containers. The advantages in this method include:

- high temperature processing at temperatures up to 150°C using high speed heat exchangers cutting down the processing time appreciably,
- a better product quality as deterioration due to over heating is prevented. The food is passed continuously through a plate or tubular heat exchanger where it is brought to sterilizing temperature almost instantaneously. The holding period is about 1–2 seconds at this temperature and the technique is known as *Ultra-high temperature* (UHT) *sterilization*. The sterilized food is then quickly cooled by another heat exchanger and enters the asceptic canning line, which consists of a tunnel maintained under sterile conditions through which sterilized cans are conveyed, filled and sealed.

Hot packing or *hot filling* technique involves the filling of previously sterilized food, while still hot, into clean but not necessarily sterile containers under clean but not necessarily asceptic conditions. The heat of the food and the holding time before cooling ensures commercial sterility of the filled product. Hot packing or hot filling is suitable for acid foods because most of the microorganisms include *C.botulinum* which do not survive at pH less than 4.5. Most fruit juices such as orange, grape, grapefruit, tomato and acid fruits and vegetables and pickled vegetables such as sauerkraut are hot packed or hot filled. For example, fruit juices are heated in the range of 77–100°C for 30–60 seconds, hot filled at about 90°C and held at this temperature for about 3 minutes with an inversion of the can before cooling.

Hot packing of low acid foods requires the presence of other preservative measures such as high concentration of sugar as in the case of fruit jams. An alternative technique 'Flash 18' process or pressure canning is used for low acid foods. Low acid foods need to be heated to temperatures above 100°C under pressure for sterilization. If the food at this temperature is filled into containers for sealing at atmospheric pressure, there will be violent boiling. In Flash 18 process, the entire canning line is placed inside a pressure chamber at about 1.1 to 1.4 atmospheric pressure and at this pressure water will not boil at temperatures below 124–127°C. Hence low acid foods sterilized at higher than 100°C are conveyed to the pressure chamber and filled into previously sterilized cans and sealed. A filling temperature of 124°C held over several minutes at this pressure gives commercial sterility to the product.

15.6.2 Pasteurization

This is a comparatively low order of heat treatment generally done at temperatures below the boiling point of water. Two primary objectives are achieved by pasteurization.

(i) Destruction of majority of but not necessarily all pathogenic and other spoilage microorganisms in liquid foods such as milk and liquid egg. In the case of milk used for cheese making, pasteurization destroys all microorganisms that would compete with the desired fermentation process at a later stage.

(ii) Extending the product shelf life from a microbial and enzymatic point of view.

The second objective is more of relevance in the pasteurization of beer, wine and fruit juice. Pasteurization will also inactivate the natural enzymes present in the food. Pasteurized food will contain many living organisms capable of growth thus limiting the storage life of the foods compared to commercially sterile foods. Pasteurized foods must be stored

under refrigerated conditions, e.g. pasteurized milk may be stored in a refrigerator for a week or so without developing significant off-flavours.

Pasteurization may be carried out in batch or continuous mode. Bulk foods such as milk and fruit juices may be pasteurized in batches in stirred, jacketed stainless steel vessels using steam or hot water. Rapid cooling of pasteurized food is necessary to limit the growth of thermophilic organisms and hence the food is passed through a cooler. Foods sealed in containers may be batch pasteurized in water or steam baths followed by water spray cooling. Continuous pasteurization of bulk foods is carried out by passing them through plate heat exchangers in four stages viz. preheating, heating, holding and cooling. The choice of temperature and time of pasteurization depends on the type of food. For example, milk is pasteurized at 62.8°C for 30 minutes or by high temperature short time (HTST) pasteurization at 72°C for 15 seconds, while whole egg is pasteurized at 64.4°C for 205 minutes.

15.6.3 Blanching

This is a kind of pasteurization generally applied to fruits and vegetables, primarily to inactivate natural food enzymes. This is a common practice when such foods are to be frozen, since mere frozen storage in itself would not arrest enzyme activity. Blanching, depending upon its severity, also destroys some microorganisms. Blanching is done by immersing the raw foods in boiling water for about 2–3 minutes or by using steam.

Questions

1. Write a note on the various factors affecting the heat resistance of microorganisms.
2. Discuss in detail the principles of thermal destruction of microorganisms explaining the term *D*-value.
3. What is the effect of temperature on thermal death time? Explain the significance of *z*-value and *F*-value.
4. How is the adequacy of heat treatment given to a food determined?
5. Describe the commercial heat preservation methods.
6. Write a note on canning of foods.

CHAPTER

16

Food Preservation by Evaporation and Drying

Evaporation involves heat transfer to a boiling liquid so that the vapour of the solvent is removed to get a more concentrated solution. Typical examples of evaporation include concentration of aqueous solutions of sugar, salt, sodium hydroxide, glycerol, glue, milk and fruit juices. In these cases the concentrated solution is the desired product and the evaporated water is normally discarded. In a few cases, water with a small amount of dissolved minerals, is evaporated to get solids free water for use as boiler feed, or as process water for special chemical processes. In some cases evaporation is practised to concentrate the solution so that upon cooling, crystallization is initiated and the desired product separates out. Evaporation is practised in food industry for pre-concentration of liquids prior to drum drying, spray drying or crystallization, to reduce the volume of liquid to minimize storage, packaging and transportation costs and to reduce the water activity and thereby aid in the preservation of foods.

16.1 FACTORS INFLUENCING EVAPORATION PROCESS

A variety of factors influence the evaporation process. These the following.

1. *Concentration of the liquid.* The liquid feed to an evaporator is relatively dilute with a low viscosity comparable to that of water. Hence heat transfer coefficients are quite high. However, as evaporation progresses, solution becomes concentrated and also

viscous, thereby decreasing the heat transfer coefficient. Circulation and turbulence can offset this disadvantage to some extent.

2. *Solubility*. The solubility limit of the solute, if exceeded, results in the formation of crystals. Thus solubility will limit the maximum concentration in solution which can be obtained by evaporation.
3. *Temperature sensitivity of materials*. Products such as foods (e.g. milk, vegetables and fruit extracts), pharmaceuticals and biological materials are sensitive to higher temperatures and undergo degradation. Hence prolonged heating at higher temperatures is to be avoided.
4. *Foaming and frothing*. Food products such as milk and some fatty acid solutions and caustic solutions produce foam or froth during boiling. The foam accompanying the vapour causes entrainment losses.
5. *Pressure and temperature*. Higher the operating pressure of the evaporator, higher will be the boiling temperature. Boiling temperature also increases as the concentration of the solution increases. Hence it is necessary to operate at lower (less than atmospheric) pressure or in vacuum.
6. *Scale formation*. Scale formation due to the deposition of solid materials on the heating surface decreases the heat transfer coefficients drastically necessitating the periodical cleaning of the evaporator.

16.1.1 Equipment

A variety of evaporators are used in the industry. The general types of evaporators consist of certain common components such as (i) a heat exchanger (to supply sensible heat and latent heat of evaporation to the feed), (ii) a separator (in which the vapour is separated from the concentrated liquid phase) and (iii) a condenser (to condense the vapour to effect its removal from the system). Industrial evaporators include the following.

1. *Open kettle or pan*. The simplest type of evaporator is the open pan or kettle in which the liquid is boiled. Heat is supplied directly or indirectly by passing steam through coils immersed in the liquid or through a jacket encompassing the evaporator. Evaporating salt water in shallow open pan solar evaporators usually crystallizes salt.

2. *Horizontal tube evaporator*. The evaporator kettle contains a bundle of horizontal heating tubes for passing steam kept immersed in the liquid to be evaporated. The vapour goes through a baffle to prevent carryover of any liquid droplets and leaves the evaporator at the top. The

evaporator can be operated continuously with the feed entering at a constant rate and the concentrate leaving at the same rate. The evaporator is used for non-viscous liquids having high heat transfer coefficients and liquids which do not deposit scale.

3. *Vertical tube evaporator*. The evaporator is also called as short tube evaporator. Vertical tubes are placed in the kettle for carrying the liquid to be evaporated while steam passes and condenses on the outer surface of the tubes. During evaporation the liquid rises in the tubes due to boiling and decrease in density causing a natural circulation. The concentrated liquid flows downward through a large central open space or downcomer while the vapour leaves at the top. The natural circulation increases the heat transfer coefficients. A variation of this type of evaporator is called the basket type where in addition to the vertical tubes, the heating element is held suspended in the body with an annular space as the downcomer. This type of evaporator is widely used in sugar, salt and caustic soda industries.

4. *Long tube vertical type evaporator*. The evaporator consists of long tubes of 3 to 10 metres carrying the liquid to be heated placed in the kettle with steam passing outside the tubes. The formation of vapour bubbles inside the tubes causes a pumping action giving high liquid velocities, an advantageous feature for efficient heat transfer. The liquid passes through the long tubes once for concentration without being recycled as in the case of manufacture of condensed milk.

5. *Falling film type evaporator*. This is a variation of the long tube type evaporator in that the liquid flows down the long tubes in a thin film with separation of liquid and vapour occurring at the bottom. This type of evaporator is useful for heat sensitive materials such as fruit juices as the heat transfer coefficient is quite high and hence the hold up time is short in the order of 5 to 10 seconds.

6. *Forced circulation type evaporator*. The heat transfer coefficient is enhanced by pumping to cause forced circulation of the liquid inside the tubes. This type is useful for evaporating viscous liquids.

7. *Agitated film evaporator*. This is a modified film type evaporator with a single large jacketed tube containing an internal agitator. Liquid enters at the top of the tube and flows downward and spread out into a turbulent film by the vertical agitator blades. Agitation increases the heat transfer coefficient of the liquid. The concentrate leaves the bottom and the vapour through a separator at the top. This type of evaporator is useful for heat sensitive viscous liquids such as rubber latex, gelatin, antibiotics and fruit juices.

16.1.2 Operation of Evaporators

The evaporators may be operated as single stage (single effect evaporators) or multiple stage (multiple effect evaporators). The single effect evaporator is used in relatively small scale operations and when the cost of steam is relatively cheap. However, it is not energy efficient as the latent heat of the vapour leaving is not used. In multiple effect evaporators the latent heat is recovered and hence they are energy efficient. In the forward feed multiple effect evaporator, the fresh feed is added to the first effect and flows to the next in the same direction as the vapour flows. This method of operation is used when the feed is hot or when the final concentrated product might be damaged at high temperatures. In the backward feed multiple effect evaporator, the fresh feed enters the last and the coldest effect and continues on until the concentrated product leaves the first effect. This method of reverse feed is advantageous when the fresh feed is cold, since a smaller amount of liquid must be heated to the higher temperatures in the second and first effects.

16.2 DRYING EFFECT OR DEHYDRATION

Drying refers to the removal of relatively small amounts of water from a material as vapour by passing hot air while evaporation refers to removal of relatively large amounts of water as vapour at its boiling temperature. Most of the foods contain enough moisture to permit the activity of native enzymes and microorganisms for spoilage, and drying is necessary to reduce the water activity and prevent microbial spoilage. Drying has been a means of preserving foods from ancient times. Grains in the field dry on the stalk by exposure to the sun. A sufficient degree of dryness in the grains (about 14% final moisture content) is achieved and no further drying is required to accomplish effective preservation. This is also true of many plant seeds and spices and also certain fruits such as dates and figs which develop high sugar content as they tend to dry on the trees.

16.2.1 Terminology

The terms *dehydration* and *drying* are used interchangeably to describe the unit operation in which all the water normally present in a foodstuff is removed by evaporation or sublimation, as a result of the application of heat under controlled conditions. Any other method that lowers the water activity in a food is also considered as a form of drying. For example, fish may be heavily salted so that moisture is drawn out from the tissues. Similarly, sugar is added in the case of condensed milk to lower the water activity.

Moisture content of a food is expressed on wet-weight basis, i.e. mass of water per unit mass of wet material, or more commonly, on a dry-

weight basis, i.e. mass of water per unit mass of dry solids. The moisture content of foods before and after drying to a level of moisture content, which would make the product stable, is given in Table 16.1.

TABLE 16.1 Moisture Content of Foods Before and After Drying

Food	*Moisture content* (%)	
	Before drying	*After drying*
Whole milk	87	5.0
Non-fat milk	90	5.0
Whole egg	74	2.9
Egg white	88	7.3
Egg yolk	51	1.1
Roasted lean beef	60	1.5
Broiled chicken	61	1.6
Boiled potatoes	80	4.0
Apple juice	86	6.2

Equilibrium moisture content of a food is the moisture content after equilibration with specified conditions of temperature and humidity. A sorption isotherm is actually the equilibrium moisture content of a food at various humidity levels at a constant temperature similar to that shown in Figure 2.2 of Chapter 2. The sigmoid shaped curve gives information about the lowest moisture content that can be achieved under a given set of conditions of temperature and humidity. The sorption isotherm of a food is of relevance in understanding the mechanism of drying and also the stability of dried food during storage. Equilibrium moisture content of a material cannot be predicted and has to be experimentally determined.

Bound water in a solid is determined from the intersection of the equilibrium moisture content of the material with the 100% humidity line. The bound water exerts a vapour pressure less than that of the liquid water at the same temperature. Substances containing bound water are called *hygroscopic materials*. If a material contains more water than indicated by intersection with the 100% humidity line, it can still exert only a vapour pressure as high as that of ordinary water at the same temperature. This excess moisture content is called *unbound water* and is held primarily in the voids of the solid. This free moisture is the moisture that can be removed by drying under the given percent relative humidity.

16.2.2 Theoretical Considerations

The basic mechanisms involved in the rates of drying of materials is yet to be understood clearly. Hence it is necessary to obtain some

experimental data on drying rates. Usually batch experiments involve the continuous monitoring of the moisture content of the sample filled in a tray so that the top surface alone is exposed to a stream of drying air. The variation of the moisture content during drying is shown in Figure 16.1. The drying cycle consists of a number of stages as shown by the variation of drying rate as a function of free moisture as shown in Figure 16.2. The stage *A–B* represents the settling down period in which the solid surface

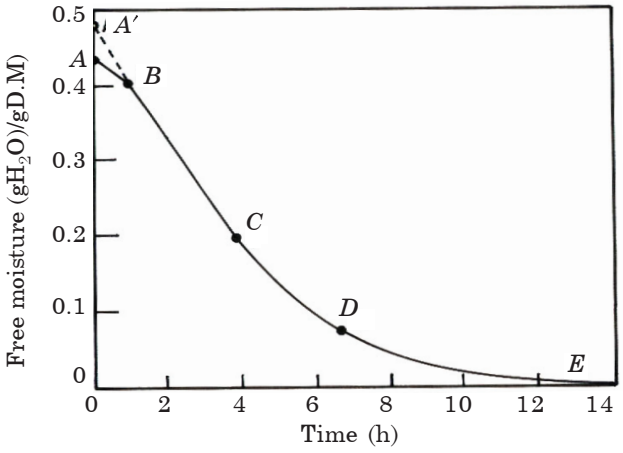


Figure 16.1 Drying rate curve.

(*Source: Transport Processes and Unit Operations*, 3rd ed., C.J. Geankoplis, Prentice-Hall of India, New Delhi, 1997, p. 537.)

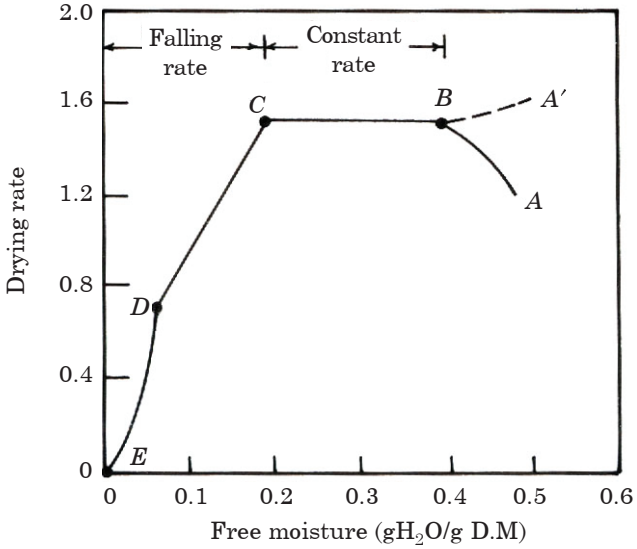


Figure 16.2 Rate of drying as a function of free moisture content.

(*Source: Transport Processes and Unit Operations*, 3rd ed., C.J. Geankoplis, Prentice-Hall of India, New Delhi, 1997, p. 537.)

comes to equilibrium with the drying air. The second stage *B–C* is known as constant drying rate period. During this stage the surface of solid remains saturated with liquid water because of the movement of water within the solid to the surface, which takes place at a rate greater than the rate of evaporation from the surface. Drying takes place by movement of water vapour from the saturated surface through a stagnant air film into the main stream of drying air. The rate of drying is dependent on the rate of heat transfer to the drying surface. The rate of mass transfer balances the rate of heat transfer and hence the temperature of the drying surface remains constant. Stage *C–D* represents the falling rate period which means, as drying proceeds, the rate of movement of moisture within the material to the surface decreases and the surface begins to dry out. The moisture content of the material at point *C* is known as the critical moisture content. From this point onwards the surface temperature begins to rise as drying proceeds. Often the falling rate period consists of two parts known as the first and second falling rate periods, i.e. *C–D* and *D–E* respectively. In the first falling rate period, the surface continues to dry out and the drying rate falls. The second falling rate period begins at point *D* when the surface is completely dry. The plane of evaporation slowly recedes from the surface into the solid and the drying rate falls further. Heat for the evaporation is transferred through the solid to the zone of vapourization. Vapourized water moves through the solid into the air stream. In the falling rate period the rate of drying is influenced by the rate of movement of moisture within the solid and the influence of air velocity decreases. In some cases no sharp break is observed at point *D* as the surface drying is gradual. The amount of moisture removed in the falling rate periods may be relatively small and the falling rate periods represent the major proportion of the overall drying time.

16.2.3 Purpose of Drying

The primary aims of drying of foods include: (i) preservation of foods, (ii) decreasing the weight and bulk of the food to economize shipping and canning costs (however, freeze-dried foods have essentially the same volume as the original food but reduced weight) and (iii) production of convenience items such as instant coffee and instant mashed potatoes, in which cases brewing or cooking steps are completed before the products are dried. The consumer simply adds water and stirs or mixes before consumption.

16.2.4 Factors of Relevance in the Control of Drying

Foodstuffs are complex and heterogeneous and many of the components of foods including proteins, fats, carbohydrates, vitamins, enzymes and

inorganic salts are strongly hydrated. The water present is in the form of solution, gel or emulsion or bound in various ways with the solid constituents. In addition, both plant and animal tissues are cellular and hold large quantities of water, further affecting the drying behaviour of foods. Food dehydration essentially involves (a) getting heat into the product and (b) getting moisture out of the product. There are various factors that influence the drying characteristics of foods.

1. *Drying temperature.* This varies with the food and the method of drying. The greater the temperature difference between heating medium and the food, greater is the rate of heat transfer. When hot air is used, it also aids the removal of moisture away from the food (mass transfer) and the higher the temperature of the air, the more moisture it will hold before becoming saturated.

2. *The relative humidity of air.* When a food material is kept in contact with air at a constant temperature and humidity it will attain a definite moisture content. A knowledge of the sorption characteristics of foods to be dried is important because the equilibrium content is the lowest moisture content that can be achieved under a given set of conditions of temperature and humidity. An understanding of the sorption behaviour is also helpful in the study of the mechanisms of drying and the stability of the dried food during storage.

3. *Velocity of air.* Higher velocity of air is more effective in drying.

4. *Drying time.* The drying time depends on the type of food and its moisture content and temperature of drying. The drying curve of a typical food (carrot dice) is shown in Figure 16.3. Drying is rapid initially, i.e. 90% of moisture in the product may be removed in the first four hours of

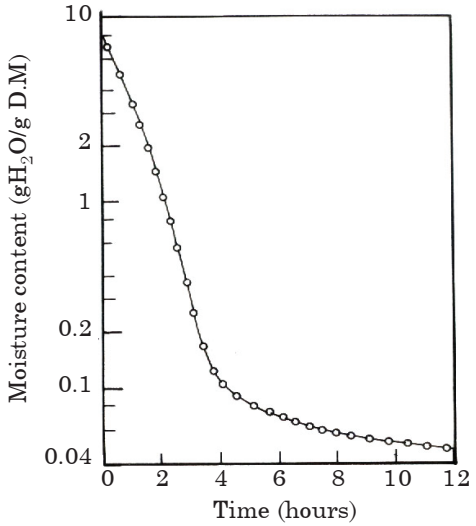


Figure 16.3 Drying curve of a typical food.

(*Source: Food Science* by N.N. Potter, AVI Pub. Co. Inc., Westport, 1978.)

drying, but the remaining 10% may require another four hours. The drying curve is asymptotic and zero moisture content is never achieved in practice.

16.2.5 Changes Brought About by Drying

Several changes occur in foods during the process of drying affecting the texture, size and appearance. These are as follows.

1. *Movement of solids*. Soluble solids will move towards the surface because of the shrinkage of the material, setting up pressure in the interior of the pieces. Migration of solids can also occur towards the centre of the pieces of food, due to a concentration gradient set up between the surface and wet centre.

2. *Shrinkage*. Animal and vegetable tissues undergo some shrinkage during drying by all drying methods except in freeze drying. Shrinkage of foodstufs during drying may influence their drying rates because of the changes in surface area and the setting up of pressure gradients within the material.

3. *Cellular structure*. Solid foods of natural tissue have a cellular structure, and moisture exists between and within the cells. When the tissue is alive the cell walls and membranes hold moisture within the cells, but when the animal or plant is killed the cells become more permeable to moisture. Blanching or cooking makes the cell more permeable to moisture. Hence cooked vegetables, meat and fish dry more easily than their fresh counterparts.

4. *Case hardening*. During drying of some fruits, meat and fish, a hard impermeable skin is often formed at the surface. This usually results in a reduction in drying rate and this phenomenon is known as case hardening. This is more common in foods that contain high concentrations of dissolved sugars and other solutes.

5. *Thermoplasticity*. Many foods are thermoplastic in nature, i.e. they soften on heating. A cellular food, such as plant or animal tissue retains its structure and some rigidity even at drying temperatures. But liquid foods such as fruit or vegetable juices lack structure and also contain higher concentrations of sugar and other solutes which soften and melt at drying temperatures. Even when all the water is removed in such foods the solid attains a thermoplastic tacky condition, sticking to the pan, which gives the impression that they still contain moisture. However, on cooling, they harden into a crystalline or amorphous glassy form and can be easily removed in the brittle condition. So belt type driers have a cooling zone, just prior to a scraper knife, to facilitate the removal of the dried food.

6. *Porosity*. Many drying techniques are aimed at making the structure more porous so as to facilitate mass transfer and increase the drying rate. Sometimes porous sponge-like structures (puffs) function as excellent insulating bodies and generally slow down the rate of heat transfer into the food. Porosity may be developed by creating steam pressure within the product and a case hardened surface through rapid drying. The escaping steam tends to puff such a product as in the case of potato puffs. Porosity may also be developed by whipping or foaming a liquid food or puree prior to drying.

7. *Chemical and other changes*. A variety of chemical changes occur during drying which contribute to the final quality in terms of food colour, flavour, texture, viscosity, reconstitution rate, nutritional value and storage stability. These changes are product specific. Browning reactions including enzymatic oxidation of polyphenols and other susceptible compounds, caramelization of sugars, Maillard browning, denaturation of proteins and loss of volatile flavour constituents occur.

16.2.6 Methods of Drying

1. *Sun drying*. This is limited to certain fruits, such as raisins, figs, apricots, dates, pears and peaches and salted fish. These fruits/fish are spread out on trays and are turned on their sides during drying. The moisture content is generally not lowered below 15% which is too high for storage stability of many other food products.

2. *Drying by mechanical driers*. This includes the methods of drying (a) by heated air, (b) by direct contact with a heated surface and (c) by the application of energy from a radiating microwave or dielectric source.

3. *Freeze-drying or lyophilization*. The moisture in the food is frozen and then sublimed to vapour, usually by the application of heat under very low pressure conditions.

16.2.7 Drying Equipment

Hot air driers. In hot air driers the food is in contact with a moving stream of hot air. Heat is supplied to the product mainly by convection. Kiln driers, cabinet tray or compartment driers, tunnel driers, conveyor driers, bin driers, fluidized bed driers, pneumatic driers, rotary driers and spray driers use hot air for drying solid foods. Liquid foods, such as milk, juices and soups may be evaporated by the use of comparatively low temperatures and low pressures in a vacuum pan. Liquid foods may also be drum dried by their passage over a heated drum with or without vacuum, or spray dried by spraying liquids into a current of dry, hot air.

The various types of driers are discussed here to give an idea of their individual characteristics.

1. *Kiln drier*. This consists of a two-floor building with a furnace or burner located on the ground floor. The heated air and the products of combustion rise through the slatted floor of the second floor, on which the wet material is spread in an even layer, 0.1–0.2 m deep. The humidified air is exhausted through a flue in the upper floor. The product is to be raked or turned regularly to promote uniform drying. The drier is mainly used in the drying of hops, apple rings and malt.

2. *Cabinet, tray or compartment drier*. The drier consists of an insulated cabinet containing an air circulating fan which moves the air through a heater and then through adjustable baffles which direct air between the trays of food. Air velocities of 25 m/s are used. Cabinet driers are cheap to build and flexible. These driers are used for drying fruits and vegetables at 1–20 tons/day.

3. *Tunnel drier*. It consists of a tunnel, which may be up to 24 m long with a square or rectangular cross-section of 2 m by 2 m. The wet food is spread in even layers on trays of slatted wood or metal mesh. The trays are assembled in stacks on trucks, which pass through the tunnel and through hot air at 2.5–6.0 m/s velocities either in cocurrent or counter current to the direction of food movement. This drier is useful for drying fruits and vegetables cut in pieces, on a semi-continuous basis at high throughputs.

4. *Conveyor drier*. This is similar to the tunnel drier in principle, but the wet material is conveyed through the system on a moving belt rather than on trucks. Cocurrent, counter current and through flow systems are used. The drier is useful for drying of fruits and vegetables of various kinds, mainly in diced form. Foam mat drying is a technique involving the production of a porous structure. The liquid material is mixed with a small quantity (1%) of edible foaming agents and air or inert gas is whipped in. The foam is spread on a belt and passed through a heated air drier. This technique is used for drying potato slurries, baby foods, etc.

5. *Vertical continuous-flow grain drier*. In case the harvested grains contain about 30–35% moisture, for safe storage for about one year, the moisture should be reduced to almost 13%. Drying of grain is usually carried out in a vertical continuous-flow grain drier. Hot air passes through a layer of grain of about 0.5 m held in the drying bin. Unheated air in the bottom section cools the dry grain.

6. *Fluidized bed drier*. Heated air is forced up through a bed of solids under fluidized conditions so that the solids are suspended in the air. The heated air acts both as the fluidizing and drying medium. The drier may be batch or continuous type. It is used for drying peas, beans, carrots, onions, potato granules, meat cubes, flour, cocoa, coffee, salt and sugar.

7. *Pneumatic drier*. The wet material is suspended in a moving stream of hot air, which conveys it through a drying system.

8. *Rotary drier*. The wet material is rotated in a cylindrical shell and heated air is passed through as the material is agitated. The drier has only limited application in food industry such as drying of meat pellets, granulated sugar and treatment of cocoa beans.

9. *Spray drier*. This drier is extensively used in food industry for drying solutions and slurries. The food material is introduced into the drying chamber in the form of a fine spray where it is brought into intimate contact with a stream of hot air. Very short drying time (1–10 sec.) and the relatively low product temperature are the main features. The droplets of the spray usually have diameter of the order of 10–200 μm thus presenting a large surface area per unit volume of material to the drying air resulting in rapid drying. The essential components of a spray drier include: (i) air heating and circulating system, (ii) a spray forming system comprising pressure nozzle, centrifugal atomiser or disc or bowl nozzles, (iii) a drying chamber and (iv) a product recovery system constituded of cyclone separator, wet scrubber, cloth filter, etc. are used to separate the product from escaping air. There are various ways of contacting the liquid droplets and hot air such as (a) horizontal co-current, (b) straight line flow, (c) simple vertical downward co-current with rotary flow, (d) complex vertical downward co-current flow, (e) vertical upward co-current flow and (f) vertical counter current flow. The drier finds use in food industry in the production of skim and whole milk powders, whey solids, ice cream mix, butter, cheese, milk based baby foods, coffee, tea, dried powdered eggs, fruit and vegetable juice powders, edible proteins, yeast extracts, wheat and corn products.

Drying by contact with heated surface. In direct contact drying, the food is brought into contact with a heated surface and heat is supplied to the product mainly by conduction. Examples include drum driers, roller driers and vacuum band driers. In such systems the necessary sensible and latent heat of evaporation is supplied to the material by conduction. In order to achieve reasonable drying times and to dry to a low moisture content, the heated surface temperature needs to be appreciably higher than 100°C and therefore, as drying nears completion the material temperature rises quite high. Thus the danger of heat damage to sensitive food materials is comparatively high. To reduce this hazard, drying by contact is often carried out under reduced pressure so that lower surface as well as material temperature may be employed.

1. *Drum drier* (film drier, roller drier): The drier consists of a single drum, double drum or twin drum with feed roller and vacuum casing. The film of feed material, as uniform as possible in depth, is applied to the drum surface. The advantages of drum drying are high rates of drying and economic use of heat. The main limitations are that it can only be applied

to foods in liquid and slurry forms and foods which can withstand relatively high temperatures for short times (of the order of 2–30 seconds). Drum driers have been used for drying of milk, soup mixes, baby foods, mashed potato, etc.

2. *Vacuum shelf drier*. Heat sensitive foods in the solid or liquid state are dried by the application of heat conduction. Drying proceeds at pressures of the order of 1–70 torr (0.135–9.45 k N/m^2; 1 atm.= 101.3 k N/m^2). The shelf temperature requires very careful control to avoid overheating of the dried portions of the food in contact with the trays. Vacuum is used only for very heat sensitive foods, e.g. fruit juice concentrates.

3. *Vacuum band drier*. The drier is mainly applicable to the drying of liquids and slurries such as fruit juice concentrates, tomato concentrates, and coffee extracts. 'Puff drying' is a new method to obtain rapid drying by so adjusting the vacuum to make the material puff up by the evolution of gases.

Drying by the application of energy from a radiating microwave or dielectric source. Microwave and dielectric sources may be used for supplying heat by radiation to the food for drying purposes. Infrared radiant driers have been used for drying breadcrumbs, starches, custard powder, cake mixes, tea, almond and spices. Dielectric heating has been used for drying of biscuits.

16.2.8 Freeze Drying (Sublimation-drying or Lyophilisation)

Foodstuffs, pharmaceuticals and biological materials which are heat sensitive even to low or moderate temperatures are freeze-dried. Freeze drying produces the highest quality food product with no loss of flavour and aroma and minimizes thermally induced degradation reactions. The method involves freezing of the material by exposure to cold air followed by sublimation of ice in vacuum from the frozen state to produce a dried product. Freezing leaves a mass with high structural rigidity, preventing its collapse during sublimation of the ice and when water is added later, the rehydrated product retains almost the original structural form. However, the method is costly due to the low rate of drying and the necessity of high vacuum equipment to maintain low pressures.

In freeze drying process, the food material is composed of a frozen core and as the ice sublimes the plane of sublimation recedes from the outer surface leaving a porous shell of the foodstuff. The heat supplied for the latent heat of sublimation of about 2838 kJ/kg of ice is conducted inward through the layer of dried material. The water vapour is also transferred through the layer of dried material. Thus heat and mass transfer occur simultaneously during freeze drying.

In most of the conventional freeze-drying systems, the vapour pressure gradient, necessary for sublimation, is attained by maintaining the total pressure in the drying chamber of the order of 0.1–2.0 torr. A condensing system is provided to remove water vapour formed and a heating system to supply the necessary latent heat of sublimation to the frozen material. In practice, it is not feasible to freeze all the liquid present in the food. The optimum rate of freezing depends largely on the material and the rate is usually determined experimentally.

Batch Freeze Driers. The essential components of a batch freeze drier are a vacuum cabinet, a vacuum system and a heating system. Freeze drier cabinets are essentially similar to vacuum shelf driers. The vacuum system must be capable of pumping down the cabinet pressure initially in a short time to prevent melting of the frozen products. In practice, this usually entails lowering the cabinet pressure between 5 and 1 torr within 10 minutes. The cabinet pressure then should be brought below 1 torr and must be held at that pressure during drying. Refrigerated condensers backed by a mechanical pumping system are commonly used for commercial purpose. Multistage steam ejectors may also be used for achieving and maintaining the required low pressure. Heat may be supplied to the frozen material by conduction or radiation or from a microwave radiator. The former two methods are used commonly, often in combination. A typical batch drier would have a capacity of 400 kg of prepared food and drying cycle of 7–8 hours. Multi-cabinet freeze driers and tunnel driers may also be used for larger throughput.

The removal of the major portion of water by sublimation results in a product with a porous structure, retaining the shape and size of the original materials. Many of the disadvantages with other drying methods are avoided in this method. Shrinkage is almost negligible, movement of soluble solids is limited and heat damage is minimized. The retention of volatile odour and flavour compounds is also high. Reconstitution characteristics both in rate and extent are high.

The disadvantages include that freezing damages the cell structure and the product becomes brittle. The high capital and operating costs involved in freeze-drying and stringent packaging requirements of the products make it very expensive.

Applications in food industry include freeze-drying of coffee, tea, meat and fish and some vegetable and fruit products.

Questions

1. What are the factors which influence evaporation?
2. Describe the types of evaporators used in food industry.
3. Describe briefly the single and multiple effect evaporator operation.

4. Explain the terms equilibrium moisture content, bound and unbound water.
5. Discuss the mechanism involved in drying of foods.
6. Give an account of the factors of relevance in the drying of foods.
7. What are the changes that occur in foods due to drying?
8. Classify the drying methods.
9. Write a note on the different types of drying equipment used in food industry.
10. Explain the principle involved in freeze-drying.

CHAPTER

17

Low-temperature Food Processing and Preservation

Freezing and cold storage are among the oldest methods of preservation. Low temperature retards chemical reactions as well as the activity of food enzymes and slows down or stops the growth and activity of microorganisms in food. The benefits of low temperature processing or preservation include: (i) preservation of food without any adverse effects on the nutritional values and flavour, colour and textural characteristics of the food, (ii) control of the rate of chemical/enzymatic changes in food due to spoilage organisms and endogenous enzymes as in aging of beef, (iii) control of the growth and metabolic activity of starter cultures of desirable food microorganisms as practised in cheese ripening, and aging of wines, (iv) enhanced ease and efficiency of unit operations such as peeling and depitting of vegetables and fruits for canning, as also cutting and slicing of bread, (v) reduced loss in flavour and associated changes during extraction of juice from citrus fruits, (vi) ease of precipitation of waxes from edible oils and (vii) increased solubility of carbon dioxide in water used for aerated drinks.

17.1 LOW-TEMPERATURE METHODS

Low-temperature processing and preservation methods may be sub-classified into two types: (i) refrigeration and (ii) freezing.

Refrigeration and cool storage is distinguished from freezing or frozen storage of foods based on the temperature of operation. Refrigeration means storage at temperatures above freezing of water in the foods usually in the range of 16°C (60°F) to –2.2°C (28°F). Refrigerators usually operate at 4 to 7°C. While pure water freezes at 0°C (32°F) most foods will

not begin to freeze until –2.2°C or a still lower temperature is reached. Refrigerated or cool storage will preserve perishable foods for days or weeks depending on the type of food.

Freezing or frozen storage refers to storage at temperatures where the food is maintained in solidly frozen condition, generally at –18°C (0°F) below. Frozen storage will preserve foods for months or even years.

17.2 MICROBIAL ACTIVITY AT LOW-TEMPERATURES

Most food spoilage microorganisms grow rapidly at temperatures above 10°C while some food poisoning organisms grow slowly even at about 3°C. Psychrotropic microorganisms grow slowly between 4.4°C and –9.4°C (15°F) provided the food is not solidly frozen. These organisms will not produce food poisoning or disease but even below –3.9°C cause the deterioration of the food quality. Below –9.4°C there is no significant growth of microorganisms in the food and there is a gradual reduction in their numbers due to slow death. But complete death of all the microorganisms does not occur merely due to low temperatures and when the food is thawed there can be a rapid multiplication of microorganisms.

17.3 REFRIGERATION AND COOL STORAGE

This is the gentlest method of food preservation as it has no adverse effects on taste, texture and nutritive value. Ideally, refrigeration of perishables should start immediately after harvest or slaughter and should be maintained throughout transportation, warehousing and storage prior to ultimate use. This is particularly true of certain metabolically active fruits and vegetables. These foods give off heat from respiration and convert initially formed metabolites in to other compounds. For example, sweet corn metabolizes its own sugar even at 0°C resulting in the loss of sweetness to the extent of 8% in a day and 22% in 4 days whereas at 20°C the loss is as high as 25% in one day. Hence a hydrocooler, where jets of cold water are sprayed on the corn soon after harvest to lower the temperature and also to inactivate surface microorganisms and wash off any pesticide residues, is used. The corn is then transported by trucks to warehouses.

Refrigeration influences agricultural and marketing practices and sets the economic climate of the food industry. Storage and shipment of refrigerated foods helps in maintaining the food supply uniformly throughout the year and to areas of deficiency. Control of prices during different seasons is also possible.

Cooling does not simply mean the placement of bulk foods into a refrigerated container. It means the removal of heat out of a body. Hence cold nitrogen gas volatilizing off liquid nitrogen is allowed to pass over the

food products to bring about quick cooling. The changes that occur in foods during cool storage are many and are influenced by factors such as growing conditions and varieties of plants, feeding practices of animals, harvest and slaughter, sanitation, damage to tissues and mixture of foods in the storage. Thus apples store well at 2.2 to 4.4°C and grapefruits at 11°C. Pigs fed on peanuts and soybeans (high unsaturated fat foods) produce soft pork and lard than animals fed on cereal grains. However, flesh of the latter keeps better in cold storage. Animals permitted to rest before slaughter build up glycogen reserves in their muscles which is converted to lactic acid after slaughter. Lactic acid is a mild preservative and enhances the keeping quality of meat in cold storage. Too low temperatures cause cold damage to fruits and vegetables. In the case of bananas and tomatoes, storage below 13°C slows down natural ripening and results in poor ripened colours. Loss of nutrients also occur even during cool storage, e.g. loss of sweetness of sweet corn and vitamins from vegetables. Loss crispness and firmness of vegetables and fruits, loss of flavour and lumping or caking of granular foods are some of the problems associated with cool storage.

Some foods should NOT be refrigerated at all. Bread is an example. The rate of staling of bread is greater at refrigeration temperatures than at room temperature. However, freezing at –22°C or below can arrest staling and frozen bread retains its freshness for many months. Cakes, cookies, waffles and pancakes are also frozen and marketed.

17.3.1 Factors of Importance in Refrigerated Storage

Factors to be considered in connection with refrigeration include: (i) control of low temperatures, (ii) relative humidity and air circulation, (iii) composition of atmosphere in the storage chamber and (iv) food variability (types of food).

Control of low temperature is achieved by the use of properly designed refrigerators and refrigerated store rooms to maintain temperatures at ±1°C of the selected temperature. The quantity of heat to be removed from food depends on the specific heat of the particular food. Fruits and vegetables produce heat due to respiration. Hence information of the specific heat and respiration rates is necessary to calculate the refrigeration load, which is the quantity of heat that must be removed from the food and the storage area to go from an initial temperature to a selected cold temperature and maintain this temperature for a specified time. Fruits and vegetables, for example, produce heat due to respiration. The amount of heat expressed in terms of British thermal units, Btu, at 0°C varies with the type of fruit or vegetable (1 Btu = 52 cal. = 1055 joules). For example, the heat given off by apples is of about 300 to 880; cabbage, 1200; carrots, 2130; onions, 600 to 1100; oranges, 420 to 1030; potatoes, 440 to 880; green tomatoes, 580 and ripe tomatoes, 1020 Btu.

Air circulation is of importance because proper air circulation helps in removing the heat away from the vicinity of the food surface to the refrigerator coils/plates. The air circulated within a cold storage room must not be too dry or too moist. Air of high humidity can condense moisture on the surface of cold food allowing the growth of molds. On the other hand, if air is too dry it will cause excessive drying out of foods. The optimum humidity to be maintained in cool storage rooms for most foods has been determined. Most foods store best at refrigerated temperatures when air humidity is in the range of 80–95%. The optimum relative humidity is 90–95% for crisp vegetables, about 70% for nuts and 50% for dry and granular products such as milk powder and eggs.

In case refrigerated storage is required for prolonged periods, precautions are to be adopted to maintain food quality. Packaging can protect foods that tend to lose moisture. Thus plastic sacks sprayed with moisture-resistant coatings are used to cover large cuts of meat. Cheeses, which are ripened for many months in cold warehouses, are protected with a wax dip. This minimizes the moisture loss and also protects against contamination and growth of surface molds. Eggs tend to lose moisture and carbon dioxide. Dipping the eggs in thin mineral oil to seal the pores of the eggshell retards these losses. Conventionally, aging of beef is done at about 2°C (35°F) for a period of several weeks. If the humidity of the storage room is much below 90% the beef dries out, and if it is above 90% mold growth occurs. Precise control of relative humidity is difficult. Hence in the accelerated aging process (Tenderay process) beef is aged in 2 or 3 days by combining high humidity with a temperature of 18°C (65°F). The accelerated aging also speeds-up microbial growth which is kept in check by irradiating the beef with ultraviolet light. However, UV irradiation dosage must be regulated because excessive exposure to UV light can cause surface fat to become rancid.

Controlled atmospheric storage is practised in cool storage. Fruits in cold storage, respire, ripen and then become overripe. Respiration depends on the availability of oxygen and carbon dioxide content in the surrounding atmosphere. Slow down in respiration and the accompanying undesirable physiological changes in such foods are achieved by the use of (i) low temperature, (ii) depletion of oxygen in the atmosphere and (iii) increase in the level of carbon dioxide. The optimum temperature, relative humidity and gas composition of the cold room atmosphere differs for different fruits and vegetables. Apples can be stored without any deterioration in quality for about six months by maintaining the temperature at 2.8°C, relative humidity at 87%, oxygen concentration at 3% (depletion from the 21%) and increase of carbon dioxide level from the normal 0.03% in air to 3% and the remaining percentage being nitrogen. Other examples of modified gas atmospheres include the use of diphenyl vapours to inhibit mold growth on citrus fruits and the use of ethylene gas to speed up ripening and colour development of citrus fruits and bananas. The term hypobaric storage refers to storage of food in

refrigerated conditions at reduced pressure (to decrease the amount of air) with high humidity (to prevent drying of food).

Food variability or types of foods that can be stored in a cold room is of relevance. Refrigerated storage permits exchange of flavours between types of foods stored near one another. For example, butter and milk absorb odour from fish, and fruits and eggs absorb odour from onions. Hence odorous foods should be stored separately or packaged properly.

17.4 FREEZING AND FROZEN STORAGE

Freezing has been a major factor in making available a variety of convenience foods to the consumer apart from preserving foods for prolonged periods. Proper freezing preserves food without major changes in size, shape, texture, and flavour of a variety of foods of plant and animal origins such as chicken pot pie, breaded fish sticks, whipped toppings and fruit pies. While dehydrated foods require reconstitution and also heating, frozen foods require only a single thawing operation before eating.

17.4.1 Freezing Characteristics of Foods

Freezing, unless properly controlled, can disrupt food texture, break emulsions, denature proteins and cause undesirable physical and chemical changes. Aqueous solutions will have lower freezing points with increasing concentrations of dissolved solids such as salts, minerals, sugar, proteins, etc. Different foods will have different freezing temperatures depending on their different compositions and solids content. Hence under a given freezing condition, they will require different times to reach a solidly frozen state. Even agricultural practices such as irrigation and fertilization practices have profound influence on freezing characteristics of foods.

A given unit of food, whether it is a bottle of milk, a cut of beef or a can of sliced apples in sugar syrup will not freeze uniformly, that is, it will not suddenly change from liquid to solid state. A progressive freezing occurs starting from the outer surface and gradually progresses to the inner core. As water in the outer sections freeze, the water in the inner portions of the food becomes more concentrated in dissolved solids and requires longer time to freze. The freezing curve for a thin section of beef is shown in Figure 17.1. The curve was obtained from experimental data of recording the temperature and water content of the sample placed at –18°C, as a function of time. Initially supercooling occurs. As time progresses, nucleation or agitation initiates the formation of the first ice crystal and latent heat of fusion causes the temperature to rise to the freezing point (slightly below 0°C). In the case of pure water, the curve

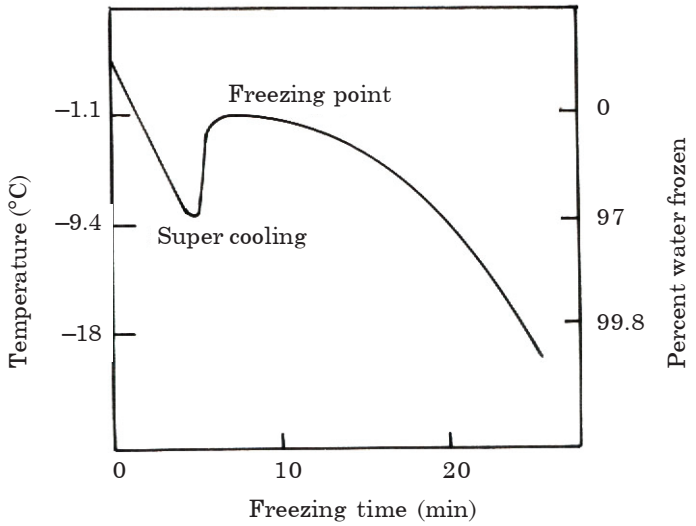


Figure 17.1 Freezing curve of a typical food.
(Source: Food Science by N.N. Potter, 3rd ed., AVI Pub. Co. Inc., Westport, 1978, p. 228.)

would not drop below the freezing point as long as liquid water remained. In the case of beef and other foods, the temperature continues to drop as more and more water is frozen out. This is largely due to increase in the concentration of dissolved solids in the remaining water which decreases the freezing point of the solution. In case of beef, about 70% of water is frozen at –4°C, 97% at –9.4°C even at –18°C not all the water is frozen completely.

The freezing curve shown in Figure 17.1 is common to most solid foods which show a super-cooling zone, followed by an inflection up to the freezing point and a subsequent drop in the temperature if there is a sufficient temperature differential between the freezing food and freezer environment. This differential provides the driving force for continued heat transfer out of the food.

17.4.2 Factors Affecting the Quality of Frozen Foods

Five factors are of importance in the maintenance of the quality of foods in frozen storage. These include: (i) solute concentration effects, (ii) ice crystal size, (iii) rate of freezing, (iv) final temperature and (v) intermittent thawing.

Solute concentration effects have an important role in maintaining the quality of most foods. The food must be frozen to a solid or nearly so to maintain good quality during frozen storage. A partially frozen zone or an

unfrozen core will deteriorate with respect to texture, colour, flavour and other properties. This is not only because of the possible growth of the microorganisms but also due to high concentration of solutes in the remaining unfrozen water. The high solute concentration in the unfrozen part of the food can damage the quality of a food in a variety of ways because of (a) precipitation or crystallization of solutes as in the case of lactose in freezing ice cream, which can impart a gritty and sandy texture to the food, (b) denaturation and precipitation of proteins due to a salting-out effect, (c) decrease in the pH value below isoelectric point of proteins due to acidic solutes causing coagulation and precipitation of proteins, (d) destruction of delicately poised colloidal suspensions, (e) supersaturation of gases ultimately forcing them out as in the case of frozen beer or soda and (f) dehydration of adjacent tissues at the micro-environmental level.

Ice crystal size also affects the quality of food in frozen storage. When water inside the cells of living tissues such as meat, fish, fruits and vegetables freezes rapidly it forms minute crystals of ice. However, if the rate of freezing is slow the ice crystal size is large and clusters are also formed leading to physical rupture of cells. Large ice crystals can disrupt emulsions such as butter, frozen foams like ice cream, gels such as puddings and pie fillings. In the case of butter, ice crystals that grow within individual water droplets in the continuous fat phase can penetrate through fat and merge resulting in water pockets and dripping during thawing of butter. Large ice crystals can puncture frozen foam bubbles of ice cream, resulting in the loss of volume on storage and often exhibiting syneresis or water separation.

Rate of freezing is of importance because rapid or instantaneous freezing produces ice crystals of small size and also minimizes concentration effects of solutes by decreasing the time of contact between solutes and food tissues and other constituents.

Final temperature and maintenance of the final temperature to an accuracy of ±1°C is important. The choice of –18°C or below as the minimum recommended temperature for freezing is based on experimental data and a compromise between maintaining quality and cost. Microbiologically, –18°C storage is not necessary since most pathogens do not grow below +3.3°C and food spoilage organisms do not grow below –9.4°C. But enzymatic reactions are not stopped but only slowed down because enzymes retain their activity even at –73°C. The reaction rates are faster in supercooled water than in frozen water at the same temperature. Long-term storage at –9.4°C where foods contain considerable amount of unfrozen water leads to enzymatic deterioration of food quality, especially by oxidative deterioration. Hence blanching of fruits and vegetables is necessary before they are frozen.

Non-enzymatic chemical reactions also do not stop entirely but only slow down at –18°C, and vegetables, fruits, nonfatty meats properly packaged and frozen at this temperature retain their quality for about 12

months, whereas at –9 to –7°C, quality may be retained only for a few days.

Damage due to intermittent thawing affects the quality of frozen foods. The different types of damage that occur during slow freezing can also occur during slow thawing. Repeated thawing and freezing during storage due to temperature fluctuations are more detrimental. As little as a 3°C fluctuation at –18°C can be damaging. Solute concentration effects set in above –12°C. Hence in general, a quick final thawing is superior to slow thawing. The refreezing curve of thawed flesh of lamb does not appear to alter the freezing point of the food mass as in the case of beef. Lamb has the characteristic in having negligible losses in fluid on thawing, and also has little structural damage by freezing. On the other hand, beef, carrots, fish and peaches show some freezing damage. Fluid loss from tissues occurs during thawing and a consequent alteration in the freezing points in such cases.

17.4.3 Methods of Freezing Food

Three methods of freezing are employed in commercial practice. These are air freezing, indirect contact freezing and immersion freezing.

In *air freezing*, cold air is used with different velocities. In 'sharp freezing' (or slow freezing), cold air between –15 to –23°C is blown over the food. 'Quick freezing' is done by air blast freezing, wherein freezing air at –18 to –34°C is blown across the food or a fluidized bed freezer may be used.

In *indirect contact freezing*, the food or food package does not come into contact with the refrigerant. The food or food package is brought into contact with a cold surface maintained at temperatures in the range of –18 to –45°C by a refrigerant.

In *direct freezing*, the food is immersed in the refrigerant (e.g. fish in brine or berries in special syrups) or sprayed with the refrigerant (e.g. cartons of fruits, vegetables, fish, shrimps and mushrooms). These foods can also be immersed in liquid nitrogen or sprayed with liquid nitrogen after packaging in cartons or aluminium cans.

17.4.4 Refrigeration Requirements for Freezing Foods

Frozen storage of foods necessitates freezing of water in the food into solid ice and to keep water in the solid form. In order to achieve this, heat must be removed from liquid water so that it can freeze and the frozen material must then be insulated to prevent it from acquiring heat. The temperature at which a food will freeze, under a set of given conditions, is dependent upon the concentration of solutes in the liquid phase. Water in most foods generally freezes at about –2°C. The refrigeration load is calculated by

determining the refrigeration requirements for freezing the food alone and the load of the freezer chamber separately and also by taking into consideration the heat generated due to movement of material and personnel in and out of the chamber and heat generated due to electric lights and equipment.

The refrigeration requirement for freezing the food alone involves bringing the temperature of the mass down to the freezing point and maintaining the food in frozen condition. The energy required to freeze a food may be expressed as:

$$H_1 = \mathrm{S} \cdot \mathrm{W} \cdot (T_i - T_f) \tag{17.1}$$

where H_1 is the energy in k.cal required to lower the temperature of the food from the initial temperature T_i to the temperature which the food freezes, T_f; S is the specific heat above the freezing point of the food and W is weight of the food in kilograms.

Once the food mass has been brought to the freezing point it is necessary to remove the heat of fusion. The energy required to accomplish this is

$$H_2 = H_f \cdot W \tag{17.2}$$

where H_2 is the energy required in k.cal to change liquid food at freezing point to solid state, H_f is the heat of fusion and W is the weight of food in kilograms.

Once the food is frozen, the temperature of the frozen mass must be lowered to frozen storage temperature which is usually –18°C. The energy required for this is given by

$$H_3 = S_s \cdot W \cdot (T_f - T_s) \tag{17.3}$$

where H_3 is the energy in k.cal to lower the temperature of the mass, W from the freezing temperature T_f to storage temperature T_s, and S_s is the specific heat of the frozen material.

The refrigeration requirement, R_f, to freeze food and bring its temperature to frozen storage temperature is the sum of the H_1, H_2 and H_3 and is usually expressed in k.cal load or in Btu (1 Btu = 0.252 k.cal). The refrigeration requirement, solely for the frozen mass is generally reported in terms of tons of refrigeration.

The next step is to establish the refrigeration requirement for the chamber to hold the frozen food in a solid state at a desired temperature. Generally, the temperature of the chamber is lower than its environment. In order to maintain a frozen mass at a desired temperature it is necessary to design a suitable chamber well insulated from its environment and create an artificial microenvironment within the chamber. Other factors to be taken into account for calculating the refrigeration requirement include: (i) prevention of heat transfer from the surroundings to the frozen food, (ii) access to the frozen chamber for movement of material and workers, (iii) heat given to the chamber by electric lights, running

electric motors in the chamber and (iv) heat given off by persons working in the microenvironment of the chamber. Usually the refrigeration requirement to maintain an empty frozen food storage chamber at a desired temperature is calculated easily using the expression,

$$H_c = \frac{K \cdot 24 \cdot SA(T_1 - T_2)}{I} \tag{17.4}$$

where H_c is the k.cal losses per 24 hour in the frozen storage, K is the thermal conductivity of the insulation material, SA is the surface area of the outer wall of the storage, T_1 is the temperature outside the chamber, T_2 is the temperature inside the chamber and I is the thickness of the material.

In addition to the work required in terms of k.cal for both food mass and the storage chamber, there are miscellaneous heat loads such as heat load in lighting the chamber, running motors, having people operating the chamber and the opening and closing of the chamber. The last one involves the changes of air due to entrances and exits from the chamber giving rise to additional heat load. Average air changes per 24 hours are dependent on the size of the chamber, its frequency of use, temperature and relative humidity.

The heat load contributed by men, H_m, working within the chamber may be calculated as

$$H_m = h \times \text{No. of man hours in chamber} \tag{17.5}$$

where h is the average heat load per man hour (approximately 3000 k.cal).

Thus the heat load to operate an empty frozen food chamber, R_c, is the sum of the individual heat loads contributing to the operation of the chamber:

$$R_c = H_c + H_e + H_m + H_a \tag{17.6}$$

where H_c, H_e, H_m and H_a are the heat loads in k.cal required to maintain the empty chamber at the desired temperature, to operate electrical equipment within the chamber, load contributed by men working in the chamber and load added by changing air within chamber during normal operating conditions, respectively.

The total refrigeration requirements R, in freezing foods and maintaining them in frozen storage can be calculated as:

$$R = R_f + R_c \tag{17.7}$$

A safety factor, in the order of magnitude of 10% is usually to the calculated refrigeration requirement for an installation. In addition, the refrigeration equipment should not operate 24 hours a day, but only for about 18 hours a day. Hence the equipment needs are calculated on the basis of having the compressor capable of handling a 24 hr load in 18 hr, i.e. the calculations of refrigeration requirement is revised upwards by a factor of 25%, which constitutes an additional factor of safety.

17.4.5 Packaging Requirements for Frozen Foods

The ice, which accumulates on the surface of a freezer plate or a coil in a frozen storage room, comes from moisture in the air in the chamber. The vapour in the atmosphere of the chamber attempts to reach an equilibrium with the materials in the chamber, as well as the chamber itself. However, it is condensed as snow on freezer plates or coils. As moisture is removed from circulation the moist materials within the chamber yield more water vapour, attempting to satisfy vapour pressure deficit within the chamber. Thus there is a steady removal of water vapour from the air in the form of ice to the coil and a consequent dehydration from the food material if it is unprotected. This leads to freezer burning of the food. In addition, unprotected food is subject to oxidation and contamination from the atmosphere of the chamber leading to substantial deterioration in quality of such foods. Hence it is necessary to package the food to maintain the quality. The package must be functional, lend to mechanical handling, economical in space and cost. Wood, metal, glass, paper and plastic materials are used successfully as frozen food containers. The simplest protective coating possible for frozen foods is a glaze (coating of ice) which has been used in fishing industry. The glaze must be replaced periodically. The most complicated packaging problems involve the processing and distribution of consumer-type individual serving units of frozen foods.

17.4.6 Effect of Freezing on Constituents of Foods

Freezing does not affect the nutritive value of proteins but it is possible to denature proteins. This may be seen in the curdling of proteinaceous materials especially during repeated freezing and thawing. Though the biological value of denatured protein need not differ from native protein, the appearance and quality of the food may be seriously altered. Enzyme activity is only retarded by freezing temperature and hence control of enzyme activity is achieved by a heat treatment (blanching) prior to freezing and storage, in the case of vegetables and fruits. Enzyme activity is stimulated in the super cooled water and the reaction rate is greater than in crystallized water at the same temperature. Hence frozen storage even at –9°C permits severe damage to the quality of the food both in the loss of nutrients and appearance. Long term storage at –6°C yields unacceptable foods.

Fats and oils undergo oxidative deterioration in frozen foods as in the case of fatty fish. The fat in frozen fish tends to become rancid quicker than in frozen animal tissues. Plant tissues are least susceptible. In the case of pork, fat may become rancid in about six months on storage at 0°C while beef retains good quality even up to two years. Rancid fats tend to lower nutritive values than fresh, sweet fats. Freezing may destabilize emulsions of oil-in-water or water-in-oil. This may be serious in prepared

precooked frozen foods and food products. But in general, fat and oil deterioration is temperature dependent and freezing offers maximum protection in preserving many fatty foods.

Vitamin losses occur during processing of foods, and during storage vitamin losses continue. Adding ascorbic acid to fruits prior to freezing is practised commercially to protect their quality. Blanching to inactivate enzymes destroys vitamin B-1 whereas it improves storage stability of carotenes (Vitamin A).

17.5 HAZARD ANALYSIS

Frozen foods cannot be inspected for safety and quality of products. High quality, safe, wholesome frozen foods are produced by determined, knowledgeable, careful, well-organized and well-implemented programs designed to anticipate and prevent problems. Frozen food processors use various control measures to ensure the production of safe and wholesome frozen foods. Hazard analysis differs from such efforts in that it provides a systematic approach for estimating the risk in producing a frozen food product. The assessment of all possible hazards, the elimination of avoidable hazards and setting limits for those hazards not possible to eliminate are the basis of a rational systematic technique.

The analysis of hazards, is based on three general characteristics: (i) a frozen product contains a 'sensitive' ingredient or ingredients assumed to be potential sources of contamination under normal circumstances; (ii) the manufacturing process employed does not contain a controlled step that effectively destroys harmful bacteria and (iii) there is potential abuse of frozen products in distribution or in handling that could render the product harmful due to the result of microbial growth. The combination of these factors is used to classify frozen foods as to the consumer's risk. If all the three hazard characteristics are present in a product, it would have a hazard classification of +++. If a hazard factor is absent, a 0 would designate this, such that 0++ = no sensitive ingredient; +0+ = product pasteurized and 000 = no hazard involved.

The hazard categories in order of decreasing risk are:

Category I. A special category of non-sterile products designed and intended for consumption by infants, the aged or the infirm, e.g. milk.

Category II. Food products subject to all three general hazard characteristics (+++), e.g. cheese (*Salmonella, Staphylococcus*), egg products (*Salmonella*), milk solids (*Salmonella*).

Category III. Food products subject to two general hazard characteristics (+0+, ++0 or 0++), e.g. butter, margarine, flour, potatoes, seasonings and vegetables.

Category IV. Food products subject to one general hazard characteristics (+00, 0+0 or 00+), e.g. spices.

Category V. Food products subject to none of the general hazard characteristics (000).

Frozen foods range in complexity from a single-component, single-product process to a multi-component series of processes and assembly operations. Single-component, single-product processes include production of frozen corn, peas, green beans and vegetables. Multi-component blended products include frozen soft-filled bakery products produced by a straight line process for a single product involving multi components or ingredients. Multi-component products with add-on involves the use of fish or battering, breading and frying operations (a multi-component product) with add-on, usually a sauce added to the sea food. Multi-component multi-process assembly operations involve frozen prepared chicken dinners.

17.6 THAW INDICATORS

The quality of frozen foods depends on the temperature of the product not only while it is in frozen storage but also during distribution. Hence it is necessary to monitor the temperature of frozen foods. Three types of thaw indicators are commonly used. These include: (i) defrost indicators, (ii) time-temperature integrators and (iii) time-temperature integrator/indicators.

Defrost indicators show a colour change when a pre-selected temperature such as –10 or 0°C is reached. Such indicators can be used for materials that should not be exposed to higher temperatures even once and hence are not much useful for frozen foods. Time-temperature integrators react as soon as they are activated at a pre-selected combination of time and temperature related to the disappearance of quality of the food. These integrators change colour gradually and develop a quantitative time-temperature equivalent. However, they do not indicate whether it has been greatly exceeded. Time-temperature integrator/indicators have a graduated scale and are activated as the activating temperature is reached at a rate proportional to the storage temperature and hence are most useful in frozen foods.

Questions

1. What are the advantages of low-temperature processing and preservation of foods?
2. Distinguish between refrigerated storage and frozen storage.
3. Write a note on the effect of low temperatures on microbial activity.

4. What are the important factors to be considered in the refrigerated storage of foods?
5. Discuss the freezing characteristics of foods.
6. Give an account of the factors which affect the quality of frozen foods.
7. What are different methods of freezing used in food industry?
8. How is the refrigeration load for a given food calculated?
9. What is the effect of low temperatures and freezing on the constituents of foods?
10. What is hazard analysis? Discuss its importance.
11. What are the packaging requirements for freezing foods?
12. What are thaw indicators? How do they help in determining the quality of the frozen food?

CHAPTER

18

Food Preservation by Irradiation and Allied Operations in Food Industry

18.1 PRESERVATION BY IRRADIATION

Radiation may be defined as the emission and propagation of energy through space or a medium. Radiation of wavelengths less than 200 nm, such as ultraviolet light, X-rays, gamma rays, alpha particles, beta particles and cosmic rays are called ionizing rays and are of importance in food preservation. These radiations have sufficient energy to ionize molecules in their path. They destroy microorganisms in foodstuffs without raising the temperature and hence the irradiation process is termed as *cold sterilization.*

18.1.1 Units of Radiation

Several units are used to express the dosage of radiation. A roentgen (r) is the quantity of gamma or x-radiation which produces one electrostatic unit of electric charge of either sign in a cubic centimetre of air under standard conditions. A roentgen-equivalent-physical (rep) is the quantity of ionizing energy which produces, per gram of tissue an amount of ionization equivalent to a roentgen. The most widely used unit of radiation dosage is the rad, which is equivalent to the absorption of 100 erg per gram of irradiated material. A megarad (Mrad) is 1 million rad. A Gray (Gy) is equal to 100 rads and a kilo Gray (kGy) is 1000 Gray.

18.1.2 Mechanism of Action of Radiations

Ionizing radiations penetrate the food to varying degrees depending on the nature of the food and the characteristics of radiations. Gamma radiations have greater penetrating power than beta particles but the latter have greater ability to ionize molecules. Electron beams of higher energy also have greater penetrating and ionizing power. When ionizing radiations pass through foods, breaking of chemical bonds occur resulting in the formation of ion pairs and free radicals. The reactions of these species with other molecules provide the mechanism of alteration of food constituents, enzymes and microorganisms.

18.1.3 Effect on Foods

Radiations produce several undesirable changes of colour, odour, taste and physical properties in foods. High dose of irradiation of meat causes a rise in its pH, destruction of glutathione and an increase in the carbonyl compounds, hydrogen sulphide and methyl mercaptan. In fats and lipids irradiation causes destruction of natural antioxidants, promotes oxidation and polymerization reactions and increases concentration of carbonyl compounds. In most foods the levels of vitamins such as thiamine, pyridoxine, B_{12}, C, D, E and K are decreased while those of riboflavin and niacin are not affected.

18.1.4 Effect of Radiation on Microorganisms

Ionizing radiations destroy all types of microorganisms. Gram negative bacteria are less resistant to radiation than gram positive bacteria and spore formers are more resistant than non-spore formers. *Micrococcus radiodurans* is one of the most radiation resistant microorganism. Other resistant types include *B.larvae*, *Streptococcus faecium* and the spores of *C.botulinum* type A. Bacteria which are more sensitive to radiation include *Pseudomonas* and *Flavobacterium* sp.

The bactericidal efficiency of a given dose of irradiation depends on several factors: (i) the kind and species of organism, (ii) the number of microorganisms (or spores) initially present in the food, (iii) the composition of the food, (iv) oxygen availability and (v) age of the microorganism. In general, microorganisms have higher resistance to radiations at lower temperatures in protein rich media and in their lag phase of growth.

18.1.5 Processing of Foods Prior to Irradiation

Foods must be properly processed prior to irradiation. Foods for irradiation must be selected on the basis of their freshness and overall desirable quality. Foods must be cleaned to make them free from debris and dirt. Finally, they should be packaged in proper containers such as cans. Glass containers cannot be used because they undergo colour changes on irradiation. Sterilizing doses of irradiation are insufficient to destroy the natural enzymes of foods. Hence to avoid post irradiation enzymatic changes blanching of vegetables and mild heat treatment of meat are necessary prior to irradiation.

18.1.6 Irradiation Process

Gamma radiation from radioisotopes of cobalt-60 or cesium-137 and electron beams from linear accelerators are commonly used for irradiation of foods. The radiation dosage depends on the nature of the food, the resistance of microorganisms and the resistance of enzymes apart from the cost of the process.

The radiation dosage required for most of the foods has been established on the basis of the logic that spores of *C.botulinum* are the most resistant species. The D_M value refers to the radiation dose necessary to reduce the existing spore population by 90% and in the case of beef, it was estimated to be 0.4 Mrad. A 12 D_M value (4.8 Mrad) of radiation dosage provides complete sterilization of the food.

Natural enzymes present in the food have higher resistance to radiation compared to that of spores of *C.botulinum*. The given D_E values for enzyme destruction (radiation dose necessary to decrease the enzyme activity by 90%) are in the range of 4–5 Mrad. The radiation dosage required for total destruction of enzyme activity would be at least 4 D_E values, i.e. 20 Mrad, which would destroy the food constituents and thus impair the safety of the food itself.

The radiation treatment given to foods is of any one of the following three types: (a) radappertization, (b) radicidation and (c) radurization.

Radappertization is equivalent to radiation sterilization equivalent to commercial sterility as practised in canning industry where the typical levels of radiation are 30–40 kGy. *Clostridium botulinum* type E spores with radiation D values of the order of 0.12 to 0.17 Mrad are most radiation sensitive while spores of types A and B have higher D values of the order of 0.279 and 0.238 Mrad respectively. In general radiation resistance of microorganisms increase at lower temperatures. To achieve 12 D treatments of meat products at about 30°C, irradiation using 41.2–42.7 kGy for beef and chicken; 43.7 kGy for pork; 31.4–31.7 kGy for ham and codfish cake are necessary. The use of a radiation of 12 D values for *C.botulinum* in meat products would not kill all the viruses and hence these must be destroyed by heating or other means prior to irradiation.

The disadvantages of radappertization is that foods that undergo colour changes or produce off-flavours cannot be irradiated. This method can be used to reduce the nitrosamines in bacon containing sodium nitrite. Addition of sodium ascorbate prior to irradiation is necessary to reduce nitrosamines.

Radicidation uses 2.5 to 10 kGy and is equivalent to pasteurization of milk. It reduces a number of non-spore forming non-viral pathogens. Fresh poultry, cod and red fish, spices and condiments have been approved for radicidation in some countries.

Radurization may be considered as equivalent to pasteurization. It enhances the keeping quality of foods by reducing the numbers of spoilage microbes. The radiation dosage levels are in the range of 0.75–2.5 kGy for fresh meats, poultry, seafood, fruits, vegetables and cereal grains.

18.2 ANCILLARY AND ALLIED OPERATIONS

18.2.1 Sanitation and Plant Hygiene

Sanitary requirements in food processing plants should be in accordance with HASSOP (hygiene and standard sanitation operating procedures). The requirements start at the selection stage of site and the factory design. Selection of a suitable site for the factory with adequate supply of good quality process water, pollution free surroundings and waste disposal facilities is a prerequisite. Hygienic design of the building, proper selection of construction material, ventilation and lighting, design and selection of equipment and equipment layout are the important criteria to be considered. Specific areas for raw material cleaning, quality grading and processing operations such as comminution, mixing, sterilization and canning/packaging should be distinct. Sterile or aseptic zones should be well separated and protected.

Dust proofing of equipment, cleaning and sterilization of process equipment and prevention of recontamination of cleaned food material by generated waste are of prime importance. Detergents, chemicals and sanitizers used for cleaning equipment, the processing zones and storage yards must be in accordance with the requirements to satisfy microbiological safety of the entire food processing plant.

18.2.2 Water Treatment

Water requirement in food industries can be broadly classified into four types: (i) water for preliminary operations such as cleaning food raw materials and equipment, (ii) process water, (iii) boiler feed water and (iv) cooling water.

Water used for the preliminary operations should be clean and free from colour, turbidity, taste and microbial and chemical contaminants. Process water should be potable and satisfy the conditions specified for a given process. The dissolved salt content should be low as high salt content can influence the palatability and texture of the food products and also lead to scale formation in heating equipment. Boiler feed water should be free from dissolved salts (zero hardness). The water quality specification for cooling water is not very stringent.

Water treatment becomes an inherent part of food engineering operations. The steps involved include physico-chemical methods and disinfection. The physico-chemical methods are settling, coagulation, filtering and membrane separation (reverse osmosis) to remove suspended solids, turbidity and colour and also to soften the water. The treated water can be used for preliminary operations and cooling purposes. Boiler feed water requires complete removal of dissolved salts, which can be achieved by passing the water through ion exchangers. Process water should satisfy bacteriological requirements and this is why disinfection is carried out using chlorination or ozonation for large scale requirements or exposure to ultraviolet light for small scale requirements.

18.2.3 Packaging

Packaging of food is also a method of preservation, which eliminates physical contact between the food and contaminants. It varies depending on the nature and type of food, the expected shelf life of preserved food, the nature of function(s) of the package and visual characteristics of the packaging material.

The packaging material from which containers are made, should be produced locally and assembled in a wide range of material combinations with possibility of moulding into different shapes. They must be cheap, easy to transport and have a maximum packaging economy for maximum product content capacity. They must satisfy environmental requirements. They should be light in weight and flexible. Paper, paper cartons, plastics in the form of films, sheets, bags, containers, bottles, glass and metals in the form of foils and cans are used as packaging materials.

The requirements of a container to be used for food packaging vary with the nature of the product. The general requirements include: (i) nontoxicity and compatibility with the specific food, (ii) ability to prevent contamination of the food by microorganisms, (iii) protection of the food from moisture, gases and light, (iv) capacity to withstand stress of handling by humans or machines, (v) tamper proof characteristics, (vi) ease of opening, (vii) transparency, (viii) pouring features, (ix) suitable size, shape and weight, (x) appearance and printability, (xi) ease of disposal, (xii) low cost and (xiii) special features for enhanced consumer convenience such as ability for dispensing aerosols, for use as stand-up

pouches, as packages that fit into the microwave ovens for instant heating and packages that can be divided into portions and resealed .

18.2.4 Storage

Storage chambers for raw food materials and processed food products should be separate. In both the cases, food spoilage can occur due to (i) microorganisms, insects and pests, (ii) chemical changes brought about by native enzymes within the food, (iii) physical changes that occur within the food such as recrystallization of ingredients, staling, destruction of emulsion, etc. and (iv) damage to packaging resulting in spilling and exposure of the food material to the surroundings.

Three main factors of food storage are of importance in preventing food spoilage. These include: (i) storage temperature, (ii) relative humidity and (iii) composition of the atmosphere within the storage chamber. The optimum conditions of temperature, relative humidity and composition of atmosphere of storage for various food products differ. Control of these parameters within acceptable limits would ensure long-term storage stability of the material.

18.2.5 Effluent Treatment

Food processing generates a large amount of waste products and waste water. A variety of waste products including plant parts, trimmings, peelings and spoiled raw materials from plant food processing units, blood, fat and grease from meat, poultry and fish processing, milk products from dairy industry, oils, fats and waxes from edible oil refineries and waste water containing detergents, sanitizing agents and soil are generated. In addition sewage from facilities used by workers also becomes part of the effluent from the industry.

The individual characteristics of the waste products and wastewater vary depending on the nature of food processing operations but the BOD load of the wastewater is very high. It is necessary to separate different waste streams based on plant operations for efficient and satisfactory treatment of the effluents. Physical, chemical and biological methods of waste treatment are practised in accordance with the requirements of pollution control agencies.

Questions

1. What are ionizing radiations? How are they effective against microorganisms?

2. What are the effects of ionizing radiations on foods, microorganisms and enzymes?
3. Give an account of the processing steps involved before irradiating food items and the irradiation process.
4. Distinguish between radappertization, radicidation and radurization.
5. Write a note on sanitation and plant hygiene in food processing industries.
6. Give a brief account of the type of water requirements for food processing industries.
7. List the requirements of food packaging materials and containers. What are the different packaging materials used in food industry?
8. What factors cause spoilage of foods during storage? How is spoilage of food prevented during storage?
9. Write a brief note on the waste disposal in food industries.

CHAPTER

19

Milk and Dairy Products

Dairy products include milk and milk products such as condensed sweetened milk, skim milk, cream, cheese, butter and ice cream.

19.1 MILK

Milk is the fluid secreted by the mammary glands of female animals and contains almost all the nutrients necessary to sustain life. The term milk is now used to refer to cow's milk exclusively while those of others are specifically named such as goat milk or sheep milk in the market. Milk is a white or yellowish white, opaque emulsion of fat globules dispersed in an aqueous solution (called whey or serum) of proteins, sugar and minerals. The specific density of milk (of cow) ranges from 1.029 to 1.039. The specific density of milk decreases with increasing fat content and increases with increasing amounts of proteins, sugar and salts. The freezing point of milk is almost a constant value at −0.53–0.55°C and is a suitable indicator for detection of dilution of milk with water. The pH of fresh milk is in the range of 6.5 to 6.75 and refractive index, 1.3410–1.3480. The redox potential of raw milk is +0.30 V and that of pasteurized milk is +0.1V. The redox potentials of dairy cheese and yoghurt are +0.05 and −0.15 V respectively.

The principal constituents of milk including fat, protein (primarily casein), milk sugar or lactose and the minerals of milk are found in varying proportions in the milk of different animals. In addition, with the exception of lactose, each of these components from different animals varies somewhat in chemical, physical and biological characteristics. For example, the fatty acids of goat's milk fat have different melting points, susceptibility to oxidation and flavour characteristics compared to those of cow's milk. Similarly, milk proteins of various species may

differ with respect to heat sensibility, nutritional properties and ability to produce allergic reactions in other species. The composition of milk from a few animals and humans is given in Table 19.1. The high degree of variability of composition of milk from different species is important in processing operations. Milk obtained from cows and buffaloes are the major sources for dairy operations. The total solids in the milk minus the fat content is generally referred to as solids-not-fat or milk-solids-not-fat or serum solids and the minerals in milk are collectively referred to as ash.

TABLE 19.1 Composition of Milk from Different Species

Constituents	*Cow*	*Buffalo*	*Goat*	*Human*
Water	87.1	82.1	83.5	87.0
Fat	3.9	7.4	6.0	3.7
Protein	3.3	4.8	4.0	2.1
Lactose	5.0	4.8	6.0	7.0
Ash	0.7	0.8	0.5	0.2

19.1.1 Milk Production and Quality Control

Milk secreted by a healthy udder is sterile but quickly becomes contaminated with microorganisms from the external body of the cow and from milk handling equipment. Hence the secreted milk is cooled immediately in a refrigerated tank where it can be stored up to 2 days before being transported to the processing plant.

Quality control tests at the process plant include test for fat and total solids content, bacterial count (especially of *E.coli*), presence and concentration of antibiotic and pesticide residues. Off-favours in milk such as rancidity, mainly due to the activity of native lipase, and oxidized flavour due to contamination of copper or iron from pipes, valves or equipment, develop in improperly processed milk.

19.1.2 Milk Processing Operations

The steps in milk processing operations carried out in a dairy plant are: (i) blending, (ii) clarification and cream separation, (iii) heat treatment, primarily pasteurization, (iv) homogenization and (v) bottling. The process steps are shown in the flow chart (Figure 19.1).

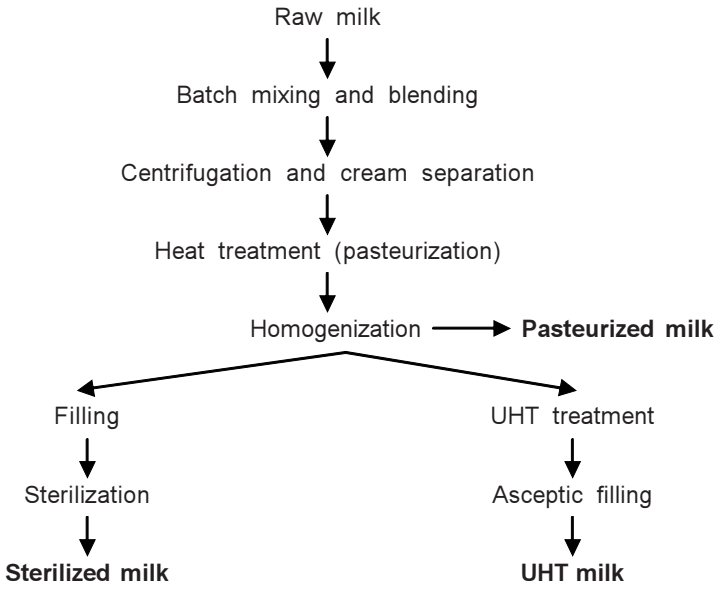


Figure 19.1 Milk processing operations.

Blending of different batches of milk in cold condition (~5°C) is carried out to obtain a specified fat content. In the second step, the blended milk is clarified by a centrifugal clarifier to remove any sediments, body cells from the cow's udder and some bacteria. A high speed centrifuge called bactofuge is used for almost complete removal of bacteria. Cream separation is often achieved simultaneously in this step. This operation is carried out at 40°C at 5000–6000 rpm.

The clarified milk is then given appropriate heat treatment depending on the requirement. Pasteurization is carried out to destroy lipase activity and other milk enzymes and also to destroy pathogenic organisms in batch or harding method by heating the milk at 62–65°C for at least 30 minutes or by HTST method (high temperature short time method) at 71–74°C for 15 seconds or in a short time process at 85°C for 2 seconds. Pasteurized milk is not sterile and hence it is quickly cooled to prevent multiplication of souring bacteria. Raw milk contains several enzymes of which alkaline phosphatase is important. This enzyme has heat destruction characteristics that closely approximate the time-temperature exposures of proper pasteurization and hence its activity is indicative of the effectiveness of pasteurization. If alkaline phosphatase activity beyond a certain level is found in pasteurized milk, it is indicative of inadequate pasteurization. The enzyme liberates phenol from phenol phosphoric acid compounds and free phenol gives a deep blue colour with certain organic compounds, which forms the basis for phosphatase test.

Ultra high temperature (UHT) treatment involves indirect heating by coils or plates at 135–140°C for 6–10 seconds or by direct heating by live

stream injection at 140–150°C for 2–4 seconds followed by aseptic packaging. Milk may also be sterilized in retail packages by heating in autoclaves at 110–120°C for 15–20 minutes.

Homogenization may be carried out before or after pasteurization. This process makes a stable emulsion of milk fat and milk serum by mechanical treatment. Homogenization is achieved by passing milk or cream through a small aperture under high pressure and velocity. The fat globules of milk with varying sizes in the range of 0.1 to 20 μm in diameter have a tendency to gather into clumps and rise due to their lower density than milk, thereby separating into a cream layer. Homogenization breaks the fat globules to uniform size of about 2 μm or less which are covered by an adsorbed layer of plasma proteins including casein micelles. This stabilizes the milk emulsion. Homogenized milk has creamy structure, bland flavour and a whiter appearance. It has greater whitening power when added to coffee and tea compared to skim milk. A soft curd is formed from homogenized milk, which is more easily digested than curd obtained from unhomogenized milk. In the manufacture of evaporated milk and ice cream, homogenization reduces the chance of separation of fat and hence provides smoother texture. Homogenization, however, accelerates lipase activity, particularly if homogenization is carried out before pasteurization leading to rancid flavour. High pressure homogenizers, low pressure rotary type homogenizers and sonic vibrators may be used for homogenization of milk. The homogenized and pasteurized milk is then cooled, bottled and marketed.

19.2 TYPES OF MILK AND MILK PRODUCTS

A variety of milk and milk products are produced in the dairy depending on consumer requirements. These include the following categories of products.

Vitamin D milk. Vitamin D content in cow's milk depends on the cattle feed and exposure of the cow to sunlight. The diet of most children is deficient in Vitamin D and hence Vitamin D enriched milk is marketed. Vitamin D activity in milk is enhanced by irradiating milk with UV light which converts the milk sterol, 7-dehydrocholesterol into Vitamin D_3. Alternatively, Vitamin D concentrates may be added to milk to bring the potency to about 400 units per quart.

Low sodium milk. Low sodium milk is required by people with high blood pressure or edema. It is prepared by passing milk through an ion-exchange resin that replaces sodium with potassium. Low sodium milk contains 3–10 mg of sodium per 100 ml compared to about 50 mg in ordinary milk.

Concentrated milk. Evaporated milk is whole milk from which most of the water has been evaporated. Raw whole milk is clarified and

concentrated in vacuum pans at a temperature of 74–77°C. It is then fortified with Vitamin D, homogenized, and filled into can and sterilized at 118°C for 15 minutes and cooled. This heat treatment gives evaporated milk a light brown colour due to sugar-protein interaction and also its characteristic flavour. *Khoa* is a semi solid obtained by evaporating milk in open pans and is used in preparing Indian sweetmeats. *Malai* is made by simmering milk to a thick layer of milk fat and coagulated proteins consumed with or without sugar.

Sweetened condensed milk. This is made from pasteurized milk by concentration and sucrose is added to sweeten the concentrate to the extent of 65% in the final product. Sweetened condensed milk is not sterile but microbial growth is prevented by the added sugar. *Kheer* is concentrated milk, obtained by evaporating milk to nearly half its original volume, resembling sweetened condensed milk.

Dry milk. Whole milk is dehydrated to the extent of 97% by spray drying or vacuum drying. Skim milk and low-fat milk are also dehydrated to manufacture milk powder. Vitamins A and D are added to enhance the nutritive value. The low-cost dry milk has a long shelf life and can be reconstituted to fluid milk by mixing with the required amount of water. Dried milk is stored in dry air tight containers.

Skim milk. Whole milk from which fat has been removed by centrifugation is called skim milk. It contains all other constituents of milk except fat and fat-soluble Vitamins A and D. These vitamins can be added to skim milk. By varying the amount of fat removed from whole milk, low fat milk (containing 0.5–2% fat) is prepared. Condensed skim milk finds extensive use in baking and confectionery industries.

Cream. Concentrated milk fat is called cream. Cream is formed as a layer of fat globules, which rise to the top in unhomogenized milk. Cream is separated from milk by centrifugation. Table cream (coffee cream) contains about 18% fat while whipping cream contains about 35% fat. Cream, used for butter making, contains about 40% fat. Sour cream, which is extensively used in bakery items as salad dressings, is prepared by pasteurizing cream with about 18% fat at 75°C for 30 minutes to kill all bacteria and then inoculating with a controlled culture of lactic acid bacteria to develop the desired acid tang in the finished product.

Yoghurt. Yoghurt (or yogurt) or dahi in India is a fermented milk product with a fine curdled gel like consistency having a sour and aromatic flavour. It is obtained by fermenting homogenized pasteurized milk with about 3% thermophylic lactic acid bacteria (mixed cultures of *Streptococcus thermophilus* or *Lactobacillus bulgaricus*) at 42–45°C for about 3 hours. During fermentation, the curdled gel-like consistency and acid flavour due to formation of carbonyl compounds such as diacetyl and acetaldehyde develop. Sugar and fruit pastes may be added to the final product to achieve distinct fruity flavours.

Sour milk. Sour milk or buttermilk is fermented fluid milk either by spontaneous souring (by *Streptococcus lactis* or *cremoris*) or by fermentation with aroma forming bacteria (*Streptococcus diacetylactis* or *Betacoccus citrovorus*) or by fermentation with pure bacterial cultures to give cultured buttermilk. During fermentation lactose is converted into lactic acid which coagulates casein at pH 4.5–5 to give the final sour tasting curdled buttermilk.

Kefir and kumiss. Kefir and kumiss are sparkling, carbonated alcoholic beverages derived from milk. The microflora such as *Torula* yeast (for alcoholic fermentation) and *Streptococcus lactis* and *Lactobacillus caucasicus* (for lactic acid fermentation) which form clotted milk particles are added to fluid milk to give kefir. Kumiss is made from goat's or mare's milk by fermentation with *Thermobacterium bulgaricus* and yeast *Candida*.

Taette milk. Milk is fermented with *Streptococcus lactis var. hollandicus* to yield a sour viscous thread-like product due to the symbiotic growth of the lactic acid bacteria and yeast. Taette is popular in Sweden, Norway and Finland.

Cheese. Cheese may be defined as the product made from the curd of the milk of cow and other animals, the curd being obtained by the coagulation of milk casein with an enzyme (rennin) or an acid (lactic acid). The curd is processed with or without further treatment with heat, pressure, salt and ripening (with selected microorganisms) to obtain cheese. Some cheese varieties are made from milk whey solids that remain after the removal of coagulated casein.

About 800 varieties of cheese are known, mostly in Western Europe, which can be classified into 18 distinct types of 'natural' cheese, reflecting the different processes by which they are made. Some of the well-known names include Cheddar, Camembert, Edam, Romano, Swiss, Roquefort, Trappist, brick and whey cheese. Cheese may also be classified on the basis of texture and kind of ripening. Thus there are hard cheese (Cheddar, Cheshire, Emmental), semi-hard cheese (Edam, Gouda, Molbo, Port Salut, Roquefort, Blue Dorset, Blue Cheese) and soft cheese (Chevre, Brie, Camembert, Limburger, Le Munster) depending on their moisture content. The varieties of cheese may be ripened by bacteria or molds, or they may be unripened (Quark, Cottage Cheese, Mozzarella, Petit Suisse). The bacteria may produce gas and form 'eyes' in the cheese as in Swiss cheese or they may not produce gas and hence no eyes as in the case of Cheddar cheese. Among the soft and semi-hard cheeses, Limburger is ripened primarily by bacteria and Camembert by a mold while cottage cheese is not ripened. Whey cheese has high lactalbumin content, the second principal protein in milk. Lactalbumin is not coagulated by rennin or the acid and so it remains soluble in whey. However, it is easily coagulated from the whey as curd by the process of heating.

The composition of some of the cheese is given in Table 19.2.

TABLE 19.2 Composition of Some of the Cheese

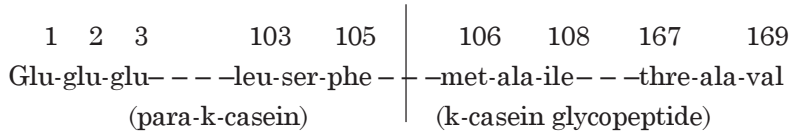
Cheese	*Moisture*	*Fat*	*Protein*	*Ash*	*Salt*	Ca	P
Camembert	50.3	26	19.8	1.2	2.5	0.68	0.50
Cheddar	37.5	32.8	24.2	1.9	1.5	0.86	0.60
Cottage	79.2	4.3	13.2	0.8	1.0	0.12	0.15
Swiss	39.0	28.0	27.0	2.0	1.2	0.9	0.75

19.3 MANUFACTURE OF CHEDDAR CHEESE

The first step in manufacturing any type of cheese is curd making. The curd is then processed through several steps to get the finished product. The different steps involved in cheddar cheese manufacture include: (i) curd making, (ii) cutting the curd, (iii) cooking, (iv) matting (v) milling and salting, (vi) pressing and (vii) curing and ripening.

19.3.1 Curd Making

The curd is formed under controlled conditions of temperature, acidity and rennin concentration to give a curd of desired moisture content and texture for subsequent processing. Pasteurized whole milk is maintained at 31°C in a vat and is inoculated with a starter culture of lactic acid producing bacteria *Streptococcus lactis*. Colouring matter, (usually carotene or annatto extract to give a pale orange colour) may be added at this stage. After about 30 minutes, a mildly acidic condition of about 0.2% acidity exists and rennin, in the form of a dilute solution (or rennet, a commercial form), is added. Rennet is a preparation of the enzyme rennin (also called chymosin) obtained from the lining of the fourth stomach (abomasum) of calves. Microbial rennets from *Mucor meihei* or proteases may also be used. Rennin specifically catalyses the hydrolysis of one particular peptide linkage in the k-casein between 105-phe and 106-met as shown.



The k-casein is split into two fragments, the glycopeptide being soluble is lost as part of the whey while the para-k-casein (precipitates in the presence of added calcium salts) remains as part of the micelle. It includes hydrophobic sections, and strong cross-linking between micelles occurs resulting in the rapid formation of curd. The mild acidity improves the

coagulating property of the rennin. Stirring is stopped and the curd is allowed to set in about 30 minutes. A custard like curd (protein gel) is formed and acid formation and rennin activity continues yielding a curd with a desirable elastic texture which, on heating or pressing, shrinks and squeezes out the trapped whey.

19.3.2 Cutting the Curd

The curd is cut into cubes of ¼ to ½ inch sides with cutting knives made of wires strung across a frame, to ensure efficient removal of whey (syneresis).

19.3.3 Cooking

The cubes are gently agitated and heated in jacketed vat using steam to a temperature of about 38°C for 30 minutes and are kept at this temperature for a further period of 45 minutes. Cooking helps the removal of whey by squeezing out from the curd cubes. Heat increases the rate of acid formation and also makes the cubes shrink in size.

19.3.4 Matting

The cooked curd is allowed to settle and the whey is drained off from the vat. The curd undergoes matting. Matting involves the fusion of curd pieces to form a continuous rubbery slab. The process of matting and the subsequent handling of the matted curd is known as cheddaring and is unique to the production of the cheddar type cheese. Cheddaring involving cutting of the matted curd into blocks, turning the blocks at 15 minute intervals and then piling the blocks on one another 2 or 3 deep. The purpose of cheddaring is to allow acid formation to continue and to squeeze out whey from the curd. The weight of the blocks on one another is a mild form of pressure. During cheddaring the vat is maintained warm. The cheddaring operation of stacking and turning the blocks goes on for about 2 hours or until the whey coming from the blocks reaches an acid concentration of 0.5–0.6%.

19.3.5 Milling and Salting

The curd blocks are milled into pieces of about 2 inch long and the milled pieces are spread out on the floor of the vat and sprinkled with salt (about 2.5 lbs/100 lbs of curd). The pieces are stirred for uniform distribution of the salt. Salt has three functions, e.g. to draw out whey from the curd by osmosis, to act as a preservative and to add flavour.

19.3.6 Pressing

The salted curd pieces are pressed in a hydraulic press at about 20 psi overnight. The extent of pressing determines the final moisture content of the finished product. If more moisture of whey is retained in the curd, fermentation makes it more acidic and this in turn decides the final texture of the cheese and the nature of the microorganism that can grow during the subsequent ripening period. Pressing also determines the final shape of the finished product.

19.3.7 Curing and Ripening

After overnight pressing, the cheese is placed in a cool drying room at 15°C and 60% relative humidity (RH) for 3 to 4 days. This causes mild surface drying and formation of a slight rind. The cheese is wax coated by dipping it in hot paraffin wax to prevent mold growth and also to prevent excessive drying during curing. The waxed cheese is placed in the curing room for ripening for at least 60 days at 2–3°C and 85% RH. For the peak flavour, ripening may be continued for 12 months or longer. During this period, bacteria in the cheese and enzymes in the rennet modify the cheese texture, flavour and colour by continuing to ferment residual lactose and other organic compounds into acids and aroma compounds. Partial hydrolysis of milk fat and further breakdown of the fatty acids and mild proteolysis of the protein also occur. In the case of cheddar cheese, these changes are relatively mild compared to those in Roquefort or Limburger cheeses.

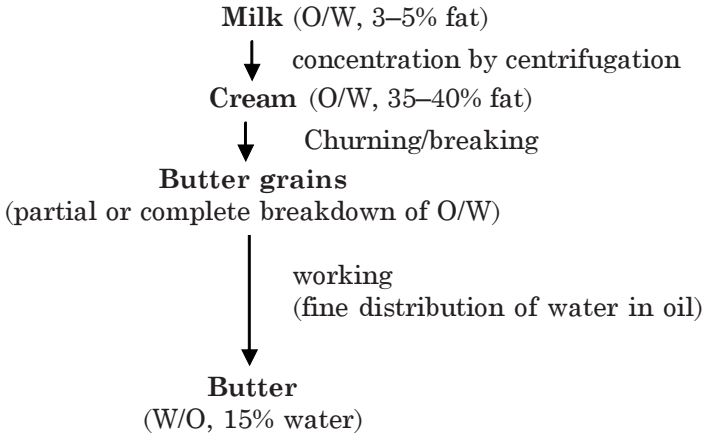
19.3.8 Other Types of Cheese

Cottage cheese is a low fat soft cheese generally coagulated with lactic acid rather than rennin. The curd is left in particulate form and is not pressed. It is not ripened and hence it is highly perishable. Swiss cheese is another hard cheese like cheddar but characterized by the formation of large holes or 'eyes' and a sweet nutty flavour. This is due to the *Propionibacterium shermanii* culture deliberately added, which converts lactic acid into propionic acid, responsible for the nutty flavour and carbon dioxide which collects in pockets within the ripening curd and forms holes during its escape.

Indian cottage cheeses include *paneer*, obtained by citric acid (1%) coagulation of buffalo milk; *channa* obtained by addition of lemon juice or acid whey (taken from earlier batches of preparation) to boiling milk; *decca* cheese and *surti* cheese made by clotting whole milk with rennin and *bandal* cheese made from cream and then smoked.

19.4 BUTTER

Butter making is an unique example of emulsion technology. The raw material, milk is an oil-in-water emulsion and the final product butter is a water-in-oil emulsion. Thus an inversion of phases occurs during butter making. The changes in the emulsion structure during butter making may be summarized as follows:



Butter is manufactured by batch or continuous process. In a typical butter making operation, the various steps involved, are: (i) separation of cream from milk, (ii) pasteurization and pre-processing of cream, (iii) churning and working and (iv) packing.

19.4.1 Separation of Cream

Cream is separated from milk by centrifugation process. Cream containing about 35–40% of butter fat with a density of 0.93 at 15°C (compared to 1.032 of milk) is produced.

19.4.2 Pasteurization and Pre-processing of the Cream

The separated cream is pasteurized at a higher temperature than that for pasteurizing milk, since the high fat content has a slight protective effect on bacteria. For the manufacture of 'sweet' butter, the fresh cream is directly pasteurized. However, cream separated in farms and stored for some time turns slightly acidic due to lactic acid produced by fermentation. In such cases, the acid in the cream is neutralized with a food grade alkali prior to pasteurization. In contrast, 'sour' butter is obtained by allowing the fermentation of the cream to proceed to produce

sufficient acid, which is standardized to about 0.25% lactic acid content by the addition of lime or sodium carbonate and the cream is then pasteurized. The flavour of sour butter may be modified by inoculating the pasteurized cream with a culture of *Streptococcus lactis* or diacetyl producing bacterial culture. A vegetable colouring material such as an extract of annatto seed or carotene may be added to deepen the yellow colour. The cream is then rapidly chilled and held at a low temperature for a predetermined time (aging). Chilling and aging are important steps in controlling the solid to liquid fat ratio and hence the texture of the final product. After aging, the cream is subjected to churning and working.

19.4.3 Churning and Working

Churning involves the agitation of the cream at 10°C to bring a partial breakdown of the O/W emulsion. During agitation in a churn, air is incorporated and a foam is produced. The fat globules in the cream accumulate at the air-plasma interface where they release a certain proportion of their liquid fat in non-globular form. Some of this fat acts as a cement causing the binding of the fat globules. As the foam bubbles form, burst and reform, more and more globules come together producing clusters or clumps of increasing size. Churning is continued until these clumps reach a predetermined size, clearly visible through inspection windows in the churn. During the process of churning, the colloidal suspension of cream containing about 35% fat in water (O/W emulsion) is inverted to butter which is an W/O emulsion containing 15% water dispersed in the fat phase. After churning, the aqueous phase called butter milk is removed and butter grains are washed free of butter milk with chilled water. Salt is added as required (2.5–13%) and working of butter is commenced. Working involves slow-speed agitation of butter grains subjecting them to kneading and folding action. The process of working has three objectives—to rupture more fat globules and produce more free fat, to disperse the remaining water through the bulk of the fat in the form of fine uniform sized droplets and to disperse salt.

Most modern churns are equipped to perform both churning and working operations. The churn made of wood or stainless steel is a cylindrical barrel rotating about an horizontal axis. Baffles are fitted internally to improve agitation. The barrel is rotated at higher speeds for churning, and at lower speeds, for working.

19.4.4 Packing

The worked butter is chilled and held in cool or frozen storage or packed as required. The butter is forced from the churn by a worm conveyor or through a shaping machine for extrusion in a continuous flow. The

extruded butter is cut into pieces by a weight controlled packing machine which wraps the butter in paper packs and cartons.

19.4.5 Ghee

Ghee is 99.5% fat obtained by boiling off the water from butter in a pan over a low fire with stirring. As the temperature increases to about 120°C, browning and pleasant aroma of caramelization occurs and the heating is stopped. The liquid is filtered to remove solid/suspended impurities to yield pure ghee.

19.5 ICE CREAM

Ice cream is a typical example of frozen dairy food. The ingredients used in the manufacture of ice cream include dairy products and non-dairy products. Dairy products include whole milk, skim milk, cream, frozen cream, butter, butter oil (contains 99% butter fat), condensed milk products and dried milk products. The non-dairy products include sweetener solids (sucrose, dextrose), stabilizers, emusifiers, egg products, flavourings, colourants and water.

The composition of ice cream consists of a total solids content of about 40% and the rest is water and incorporated air. The solids include milk fat (16–20%), MSNF (Milk-solids-not-fat) (5–8%), sugar (12–17%), stabilizers and emulsifiers (0.2–0.4%), flavouring and colouring matter. The mixture of constituents prior to incorporation of air is known as ice cream mix. Ice cream in the final form contains a large amount of air as air is whipped into the mix. Whipping of air is necessary, as small uniform air cells are necessary to prevent the ice cream from being too dense, too hard and too cold in the mouth. The increase in volume caused by whipping air into the mix during freezing process is known as 'overrun'. The usual range of overrun in ice cream is about 70 to 100%. If ice cream has 100% overrun, then it has a volume of air equal to the volume of mix that was frozen. The percentage of overrun is arrived at by the relationship

$$\% \text{ overrun} = \frac{\text{volume of ice cream} - \text{volume of mix}}{\text{volume of mix}} \times 100$$

19.5.1 Functions of the Ingredients

Each of the solid constituents of the mix has distinct functions. Milk fat gives rich flavour and improves the body and texture. It is also the concentrated source of calories and contributes to the energy value of ice cream. MSNF contribute to the flavour, body and also to get a desirable

texture. Higher levels of MSNF permit higher overruns without textural breakdown. Sugar gives the sweetness to the product and lowers the freezing point so that the mix does not freeze to a solid in the freezer. Sucrose from cane or beet, or dextrose or corn syrup may be used. Stabilizers are gums such as seaweed gums, pectin or carboxymethyl cellulose. These form gels with water and improve body and texture. They also bind water and prevent the formation of large sized ice crystals during freezing. Emulsifiers such egg yolk, lecithin or mono- and diglycerides help the dispersion of the fat globules throughout the mix and prevent them from clumping together. Emulsifiers also improve the whipping or air incorporation properties of the mix to reach the desired overrun and make the ice cream dry and stiff. Flavours such as vanilla, chocolate, strawberry or coffee flavours of synthetic type and fruits and nuts are added.

19.5.2 Manufacture of Ice Cream

The steps involved are: (i) preparation of the mix, (ii) pasteurization, (iii) homogenization, (iv) aging, (v) freezing and (vi) hardening.

Preparation of the mix. The liquid ingredients are mixed and heated to about 43°C. Sugar and dry ingredients are added to the warm mix and dissolved. Gross particulates such as nuts or fruits are not added at this stage but added during the freezing step. The formulated mix should have proper viscosity, stability and handling characteristics.

Pasteurization. The mix is pasteurized by batch or continuous heating process in equipment similar to those used in milk pasteurization. HTST (79–82°C for 25 seconds) or batch method (68–71°C for about 30 minutes) may be used. Pasteurization renders the mix free from harmful bacteria, brings into solution and aids in blending all the ingredients of the mix, improves flavour and keeping quality and also produces a more uniform product.

Homogenization. The mix is homogenized at the same temperature as it comes out of the pasteurizer. A two-stage homogenizer (2500 psi first stage and 600 psi at second stage) is used to breakdown the fat globules and clumps so that butter or cream formation and separation are prevented. Homogenization also blends the ingredients thoroughly, improves the texture and palatability of the ice cream. It also reduces the aging time, aids in obtaining the desired overrun and produces a more uniform product.

Aging. The homogenized mix is cooled rapidly to about 4°C to prevent bacterial growth and held at this temperature for about 3 to 24 hours. Aging provides time for the melted fat to solidify, the gelatin or other stabilizers to swell and combine with water. The milk proteins also swell

with water and the viscosity of the mix is increased. These changes lead to quicken the whipping process to desired overrun in the freezer to give a smoother ice cream with desirable body and texture and also to slow down ice cream melting.

Freezing. The mix is now ready to be frozen. The cold mix at about 4°C is pumped to a batch or continuous freezer. The mix and air enter the freezing cylinders which are chilled by circulating refrigerant between double walls. The twin purposes of freezing operation are: (i) to freeze the mix to about –3°C and (ii) to beat in and subdivide air cells. Freezing must be quick to prevent the growth of large sized ice crystals that would coarsen the texture and air cells must be small in size and evenly distributed to give a stable frozen foam.

Scrapped surface freezer provided with a mixing element or dasher is used. The rotating dasher with its sharp scraper blades removes the layers of frozen ice cream off the inner freezer wall as it is formed. This prevents build-up of an insulating layer which would otherwise decrease the freezing efficiency of the freezer wall. The ice cream scrapings are mixed with the remaining mix in the freezer cylinder and thus serve as seed material for forming small ice crystals which speed-up the freezing process. The dasher's rods also beat air into the freezing mix.

The mix passing through the freezer cylinder is frozen and whipped to a temperature of about –3°C in about 30 seconds. At this temperature, all the water in the mix is not frozen and ice cream is semi-solid, in which condition it is easily pumped out of the cylinder as a continuous extrusion by the incoming unfrozen mix and the propelling action of the dasher. Fruit and flavouring nuts are added to soft ice cream coming out the freezer. The soft ice cream goes to a packaging machine where it may be packed in bulk containers or small packages for retail sale.

Hardening. The cartons are sent to the hardening chamber where the freezing process is completed. The temperature is maintained at –23 to – 34°C and the remaining water freezes, making the ice cream firm within 24 hours. The ice cream is now ready for storage or delivery.

Physical structure of ice cream and possible defects. Changes in the physical structure are the causes of several common defects in the finished product. Ice cream is a foam containing air cells which constitute the overrun. The overrun gives the ice cream approximately twice the volume of the original mix. In the frozen ice cream the foam forms a film of the mix surrounding the air cells. The fat globules and frozen ice crystals are dispersed within the film or layer of the mix. As ice cream ages, the foam shrinks and the weakened film of mix collapses and thereby the ice cream loses volume. This can be excessive if the mix is low in solids content and constitutes a serious defect.

Fluctuating storage temperature permit repeated partial thawing and freezing and the ice cream becomes coarse and icy due to the formation of bigger sized ice crystals. Lactose, if present in excess, crystallizes out

and the ice cream becomes grainy or sandy in texture. Excessive shrinkage occurs if the storage temperature is higher. Shrinkage due to mechanical compaction when ice cream is dipped from tubs to make cones results in dipping loss. Other textural defects include gummy, crumbly, curdy and watery characteristics due to poor mix formulations. Flavour defects include oxidized flavour and cooked flavour occur due to poor quality of the ingredients.

Questions

1. Describe in detail the milk processing operations.
2. Give an account of the different types of milk and their characteristics.
3. How is Cheddar cheese manufactured?
4. Describe the steps involved in butter making.
5. Give a detailed account of the ingredients of ice cream and their functions.
6. Write in detail the steps involved in ice cream manufacture.

CHAPTER

20

Vegetables and Fruits

Vegetables and fruits are parts of plants, which can be consumed either raw or processed in some way. They have many similarities with respect to their composition, methods of cultivation and harvesting, storage and processing characteristics. Fruits are those portions of the plants, which house seeds. Vegetables such as tomatoes, cucumbers, peppers, eggplant, lady's finger and sweet corn are in fact seed bearing fruits. Based on consumer acceptance and usage, vegetables are those plant parts consumed with the main course of a meal while fruits are commonly consumed as dessert.

20.1 VEGETABLES

Vegetables, apart from providing nutrition, contribute to the appetizing colour, texture and flavour to the food. Vegetables are classified into groups based on their growing season, or based on the parts of plants consumed and shape or appearance, but none of the classifications is either complete or satisfactory.

Root vegetables include carrot, radish, beetroot, parsnip and turnip. Carrot is rich in carotene a precursor for Vitamin A. Beetroot (garden beet) is similar to sugar beet with a high content of sugar. Radish (mooli) in white or red colour has a pungent flavour. These vegetables are consumed as raw vegetables as salad, cooked or pickled in vinegar. *Fruit vegetables* include tomato, bell pepper (capsicum), okra (lady's finger), egg plant (brinjal) and chow-chow. *Cole crops* grow best in cool moist conditions. These vegetables are rich in calcium and Vitamin C. This group includes cabbage, knol-khol, cauliflower and Brussels sprouts. Cabbage (ghobi) consists of thick overlapping leaves attached to a short thick stem to form a spherical or oval shaped head. It is consumed as raw shredded cabbage, in the form of salad, as cooked curry or pickled (Sauerkraut). Cauliflower (phul ghobi) has a single stem bearing a large

rounded flower head of creamy white flower buds. It is used in soup, curry and in mixed vegetable pickles. *Tubers* are rich in carbohydrates and can be stored for long periods. Potato, sweet potato, tapioca (cassava), yam and arrowroot belong to this group. *Bulbs* are underground buds having fleshy leaves and include onion, garlic and leek which have a characteristic pungent flavour. *Curcurbits* include cucumber, pumpkin, gourds such as bitter gourd, ribbed or ridged gourd, sponge gourd, snake gourd, bottle gourd and white or ash gourd. Cucumber is consumed as raw vegetable, in salads and in pickled form. Gourds and pumpkin are usually consumed mostly as cooked curries. *Leafy vegetables or greens* from many plants are part of Indian diet. These include agathi, amaranth, spinach, spinach beet, celery, curry leaves, fenugreek, lettuce, parsley and purslane. Other vegetables widely consumed in India include drumstick, leguminous vegetables such as peas and beans and mushroom.

Most vegetables are high in water (>80%) and low in protein (~3–3.5%) and fat contents. Legumes such as peas and beans have relatively higher protein content and sweet corn has a higher fat content. The dry matter content of most vegetables vary between 10 and 20%. Carbohydrates (3–20%), nitrogenous compounds (1–5%), minerals (~1%), crude fibre (1%) and fats (0.1–0.3%) constitute the dry matter. They also contain vitamins and flavour substances. Glucose and fructose (0.3–4%) and sucrose (0.1–12%) are main carbohydrate constituents. Starch occurs as a storage polysaccharide particularly in tuber vegetables. Other polysaccharides such as cellulose, hemicellulose and pectins contribute to the firmness of the tissues. The major constituent of nitrogenous substances include proteins, particularly enzymes such as oxidoreductases (lipoxygenases, phenoloxidases, peroxidases), hydrolases (proteases, glycosidases, esterases), transferases (transaminases), lyases (glutamic acid decarboxylase) and ligases (glutamine synthetase). Besides, free amino acids, peptides and enzyme inhibitors are also present. Malic and citric acids occur in most vegetables while oxalic acid is present in relatively large amounts in green beans, rhubarb, Brussels sprouts and red beet. Most of the vegetables contain ascorbic acid, riboflavin and thiamine. The main minerals include potassium followed by calcium, iron, sodium and magnesium with anions such as phosphate, chloride and carbonate. Phenolic compounds, hydroxy acids, flavones and flavonols also occur in vegetables. Aroma of vegetables is due to a single character impact compound as in the case of potato (2-isopropyl-3-methoxypyrazine), cabbage (dimethyl sulphide), onion (thriopropanal-S-oxide), garlic (di-2-propenyldisulphide), red beet (geosmin) or mixtures of flavour compounds. Pigments include chlorophyll, betalains and carotenoids.

20.2 HARVEST

Vegetables during their growth and maturation on the plants undergo

changes in their colour, texture and flavour day by day. Harvesting is timed so as to capture the peak qualities in vegetables. Freshly harvested vegetables have the desired plumpness, succulence, crispness attributed to a high degree of turgor. Cell turgor in vegetables and fruits is due to osmotic force, which influences the texture. The cell walls in the living or freshly harvested vegetables and fruits have varying degrees of elasticity and selective permeability to water and small molecules due to osmotic pressure within the cell. The selective permeability and elasticity are responsible for the desirable textural characteristics. After harvest, the vegetables lose their peak quality quickly depending on post harvest practice of storing and further processing. For example, sweet corn loses its sweetness by about 25% in one day after harvest, partly due to conversion of sugars into starch and partly due to continued respiration wherein sugars are utilized for energy. A similar loss of sweetness is observed in peas and lima beans. Some vegetables liberate heat, even after harvest, affecting their flavour and texture characteristics as well as facilitating the growth of microorganisms. Harvested vegetables lose water due to transpiration, respiration and physical drying resulting in loss of weight, wilting of leafy vegetables, loss of plumpness of fleshy vegetables. Therefore, cooling of vegetables by cold water spray soon after their harvest slows down the degradative changes in vegetables.

20.3 STORAGE OF VEGETABLES

Most vegetables are consumed fresh. However, vegetables can be stored for prolonged periods depending on their type. Root and tuber vegetables such as potatoes, carrots, celery, onions, cabbage and turnips can be stored for months while leafy vegetables such as lettuce and spinach, and others such as beans, peas, cauliflower, cucumbers, tomatoes and asparagus can be stored for a few days only. Storage of vegetables brings about changes in their composition mainly due to storage temperature. The sugar content quickly decreases in most cases. Storage temperature is chosen depending on further processing requirements. For example, potatoes meant for dehydration and chips manufacturing should have low sugar content so as to minimize Maillard browning. Hence they are stored at temperatures above 10°C. In contrast, potatoes stored at below 10°C continue to build up high levels of sugars.

Cold storage of vegetables at –1 to +4°C with high humidity in the range of 80–95% is the most appropriate method. During storage, weight loss to the extent of about 2–10% can occur. Ascorbic acid and carotene contents decrease besides starch and pectin degradation, while in some vegetables such as cauliflower, lettuce and spinach the free acid content increases.

Deep freezing or frozen storage of vegetables is practised in the case of beans, peas, pepper, Brussels sprouts, carrot and edible mushrooms.

The vegetables selected for frozen storage are of high quality. The vegetables are washed to remove adhering soil and pesticide, fungicide or insecticide sprays and to reduce the surface microbial load. Some vegetables such as onions and peppers require skin removal. The cleaned vegetables undergo cutting, trimming, pitting or coring. For example, asparagus spears are cut to precise lengths leaving behind clippings of the base of the more fibrous stalk which finds use in making soups and other heat processed products. Green beans are cut into different shapes along the length while olives are pitted. The vegetables are then blanched in boiling water for 2–4 min or by steam for 2–5 min and immediately cooled to about −40°C in plate or air freezer and then stored at −18 to −20°C. Most of the nutrients and vitamins are preserved by deep freezing. However, irreversible changes occur in the texture which include softening, ductile stickiness, looseness or flaccidity or soggy appearance.

20.4 VEGETABLE SALADS

The term includes uncooked, raw or green vegetables sometimes seasoned with salad dressing. Salad is prepared by mixing sliced or diced carrots, onions, cabbage, cauliflower, lettuce, spinach, parsley, celery, cucumber, beetroot and served as appetizer or first course or as a separate course of meal.

20.5 VEGETABLE PRODUCTS

These include dehydrated vegetables, canned vegetables, pickled vegetables and vegetable pastes and juices produced to preserve the vegetables and also to provide new instant or convenience products.

20.5.1 Dehydrated Vegetables

Dehydration reduces the moisture content of vegetables below the required level for the growth of microorganisms and at the same time preserves the flavour, aroma and appearance. The dehydrated vegetable on the addition of water regains the original shape or appearance. Significant changes in the composition of nutrients occur due to dehydration. Concentration of proteins, carbohydrates and minerals occur along with some chemical changes. Fats are oxidatively degraded, accompanied with decrease of odour or flavour. Maillard reaction is facilitated resulting in darker colour and development of new aroma substances. Vitamin levels decrease and original volatile flavour and aroma compounds are lost to a major extent.

Vegetables for dehydration are washed, peeled, cleaned and may be sliced or diced. Blanching for 2–7 minutes using hot water or steam or sulphur dioxide inactivates the native enzymes. The blanched vegetables

are then dehydrated in a conveyor or tube drier maintained at 55–60°C to a final moisture content of 4–8%. Liquid or paste forms such as tomato or potato mash are dried in a spray or drum drier. Dehydration may be carried out by freeze-drying to yield high quality products as in the case of peas and cauliflower used for the production of soup powders. Dehydrated vegetables are light, air and moisture sensitive and hence require careful packaging under nitrogen or vacuum in wax-impregnated paper or cardboard, multilayer foils, metal cans or glass jars.

20.5.2 Canned Vegetables

Canning, which involves heat sterilization, is an important method of preservation of vegetables. The freshly harvested vegetables are cleaned, trimmed, sorted, graded and blanched. Blanching inactivates the native enzymes and also removes undesirable flavour compounds as in the case of cabbage and the air inside the plant tissues. Moreover, it softens and shrinks the product and thereby increases the packing density in cans. Brine (1–2%) is used as the filling liquid in the cans. Sugar solution in the case of peas, tomato, sweet corn and red table beet or citric acid (0.05%) in the case of cauliflower, horse beans and celery or calcium salt in the case of tomato or cauliflower or monosodium glutamate (100–150 mg/kg) may be added to round off the flavour. Sterilization is carried out in autoclaves at different temperatures for different duration depending on the type of vegetable. For example, heating at 118°C for 18 min sterilizes a kilogram of canned peas while cauliflower requires heating to 116°C for 20 minutes. The filled sterilized cans are then cooled rapidly.

The nutritional and physiological value of the sterilized vegetables remain largely unaffected as protein and carbohydrate constituents remain undiminished. However, vitamin losses are quite considerable.

20.5.3 Pickled Vegetables

Spontaneous lactic acid fermentation of vegetables such as cabbage, cucumber and beans yields pickled vegetables. Fermentation lowers the pH, inhibits the growth of spoilage microorganisms and simultaneously brings about enzymatic softening of cells and tissues improving the digestibility and wholesomeness of the vegetables. Salt is used as a preservative and also to facilitate fermentation.

Cucumbers (unripe ones) after addition of dill herb or other flavouring spices are placed in salt solution (4–6%) and glucose may be added and then allowed to ferment at 18–20°C. Initially, fermentation is primarily due to heterofermentative (*Leuconostoc sp*) bacteria followed by homofermentative *Lactobacillus plantarum* and yeasts. Lactic acid, ethanol, volatile acids, carbon dioxide and small amounts of aroma

substances are formed. Lactic acid (0.5–1%) is metabolized by film yeasts partially.

Sauerkraut is pickled white cabbage heads obtained by lactic acid fermentation in the presence of salt (1.5–2.5%) at 18–24°C for 3–6 weeks. Lactic acid content in the final product is not less 6 g/l with a final pH of 3.6. The sauerkraut may be canned by filling the cans at 70°C, followed by exhausting, sealing and sterilizing at 95–100°C.

Vinegar pickled vegetables are prepared by pouring preboiled and still hot vinegar (2.5%) onto unblemished vegetables such as cucumbers, red table beets, onions, pepper or mixed vegetables including carrots, cauliflower, onions, peas and mushrooms. Salt, spices, herb extracts, sugar and chemical preservatives are usually added.

20.5.4 Vegetable Pastes, Juices and Powders

Vegetable puree or paste is a finely dispersed slurry of the vegetable such as tomato. After removing the skin and seeds the vegetable is passed through a pulping machine. Vegetable juices of tomatoes, cucumbers, carrots, radishes, red beet and cabbage are obtained by disintegrating the cleaned and blanched vegetables. The juice is filtered or centrifuged and salt (0.25–1%) is added. The juice is pasteurized for improving storage. Vegetable powders are produced by spray drying, vacuum drum drying or freeze drying the vegetable juice with or without the addition of a drying enhancer such as starch to a residual moisture content of 3%. Tomato powder is the most important product. Spinach and red beet powders are used for food colouring.

20.6 FRUITS

Fruits are juicy edible products of plants and trees and contain the matured ovary including the seeds. Fruits are commonly classified as (i) pomaceous fruits, (ii) stone fruits, (iii) berries, (iv) tropical and subtropical fruits, (v) citrus fruits, (vi) shell (nut) fruits (cashew nut, pea nut, almond, pistachio, walnut) and (vii) wild fruits (rose hips, elder berry).

Berries are fruits with layers of fruit coat or pericarp and mostly homogeneous inside. The pericarp layers contain seeds embedded in a juicy pulp. Strawberry, blackberry, gooseberry, raspberry, blueberry, grapes and currant are some of the well known berries. The berries have appreciable amounts of organic acids (citric, malic and benzoic acids), rich in flavour and colour and are consumed as dessert fruits and also used in the preparation of jams, jellies and baked products such as fruit pies.

Citrus fruits include orange, grapefruit, lemon, lime, citron and tangerine. They are brightly coloured with a pleasing flavour and

sweetness. Citrus fruits are rich in Vitamin C, citric acid and pectin. The skin of the fruit and flowers and leaves also contain essential oils which are extracted for use in perfumes. Citrus fruits are mostly served as juices or slices.

Stone fruits or *drupes* have a thick skin and juicy flesh enclosing a single seed (stone). Apricots, cherries, peaches and plums belong to this group.

Pomme fruits include apple and pear. The receptacle surrounding the ovaries in the flower enlarges into edible mass enclosing the cells containing the seeds.

Tropical and subtropical fruits include amla or Indian gooseberry, avocado or butter fruit, banana, carambola, dates, guava, jackfruit, jamun fruit, mango, papaya, passion fruit, pineapple or *ananas*, pomegranate, sapota and *seetaphal*.

Fruit composition varies with the type and ripeness of the fruit. Fruits contain water to the extent of about 70 to 90% by weight. The dry matter content varies between 10 and 20%. The major constituents include sugars, polysaccharides and organic acids. Minor constituents include nitrogen containing compounds, lipids, vitamins, minerals, pigments and aroma compounds. Nuts contain about 20% nitrogen containing compounds and lipids as high as 50%. Avocado (butter fruit) is rich in fats.

Among the carbohydrates, glucose and fructose are the major monosaccharides with only traces of other monosaccharides. Sucrose is the dominant oligosaccharide. Sugar alcohol D-sorbitol is present in pomme fruits and stone fruits but not in pineapple, banana, berries and citrus fruits. Meso-inositol occurs in oranges. All fruits contain cellulose, hemicellulose, pentosans and pectins. Starch is present in unripe fruits and its content decreases to negligible levels as ripening proceeds. However, in ripe banana the starch content is about 3% and it is more in cashew nuts.

Other nutrients include proteins (0.2 to 1%), fats (0.1 to 0.4%, about 25% in butter fruit) and vitamins (mainly ascorbic acid particularly in citrus fruits such as orange, strawberry, guava and lime). Pigments (carotenoids in yellow and orange coloured fruits, anthocyanins in red, blue and purple coloured ones) and flavour precursors and compounds are also present. Organic acids such as malic and citric acids occur in most fruits, tartaric acid is found in grapes and oxalic acid as calcium salt is also present in a large variety of fruits.

20.6.1 Ripening of Fruits

Endogenous enzymes bring about a variety of changes in texture, flavour and colour of fruits during ripening. As the fruit ripens, most of the pectin is converted to pectic acid which softens the edible fruit tissue; sweetness

increases due to the formation of sugars; free acid content decreases reducing the tart or sour taste; attractive colour and appetizing flavours also develop. These enzymatic changes continue even after complete ripening of the fruit resulting in overripening, development of off-flavours and spoilage of the fruit.

Ripening involves complex changes in physical and chemical properties of fruits. Softening, increasing sweetness, changes in aroma and colour are the most striking phenomena. With the onset of the ripening process, an increased disorganization occurs in chloroplasts and other cell organelles. Photosynthetic activity and other synthetic activity decrease and finally stop completely. Catabolic processes catalysed by cytoplasmic enzymes become dominant. Respiration rate increases with growth followed by a slow decrease until the fruit is fully ripe. In many fruits, ripening is associated with renewed rise in respiration rate which is denoted as climacteric rise. Maximum production of carbon dioxide occurs in this stage. Fruits may also be classified on the basis of their respiratory behaviour after harvest into two types: (i) *climacteric type* fruits (apple, apricot, banana, pear, mango, papaya, peach, tomato, plum and passion fruit) and (ii) *non-climacteric* type fruits (pineapple, orange, strawberry, fig, grape fruit, cucumber, grape, lemon and cherry). Non-climacteric fruits generally ripe on the plants and contain no starch. Climacteric condition is an indication of the natural end of a period of active synthesis and maintenance and the beginning of the actual decay of the fruit.

Changes in metabolic pathways also occur during ripening of fruits. During ripening of banana, aldolase and carboxylase activities increase while in apples, malate and pyruvate decarboxylase activities increase. As ripening proceeds, production of succinate from citrate in apples decreases to negligible levels.

Individual constituents of fruits also undergo significant changes during ripening. For example, the starch content in apples increases during growth on trees and decreases to negligible level by the time of harvest. Sugar content increases during the same period. Hemicellulose content decreases. Pectin fraction also undergoes changes during ripening. The molecular weight of pectins decreases and there is a decrease in the degree of methylation, particularly in bananas, citrus fruits, strawberries, mangoes and melons. Insoluble protopectin is transformed into soluble form. The organic acid content in oranges and grapefruits decreases, while in lemons, it increases. The protein content in some fruits increases due to biosynthesis of enzymes. The formation of typical aroma compounds occurs during ripening. For example, in bananas, volatile aroma compounds are formed only after about 24 hours after the climacteric stage has passed.

20.6.2 Effect of Chemicals on Ripening

The regulation of plant growth is caused by phytohormones such as auxins, gibberellins, cytokinins, abscisic acid and ethylene. Many chemicals either imitate the activities of phytohormones or inhibit their activities. Hence these compounds find use for a variety of purposes such as to induce blossoming, enhance yield, facilitate harvesting, promote fruit ripening, prevent sprouting or germination and to retard aging and thus enhance shelf life and stability of fruits. For example, kinetin and N-benzyladenine retard chlorophyll degradation and aging of leafy vegetables such as cucumber, spinach and beans. N-benzyladenine improves yield of grapes, melons, and other fruits. Gibberellin A_3 can be applied before harvest to oranges or lemons, or after harvesting bananas, guava and tomatoes to retard fruit ripening. Application of gibberellin antagonists such as N,N-dimethylsuccininc acid amide and N-pyrrolidinyl succinic acid amide improves the firmness and colour of fruits, accelerates ripening of cherries, peaches and improves formation of grapes. Maleic acid hydrazide is used as sprouting inhibitor of potatoes, carrots or onions, to retard ripening of mangoes and tomatoes and to maintain the firm texture of apples during storage. Compounds which bind ethylene (a ripening promoter) retard ripening. For example, bananas sealed in *polyethylene bags* have a prolonged shelf life in the presence of potassium permanganate coated on silica.

Fruit ripening is coupled with biosynthesis of ethylene. Ethylene increases the membrane permeability and thereby probably accelerates metabolic and ripening activities. Ethylene stimulates oxidative and hydrolytic enzymes such as catalase, peroxidase and amylase, and inactivates inhibitors of these enzymes before the climacteric stage in mangoes. Climacteric and non-climacteric fruits respond differently to the presence of external ethylene. The increase of respiratory rate of unripe climacteric fruits sets in earlier due to the presence of external ethylene. In contrast, the extent of increase of respiratory rate of unripe non-climacteric fruits depends on ethylene partial pressure in the surrounding atmosphere. Ethylene and compounds such as 2-chloroethylphosphonic acid, which release ethylene, are used commercially for enhancing ripening process. Ethylene is used before picking of fruits such as figs, mangoes, melons, tomatoes and pineapples to obtain more uniform ripening. In the case of bananas, citrus fruits and mangoes, ethylene can accelerate ripening. Ethylene also induces blossoming of pineapple plant, accelerates detachment of stone fruits and olives and defoliates vines. Carbon dioxide is an ethylene antagonist and retards ripening.

20.6.3 Fruit Harvesting

Fruit harvesting is mostly carried out by hand picking. The proper time

for picking fruits from trees depends on several factors such as, variety of the fruit, weather, ease of picking, maturity and ripeness and the purpose for which the fruit will be used. Maturity is the condition when the fruit is ready to eat or, if picked, will become ready for consumption on further ripening. Ripening is that optimum condition when colour, flavour and texture have developed to their peak levels. Very soft fruits such as cherries and peaches are picked when they are mature but not yet ripe. Similarly, oranges are allowed to ripen on the trees to allow the increase of sugar content, decrease of acid content for consumption as whole fruit, or juice production.

The harvested fruits are washed to remove adhering soil and pesticide residues and also to reduce the microbial load on the surface. The cleaned fruits are sorted and graded before marketing.

20.6.4 Blanching

Fruits meant for cold storage are blanched to inactivate endogenous enzymes and prevent continued ripening. Heat blanching of fruits for use in baking industry for making fruit pies is carried out. However, heat blanching of fruits for cold storage is not preferred since the turgor and texture are drastically affected; chemical inactivation of enzymes is usually adopted. An ascorbic acid dip or sulphur dioxide dip prevents overripening. Ascorbic acid functions as an antioxidant preventing the oxidation of polyphenols and tannins. Ascorbic acid dissolved in sugar syrup (0.05–0.2%) together with citric acid is used. Citric acid retards oxidative colour changes and also chelates metal ions which catalyse the oxidation reactions. Sulphur dioxide (0.25%) or its equivalent of sodium sulphite or sodium metabisulphite in water may be used to prevent oxidation reactions by dipping the fruit slices for about 2 hours.

20.6.5 Storage of Fruits

Fruits can be stored for a few months either in cold storage or storage under controlled atmosphere. The suitability, duration and conditions of storage depends on the variety and quality of fruit. Temperature of –1 to +2°C and 80–90% relative humidity are normally used for storing fruits. Apples can be stored for about 4–8 months, pears for about 2–6 months, grapes for about 2–3 months, strawberries and raspberries for 1–2 weeks and cherries for about 4–5 days. Aeration coupled with purging to remove ethylene is necessary. During cold storage moisture loss results in weight loss to the extent of 3–10%.

Fruits may be stored in frozen condition at –18 to –24°C for periods up to 2 to 4 years. The choice of suitable varieties of fruit at optimal ripening stage is important. Pineapples, apples, apricots, grapefruit,

strawberries and dark coloured cherries are highly suitable for frozen storage. In contrast, light coloured cherries, plums and grapes as well as tropical and subtropical fruits are normally not suitable. Some fruits are blanched (pears, apples, apricots and peaches) and some are immersed in 30–50% sugar solution or covered with solid granulated sugar in the absence of oxygen prior to freezing. Rapid chilling to about –30°C of air temperature (freezing time of about 3 hours) is important to prevent microbial growth.

Fruits may also be stored under modified or controlled atmospheric conditions by lowering oxygen content to about 3% (from the normal 21% in air) and enhancing carbon dioxide concentration to about 5% and maintaining the temperature in the range of 0–5°C. For each fruit variety the optimal conditions vary. However, high concentrations of carbon dioxide can lead to aroma defects in mangoes, oranges, bananas and strawberries. Discolouration of fruits may also occur.

20.7 FRUIT PRODUCTS

A variety of processes have been developed to provide new fruit products as well as to store surplus fruits for prolonged periods.

20.7.1 Dried Fruits

These are produced by removal of moisture to prevent microbial growth. Fruits such as apples, apricots, peaches, pears, dates, figs and raisins are washed and trimmed and pretreated in various ways before drying. Pomme fruits (apples and pears) are initially peeled and freed from the core and calix (seed compartment). Apples are cut into thin slices and given sulphite treatment to prevent enzymic and non-enzymic browning, stabilize vitamin C content and also to prevent microbial activity. Dilute solutions of citric acid may also be used to prevent browning. Whole or sliced pears are heated with steam to achieve a translucent appearance. Stone fruits (plums) are dipped into a hot dilute solution of sodium hydroxide or potassium carbonate and rinsed with water. Apricots and peaches are washed with cold and hot water alternately and halved to remove the stone seeds. Grapes are the most commonly dried berry fruits. Raisins are dark coloured dried grapes which contain seeds while sultana raisins are seedless, light coloured dried grapes. The pretreated fruits are usually dried in sunshine followed by drying in drying chambers or tunnel dryers at 65–75°C maintained at a relative humidity of 15–20%. Vacuum drying at 60°C is a gentle method.

20.7.2 Canned Fruits

Stone fruits such as pears, pineapples and apples are canned by heat sterilization of undamaged, aroma-rich, and not overripe fruits. Sterilization is carried out in cans and glass jars. The stone seeds are removed from the fruits. Apples and pears are peeled and sliced. Apricots are readily peeled after alkali treatment at 65°C. To avoid loss of aroma, fruits such as cherries, yellow plums, strawberries and gooseberries which shrink on heat treatment are dipped prior to canning into a hot 30% sugar solution and then covered with sugar solution in cans. The cans are vacuum-sealed at 77–95°C for 4–6 minutes and then heat sterilized. The sterilization time and temperature vary with the type of fruit species.

20.7.3 Rum Fruits and Fruits in Sugar Syrup

Rum fruits are prepared by steeping the fruit in dilute spirits in the presence of sufficient sugar. Pears and plums are poached in wine vinegar sweetened with sugar and spiced with cinnamon and cloves. Fruits in sugar syrup are prepared by treating raw or cooked portions of fruits (usually lemon or orange peels) with highly concentrated sucrose syrup mixed with starch syrup. Starch syrup is added to enhance translucency, smoothness and tractability of the product.

20.7.4 Jams, Jellies and Marmalades

Jams may be made from one kind of fruit or mixed fruits. Fresh fruits, slices and fruit pulp in the form of slurries are boiled in an open kettle with addition of sugar. Gelling agents, starch syrup and tartaric or citric or lactic acid are added and the product is thickened with constant stirring. Jellies are prepared by mixing the fruit sap with half its weight of sugar and thickened by boiling in an open kettle or in a vacuum kettle. The scum is carefully skimmed off and the mixture is boiled further till a moisture content of about 42% is reached. Marmalades are thick spreadable fruit slurries obtained by boiling and thickening of fresh fruits or fruit pulps and slurries by adding sugar. The hot marmalade is then poured into appropriate containers.

20.7.5 Fruit Beverages

Fruit beverages are manufactured mainly for preparing pleasantly flavoured soft drinks and as a supplement and source of Vitamin C to the diet.

Fruit juice is the natural liquid expressed by pressure or other mechanical means from the edible portion of the fruit. The juice is turbid or cloudy containing insoluble cellular components and finely divided tissue. The solid content varies between 5 and 20% by weight. The composition of the juice is unaltered during preparation and preservation. It is consumed as such without clarification to retain the natural flavour and colour, e.g. orange juice.

Importance of fruit cell composition and structure in juice making. The suitability of a fruit species for juice making depends on the fruit composition, which is influenced by genetic make-up, growth characteristics, climate, maturity, nutritional characteristics and post harvest changes. Among the chemical components the free acid and sugar content, the type and amounts of phenolic components and availability of vitamins determine the suitability of fruits for juice making.

All fruit cells contain a vacuole enclosed in a semi-permeable lipoprotein membrane, which contains water soluble constituents. The cell wall consists of three layers, primary wall, middle lamella and secondary wall. The middle lamella acts as intercellular binding substance of cell and is composed mainly of pectin. Pectin or pectic substances are a group of closely related polysaccharides. Pectin can be extracted from cell wall by hot water, weak acids or chelating agents. The soluble pectin so obtained has high viscosity. Chemically, pectin is a linear polygalactouronic acid backbone with varying proportion of carboxyl groups existing as methyl esters. When the methyl ester content is less than 10%, pectin is called pectic acid or pectate. Complete removal of pectin causes separation of cells and maceration of tissue. Microorganisms produce pectin digesting enzymes and hence cause 'soft rot' of fruits and vegetables which appear mushy. Secondary wall contains pectin and lignin. Primary wall consists of cellulose fibres called microfibrils embedded in a matrix of pectins, hemicellulose and proteins. Enzymes do not easily degrade pectin within the primary and secondary cell walls. Hemicelluloses are polysaccharides that can be extracted by strong alkali and consist of four major substances called arabinans, galactans, xyloglucans and xylans. Cellulose is a linear β-1,4-polymer of glucose and is responsible for the rigid structure of plant cell wall.

Fruit juice production. The traditional process of making fruit juices involves two main stages, namely (i) pulping and (ii) extraction or pressing. Pulping involves mechanical disruption of the washed and cleaned fruit by crushing or grinding to give the pulp which is a mixture of disrupted cells and the cellular fluids. It also contains oily or waxy material and pigments of the skin or rind of the fruit. The quantity and fineness of the pulp are adjusted by altering the crushing/grinding process. In the second stage the raw juice is separated from the pulp by the use of sieves or filter presses or centrifuges. Pectinases may be added as pressing aides to enhance juice yield.

The raw juice may be either cloudy or clear depending on the pressing conditions and nature of juice required. The cloudy juice may be consumed as such without clarification as in orange juice. Some fruits such as mango, tomato and apricots are not suitable for direct processing into juice because of their flavour and colour. The pigments are bound to the pulp particles which are too thick to drink. In such cases the whole fruit is disintegrated to give purees which are then further processed to give clear or cloudy juices as required. The stability of the cloud is of importance in cloudy juices and nectars. In contrast, the juices of fruits such as apple, berries and grapes are consumed in clear condition. Hence the raw juice requires clarification to remove all the suspended solids and it is necessary to maintain them in that condition throughout their storage life. Raw fruit juices are not suitable for direct consumption in the case of certain fruits such as passion fruit, because of its strong flavour and black currant and sour cherry because of their acidity. In such cases cloudy juices called nectars are prepared by adding water, sugar or acid to the pulp of fruits. The raw juice may be viscous due to the presence of soluble pectin particularly in the case of bananas, currants, berries and cherries. Heating the juice may lower viscosity but may affect the quality of the juice.

The raw juices are generally unstable because of enzymatic and non-enzymatic changes and also due to the attack by microorganisms. Fruit juices are also susceptible to the presence of oxygen. Dissolved oxygen can destroy Vitamin C and decrease the nutritional value. Hence the juice is usually deaerated by spraying into vacuum chambers. Heat treatment or refrigeration of juices can restrict the growth of microorganisms. Usually the juices are either (i) chilled aimed at short storage life, (ii) pasteurized in single strength in cans, brick packs, bottles or polythene pouches for direct marketing and consumption, (iii) concentrated and frozen for longer storage life and as feedstock for soft drink manufacture or (iv) dehydrated into powder.

The modern method of juice making involves the use of enzymes at different stages. The enzymes are used for maceration, juice extraction, liquefaction, clarification and haze reduction.

Enzymes for maceration. Macerated fruits and vegetable purees are used as constituents in the manufacture of nectars, dairy products and baby foods. The conventional process of maceration is based on mechanical crushing/grinding and heating which can lead to browning and discolouration of the puree, loss of vitamins and production of off-flavours. Blanching the fruit prior to enzyme maceration inhibits the activity of native PG and PE (pectin esterase, EC 3.1.1.11). Hence enzymes endo-polygalactouranase (endo PG; EC 3.2.1.15) and pectin lyase (PL; EC 4.2.2.10) are used for maceration to yield a product of smooth consistency and at the same time retaining the cell wall intact.

Enzymes for juice extraction. Pulps of soft fruits such as black currants, strawberries, raspberries and bananas are difficult to press. Hence treatment of the pulp at 15–30°C for about 60 minutes with pectinases at concentration in the range of 50 to 200 mg/l facilitates pressing and ensures high yields of juice and pigments.

Enzymes for liquefaction. The cell walls of fruit pulp can be degraded completely and liquefied by the use of a mixture of pectic, hemicellulolytic and cellulolytic enzymes. Enzymatic liquefaction of tropical fruits and vegetables results in juice yields of about 90%. Depending on the accessibility of cell wall components to the enzymes, clear juice in the case of papaya, cloudy or pulpy juices from peaches and carrots may be obtained. The enzyme liquefied juices are clarified usually by centrifugation. During apple juice production the juice is extracted by pressing and the residual solid material is called *pomace*. The solid is treated with a mixture of enzymes of pectinases and endo- and exo-1,4-glucanases to liquefy and extract additional juice. This treatment releases flavour precursors and flavour releasing enzymes within the cell walls. However, the juice is susceptible to undesirable non-enzymatic browning during heating or inadequate cool storage, increased acidity and susceptibility to haze formation.

Enzymes for clarification. Clarification involves reduction of viscosity and removal of cloudy material from extracted fruit juices (mostly applicable to apple juice which is usually consumed as a clear liquid). Fungal pectinases and pectin esterases (PE) and endo-PG degrade methoxy pectins in apple juice. Pectin lyase (PL) may also be used. Pectinases remove the pectin coating on the cloud particles partially and thereby reduce electrostatic repulsion between colloidal particles. This induces coagulation and precipitation of colloidal particles enhancing the clarity of the liquid. Ultrafiltration is also adopted to remove microorganisms during clarification of juices.

Enzymes for haze reduction. Concentrated apple or pear juice is susceptible to haze formation because of enzymatic liquefaction or heat treatment. The normally soluble hemicellulosic arabinans are partially hydrolysed during heat treatment or due to the activity of native enzymes resulting in linear arabinans which can associate to form insoluble crystals or haze. Arabinases (arabinosidase EC 3.2.1.55) may be used to break down the arabinans and prevent haze formation.

Fruit juice concentrate. It is concentrated fruit juice obtained by either heating or freezing. The concentrate is relatively more stable chemically as well as microbiologically. The dry matter content is about 60–75%. Concentration by evaporation along with a recovery step for aroma constituents is the preferred industrial method. Evaporation in a 3- or 4-stage gradient evaporator over a short residence time of 3–8 min at 100°C in the first stage and at 40°C in the final stage is adopted. The

concentrate is cooled to 10°C. The aroma constituents of the fruit juice are recovered by rectifying the condensate in the first stage. The aroma constituents are enriched by counter current distillation and added to the evaporated juice only at the final dilution stage.

Concentration by freezing is adopted for juices with sensitive aroma constituents such as orange juice. The juice is cooled continuously below its freezing temperature in scraper cooler. The ice crystals are removed by filtration or by centrifugation to yield a concentrate with about 40–50% solids content. Yet another method of concentration is ultrafiltration of the juice under a pressure of 5–10 bars to get a concentrate of 25% solids content.

Fruit nectar. Fruit nectar is a viscous and turbid substance produced from fruit slurries or whole fruits such as apricots, pears, strawberries, peaches and sour cherries. The cleaned fruits are disintegrated and heated to inactivate the enzymes. The fruit mash is treated with pectin and cellulose degrading enzymes to give a macerated product. The disintegrated product is filtered and the filtrate is then saturated with sugar and additives such as citric or ascorbic acid followed by homogenization and pasteurization. Fruit nectars of berries and stone fruits are commonly denoted as sweet musts.

Fruit syrup. Fruit syrup is made by boiling the fruit with an excess of sugar or by directly treating fresh fruit or fruit juice with sugar to yield a thick fluid. Fruit syrups are rapidly cooled to prevent aroma losses and crystallization of sugar. The sugar is partially converted to invert sugar during heat treatment and caramelization of sugar is thus prevented. Fruit syrups of citrus fruits may contain small amounts of peel aromas. Low acid fruit syrups are treated with tartaric or lactic acid for preservation.

Fruit powders. Drying of the fruit juices or concentrate mixed with large amounts of glucose, maltose or starch syrup yields hygroscopic fruit powders with 3–4% moisture content. The added sugars prevent clumping or caking. Drying is carried out by spray-drying or vacuum foam drying. Freeze-drying may also be used.

Fruit beverages. A variety of fruit beverages are marketed. These include:

Fruit drink is the liquefied whole fruit containing at least 10% of whole fruit by volume.

Fruit squash is fruit juice containing appreciable quantities of fruit pulp and sugar added for sweetening, e.g. orange, lemon or mango squash.

Fruit cordial is a clear liquid obtained by removing all suspended solids from fruit juice by clarification, e.g. lime juice cordial.

Fruit punch is made by mixing desired fruit juices at the time of serving.

Sherbat is a cooling drink made by diluting and sweetening fruit juice.

Questions

1. How are vegetables stored?
2. Give an account of processed vegetable products.
3. How is fruit juice manufactured by traditional method?
4. Describe the use of different enzymes in fruit juice manufacture.
5. What are the chemical changes that occur during ripening of fruits?
6. How do chemicals influence ripening?
7. Write briefly on the preparation of jams, jellies and marmalades.

CHAPTER

21

Cereals, Legumes and Nuts

21.1 CEREALS

Cereals are the cheapest source of food energy and protein content and constitute the bulk of the food consumed by humans. The principal cereal crops include rice, wheat, corn (maize), sorghum (jowar), millets, barley, oats and rye of which rice, wheat and to some extent, maize are relatively more important.

21.1.1 Rice

Being the staple food of more than half the world population, rice is the most important food crop which is cultivated as paddy. The rice grain is covered by a hull and bran, which are indigestible, by humans. The rice endosperm, obtained after removing the outer layers of hull and bran of the paddy and the germ, is mainly starch and is consumed as cooked intact grains along with vegetables, pulses, meat or fish. Rice is also consumed in other forms such as parched rice, puffed rice and fermented preparations such as *idli* and *dosa*. The rice grain contains almost 80% starch, 5–8% proteins, 10–14% of moisture and less than 1% of minerals, particularly phosphorus. Starch is the major constituent of carbohydrates with smaller amounts of free sugars such as glucose, sucrose and dextrin. Glutelin is the principal protein with small quantities of albumin, globulin and prolamin. The rice proteins are rich in arginine but deficient in lysine and threonine. Enzymes in rice include amylases, proteases, lipases, oxidases, peroxidases and phenolase. About a fourth of protein content of the whole rice and minerals are lost in bran and germ on milling the paddy and polishing the rice.

Processing of rice. *Raw rice* is obtained by milling paddy. Paddy is fed to machines called shellers or hullers consisting of rubber discs rotating at different speeds. The grains are not crushed but the hull is rubbed off from the underlying kernels. The hulls are separated from the kernels (known as brown rice) by jets of air. The brown rice is milled in a milling machine consisting of a ribbed rotor, to remove the outer layer of bran and germ and yield the unpolished rice. The unpolished rice develops rancidity and hence is polished using a brush machine to remove the aleurone layer. The polished rice may be further treated with a coat of sugar and talc to a glossy white finish. The milled rice contains mainly unbroken kernels and some amount of broken kernels which are separated by screening.

Parboiled rice is obtained by parboiling paddy before milling to reduce grain losses during milling. Parboiling involves soaking paddy in water for a short time followed by heating in steam and drying before milling. Parboiled rice is relatively more nutritive compared to raw rice as vitamins and minerals are solubilized from the hull and bran and deposited on the endosperm during soaking and steaming. In addition parboiled rice is more resistant to fungus and insect attack. However, parboiled rice is yellowish in colour and sometimes has an unpleasant off-flavour and odour.

Rice products. A variety of rice products are produced in order to enhance the storage capability, facilitate easy cooking to almost instant cooking and provide new foods with better nutritive value and flavour. The different rice products include parched rice, puffed rice, flaked rice, rice flour and rice starch. *Parched rice* is prepared by throwing parboiled rice into sand heated to high temperature. The rice kernels begin to swell and crackle. The mixture is sieved to separate parched rice from sand. The crisp parched rice is consumed as salted, unsalted or mixed with milk or buttermilk. *Puffed rice* is made from sun dried paddy. The paddy is moistened with hot water and after 2–3 minutes, water is decanted off. The paddy is set aside in moistened condition for about 10 hours, sun dried for a short time and parched in hot sand. The paddy grains swell and burst to release the soft white puffed rice. The puffed rice is separated from sand by sieving and from husk by winnowing. *Flaked rice* is made from parboiled rice. Paddy is soaked in water for about 3 days to soften the kernel, which is then boiled in water for a few minutes. After cooking, the water is drained off and the paddy is heated in a shallow vessel to break open the husk. The paddy is pounded with wooden pestle to release the husk and to flatten the rice kernel. The husk is removed by winnowing to obtain whitish thin papery flaked rice used in several food preparations with added flavouring and sweetening agents. Flaked rice can be stored for several months without deterioration. *Rice flour* is made by pulverizing the broken rice kernels obtained during milling. It is used in the manufacture of biscuits (to prevent sticking), pancake mixes and

waffles for its water absorbing characteristics, in baby foods as thickener and as the main raw material in vermicelli, papad and a variety of other preparations. People who are allergic to wheat flour, use rice flour. *Rice starch* is prepared from broken rice kernels. The kernels are steeped in 0.3% caustic soda solution for about a day to facilitate the separation of the starch granules embedded in the protein matrix. The caustic soda treated kernels are washed with water, dried and ground into flour. The flour is mixed with caustic soda solution to dissolve the gluten proteins in the flour, leaving behind starch which is collected, washed and dried. Rice starch is used in food preparations such as puddings, ice creams and pies. It is a raw material for the manufacture of glucose, dextrins and adhesives. It is also used as a stiffening agent in laundry, as a thickener in calico printing and in finishing textiles and in cosmetics.

Byproducts of rice include rice bran, rice polishings, rice bran oil and husk. Rice bran and rice polishings are used as animal feed. Rice bran oil is obtained from rice bran by solvent extraction. It is rich in unsaturated fatty acids and its stability is due to natural antioxidants. The oil is refined and used as cooking oil or salad oil. Rice husk contains about 20–22% ash which is mostly silica. Rice husk is used as an abrasive and as a fuel.

21.1.2 Wheat

Wheat grain is enclosed in an outer covering of bran of several layers. Wheat bran is largely made of cellulose and contains a high proportion of B-group vitamins and nearly 50% of the mineral content of the whole wheat. It is largely indigestible by humans. The germ situated at the base of the grain is rich in fats, protein, vitamins B and E, and iron. The endosperm is mainly starch and is separated from the germ by a membrane called scutellum. The endosperm is the largest component of wheat grain and contains mainly starch granules embedded in a matrix of protein. The periphery of the endosperm is composed of a single layer of cells called the aleurone layer rich in proteins than the endosperm as a whole, but is removed with the bran during milling of the wheat.

Processing of wheat. Wheat is consumed as a variety of products such as bread, bakery products and in India, as *chapati* prepared from wheat flour. Processing of wheat into flour is a two-stage operation involving drying and milling.

Bread wheat should have a moisture content less than 20% before it is harvested. Wheat with greater moisture content is susceptible for spoilage by bacteria and moulds. Typically, wheat is dried at temperatures of 40–50°C to a moisture content less than 13%. Higher temperatures tend to denature the natural enzymes.

The milling process of whole wheat consists of cleaning operation in the first stage, involving the use of (a) rubble separator to remove gross

contaminants, (b) disc separator to remove weed seeds and broken grains, (c) magnetic separator to remove pieces of metal and (d) washer and whizzer device where wheat is rapidly sluiced round at high speed in a stream of water to remove dirt. The washed wheat then undergoes conditioning or tempering which is a complex process. The wheat is said to be conditioned when the grains retain certain percentage of water, with water penetrating the grain structure in an appropriate way. The percentage of water for conditioning varies with the variety of wheat. Most wheats are best conditioned if they are held in a holding container (conditioner) at 20–25°C for about 48 hours at a moisture content of 15.5%. Conditioning is accelerated at higher temperatures, very rapid conditioning occurring in a few seconds at 70–75°C.

In the second stage of milling, the conditioned wheat is subjected to flour milling to obtain the maximum proportion of flour of precise specification of quality, from the wheat. The desired quality of wheat flour is obtained by efficient separation of the starch of the endosperm from the hard and indigestible outer husk of the wheat aleurone layer and the germ. The separation is achieved by passing the grains first between horizontal rollers called 'break rolls' and passing the grit through sieves called 'plan-sifter'. The break rolls convert the grains into 'semolina' which are small granules made up largely of the endosperm. The outer husk is sifted out as bran or coarse wheat feed. The semolina is purified by passing through a series of sieves to remove any husk and then passed through a 'gradual reduction system' consisting of smooth rollers called 'reduction rolls' to get a fine flour of starch. The extraction rate of flour, i.e. the percentage of the original weight of wheat entering the mill that is recovered as flour, varies from 95% at the break roll stage to about 80% of creamy flour and finally, to about 70% superior quality white flour suitable for bread making. Different extraction rates of flour is achieved by adjusting the sieves in the reduction system.

Wholemeal flour is obtained by milling the whole wheat without the separation of bran, germ, endosperm and the scutellum. It is dark in colour and the bread made from it is coarse. Self-raising flour is made from wheat of low protein content which gives weak dough used for making cakes, pastries and puddings. Weak dough is not fermented with yeast but leavened with additives such as sodium bicarbonate and acidic calcium phosphate.

Bread Manufacture. Wheat and rye are the only cereals capable of yielding a flour suitable for baking bread. The composition of wheat depends on the variety of the seed, the nature of the soil and the climate. The bread wheat is 'hard' spring-sown wheat with a protein content in the range of 10–14%, yields a 'strong' flour and is grown mainly in dry climates of North American continent, particularly Canada. The 'soft' winter wheat of Europe, grown under more humid conditions with a protein content in the range of 8–10%, yields a 'weak' flour best suited for biscuits, cakes and pastry.

The main steps involved in the manufacture of bread include: (i) dough mixing and fermentation to convert the starch of wheat flour into sugar capable of being fermented by yeast to yield a porous dough and to form an open honeycomb texture of the finished bread, (ii) moulding and proving the dough to give a fine silky structure which gets fixed at the next stage and (iii) baking which is a heat conversion process to yield the finished bread as well as to develop flavour from the alcohol and other products of yeast fermentation by thermal reactions.

Dough mixing and fermentation. The flour is mixed with salt, fat, a small quantity of baker's yeast and water in a mechanical mixer to form the dough. The dough is covered and fermented in bulk at a controlled temperature of 24–26°C for about two to three hours. The flour enzymes convert the starch into maltose, the main nutrient for yeast. The mechanical mixing stretches the gluten fibres. Small amounts of carboxylic acids, alcohol and carbon dioxide are formed as fermentation products. The carbon dioxide lowers the pH and also acts as a leavening agent giving ultimately the honeycomb structure of bread. The wheat proteins, gliadin and glutenin become hydrated and form an elastic complex called gluten when the flour is mixed and kneaded with water. The presence of elastic gluten makes the preparation of bread possible, because it forms an interconnected network trapping carbon dioxide in pockets. As gas production continues, the gluten strands are stretched and it is believed that bonds between adjacent protein molecules are broken and reformed to produce an elastic, gas retaining, three-dimensional network. A ripe dough that is ready for baking is springy and elastic. It can be easily stretched out and it recovers its shape on releasing. In contrast, an unripe dough can be stretched but is not elastic and an over-ripe dough cannot be stretched far without breaking and has lost its capability to retain the gas. After initial fermentation, the dough is mixed and kneaded for a few minutes to expel some of the gas and tighten up the dough. Thereafter, fermentation is allowed to continue for a further period of about one hour.

In the Chorleywood bread process of Britain, the fermentation time is shortened considerably by high speed mixing of the flour mixed with about 75 ppm of a mixture of ascorbic acid and potassium bromate. Ascorbic acid reacts with endogenous glutathione converting it to disulphide form during dough kneading. Azodicarbonamide, a fast acting flour improver may also be used in addition to the two compounds. The dough formation is brought about by a powerful mixing machine equipped with a meter to indicate the energy expended during the development period. The mixing and development are completed in about five minutes. The powerful mixing enables rapid stretching of the gluten which normally stretches out slowly, aided by gas formation during fermentation in the conventional method. The dough made by this process is known as mechanically developed dough.

Activated dough development also enables to reduce the time required for fermentation. The process uses flour containing about 50 ppm of L-cysteine instead of the ascorbic acid/potassium bromate mixture used in Chorleywood process.

Moulding and proving. The fermented dough is sent to a divider which cuts it into accurately weighed pieces from which the individual loaves of bread will be produced. The pieces are rolled into balls in a moulder. The gluten fibres stretch out during this operation. The balls are sent to the first proving stage in a temperature controlled chamber at about 25°C where the stretched gluten fibres are allowed to relax while the dough is still undergoing fermentation. The first proving stage lasts about 10–15 minutes. The dough balls are moulded into sausage shaped pieces, which are placed in individual baking tins and sent through a second proving chamber maintained at 35–38°C over a period of 40–50 minutes.

Baking. Baking the proved dough pieces in individual baking tins is carried out over a period of 40–50 minutes in an oven maintained at 250–260°C. During baking, the dough rises rapidly because the pockets of carbon dioxide in the loaf expand as the temperature increases. The activity of yeast also is enhanced resulting in the production of more gas until the temperature reaches about 55°C, when fermentation ceases. At about 70–75°C, water present in the dough causes the starch granules to swell and gelatinize. Starch probably abstracts water from gluten and the gelatinized starch supports the structure of the loaf. Hot gluten is soft, devoid of its elasticity and it begins to coagulate. The coagulation continues till the end of baking. The temperature of the interior of the loaf never exceeds 100°C and water and alcohol formed during fermentation escape giving bread the characteristic aroma of baking. Considerable dextrin formation occurs at the outer surface of the loaf due to the action of heat and steam on the starch. The sugars formed are converted to caramel giving the attractive brown crust of the bread. Maillard reactions also contribute to the brown colour of the bread. The baked bread is cooled, sliced and packed.

Types of bread. Wholemeal bread is made from flour obtained by milling the whole wheat grains including the bran and the germ. No other cereal is used in making this bread. Flour improvers are not used but other additives may be used. Brown bread contains about 0.6 wt.% of crude fibre. Flour other than wholemeal flour may be used. White bread contains flour from other cereals in addition to wheat and may contain a wide variety of ingredients such as milk, milk products, liquid or dried eggs, sugar, fats and oils.

Biscuits, quick breads, cakes, pastries and cookies. Biscuits are made from wheat flour with the addition of other ingredients such as salt, fat, sugar, baking powder, milk and flavouring agents. The biscuit dough is kneaded for an optimum time of about 15–20 minutes, rolled into thin

sheets, cut into a variety of shapes and quickly baked in ovens at 210–230°C for about 15 minutes. Quick breads include a variety of wheat flour based products containing baking powder for quick leavening. The dough is baked immediately after mixing the ingredients and served warm. Quick breads include muffins, waffles, pancakes and biscuits.

Cakes, cookies and pastries are baked products made from wheat flour shortened with butter or margarine containing other ingredients such as eggs, sugar and baking powder as the leavening agent. Unshortened cakes do not contain fat. The consistency of the doughs and content of fat and other ingredients vary for cakes, cookies and pastries. In cake making, the fat is creamed into a plastic mass to which sugar is added with continued creaming. Egg or egg yolk is then beaten into the creaming mixture with whipping to incorporate air. Wheat flour is sifted together with baking powder, flavourings and salt. Other ingredients such as milk, are added to make the cake batter. The cake batter is baked in ovens at 175–190°C for about 15 minutes.

Macaroni products. Macaroni products also known as pasta products include macaroni, spaghetti, vermicelli, noodles and ravioli. These are made from the semolina of protein rich hard wheat called durum wheat. The semolina and water is made into a stiff dough from which various pasta shapes such as ribbons, tubes, spirals and sheets are made by extrusion through appropriate dies. Pasta products are ready-to-cook foods and on boiling with water, absorb up to three times their weight of water to form soft food.

21.1.3 Maize

Maize or corn grains consist of the outer hull or bran with a high fibre content, embyro (germ) rich in oil and the endosperm rich in starch. Whole maize contains about 11% prtoein, 4% fat, 3% fibre, 65% of starch and other carbohydrates and 1.5% of minerals. Maize is deficient in vitamin niacin. Maize is milled to separate the outer layer and germ from the endosperm. The germ is recovered to obtain germ oil, a valuable product used as salad oil. Maize bran and the oil cakes are used as animal feed. The starchy endosperm separated during milling is used to make maida flour and suji. Larger grits obtained by screening are used for making corn flakes and porridge. Corn starch is hydrolyzed to give glucose and high fructose corn syrup.

21.1.4 Millets

Millets are small seed grasses and include mainly two varieties: finger millet (*ragi*) and pearl millet (*bajra*) that constitute the food of rural and economically weaker population in India. Other millets include foxtail millet, kodo millet and proso millet.

Ragi has a better nutritive value compared to rice with a composition of proteins (7%), carbohydrates (75%), fat (1.5%) and minerals (2.5%). It is rich in calcium, phosphorus and iron and contains B-group vitamins (but poor in riboflavin). Ragi is milled into flour for use in preparations such as *chapati*, porridge and *dosa*. Ragi grain is malted and malted ragi flour is a nourishing food for infants and is used in the preparation of milk beverages ('ragi malt').

Bajra grain has a composition of proteins (12%), carbohydrates (67%), fibre (1.5%), fat (5%) and minerals (3%). The rich minerals found in bajra are calcium, iron and phosphorus, and vitamins of importance include thiamine, riboflavin and niacin. Bajra is milled to remove the husk and cooked in the same way as rice or ground into flour for making chapaties.

21.1.5 Sorghum

Sorghum (jowar) grains contains protein (10%), carbohydrates (73%), fibre (2%), fat (2%) and minerals (1.5%). Vitamins include riboflavin, pyridoxine, pantothenic acid, nicotinic acid and biotin. Sorghum starch obtained by milling to remove the bran and germ is similar to corn starch and is used interchangeably with corn starch in food industry. Sorghum malt is used in the preparation of infant foods and for brewing kafir beer containing 3% alcohol. Malt extracts of sorghum are used in pharmaceuticals.

21.1.6 Barley, Oats and Rye

Barley is one of the world's oldest cultivated cereals for bread making. Barley is mostly used for malting and brewing and also as cattle feed.

Oat is a nutritious cereal but contains about 30% hull which is difficult to separate from the grain. Oatmeal is the common breakfast cereal containing proteins (14%), carbohydrates (63%), fat (8%) and is relatively rich in B-group vitamins.

Rye is the only cereal, apart from wheat, from which bread can be made. It used to be the major food crop in Eastern Europe. Rye is used as cattle feed and for the production of distilled liquors.

21.1.7 Breakfast Cereals

Breakfast cereals include ready-to-eat cereals and ready-to-cook convenient food products. Ready-to-eat cereals include puffed, flaked, shredded and granular products made from wheat, maize, oats, barley or rice. These may be enriched with sugar, honey or malt extract and fortified with vitamins, minerals and other nutrients. Ready-to-cook cereals are mainly wheat products. Making *uppma* (in India) or *farina* involves boiling/cooking semolina with water. The preparations may be

made with vegetables, nuts, dal and flavoured with spices. *Bulgar* is yet another popular wheat based preparation in West Asian countries involving multi-stage cooking.

21.2 LEGUMES

Legumes or pulses are the edible fruits or seeds of pod-bearing plants. The protein content of legumes is quite high, in the range of 20–40% and hence legumes are an important nutritional constituents of diet. In India, the term 'gram' is used to denote dry legumes while the split decorticated grains are called *dhal*. The principal legumes in India include Bengal gram (chana), red gram or pigeon pea (tur dhal), black gram (urad), green gram (mung), peas, lentil (masur), horse gram, cow pea, cluster bean or guar, Indian bean, French bean or kidney bean and soybean.

The composition of pulses depends on the species and all of them are good sources of vegetable proteins. The proteins in pulses are mainly globulins and to a smaller extent, albumins. Pulse proteins are deficient in sulphur containing amino acids, particularly methionine, and in tryptophan. Only soybean protein is the most satisfying pulse protein with respect to nutrition. Though the pulses are high in protein content, the digestibility and hence the nutritional value is relatively low because of the high molecular weight. In addition, some of the proteins exist as complexes of carbohydrates, phytin and polyphenols. Pulses also contain 50–60% carbohydrates including starch as the major constituent besides soluble sugars, fibre and unavailable carbohydrates. Oligosaccharides such as raffinose, stachyose and verbiscose constitute the unavailable carbohydrates as α-galactosidase is not available in the human and animal digestive system for digesting these carbohydrates. These carbohydrates are utilized by the microflora of the lower intestines resulting in the formation of large amounts of carbon dioxide, hydrogen and a small amount of methane. These gases give rise to flatulence. Processing of pulses such as soaking, germination, cooking, autoclaving or fermentation reduces the oligosaccharide content to a large extent. The fat content in pulses is relatively small (~1–2%) except in soybeans and winged beans. Minerals such as calcium, magnesium, iron, zinc, potassium and phosphorus are available in pulses. Phosphorus exists as phytate phosphorus and phytin is responsible for complexing proteins and minerals and thus rendering them indigestible. Phytin is degraded by cooking, soaking, germination or fermentation. Pulses have small amount of carotene, while peas have Vitamin A activity in higher proportions. Pulses also contain thiamine, and niacin but have a poor riboflavin content.

Pulses also contain toxic substances such as inhibitors of enzymes (trypsin, chymotrypsin and α-amylase), hemagglutinins, goitrogens, cyanogenic glycosides and alkaloids. The inhibitor of trypsin is a protein which suppresses the hydrolytic release of amino acids and affects the

growth of animals. Hemagglutinins are also proteins and they combine with digested products in the gut and prevent their absorption. Goitrogens are substances naturally present in beans, bind iodine and cause iodine deficiency in the thyroid ultimately leading to the disease goitre. Saponins are a group of glycosides of high molecular weight found in soybean, jack bean and sword bean. On shaking with water, they produce lather or foam. Toxic saponins cause nausea and vomitting. Cyanogenic glycosides on hydrolysis by β-glucosidase produce the toxic hydrogen cyanide. In addition to the aforementioned ones, toxic substances in *khesari dhal* produce lathyrism, an incurable paralytic disease affecting the lower limbs. Another fatal disease associated with eating broad beans or by inhaling the pollen of the flower is favism or haemolytic anaemia.

21.2.1 Pulse Processing

Pulses are processed by a variety of methods in order to improve their nutritive value. These methods include soaking, germination, decortication, cooking or fermentation. Soaking in water reduces the oligosaccharides and the phytin content in the pulses. Germination and sprouting improve the nutritive value due to increase in riboflavin, niacin, choline and biotin content. Folic acid content decreases while pantothenic acid content remains unaltered. Germination converts starch into sugar, and sprouts are more delicious as a salad or a vegetable. Germination also reduces the toxic and anti-nutritional factors in the pulses. Decortication involves the removal of the bitter tasting and indigestible seed coat or husk. Split pulses, after removal of the husk, are called dhal having better keeping qualities and digestibility. The seed coat is easily removed by rubbing as it absorbs more water when the pulse is soaked in water. Alternatively, the soaked grains are dried and the husk is removed by pounding and winnowing. Roasting also removes the seed coat in the case of Bengal gram and peas. Cooking destroys enzyme inhibitors and improves the nutritional quality and digestibility. Fermentation is an important method of processing the soybeans because they cannot be directly used as food due to the presence of toxic substances. Fermented soybean products include soybean sauce, soybean paste (*miso*), *tempe, natto* and *hamanatto*.

Soybean products. A variety of soybean products are common in Asian countries. The readily digestible proteins are separated from the indigestible ones in the form of soybean curd or Tofu and soybean milk. Tofu is prepared by grinding soaked beans into an emulsion of a milky fluid, cooking the emulsion and straining. The curd is precipitated from the fluid by adding calcium sulphate. The precipitate is allowed to settle, collected, washed and dried. The off-white colour curd has soft texture and can be cut into slabs or slices and marketed. The fresh curd has protein content in the range of 6–18%. It is stored under refrigeration to prolong its shelf-life.

Soybean milk is prepared by grinding the soaked beans in a stream of water to get an emulsion. The emulsion is cooked for about 20 minutes and additives such as sugar, salt, lime, malt and margarine are added. The cooked milk is then homogenized or emulsified. The milk is consumed fresh or may be spray dried. Soybean curd and milk are nutritious products especially for children who cannot digest cow's milk. Soybean milk contains lower amount of sodium compared to cow's milk and is good for high blood pressure patients.

Fermented soybean products. Fermentation is an important processing method for converting the non-digestible soybean into nutritious, easily digestible food products. *Shoyu*, (soya sauce), *miso* (soybean paste), *tempeh, natto* and *hamanato* are some of the well-known soybean based fermented oriental food products. *Shoyu* is prepared by *koji* fermentation using cooked soybeans mixed with ground wheat and inoculated with *Aspergillus oryzae*. After fermentation salt is added and the product is ripened or matured over 6 months to 3 years to leave the sauce with 5-6% protein content. *Miso* is prepared by fermenting cooked soybeans mixed with steamed rice or barley and inoculated with *Aspergillus oryzae*. Fermentation occurs over several days, the rate of fermentation being controlled by adding salt. The final product with a 10% protein content has a paste-like consistency and is used in the preparation of soups or served with rice as dressing or side dish. *Tempeh* is a highly nutritious food rich in protein and amino acids with a strong smell. It is produced by fermenting cooked and mashed soybeans with *Rhizopus oryzae*. *Natto* with a mushy flavour is similar to *tempeh* and is fermented using *Bacillus subtilis*. *Hamanatto* is obtained by fermenting steamed soybeans using a fungus and the fermented product is sun dried. The dry product is treated with salt solution and allowed to ferment over another 3 to 12 months. It is again sun dried before making it ready for consumption.

21.3 NUTS

Nuts are seeds or fruits consisting of fat rich kernel usually surrounded by a hard, brittle shell. Coconut, groundnut (peanut), chestnut, cashew nut, almond, pistachio and walnut are some of the well-known varieties of nuts. Nuts may be classified into three major groups based on their nutrient composition: (i) those with high carbohydrate content (e.g. chestnut with 50% content of carbohydrates and relatively poor in fat and protein content), (ii) those with high protein as well as high fat content (e.g. groundnut, almond and pistachio) and (iii) those with high oil or fat content (e.g. cashew nut, walnut).

Nuts are concentrated sources of nutrients and are consumed in raw, roasted, salted or cooked state. The palatability, flavour and digestibility of nuts depend on the type of nut and its treatment. For example, almonds are equally palatable in raw and cooked condition, while groundnuts have

better flavour on cooking and dessert nuts such as cashew nuts are roasted and salted. The flavour of nuts depends on the oils present in them and these undergo changes during cooking. Nuts are used as ingredients in cakes, biscuits, sweetmeats, ice creams, puddings, processed foods and in weaning foods for children.

21.3.1 Processing of Nuts

Nuts perish quickly and require proper processing for long-term storage. Most nuts are enveloped in a fleshy or fibrous husk and as they ripen and fall off the trees, the husk may be detached in some cases. Dehusking is carried out as soon as the nuts are collected in order to prevent changes in the colour and quality of the nuts. Nuts, after collection, are washed and sun dried or dried in mechanical dryers. Nuts may be stored for long duration or marketed in their shells after bleaching the shells with sulphur dioxide to remove stains and give a uniform colour. Nuts are also marketed as kernels whose shelf life is rather limited and hence moisture-proof containers are necessary for long-term storage (up to two years) to prevent the loss of flavour, colour or texture. Nuts marketed in cooked or salted condition are usually vacuum packed to prevent absorption of moisture.

21.3.2 Peanut Butter

Peanut butter is a popular table spread made from groundnut kernels. The nuts are roasted to a moderate degree and the skin and germ are removed. The kernels are finely ground with the addition of salt and hydrogenated groundnut oil to give the final product containing about 48% fat content.

Questions

1. How are raw and parboiled rice processed?
2. How is wheat processed and milled for making bread?
3. Write in detail the various stages involved in the manufacture of bread.
4. What are quickbreads? How are they produced?
5. What are macaroni products?
6. Write a note on breakfast cereals.
7. What are legumes and pulses? How are they processed?
8. Write briefly on fermented soybean products.
9. How are nuts classified? How are they processed?

CHAPTER

22

Meat and Meat Products

22.1 MEAT

The term meat refers mainly to skeletal muscle and adhering fat obtained from warm-blooded animals, though other parts including internal organs which are fit for human consumption are also considered as meat. Apart from skeletal muscle, meat includes heart muscles and smooth muscles of spleen, lymphatic glands, epidermis, intestinal tract and mucous linings. The muscle tissue contains moisture (75%), nitrogenous substances (22%), fat (1.5–2%), minerals (1%) and small amounts of carbohydrates (0.05–0.2%). The nitrogenous substances are mainly proteins which belong to three groups, proteins of the contractile apparatus which are extractable with salt solution, water soluble proteins and water insoluble proteins. The contractile apparatus proteins constitute the major amount of proteins and include myosin, actin, tropomyosin and troponin. The soluble proteins include myoglobin, hemoglobin and enzymes. The insoluble proteins are the proteins (mainly collagen and elastin) of the connective tissue and lipoproteins of membranes. Other nitrogenous substances include free amino acids, peptides, amines, guanidine compounds, purines, pyrimidines and quaternary ammonium compounds. The glycogen content of muscle varies depending on the age and condition of the animal prior to slaughter. Sugars constitute only to the extent of 0.1%. Lactic acid, glycolic acid and succinic acid are also present in the muscle. Muscle tissues are rich in vitamins such as thiamine, riboflavine, nicotinamide, pyridoxine, pantothenic acid, folic acid, biotin, cyanocobalamin, and minerals such as potassium, sodium, magnesium, calcium, iron, zinc, phosphorus and chloride. The red colour of meat is due to myoglobin and meat cured with nitrates remain pink for a long time.

22.1.1 Post Mortem Changes in Meat

As soon as the animal is slaughtered blood circulation stops and anaerobic conditions set in and energy rich phosphates such as creatine phosphate, ATP and ADP undergo degradation. Glycogen remains the sole energy source and glycolytic reactions continue for some time till the pH of the muscle decreases to about 5.5 due to the formation of lactic acid, or till glycogen reserve is available. As glycolysis comes to a stop, ATP generation also comes to an end and the muscle tissue becomes stiff and rigid or the state of rigor mortis sets in. The onset of rigor mortis occurs at different times in different species and also depends on the age, state of activity of the animal before its slaughter and feeding practices. In general, rigor mortis sets in within 10–20 hours in beef, 4–18 hours in pork and 2–4 hours in chicken.

The rate of decrease in pH and the final pH of the muscle influence the water holding capacity of meat and are of significance in determining the quality of meat. Rapid decrease in pH and ATP level causes pork muscle to become pale, soft and undergo extensive drip loss because of lower water holding capacity. Such a PSE (pale, soft and exudative) meat has a low tensile strength, loses substantial amount of weight on hanging, and on thawing, drip losses continue. Similarly, the occurrence of DFD (dark, firm and dry) meat is characteristic of a stress impaired hog. DFD meat has high pH due to low level of lactic acid. The PSE effect is not significant in beef muscle as fat oxidation slows down glycogen breakdown during post mortem. Rapid and intense cooling of the muscle tissue leaves the meat with a greater water holding capacity and makes it of better quality compared to muscle tissue cooled slowly.

22.1.2 Aging or Ripening

Rigor mortis in beef muscle resolves in about 2–3 days. The meat again becomes soft and tender due to aging. The connective tissue proteins breakdown due to the action of native proteolytic enzymes during aging. Tenderness in meat is a desired quality. Aging improves tenderness and allows time for the development of aroma depending on the temperature of storage. Beef requires aging for 14 days at 0°C, 6 days at 8–10°C or 4 days at 16–18°C. A slight increase in pH and increase in water holding capacity occur during aging.

Muscle tissue contains about 20–25% protein and about 75% water; i.e. about 350–360 g of water per 100 g of protein. About 95% of this water is held by capillary forces between the thick and thin filaments and only 5% is bound directly to the hydrophilic groups on the proteins. The extent of water holding by the protein gel network depends on the abundance of cross-linking between the peptide chains. A decrease in the number of cross-linkages results in swelling of the muscle while increase in the

cross-linkages shrinks the protein gel causing syneresis. The muscle is swollen, soft and juicy when it holds a large amount of water. As rigor mortis sets in and ATP level decreases, the water holding capacity also decreases. Addition of salt and increase in pH generally improves the water holding capacity of meat. Meat swelling and water holding capacity are low at pH 5.0–5.5.

Aging is adopted only in the case of beef and occasionally in the case of lamb and mutton. Pork is never aged because of its high fat content.

22.1.3 Tenderizing Meat

Tenderness in meat depends on the extent of cross-linkages between muscle fibres and collagen, and decreases with increasing age of the animal. The meat of aged animals is tough and difficult to chew. In addition, during cooking, collagen is solubilized and tenderness of meat decreases. Aging improves tenderness of meat. Meat is tenderized by mechanical methods such as cutting, pounding, grinding or needling, use of ultrasonic vibrations to break the meat fibres or by enzymes. Papain (proteolytic enzyme from papaya) or bromelain (from pineapple) is rubbed into the fresh meat or sprinkled in the form of a powder or solution. The enzyme is also injected into the veins of living animals just before their slaughter. The protein fibres are hydrolyzed by the enzyme thereby improving tenderness of the meat.

22.1.4 Types of Meat

Beef is bright cherry red or dark red lean meat or meat marbled with fat obtained from the muscle fibres and associated connective tissue of cows and bulls, the best quality being obtained from young bulls and 4–6 year old oxen. The meat of old cows and discarded dairy cattle of more than 12 years old is of lower quality. Beef from the carcasses are hung for about a week before being cut up for soup meat and for about two weeks for roasts and steaks. *Veal* is white to pale red meat with tender, limp and sticky muscle fibres of 6–14 weeks old young calves. The meat is hung for about a week before use.

Mutton is meat with associated fat from sheep older than an year, the best quality being obtained from 2–4 year old sheep slaughtered in autumn. *Lamb* is meat from younger sheep with mild flavour and taste. *Goat meat* is lighter in colour than mutton. *Ram meat* (from male goat) is also used.

Pork is meat from pigs with fine muscle fibres, soft and tender, interspersed with as much as 20–40% fat and is grey-white to pale pink in colour. The meat is hung for about 4 days before butchering. Young *horse meat* is bright red compared to dark red colour of older animals consisting of firm muscle fibres and a characteristic sweet taste and

flavour due to high glycogen content. The linolenic acid content of horse meat is relatively greater compared to that in beef or pork. The small quantity of white fat which melts at about 30°C during cooking, appears as drops on the surface.

Poultry meat includes meat from chicken, ducks, geese, pigeons, turkeys and peacocks. The quality and colour of the meat vary with the breed, age and feeding habits as well as the body parts. The meat of wild game such as wild boar and rabbit, birds or fowl such as pheasant and partridge have distinct characteristics depending on the breed.

Organ meat or *variety meat* includes organs of various animals such as tongue, heart, liver, kidney, brain, spleen, bladder, stomach and intestines. These may be consumed as such (as in the case of liver, kidney, or heart) or minced and then filled into sausages (spleen, tongue, cow's udder, beef stomach, etc.).

Blood, which drains from the slaughtered animal, is used for making sausages. Glands of animals such as adrenal, pancreas, mammary, ovary, pituitary and thyroid are used for extracts or pharmaceutical products.

22.1.5 Grades of Meat

Meat sold in the market must be from healthy living animals, slaughtered hygienically and certified as fit for human consumption by a competent authority. Meat after proper inspection is graded based on maturity and marbling which influence the quality and palatability of meat. The maturity of the animal affects the texture, firmness and colour of the meat. Fine structural lean meat is tender compared to a meat with a coarse texture. The tenderness of the meat of a young animal increases as it matures to an optimum age. Marbling refers to the amount and distribution of fat in the meat which appears as white flecks within the lean meat and is considered as an index of quality. In addition to these, high juice content and bright red or pink colour also indicate the freshness of meat.

22.1.6 Preservation

Meat may be preserved by cold storage, drying, pickling or smoking.

Refrigeration and freezing. The slaughtered animal carcasses are skinned and the fresh meat is preserved in the form of sides or quarters by refrigeration or freezing. Cooling process for preservation of meat is carried out in a step wise manner with a blast of air (3.5 m/s) at –10°C for about 3 hours, with a blast of air (1.2 m/s) at 2°C for about 18–19 hours and over 18 days with air at 4°C. The shelf life of meat at 0°C is about 3–6 weeks. Freezing and storing at –18 to –20°C prolong the shelf life substantially to about 12–15 months. Freezing may be carried

out in a single step of direct freezing or two steps of initial cooling and freezing using an air blast at –40°C. During cold storage, water holding capacity of meat increases. Lipid oxidation is very slow in large cuts. Freezing meat immediately after slaughter while the carcass is still warm results in meat which after thawing loses a large amount of fluid (called thaw rigor) due to a sudden high rate of ATP breakdown. The thaw rigor can be avoided if the warm meat is disintegrated in the presence of sodium chloride before freezing it.

Drying. Drying is yet another method of meat preservation. Drying in a stream of hot air at 40–60°C, drying in vacuum or freeze-drying may be used. The moisture content of the dried meat is about 3–10%. The shelf life is limited due to oxidative degradation and discolouration due to Maillard reaction. Dried beef and chicken are used for making soup powders. Meatballs and minced meat pieces are also dried.

Salt curing and pickling. Salting meat with sodium chloride or curing and pickling by the addition of sodium nitrite and or sodium nitrate is common practice. Sodium chloride in high concentrations inhibits microbial spoilage of meat, causes swelling (at >5%), and induces shrinkage (10–20%) in meat. Salted meat retains its natural dark red colour and on cooking, becomes grayish brown. Salt curing is done either by dry curing, i.e. rubbing salt on the meat surface or by wet pickle curing by submerging meat in 15–20% brine or by rapid pickling involving injection of salt solution into blood vessels. Curing with sodium nitrite and nitrate produces meat of highly stable colour. Additives, such as sugar or spices may be added to pickling salts to improve colour, flavour and formation of meat aroma. Aroma formation and stabilization of the red colour of pickled meat are also enhanced by *Micrococcus* and *Achromabacter* species in the pickling brine.

Smoking. Smoking of meat is usually associated with salting. Volatile compounds in sawdust smoke or woodsmoke have bactericidal and anti-oxidative properties, and they deposit on the surface and also penetrate into the meat.

Heating. This is an important finishing process and is also used for the production of canned meat. Heat treatment results in protein coagulation, release of juices, increase in pH, development of cooked or roasted meat aroma, softening and shrinking and also partial conversion of collagen of the connective tissue into gelatin.

22.1.7 Cooking of Meat

Cooking brings about changes in the meat texture by melting the fat, thermal breakdown of gelatin into soft gelatin and its solubilization, tissue softening and muscle fibre separation. Overheating causes extensive

contraction, shrinking and toughening of meat. Generally, cooking at lower temperatures for a longer time yields cooked meat with minimum drip loss and shrinkage and increased juiciness and uniform colour than cooking at higher temperatures for short time. Meat may be cooked by moist heating or dry heating methods. Moist heating methods of cooking include braising, stewing and pressure cooking. Meat may be cooked to rare, medium or well-done stages depending on temperature. Braising involves cooking the broiled or fried meat cuts with small amounts of water in a closed vessel. Stewing is adopted to cook large pieces of tough cuts of meat in simmering water. Less tender meat cuts are usually cooked by moist heating to convert the collagen into gelatin and increase the tenderness of meat. Dry heating methods such as roasting, broiling or frying are used for tender meat cuts. Roasting usually in an open pan at 160–170°C yields a cooked meat with adequate surface browning with a good flavour and appearance as heat is transferred from the outer surface to the interior. Roasting produces juicy and tender meats compared to other dry heating methods. Broiling involves cooking meat by direct radiant heat such as open fire of gas flame, live coals or electric oven. Broiling is used to cook tender meat cuts which are at least 3 cm thick, as thinner cuts will become too dry on broiling. Broiling is carried out by placing the meat in a rack and heating to about 175°C till the heated surface becomes brown. The broiled surface is salted and turned and broiled on the other side. Frying is carried out either by pan frying with small amount of fat added or deep fat frying where the melted fat is deep enough to cover the meat.

A number of changes occur during cooking of meat which include changes in colour, texture and flavour of meat due to denaturation of proteins and conversion of collagen into gelatin. Nutritional value of cooked meat remains high, though loss of B-vitamins occurs. Meat aroma of raw meat is weak. Heat treatment intensifies the aroma of meat and the character of the aroma depends on the type of meat and also on the method of heat treatment such as cooking, pressure cooking, roasting, stewing, broiling or barbecuing. The differences in the nature of lipid degradation products in different types of meat at different temperatures and concentrations of degradation products are primarily responsible for the characteristic aroma variations. Meat aroma consists of volatile aroma compounds, non-volatile flavour compounds and flavour enhancers. The volatile aroma compounds are formed by Maillard and Strecker degradation reactions and include alcohols, aldehydes, ketones, carboxylic acids, benzene derivatives, furans, lactones, pyrans, pyrroles, pyrimidines, pyrazines, thiols, sulphides, thiophenes, thiazoles and trithiolanes. Sulphur compounds play a significant role in the meat aroma. Meat like aroma can be generated by heating cystine, cysteine, methionine or thiamine with reducing sugars or by reaction of hydrogen sulphide with alkenals and hydroxydihydrofurans or by other reactions. The non-volatile flavour compounds include amino acids, peptides and lactic acid.

Flavour enhancers in meat are glutamic acid and inosine-5'-monophosphate (IMP).

22.1.8 Meat Products

Commercially available important meat products include canned meat, ham, bacon, sausages and pastes. Canned beef, pork, corned beef, cooked sausages, jellied meat, pickled ham are canned under sterile conditions after appropriate heat treatment. Ham is salt-cured and pickled, smoked hind leg of pig. The hind leg section extending from the knee to the hip is cut either long or short . The long cut includes the whole hip and a long part of the shank, i.e. part of the leg above the knee. The short cut ham consists of a part of the hip and a short section of the shank. Fresh or green ham is unprocessed ham used for preparing dry country ham with ethnic flavour. The prime cuts or halves of ham include butt (part of the hip), center (bone-in) and shank (bone-out). Boneless ham and bone-in ham are either dry or wet cured, matured for 3 weeks and smoked or cooked. Bacon is either dry cured or pickled pork belly. The belly portions are then matured for about 2 weeks and smoked or cooked. Sausage making consists of grinding, mincing or chopping the muscle tissue and other organs and blending them with fat, salts, seasonings (herbs and spices) and binders or extenders. The sausage mix or dough is stuffed into cylindrical natural casings such as hog or sheep intestine or hog's bun (for liver sausage) or synthetic casings made of cellulose and sold as raw or cooked or smoked sausages. Depending on the type of processing, sausages are classified as: raw or coarsely ground or fermented and emulsified pork or beef, water cooked luncheon and breakfast sausage, heat treated frankfurters (hot dogs) and wieners or mortadella. Meat pastes are spreadable pastes containing delicately cooked high quality meat products (free from slaughter scrapings) made from meat and fat of calves, hogs, poultry or wild animal meat. Beef extract is a concentrate of water soluble beef ingredients without fat and proteins. Finely comminuted beef is extracted with water at 90°C by countercurrent extraction. Fat is removed by separators and the filtered extract containing about 5% solids is concentrated to about 60–65% solids content by multiple stage vacuum evaporation with decreasing gradient from 92 to 46°C. The final evaporation to 80–85% solids is carried out under atmospheric pressure and 65°C or higher. Chicken extract is obtained by evaporation of chicken broth or by extraction of chicken halves with water at 80°C followed by concentration in vacuum to get a finished product containing 75–80% solids content. Yeast extract is made from yeast cells of *Saccharomyces* and *Torula* species. The cells are forced to undergo plasmolysis by the addition of salts or autolysis by exposing to steam and then extracted with water. The extract is concentrated to a brown paste which is rich in B group vitamins.

22.2 POULTRY

The term *poultry* refers to domestic fowls such as chicken, ducks, turkey, pigeons, geese and guineas of which chicken and turkey are commonly used for their meat. Poultry is usually classified on the basis of age, since age influences the tenderness of the meat and fat content and determines the cooking method. Broiler or fryer chicken is about 8–10 weeks of age with tender meat, a soft, smooth textured skin and flexible breastbone cartilage. A rooster chicken is 12–20 weeks of age with similar characteristics as the broiler but less flexible breastbone cartilage. A stag is a male chicken of less than 10 months of age with a coarse skin, toughened and darkened flesh and a hard breastbone cartilage. A cock is a mature male chicken over 10 months of age with similar characteristics of a stag but more hardened breastbone.

22.2.1 Poultry Processing

Generally, birds meant for slaughter are not fed for about 12 hours prior to ensure their crops are empty to facilitate cleaner operation. The live bird is stunned and made unconscious and jugular vein is cut to bleed the bird completely in about 3–4 minutes. The bird is then scalded by dipping in hot water to facilitate defeathering. After removal of feathers the bird is eviscerated, i.e. the entrails (stomach, intestine, lungs, kidney, head, feet and oil gland) are removed. The eviscerated bird is washed and chilled rapidly to about 2°C. The chilled chicken is graded and packed in films or bags which have high resistance to transmission of moisture and air. The packed poultry may be stored in a refrigerator for a few days or stored up to a few months in a deep freezer (–22 to –18°C).

Poultry goes into rigor mortis and relaxes rapidly compared to meat. Hence poultry is ready for cooking only about 5 hours after slaughter. The dressed chicken may be cooked by broiling, frying, or roasting. Birds with tough meat may be braised or stewed. Chicken meat has a high protein to fat ratio with a protein content of 20–25% and very low fat content. The fat contains relatively more of unsaturated fatty acids. The flesh is also rich in B group vitamins and minerals.

22.3 SEAFOODS

Fish, whales, crustaceans and molluscs are a rich source of biologically valuable proteins, fats and fat-soluble vitamins. Out of 25,000 species of fish, only about 250 species are considered as edible ones. Edible fish may be classified into finfish and shellfish on the basis of their anatomy. Finfish are the type of fish with vertebrae and fin-appendages and are covered with protective scales. Finfish are found in salt water (shark,

salmon, mackerel, tuna, herring, cod, ribbon fish, etc.) as well as in fresh water (sardine, mullet, catfish, carp, trout, etc.). Finfish such as salmon and eels can live in both salt and fresh waters. Saltwater fish have a more distinctive flavour than freshwater fish. Shellfish are without a skeleton but covered with a hard shell and may be sub-classified into molluscs and crustaceans. Molluscs have a soft unsegmented body enclosed partially or wholly in a hard shell. Molluscs include oysters, clams, scallops and mussels. Crustaceans have a segmented body covered with crustlike shells, e.g. crabs, lobsters, shrimps and crayfish. Edible shell-fishes are mainly salt water fishes.

Fishes are also classified on the basis of their fat content as lean fish with less than 2% fat (e.g. cod, haddock, sole, etc.) medium with 2–5% fat (e.g. sardine, halibut, etc.) and fat fish with more than 5% fat (e.g. salmon—14%, eel—26%, mackerel—12%, tuna—16%, herring—18%, etc. Fat fish taste better than other types.

Fish protein content varies between 13 and 22%. Other main constituents include lipids, Vitamins A, D, E and K and minerals particularly calcium, magnesium, iron, copper and phosphorus.

22.3.1 Storage and Processing of Fish and Fish Products

Fish undergo decomposition and spoilage readily and rapidly even at temperatures close to 0°C. Hence they are cooled or frozen, or dried, salted and smoked followed by pickling in vinegar or in gelatin with vinegar added.

Refrigeration or freezing of fish preserves freshness, wholesomeness and the nutritive value. Fish are packed in ice on the ship itself almost immediately after the catch. Fish such as tuna, mackerel, cod, haddock, red fish, etc. may also be frozen on ships or on land either as whole fish, or gutted, with or without head. Quick freezing to about –30 to –40°C is adopted to quickly pass over the critical temperature range of –0.5 to –5°C. Liquid nitrogen or dichlorodifluoromethane is used as coolant. The whole fish are sprinkled with water so as to form a glaze of ice to protect from oxidation. Alternatively, fish may be frozen in an alginate jelly or coated with latex film. Glass jars, plastic films or wax impregnated cartons are used for packaging fillets. Fillet sides are strips of flesh cut parallel to the central bone, from which main bones, fins, belly flap and skin removed. The fillets may be sprinkled with egg or bread crumbs. The residues from hygienically cut fillets are processed into fish balls.

Refrigerated and frozen fish must be consumed soon after thawing, otherwise juices drip out and meat decays rapidly. Since fish muscle enzymes retain some activity even at –10°C prolonged storage or insufficient cooling leads to rapid deterioration.

Drying using solar radiation or driers is adopted for non-fatty fish such as cod, haddock, ling, tuck or saithe. The head is removed and the

fish is split, gutted and dried. Alternatively, machine-cut, headless belly clipped fish are salted and dried.

Salting is one of the oldest methods of preservation. Salting is carried out by rubbing, sprinkling or steeping in brine the whole fresh or frozen fish or parts of fish. The salt content may be about 12–20% of the tissue. Salting provides a finished or semi-finished product amenable for further processing.

Smoking is carried out with fresh or frozen fish by exposing the same to freshly generated sawdust smoke. Cold smoking at 20–25°C over 2–4 days of salted fish or hot smoking at 100–120°C for 2–4 hours of whole, gutted or descaled fish with limited shelf life of about 3–4 days is quite common.

Marinated fish of limited shelf life are marketed in cans or jars and is carried out by soaking or steeping spiced and salted fish in marinade which consists of vinegar, wine or a mixture of both. Fried and cooked fish and fish products are also marketed in cans. Fish with extended shelf life up to one year without cold storage, are made by steam retorting of fresh or frozen fish or fish portions followed by packaging in vacuum sealed air-tight containers. Lacquer coated tin cans or inert aluminum cans are used when fish is canned with tomato or mustard sauce, vinegar or lemon juice.

Whales and crustaceans. Whale (a sea mammal) meat is similar to that of game meat or beef with long coarse muscle fibres. The fresh meat has a gray-reddish colour and a pleasant flavour but deteriorates rapidly due to fat oxidation. Frozen whale meat is dark blackish brown in colour with a rough, firm texture. Whale meat has a very short shelf life.

Crustaceans have no backbone but the body which is divided into sections is protected by a hard shell. Crustaceans include shrimp, cray or crab fish, crabs and lobster. Shrimps are marketed live with shell or fresh with or without head and shell. Canned shrimps are pasteurized at 80–90°C and have a short shelf life. Crabs and lobsters are usually marketed live or frozen fresh and canned. Crab meat, crab paste, lobster meat or paste are other processed products.

Molluscs. These include clams, oysters, mussels and scallops. The best meat is obtained from a 3–5 year old oyster. Mussel is consumed as cooked meat, fried or marinated. The meat is rich is protein (17%) and in Vitamins A and B.

Turtles and Frogdrums. Turtle or tortoise is consumed mostly as a soup or stew. Frogdrums (thigh portion of frogs) have a soft texture and white colour, are eaten as cooked, roasted or stewed.

22.4 EGGS

Hen's egg consists of three main parts, the shell, the egg white and the egg yolk. The shell consists of calcite crystals embedded in a matrix of

proteins and polysaccharide complex. Inside the shell the viscous colourless liquid called the egg white accounts for about 60% of the total egg weight. It is an aqueous solution of albumen of which the main protein is ovalbumin (55%), and a phosphoglycoprotein composed of three components A_1, A_2 and A_3. Ovalbumin is readily denatured by shaking or whipping its aqueous solution but is resistant to thermal denaturation. Other egg white proteins include conalbumin (13%), ovomucoid (10%), ovomucin, lysozyme (ovoglobulin G_1), ovoglobulins G_2 and G_3, avidin, flavoprotein and ovoinhibitor. Other constituents of egg white include carbohydrates (1%), minerals (0.6%—mainly sulphur, sodium, potassium, phosphorus and calcium) and minor quantities of lipids and vitamins. The third component of egg is the egg yolk with a solid content of 50%. It is a fat-in-water emulsion containing lipids (about two-thirds of dry matter content) and proteins (about one-third). Egg yolk separates into granules and a clear supernatant plasma on high speed centrifugation. The granules contain egg yolk proteins, lipovitellins (high density lipoproteins), phosvitin (a phosphoprotein), livetin and low density lipoproteins. The egg yolk lipids include triacylglycerols, phospholipids, cholesterol and cholesterol esters. Carbohydrate content in egg yolk is about 1%. The egg yolk is rich in vitamins panthothenic acid, A, riboflavin, thiamine, B_6, tocopherols and folic acid. The yellow colour of egg yolk is due to carotenoids and xanthophylls.

Eggs have high nutritional value. Moreover, eggs may be used as thickening agents, binding and coagulating agents, coatings, foaming agents, emulsifiers, shortening agents, flavouring agent and colourant in a variety of food products.

Eggs stored at room temperature lose their prime quality in about a week. Refrigerated storage enhances the storage stability to about two weeks. Eggs may be stored for about 6 months in cold storage at –1°C and relative humidity of 80%. Eggs may also be stored by spraying cleaned eggs with mineral oil to close the pores in the shell, thereby retarding the loss of moisture and rise in pH due to loss of carbon dioxide.

22.4.1 Egg Products

Egg products include dried eggs, frozen egg products and liquid egg products. *Dried egg powder* is used in the preparation of egg based products. The whole liquid content of the eggs or the separated egg white and egg yolk is homogenized by mixing or churning and then pasteurized. Sugars present in the liquid egg are removed by microbial fermentation (*Streptococcus sp, Aerobacter* or yeasts) or by glucoseoxidase/catalase enzymes (to oxidize glucose to gluconic acid) to prevent reaction with amino components resulting in undesirable brown discolouration and faulty aroma. The liquid egg is then spray dried with jet or centrifugal spray drier and rapidly cooled. The shelf life of dried egg white is

unlimited while whole egg powder devoid of sugar is stable up to one year at room temperature. Sugarless yolk powder has a shelf life of 8 months at room temperature. *Frozen egg powder* is prepared by homogenizing, pasteurizing whole liquid or separated egg white and egg yolk and then frozen to –23 to –25°C. It has a shelf life of about 8 months at storage temperature of –18°C. *Liquid egg product* is obtained by homogenizing the liquid and pasteurization at 65°C for about 3 minutes. Pasteurization alone however, cannot protect the product from microbial attack, particularly from *Salmonella sp*. Hence preservatives such as sorbic acid or benzoic acid are added.

Questions

1. What is rigor mortis? How does it affect the quality of meat?
2. What is tenderization of meat? How is it carried out?
3. Write a note on the different types of meat.
4. Give an account of the methods used for preserving meat.
5. How is poultry processed?
6. What are sea foods? Give examples.
7. How is fish preserved?
8. How are egg products obtained?
9. Write a note on the different processes of cooking meat.

CHAPTER

23

Fats and Oils

23.1 INTRODUCTION

Fats and oils may be of vegetable, animal or marine origin. The term 'fat' designates a solid at room temperature, while oil is a liquid at the same temperature. Examples of solid fats include butter, peanut butter, cocoa butter, lard and tallow and those of liquid oils include corn oil, cotton seed oil, olive oil, pea nut oil, sesame seed oil, sunflower seed oil, soybean oil, cod liver oil and whale oil. Edible beef fat (tallow) is obtained from bovine adipose tissue covering the abdominal cavity and surrounding the kidney and heart and from other compact undamaged fat tissues. Sheep tallow is harder and more brittle or fryable than beef tallow. Lard is hog (swine) fat obtained from the fat tissue covering the belly and other parts of the body. The back fat is mainly utilized for manufacturing bacon. Sea mammals such as whales, seals and herring fish are sources of marine oils. Some of these fats and oils have special flavour attributes, e.g. lard contributes a meaty flavour while olive oil has a distinctive flavour on salads and butter fat has a desirable buttery aroma and flavour.

23.1.1 Functional Properties

Fats and oils are important because of their nutritional and flavour characteristics. In addition, fats are used as shortenings, tenderizers, lubricants, frying media, emulsions, whipping agents and for other special requirements in food processing. For instance, butter and table margarine are used because of their plastic texture which does not become too hard on refrigeration or too soft in summer that runs off as a liquid. Similarly, salad oils should be clear, easy to pour and should not become cloudy or crystallize on refrigeration.

23.2 PRODUCTION OF FATS AND OILS

The fats and oils are obtained from animal tissues by rendering while oils from plant sources are obtained by pressing or solvent extraction.

Rendering. Animal fats of commercial importance include tallow from beef fat and lard from hogs. These are obtained by the process of rendering. The process involves heating meat scraps to release the fat from adipose tissue. The fat expands with heating, tearing the adipose tissue cell membrane and flows freely. Rendering may be carried out in dry condition or as wet rendering. In dry rendering, dry heat is used to cook the tissue under vacuum to remove moisture while in wet rendering, water or steam is used. Low temperature rendering produces a fat of lighter colour. Higher temperature is used when fat of more meaty flavour is desired. Rendering is also used to recover oil from whale blubber, seals and fish (herring, sardines, etc.) tissue.

Edible beef fat is light yellow in colour due to carotenoids derived from animal feed. Prime beef fat is obtained by melting the fresh and selected beef fat trimmings in water at 50–55°C. This fat on heating 30–34°C yields two fractions—the liquid oleomargarine and the solid oleostearine. Oleomargarine, is a soft fat with a consistency similar to that of melted butter and is used by the margarine and baking industries. Oleostearine, also called pressed tallow, has a high melting point (50–56°C) is used in the production of shortenings. Lower quality tallow is used as raw material for soap and detergent industries.

Pressing or expelling. Oils of plant origin include those of fruits and seeds. All edible oils are of plant origin and consumed as pure oil from a single fruit plant or oilseed plant. The two fruits having economic importance in oil production are those of olive and palm. Seed oils include those of soybean oil, sunflower seed oil, etc. Olive oil is obtained by kneading the disintegrated fruit occasionally by adding common salt. The oil is then pressed out or separated by gravity decantation. Palm fruits provide two different oils: one from the pulp and the second from the seeds. The fruits are steam treated to inactivate the high lipase activity and to separate the pulp from the seed. The oil is recovered by pressing the pulp. Seed oils are also released by pressing or expelling, using mechanical presses or expellers to squeeze out the oil from seeds. Expellers are continuous screw type presses. Seeds are usually first cooked slightly to partially breakdown the cell structure and to melt the fat for easier release of oil. The seeds may also be ground or cracked to facilitate the release of oil. The heat from cooking or grinding the seeds should not be excessive or it may darken the colour of the oil.

Solvent extraction. The ground seeds are rolled into thin flakes by passing them between smooth steel rollers so as to enhance the surface area for efficient solvent extraction. Non-toxic fat solvent petroleum ether,

i.e. technical hexane which contains 2- and 3-methyl pentane and 2,3-dimethylbutane and is free from aromatic compounds is used for extraction of oil. The solvent percolates through the seeds and after the oil is exhausted, the solvent is distilled off to recover it from the oil-solvent mixture called miscella. The maximum amount of solvent remaining in the oil is 0.1%. The oil-free flakes are steamed to remove the solvent (desolventizing) and sold as protein rich feed meal for cattle.

The crude oil obtained by rendering, expelling or solvent extraction is not suitable for immediate consumption because it contains polar lipids, free fatty acids, waxes, pigments, sulphur compounds, phenolic compounds, trace metal ions, autooxidation and degradation products and contaminants. Hence it has to be refined.

23.3 REFINING

The edible oil refining process involves the following steps: (i) vegetable lecithin removal, (ii) degumming. (iii) free fatty acid removal (deacidification), (iv) bleaching and (v) deodourization.

(i) *Removal of vegetable lecithin.* This is an essential step particularly in rapeseed and soybean oil processing. Water, about 2–5% is added to crude oil and the emulsion formed contains the phospholipids including lecithin. The emulsion is separated or clarified by centrifugation and lecithin is recovered from aqueous layer by evaporation of water in vacuum.

(ii) *Degumming.* To remove protein, residual phospholipids and carbohydrates, degumming is done by the addition of phosphoric acid (0.1%) to the oil. A filtering aid is added to the oil and the oil is filtered to remove the coagulated matter.

(iii) *Deacidification.* Removal of free fatty acid is carried out by alkali refining in which the oil is treated with a 15% solution of sodium hydroxide to form sodium salts of fatty acids (sodium soap). The sodium soap is removed by hot water washing and the oil is then vacuum dried. The free fatty acid content in deacidified oil is about 20 to 60 ppm. Deacidification of oils which contain low content of unsaturated lipids such as palm or coconut oils, may be carried out by steam distillation (physical refining) at low pressures of about 1–10 mbar and at temperatures of 210–270°C. Physical refining destroys carotenoids due to thermal bleaching as in palm oil. Physical refining is often combined with an oil deodourization step. Deacidification may also be carried out by a solvent extraction. Ethanol extracts free fatty acids and this procedure is preferable with oils containing high amounts of free fatty acids. Furfural extracts polyunsaturated triacylglycerols, while propane under pressure preferentially extracts saturated triacylglycerols leaving behind the unsaturated ones and unsaponifiable matter. Pressurized propane is utilized in marine oil fractionation to recover Vitamin A concentrates.

(iv) *Bleaching.* The plant pigments of chlorophyll, carotenoids and autooxidation products in the oils are removed by stirring the heated oil with activated carbon or adsorbent clays for about 30 minutes in vacuum. The bleached oil is removed from the adsorbent by filtration. The oil retained by the adsorbent is recovered by hexane extraction and recycled into the refining process. The residual alkali soaps, gums, unsaponifiable matter and heavy metal are also removed during the bleaching process. Animal fats are bleached by heat alone.

(v) *Deodourization.* Natural oils and fats contain volatile odorous compounds. Some of these are desirable as in the case of olive oil, cocoa butter, lard and fresh butter fat. But other oils such as fish oil and several seed oils have disagreeable odour. The undesirable compounds are removed by vacuum steam distillation at 190–210°C at 0.5–10 mbar. Deodourization requires anywhere between 20 minutes and 6 hours depending on the type of fat or oil. This refining step also removes the highly desirable aroma compounds which are important for the individuality of cold pressed oils such as olive oil. After refining, the oils may be packed into containers for bulk or retail marketing.

23.4 HYDROGENATION

The greater demand for solid or semi-solid fats at room temperature is satisfied by a process called fat hardening, which is total or partial hydrogenation using nickel metal supported on kieselguhr or alumina as catalyst. Hydrogenation of oil is carried out in an autoclave at a temperature of 150–250°C and a hydrogen pressure of about 1–3 bar. After hydrogenation the fat is filtered, deacidified, bleached and deodourized.

The unsaturated triglycerides are fully hydrogenated for producing high melting fats for cooking, frying and baking fats or partially hydrogenated to produce fats for different end uses. Oils rich in fatty acids with one double bond are more stable against autooxidation and have a longer shelf life similar to olive oil and find use as salad oil or for the manufacture of shortenings. Linolenic acid in soybean oil is hydrogenated selectively without affecting linoleic acid to increase its stability against autooxidation. Hydrogenation is also carried out to give fats melting around 30°C with a plastic and spreadable consistency.

23.5 INTER-ESTERIFICATION

Natural oils and fats are subjected to inter-esterification to rearrange or randomize the fatty acid residues in the triglycerides to produce fats and oils with new properties. Such tailored fats have melting characteristics and consistency desirable for specific end products. Sodium methylate is

used as the catalyst. The dried fat or oil is stirred with the catalyst (0.1–0.3% by weight) at 80–100°C to carry out the inter-esterification. When the reaction is complete, the catalyst is destroyed by adding water and removed by repeated washing with water. The inter-esterified oil is bleached and deodourized.

Inter-esterification of partially hydrogenated palm oil gives a fat with better baking characteristics such as volume and softness of the baked items. Different varieties of margarines are produced by inter-esterification of different oils. For example, vegetable margarine 30% w/w of diunsaturated C-18 fatty acid is produced by inter-esterification of partially hydrogenated sunflower oil blended with its natural liquid oil.

23.6 FRACTIONATION

The demand for special fats with standardized properties is met by fractional crystallization of the triglycerides with different melting points or by hexane solubilization of fractions. In winterization of rapeseed (Canola), cottonseed, or sunflower oil, small amounts of high melting triglycerides and waxes are removed by slow cooling to yield salad oils which would otherwise cause turbidity during cold seasons or refrigeration.

23.7 COMMERCIAL FAT PRODUCTS

23.7.1 Margarine

Margarine was originally produced from beef fat to imitate the costly dairy butter. Now it is manufactured using different natural oils and blended oils. The properties of margarine including nutritional value, spreadability, plasticity, shelf life and melting properties resemble those of butter and depend on the fat ingredients. Margarine is an water-in-oil, (W/O), emulsion containing about 80% fat (about 40% in diet margarine) and 18% water. The other constituents include a mixture of mono- and di-glycerides and lecithin to stabilize the emulsion, skim milk to achieve a protein content of 1–2%, citric or lactic acid in the aqueous phase for flavour and also to function as antimicrobial agents and sequestrants of heavy metal ions. Aroma compounds such as diacetyl, butyric acid or lactones may be added. Sodium chloride (0.2%) is used to round off the flavour. Margarine is coloured by the addition of β-carotene or unbleached palm oil and may be enriched with Vitamins A and D_2.

Margarine manufacture involves three steps: (i) emulsification of water in the continuous oil phase, (ii) chilling and mechanical handling of the emulsion and (iii) crystallization to preserve the type of W/O emulsion by efficient removal of the released heat of crystallization.

23.7.2 Mayonnaise

Mayonnaise is a semisolid oil-in-water (O/W) emulsion of edible vegetable oil (50–85%), egg yolk or whole egg (5–10%), vinegar, salt, glucose and seasonings. The emulsion is stabilized by egg yolk phospholipids.

Questions

1. Describe the processes of rendering, expelling and solvent extraction with reference to extraction of oils.
2. Describe the steps involved in the refining of edible oil.
3. Write a note on the commercial fat products used in food industry.
4. What is the significance of hydrogenation and inter-esterification of oils? How are they carried out?

CHAPTER

24

Beverages

Beverages are consumed mainly for their thirst quenching properties or for their stimulating effects. Three groups of beverages are commonly consumed. These include: (i) carbonated non-alcoholic beverages or soft drinks, (ii) non-carbonated non-alcoholic stimulating beverages such as coffee and tea and (iii) alcoholic beverages.

24.1 CARBONATED NON-ALCOHOLIC BEVERAGES

The most popular soft drinks include those based on cola (extract from the cola tree), orange, root beer, ginger, lemon and lime. Soft drinks may be classified into three types as carbonated, fruit flavoured and sparkling soda water. The carbonated beverages, may contain artificial flavour or natural fruit juice.

The major ingredients of soft drinks include water, sugar and/or sweeteners, flavour emulsions and cloudifiers, colouring agents, preservatives and acids and carbon dioxide. Water constitutes the largest component to the extent of 92–93%. It must be free from suspended matter, colouring matter, objectionable odours and minerals which may interfere with the flavour and colour of the soft drink. Sucrose is the most commonly used sweetener. It contributes to the calorific value of the drink. High fructose corn syrup (HFCS) is a relatively cheaper sweetener with lower calorie content. Fruit extracts, blends of natural fruits and synthetic flavours may be used for flavouring the beverage. Flavouring agents are generally used in the form of emulsions (in vegetable gums). Caramel and synthetic colouring matter are used as ingredients to get desired colours. Acids such as citric acid (adopts well to nearly all light and fruity flavours), lactic acid, malic acid, tartaric acid (for grape flavours), phosphoric acid in soft drinks with leaf, root, nut or herbal flavours are used as mild preservatives and also to enhance the flavour and taste. Sodium benzoate is the most commonly used preservative. Pure

carbon dioxide gas enhances the flavour, gives a sparkle and extends the life of the drink.

Carbonated soft drinks are packaged in glass or polyethylene terephthalate (PET) bottles or cans with pull-ring top.

24.2 STIMULATING BEVERAGES

Both coffee and tea have no food value but are consumed entirely for their refreshing and stimulating beverage characteristics due to the presence of caffeine to the extent of about 1% in coffee and about 4% in tea.

24.2.1 Coffee

Coffee is an evergreen shrub or small tree native to tropical regions of central Africa and Asia. Three species of coffee are of commercial importance, *Coffea arabica* yielding best quality of coffee beans, *Coffea robusta* and *Coffea liberica* yielding lower quality beans. The coffee flowers are white and sweet smelling, which develop into green fruits or berries, turning red on ripening. The ripe fruits are referred to as cherries. Each cherry usually contains two greenish grey seeds or beans at the centre. Sometimes a single pea shaped bean exists and hence the name peaberry. Each seed is covered by a thin membrane and a silver skin both of which are covered with a parchment like membrane (hull) which is further surrounded by mucilaginous pulp. The ripe cherries are picked and both pulp and hull are removed before the coffee beans are roasted for brewing the decoction. The steps involved include: (i) coffee processing, (ii) roasting, (iii) grinding and (iv) brewing the decoction.

(i) *Coffee processing.* The processing step involves removal of the skin, pulp, parchment and silver skin. Either of the two methods, viz. dry method or wet method is employed. In the dry method of curing, the cherries are hulled and air-dried in the sun or artificially. In the wet or washed-coffee method, the cherries after removal of the outer skin are soaked in water and passed through pulping machines to separate the pulp from the rest of the bean. Pulping leaves behind a mucilaginous coating of the beans. The mucilage is removed by spontaneous natural fermentation primarily by pectinolytic bacteria, mostly coliforms, along with pectinolytic bacilli and fungi or by added pectin digesting enzymes. Sometimes an acid fermentation by lactic acid bacteria may follow the natural fermentation. After the pulp and its residues have been washed away, the beans are dried to reduce the moisture content to about 10–12%. Drying must be uniform because during the process of drying, the colour and flavour attributes are modified. Overdrying or fluctuations in temperature will affect the quality of coffee bean. The dried beans are

hulled to remove the thin layer of parchment yielding bluish-green coffee bean (green coffee). The green coffee beans are sorted by colour and defects and then graded for size, colour and by cup test to determine their potential brewing quality. Each variety of coffee has its own characteristic flavour. Commercially, different varieties of green coffee beans are blended to give the desired flavour, aroma, colour and strength or body of the beverage from the roasted beans.

(ii) *Roasting.* The green coffee beans have no flavour or aroma and have an unpleasant taste. All the desirable attributes of the beverage such as flavour, aroma, colour and taste are due to proper roasting of the green beans. Roasting is typically carried out at about 260°C for about 5 minutes when the bean temperature reaches about 200°C. Physical and chemical changes occur during roasting. The beans swell in size, become brittle and change colour to deep brown. Moisture and a small percentage of volatiles are lost and carbon dioxide is produced in relatively large amounts, part of it escaping while the rest is absorbed by the roasted bean. Carbohydrates decompose, caramelize and in combination with other constituents contribute to the flavour and aroma. Fatty substances and proteins are degraded. However, caffeine remains unaffected during roasting. Freshly roasted coffee has the best flavour and aroma, which deteriorate on standing. Similarly, coffee exposed to air undergoes oxidative changes and becomes stale. The staling of coffee is prevented by the presence of carbon dioxide in the roasted bean.

(iii) *Grinding.* The roasted beans are cooled and ground to a size depending upon its intended end use such as home brewing, vacuum extraction, percolator or vending machine. In each case the particle size and size distribution affect the brewing time, turbidity in the extract and other characteristics. Ground coffee loses its flavour and aroma quickly and hence freshly ground coffee is used for brewing or the ground coffee may be packed in hermetically sealed cans or jars under vacuum or inert gas.

(iv) *Brewing.* Freshly ground coffee is contacted with hot water in suitable vessels to extract the soluble caffeine and flavouring materials. Vacuum coffee is made in a two part container. The upper compartment has the ground coffee and contains an open tube that extends to the bottom compartment containing water. The water in the lower compartment, on heating, exerts sufficient pressure to force the hot water through the open tube into the upper compartment and contacts the ground coffee. When the heat is reduced in the lower compartment the coffee extract is sucked into the bottom compartment. Drip coffee is made in coffee filter—a two compartment container, the perforated upper compartment holding the ground coffee. Boiling water is poured which percolates down through a bed of ground coffee and the brew is collected in the lower compartment. The strength and flavour of the beverage

depends on ratio of coffee to water, particle size, temperature of water, time and any mixing action. An optimum degree of extraction is necessary to get the best flavour and aroma. In vending machines, the ground coffee powder for individual cup amounts are packed into water permeable pouches and on demand brewing is carried out for a cup at a time in a few seconds and the brew is dispensed into a cup.

Instant coffee is made by spray drying or freeze drying the strong coffee brew into a powdered or granulated water-soluble solid form. Instant coffee has a flavour similar to freshly brewed coffee but lacks the aroma.

24.2.2 Tea

The leaves and the buds of the tea shrub *Camellia sinensis* are used in the manufacture of tea. The shrubs are regularly pruned to obtain maximum leaf production. The slow-growing smaller shrub with smaller leaves yield 'China' tea while the 'Indian' or 'Assam' tea shrub is a fast growing one with large drooping leaves. The tea leaves are plucked by hand 5 or 6 times during the season lasting 8–9 months at intervals of about one week. The terminal white haired bud and two terminal leaves from the end of each shoot are plucked for producing high quality tea. Plucking of longer shoots containing three or even four to six leaves is also common, though the quality decreases with longer shoots. Processing of the plucked leaves and the buds yields the three commercially available varieties of tea, viz. black tea, green tea and oolong tea.

Black tea is the most popular variety. Processing steps include withering, rolling, fermentation, drying or firing, grading and packing. The plucked leaves and buds are spread thin and withered in trays or drying racks in drying rooms or drum dried. During this withering step, which lasts over 4 to 18 hours, partial drying of leaves occurs reducing the moisture content of the fresh leaves from about 75% to about 60–65% so that the leaves become flaccid and are ready for the next stage of processing. During withering, proteins are hydrolyzed to amino acids, a part of which is transaminated to keto acids. These acids are the precursors for aroma substances.

The next step after withering is rolling. In the rolling step the withered leaves and buds are conditioned in rollers to break open the cells and release the juices and enzymes. During this conditioning, a uniform distribution of polyphenol oxidase in leaves is achieved and the substrate and enzymes are brought together. A variety of rolling techniques are used which determine to some extent the flavour characteristics of the different tea varieties. In the final rolling step, the tea leaves are macerated by crank rollers under pressure. The rolling step may be regarded as the first stage in fermentation. After rolling the leaves are spread out thinly in layers of 5–7 cm thick and fermentation occurs over about 1–4 hours at

35–40°C. During fermentation the enzymes, particularly polyphenol oxidase, bring about the oxidation of the polyphenolic compounds present in the juices, resulting in a change of colour from green to bright coppery red and development of an odour similar to that of sour apples. The formation of pigments and aroma substances occurs during this step. In addition, oxidation of amino acids, carotenoids, and unsaturated fatty acids also occurs. The enzymatic oxidation of flavanols via the corresponding o-quinones provides condensation reactions leading to formation of theaflavins, epitheaflavic acids, benzotropolone derivatives and thearubigins, a group of compounds responsible for the characteristic reddish-yellow colour and astringent taste of black tea extract. Aroma development during fermentation is accompanied by an increase in the volatile components content due to Strecker degradation reactions of amino acids with oxidized flavonols. Important tea aroma constituents such as dihydroactinidiolide, theaspirone, linalool and others such as β-ionone and its derivatives are formed. In addition, biosynthesis of aroma compounds such as nerolidol also occurs. Firing or heating the fermented leaves in ovens at 85–95°C for about 20 minutes stops the fermentation process. The firing step reduces the moisture content to 3%, fixes the tea aroma and changes the colour to black (hence black tea). The steps of rolling and fermentation may be carried out continuously in CTC machines (cutting, tearing and curling) reducing the total time to about 1–2 hours. During the firing step, the enzyme activity initially rises and about 15% of the theaflavins are formed during initial stages and finally the enzymes are inactivated. The tea becomes black due to the conversion of chlorophyll to pheophytin at high temperatures in acidic environment. The undesired brown colour develops at higher pH values. The astringent character of tea decreases due to the formation of complexes between phenolic compounds and proteins. The firing step also influences the balance of volatiles, with a loss of some of the volatile compounds while simultaneous build-up of other aroma constituents such as β-ionone, dihydroactinidiolide and theaspirone occurs. In addition, pyrazines, pyridines and quinolines are formed due to sugar-amino acid interactions. The various constituents in black tea include oxidized polyphenolic compounds (~25%), proteins (15%), caffeine (4%), fibre (25%), other carbohydrates (7%), lipids (7%), amino acids (4%) and minerals (5%).

The dried tea is cleaned and graded into various commercial varieties according to the quality. The desirable quality of black tea is related to the polyphenol and enzyme content of the first two leaves and buds used in processing, which are maximum at about 27–28%. The 'leaf grade' tea is of highest quality followed by 'broken grade' which contains smaller and cut leaves. The leaf and broken grades are further categorized into orange pekoe, pekoe and souchang. The first two categories refer to the size of the leaf only, the orange pekoe having the largest size is the best quality tea. Souchang tea quality is the lowest. Apart from these, lesser grades comprising of waste products as fannings and dustings are also marketed.

Tea is generally blended and packed into aluminum foil lined wooden boxes or paper cartons of smaller sizes.

Green tea is made similar to the black tea without involving the withering and fermentation steps. The tea leaves are heat treated by pan firing or steaming to inactivate the enzymes, rolled and dried. Green tea is a light yellow-green beverage with a distinct flvour and aroma.

Oolong tea is an intermediate between black and green tea in colour and taste, and is obtained by partial withering and light fermentation steps before drying the leaf.

24.2.3 Cocoa and Chocolate

Cocoa is consumed as a suspension in water and in large amounts to get the stimulating effect in contrast to a clear extract as in the case of coffee and tea. The primary stimulants in cocoa products are the alkaloids, particularly theobromine. Cocoa products provide nutrients such as carbohydrates, fats and proteins.

Cocoa beans are the seeds of the tropical cacao tree *Theobroma cacao*, originating in the northern part of South America. The tree flourishes in warm (24–28°C) and moist conditions. The tree is sensitive to sunshine and wind and hence is often planted and cultivated under shade trees (cacao mothers). The tree can grow up to 10–15 metres but in plantations it is pruned to about 2–4 metres. The tree blooms throughout the year with small red or white flowers giving about 20–50 ripe fruits per tree or about 0.5–2.0 kg of fermented dried beans. The ripe fruit or pod contains the sweet pulp (glucose and fructose content is ~10%) surrounding 20–40 almond shaped seeds or cacao beans each weighing about one gram. The ripe fruit is harvested throughout the year but mostly twice a year. The summer harvest is larger and of higher quality. The two major groups of *Theobroma cacao* are the Criollo tree which is sensitive to climatic changes and to attack by diseases and pests while the Forastero tree is more resistant. The flavour characteristics of Forastero beans are relatively less though the harvest from the tree is higher. Other varieties of commercial importance include the Calabacillo and Ecuadorian Amelonado. Cacao beans are classified based on their geographical origin (flavour beans come from Ecuador, Venezuela, Trinidad, Sri Lanka and Indonesia, while commercial beans come from Ghana, Nigeria, Cameroon, Ivory Coast, Brazil and Dominican Republic), grade of cleanliness and the number of steps they are subjected to for processing. The various processing steps include harvesting, fermentation, drying, roasting, winnowing, alkalization, butter removal and grinding to get cocoa powder. The cocoa butter is used in the manufacture of chocolates and confectionery while the cocoa powder after the removal of cocoa butter is used for preparing the cocoa beverage.

Cocoa processing. The fully ripe pods are carefully cut from the trees, cut open and the bean scooped out with the surrounding pulp or mucilage. Most of the beans are fermented (except Arriba and Machala varieties of South America) over a period of 2–8 days in heaps, baskets, perforated barrels or on the floor with occasional mixing to facilitate accessibility of air. The temperature of the beans increases to 45–50°C and pH decreases to acidic region and the beans lose their germinating capacity. Flavour and colour formation and partial conversion of astringent phenolic compounds occur during fermentation. Enzymatic decomposition of the pulp to a liquid (fermentation juice) also occurs, which drains off.

Well-fermented cocoa beans are uniformly dark brown in colour and are readily separated into their cotyledons while inadequately fermented or unripe beans are smooth in appearance and of low quality. During fermentation the pulp sugar is converted to alcohol, carbon dioxide and to a small amount of lactic acid by yeast. Proteases and glycosidases hydrolyze the proteins and polysaccharides in the pulp and pulp liquefies and drains away facilitating aeration. Acetic acid bacteria oxidize alcohol to acetic acid and the pH decreases from 6.5 to 4.5 and the temperature increases to 45–50°C. The seed cell walls become permeable and the living cacao seed is killed. Oxidation and condensation reactions of the phenolic compounds commences with the formation of brown or brown-violet colour due to phlobaphenes and the harsh astringent flavour mellows as soluble phenolic compounds are removed. Fermentation should be carried out carefully for the formation of cocoa aroma and to prevent the growth of detrimental microorganisms such as molds, butyric acid bacteria and putrefaction inducing bacteria.

Fermentation is terminated by drying to a final moisture content of 8%. After fermentation the beans may be washed and dried to a moisture content of 6–8%. The dried beans are cleaned and separated by size and roasted in a two-step process. Roasting at temperatures not exceeding 150°C, reduces the moisture content to about 2% and facilitates the removal of acetic acid, volatile esters and other undesirable aroma components. The aroma of the beans is enhanced, the colour deepens and the seed hardens and becomes brittle facilitating the easy removal of the shell. After roasting the beans are cooled immediately to prevent over-roasting. Roasting is carried out in batch or continuous roasters using a stream of hot air or directly through heated surfaces and lasts anywhere between 10–35 minutes.

The roasted beans are winnowed to remove the shells and germ-rootlets. During winnowing, the beans are lightly crushed to preserve the cotyledons or nibs and the shells in larger pieces and to avoid dust formation. The whole nibs of about 80% in yield are separated from the shells, seed coat and germ. The nibs are used for the manufacture of cocoa and chocolate. About 4% fine cocoa particles are also separated by winnowing. The cocoa shell is used for the recovery of theobromine, production of activated carbon, as an ingredient in cork manufacture and after extraction of fat as fertilizer or fuel.

The composition of the fermented and air dried cocoa nib includes fat to the extent of 54%, proteins (11%), starch and other carbohydrates (17%), theobromine (1.2%), caffeine (0.2%) polyhydroxyphenols (6%), moisture (5%) and organic acids, mainly acetic, citric and oxalic acids (1.5%). Cocoa aroma depends on harvesting, fermentation, drying and roasting steps. The fresh beans have the taste and odour of vinegar. The characteristic bitter and astringent taste and the residual sweet taste of fermented beans may be impaired due to improper processing or use of unripe or overripe fruits. More than 400 volatile compounds contribute to the aroma of the roasted beans. The sweet, flowery green aroma of cocoa has been identified with linalool and a series of aldehydes such isopentanal, phenylethanal and their condensation product, 5-methyl-2-phenylhex-2-enal. The roasted aroma is derived from pyrazines and pyridine derivatives and bitter taste is mainly due to purines, therobromine and caffeine.

After roasting and drying, the cocoa nibs are disintegrated and milled to rupture the cell walls and expose the cocoa butter. Crushing rolls or knife-hammer mills are used for disintegration followed by roller ball or disc attrition mills which are used for fine disintegration. The disintegration of cocoa particles produces a homogeneous paste of flowing cocoa mass called the cocoa liquor.

The cocoa mass is subjected alkalization process to neutralize the free acids partially and also to mellow the flavour, improve the colour, enhance the wettability and dispersability and enhance the suspension holding ability so that sediment formation is prevented in the cocoa drink. The alkalization process involves treating the roasted nibs with a dilute 2–2.5% alkali solution (magnesium oxide or hydroxide, sodium or potassium carbonate or hydroxide) at 75–100°C, neutralization, if necessary with tartaric acid and drying to moisture content to ~2% in vacuum dryer or by kneading the mass at a temperature of 100°C. The alkalization process yields a cocoa mass with a spongy and porous cell structure due to the swelling of starch. The alkalized cocoa mass contains butter (52–58%) and up to 5% ash and 7% of alkalized mass or liquor.

The alkalized cocoa mass/liquor is converted into cocoa powder by reducing cocoa butter content of mass from about 54% of nib weight by hydraulic or mechanical pressing of the mass at 400–500 bar and a temperature of 90–100°C. The expelled cocoa butter is passed through a filter press to remove any debris and then molded and cooled. The bulk of cocoa butter so produced is used in chocolate manufacture. The residual cocoa press cake after expelling the cocoa fat contains about 10–24% cocoa fat. It is disintegrated by a cook breaker (rollers with intermashing teeth), then ground in a peg mill and segregated into fines and coarse cocoa powder by an air sifter. The coarse particles are recycled and milled repeatedly into fines. Depending on the fat content the cocoa powder is classified into lightly defatted powder (20–22% residual fat), and extensively deffatted powder (butter content varying between 10 and 20%). The cocoa powder finds use in the manufacture of cake fillings, icings, pudding powders, ice creams and cocoa (chocolate) beverages.

24.3 ALCOHOLIC BEVERAGES

Three classes of alcoholic beverages such as malted beverages (beer and ale), fermented fruit juices (wine and cider) and distilled liquors (brandy, whisky, rum and vodka) are common.

24.3.1 Beer

Beer and ale are the principal malt beverages. Beer is made from barley malt to which hops (dried flowers of the hop plant) and cereal or malt adjuncts (starch or sugars obtained from corn, rice, wheat, soybean, potato, cassava or barley) are added and fermented. Beer is manufactured by the brewing process. The steps involved are: (i) malting, (ii) mashing to prepare wort (unfermented beer), (iii) boiling the wort with hops, (iv) fermentation, (v) maturation or aging and (vi) finishing.

(i) *Malting*. Malting is controlled germination of cereal grains (barley) to activate the enzymes of the resting grain in order to convert the starch to fermentable sugars. Barley for brewing should have low nitrogen content compared to barley used in distilleries. The barley grains are kiln dried to control the moisture content to about 10–14% and stored for about three weeks to allow secondary ripening. The stored grains are then soaked or 'steeped' in water at 10–16°C, germinated at 16–21°C for about five to seven days and kiln dried to arrest the enzyme activity without destroying it. Most of the sprout or germs are removed to yield the remaining mass called the malt. The malt is a rich source of amylases and proteases required for the brewing process.

(ii) *Mashing*. Mashing is carried out to initiate hydrolysis and solubilize the polysaccharides. The malt is crushed and mixed with water at temperatures in the range of 38–50°C. Starchy cereal or malt adjuncts are cooked at about 100°C under steam pressure and added to the malt mash. The addition raises the temperature of the mash to about 65–70°C at which saccharification (production of sugars from starch) occurs rapidly. The mashing conditions are regulated with respect to concentration of malt, malt adjuncts and temperature to vary the proportion of various carbohydrates. The temperature of the mash is then raised to 75°C to inactivate the enzymes. The mixture is then allowed to settle the insoluble solids which are filtered off to yield a clear liquid called wort.

(iii) *Brewing*. This involves boiling the wort with hops in a brew kettle. The various objectives of this step include concentration and sterilization of the wort, inactivation of enzymes, extraction of soluble substances from hops for imparting flavour, bitterness, stability and head retention to beer, caramelization of sugars to a slight extent, coagulation and precipitation of proteins and other insolubles such as tannins to

prevent haze formation of beer, and contribution of antiseptic substances (effective against gram positive bacteria) called *alpha resins* such as humulone, cohumulone and adhumulone.

(iv) *Fermentation.* A special beer yeast of bottom type is *Saccharomyces carlsbergensis* is used for inoculation or 'pitching' of the cooled wort. The wort temperature is maintained at 3–14°C and fermentation is continued for about eight to fourteen days. During fermentation, yeast converts sugars mainly into alcohol and carbon dioxide and small amounts of glycerol and acetic acid. Higher alcohols and acids are formed from proteins and fats. The acids and alcohols combine to yield esters contributing to the aroma. As the carbon dioxide evolution increases foaming increases and finally the bottom yeasts 'break' by flocculating and settling to the bottom of the tank. The mixture is filtered to yield 'green' beer.

(v) *Maturing.* The young or 'green' beer is chilled and stored or 'lagered' in vats at about 0°C for several weeks and even to months. During aging, proteins, yeast, resins and other undesirable substances precipitate out and the beer becomes clear and mellowed or matured. Esters and other compounds are produced to contribute to the taste and aroma. The body changes from harsh to smooth.

(vi) *Finishing.* The lager beer obtained by aging is carbonated to a carbon dioxide content of about 0.5% by means of gas collected during fermentation. The beer is then cooled, clarified or filtered and packaged into barrels or smaller bottles and cans. The beer for cans and bottles may be pasteurized at 60°C or filtered through membranes to remove all the remaining yeasts. The alcohol content in beer is about 3.8%.

Beer defects and diseases. Beer defects are certain undesirable characteristics such as turbidity due to unstable proteins, tannins and resins and development of off-flavours due to poor quality ingredients or contact with metals. Beer infections or diseases are primarily due to growth and activity of undesirable microorganisms during fermentation. Butyric acid fermentation by *Clostridium sp.* or lactic acid fermentation by lactics may result in off flavours. Yeasts grow in beer and cause turbidity or cloudiness and bitterness, e.g. *Saccharomyces pastorianus.* Bacteria of genera *Pediococcus, Lactobacillus, Flavobacterium* and *Acetobacter* cause beer diseases resulting in sourness, turbidity, ropiness and bad odour akin to hydrogen sulphide. However, most of the yeasts and bacteria are killed by boiling the wort and hops and maintaining asceptic conditions during subsequent stages of brewing.

Other beer types and related beverages. Variations in beer types are manufactured by varying the concentration of malt and hops, aging time, initial concentration of fermentable solids and temperature of fermentation to yield beers differing in alcohol content, colour and

residual fermentable carbohydrate content. For example, *malt liquor* has a higher alcohol content while *bock beer* is a dark beer with higher alcohol content than the normal beer. Low calorie beers are light in colour and contain no carbohydrates manufactured from prehydrolyzed wort using fungal glucoamylases and amylases, e.g. *Pilsener beer*. *Ale* is made using a top yeast, a strain of *Saccharomyces cerevisiae*.

Sake is a beer of Japanese origin with an alcohol content of 14–17%. *Aspergillus oryzae* is grown on soaked and steamed rice mash to yield a koji, a starter culture, rich in amylases and proteases. The koji is mixed with more rice mash and fermented for about 10 days. During fermentation several yeast species ferment sugars to alcohol. The mixture is filtered to yield the clear liquor, sake.

Sonti is a rice beer of Indian origin. The mold *Rhizopus sonti* and yeasts are active in the alcoholic fermentation.

Ginger beer is an acidic and mildly alcoholic beverage made by the fermentation of sugar solution flavoured with ginger.

24.3.2 Wine

Wine is principally fermented grape juice. It may also be produced by fermentation of juices of fruits such as apple (cider), peaches, apricots, plums, pears, cherries and berries and honey (mead or honey wine). Grape wine is either red or white depending on whether the skins of purple or red varieties of grapes have been fermented to impart the red colour of the pigment, or white grapes or expressed juices of other grapes without the skin have been fermented. The steps involved in the manufacture of grape wine include: (i) juice extraction, (ii) fermentation and (iii) aging.

(i) *Juice extraction*. Grapes, specially adapted to wine making, are harvested at a stage when the sugar content is about 15–25% depending on the variety and ripeness. The composition of grapes varies due to the climatic conditions prevailing during their cultivation and hence the quality of wine varies from one year to another. Thus the practice of vintage dating of wine is followed to indicate the year of the crop. The harvested grapes are stemmed and crushed to yield the expressed juice called 'must'. The must is then treated with sulphur dioxide at 75–200 ppm concentration or equivalent amount of potassium bisulphite to inhibit the growth of undesirable microorganisms.

(ii) *Fermentation*. The must is inoculated with a 'natural inoculum' of yeast present on the grapes or a special wine yeast strain of *Saccharomyces cerevisiae var. ellipsoideus*. The primary or active fermentation is initiated by stirring the mixture with aeration to facilitate the growth of yeast and extraction of the pigment from the skins. Alternately the red pigments may also be extracted separately and added to the must. Later mixing is stopped to facilitate anaerobic fermentation

to yield alcohol. The primary fermentation occurs over a period of 3–5 days at an optimum temperature of 24–27°C for red wines while for white wines the fermentation period is about 7–14 days at a temperature of 10–21°C. Heat is liberated during fermentation and hence cooling the must is necessary.

After the primary fermentation advances sufficiently the fermented juice is drawn off to separate it from the residue (pomace) and is allowed to undergo secondary fermentation in a tank under a light pressure of carbon dioxide. During the secondary fermentation which occurs over a period of 7–11 days at 21–29°C the remaining sugar is fermented to yield a dry wine. Clear wine is syphoned off from the sediment at the bottom of the tank and filled or'racked' into barrels for aging.

(iii) *Aging.* The wine is cooled for several days, filtered and trasferred into wooden (white oak or redwood) or plastic lined concrete tanks for aging. The tanks are filled completely and sealed to keep out air. The wine is periodically racked off from the bottom sediment. Final aging in bottles for long periods of months to years produces desirable changes in body and flavour of wine. The characteristic aroma and bouquet develop due to the various esters and alcohols. In addition, malic acid of grape juice is fermented by lactobacilli during aging to give lactic acid and carbon dioxide and also decrease the acidity. After the process of aging is complete, the wine is filtered or clarified, bottled and pasteurized. The final alcohol content varies between 8 and 13% by volume.

Types of wines. *Still* wines do not retain any carbon dioxide produced during fermentation whereas *sparkling* wines contain considerable amounts. Wines may also be carbonated artificially. *Dry* wines do not contain any fermentable sugars and *fortified* wines contain added wine spirits or brandy to increase the alcohol content to about 20% by volume compared to the normal 12–15% in dry wines. *Table* wines have a lower content of alcohol with little sugar while *dessert* wines are fortified sweet wines.

Wine defects and spoilage. Wine defects include turbidity, cloudiness, precipitation and loss of colour or colouration due to metals such as iron, tin and copper and their salts. Microorganisms, both aerobic and facultative, cause wine spoilage by imparting cloudiness, bitter characteristics, ropiness, undesirable flavours, high volatile acidity and low alcohol content. Wild yeasts, molds and bacteria of the genera *Acetobacter, Lactobacillus, Leuconostoc* and *Pediococcus* cause spoilage of wine.

24.3.3 Distilled Liquors

Distilled liquors or spirits are alcoholic beverages with high alcohol concentration obtained by distilling fermented sugar containing liquids.

Distilled liquors include whisky (Scottish and Canadian brands) and whiskey (American and Irish brands) made by fermentation of cereal grains followed by distillation of the liquor. Brandy is obtained by distilling wine and rum from fermented molasses. Other distilled liquors include gin, vodka and even pure alcohol. An essential stage in the manufacture of distilled spirits is distillation. After fermentation of the cereal grains for about 48 hours the fermented liquid containing about 6–10% ethanol is distilled off along with other volatile constituents. The crude alcohol obtained by first distillation contains acetaldehyde, methanol, higher alcohols, esters and furfural. The crude alcohol is diluted and rectified by repeated distillation to remove the fusel oil and yield almost pure alcohol of about 96% purity. This alcohol is used for making alcohol fortified beverages. Distillation is carried out in pot stills, or in continuous distillation stills in a modern distillery. Pot still distillation yields alcohol with many volatile components which contribute to the characteristic, subtle flavours of the finest varieties of whisky, brandy and other renowned spirits. Cognac, considered the best brandy, is restricted to brandy made in the region of Charente in France. It is made by double distillation (distilling twice) in pot stills.

In the manufacture of *whisky*, different cereal grains such as wheat, rye, oats, barley, corn and millet are used. The cereal grains are ground, mixed with acidified water and made into an uniform mash by starch gelatinization. The mash is then saccharified by mixing with malt and stirring at 56°C. The malt diastase enzymes are responsible for the saccharification of the starch. The enzymes are then inactivated by heating the mash to 62°C. The sweet mash is quickly cooled to 20–23°C and fermented by inoculating high alcohol yielding special yeast. The fermented liquid is distilled to yield about 30–35 litres of alcohol per 100 kg of the cereal grain. The flavour of the final product depends on the starting material, different processing steps used and maturing or aging. For example, Scotch whisky is made from barley malt which has been exposed to meat moss smoke during kiln drying. The smoked malt is mashed at 60°C and filtered to give a wort which is fermented at 20–32°C. The distillation of the fermented liquid is carried out in two steps in simple pot stills. The harsh, raw liquor from the first distillation is distilled again in the second step to remove the harsh components known as the head and tail fractions. The distillate with about 63% alcohol is stored/aged in wooden casks to develop aroma. The alcohol content is then reduced to drinkable level of about 40% and bottled. Scotch whisky is also made from grain (Scotch grain whisky). American whiskey is made from corn, rye or wheat by saccharification with malt enzymes, fermentation of the wort followed by double distillation. Irish whiskey is made from malted barley alone or with a mixture of unmalted barley, wheat, rye and oats.

Brandy is distilled wine which contains at least 38% by volume of alcohol. Fermented grape juice (must) is distilled in copper pot stills on an

open fire often without prior removal of yeast. The first distillate with a harsh unpleasant odour is refined by repeated distillations. The distillate is then aged in oak casks in damp storage yard for periods upto 20 years to obtain brandy. During aging, the wine distillate extracts phenolic compounds and colours of the wood thus acquiring the typical golden yellow or greenish yellow colour of brandy. Simultaneously, oxidation and esterification reactions occur mellowing and polishing the flavour and aroma of the distillate. To improve the quality, an essence prepared by wine distillate extraction of oak wood, plums, green walnuts or deshelled almonds is added. The brandy may be sweetened by the addition of sugar, burnt sugar or 1% dessert wine. The brandy is treated with clarifying and filtering agents and diluted with water to obtain the desired alcohol content in the final product.

Fruit brandy is made from cherry (Kirschwasser in Germany), plum (Chriesewasser in France and Switzerland) or raspberry. Whole ripe fruits are partly crushed together with or without seeds and pounded into a pulp. The pulp is left to ferment over several weeks, using a pure yeast culture. The fermented mash is distilled in a copper still over an open fire. Usually, the first and the last fractions of distillation are separated from the main distillate. The main distillate containing about 60% alcohol to diluted to about 40% and marketed as clear, colourless brandy.

Rum is produced from sugar syrup mixed with molasses and distiller's wash (residue left from a previous distillation) by spontaneous fermentation at 36°C followed by distillation using pot stills. Parts of aromatic plants may be added to the fermenting mash to increase its aroma. The distilled rum is aged in oak wood casks during which esters and other aroma constituents are formed. Jamaican rum is considered the best rum and marketed in different grades for drinking and blending.

Arrack is produced from rice, sugar cane molasses, sweet coconut or palm extract or sugar enriched plant juices by fermentation and subsequent distillation. Dates are used in arrack preparation in the Middleeast.

Vodka, which is a popular drink for the Russians, is made from grain alcohol with an alcohol content of at least 40%, and it has a characteristic smoothness and neutral flavour. *Gin* is obtained from alcohol by addition of distillate from the berries of *Juniper communis* and spices. It contains at least 38% alcohol.

Liquears (Cordials) are alcoholic beverages with 20–35% alcohol and contain sugar (200–500 g/l) or starch syrup and flavoured with fruit, spices, extracts or essences.

Questions

1. What are soft drinks? How are they made?
2. Give an account of the steps involved in coffee processing.

3. How is instant coffee manufactured?
4. How is black tea processed?
5. Write in detail the steps involved in chocolate manufacture?
6. How is cocoa processed to get cocoa butter?
7. How is beer manufactured?
8. Write briefly on beer defects and diseases.
9. What are beer related beverages? How are they made?
10. Describe in detail the steps involved in the conversion of grape juice into wine.
11. What are wine defects?
12. What are distilled liquors? How is brandy manufactured?

CHAPTER

25

Sugar, Sweeteners, Honey and Confectionery

25.1 SUGAR

Sucrose is available in a variety of plant sources. However, the two most important sources of sugar for commercial production include sugar cane (*Saccharum officinarum*) and sugar beet (*Beta vulgaris*). In India, sugar is manufactured from sugar cane in the form of jaggery (*gur*), open pan sugar (*khandasari*) and vacuum pan sugar (*white sugar*).

25.1.1 Sugar Obtained from Sugar Cane

Jaggery is obtained mostly from sugar cane and also from palmyra, date palm and coconut. The harvested sugar cane is crushed within a day to avoid loss of weight and loss of sucrose due to inversion. The sugar cane juice so obtained is freed from coarse suspended impurities and boiled in open pans. During heating, vegetable and chemical extracts are added to the juice to flocculate colloidal impurities. The juice is clarified, and boiled vigorously to 115–117°C with constant stirring to yield a concentrate semisolid mass which on cooling solidifies into jaggery. The clarified juice may be decolourized with sodium hydrosulphite or activated carbon before boiling. Jaggery is marketed in lumps of various sizes and shapes. Jaggery has a light colour, good flavour, hardness and crystalline structure with good keeping qualities. It contains about 65–85% sucrose, 10–15% invert sugar and 2.5% ash. Jaggery finds use in the preparation of non-crystalline candies and a variety of sweets.

Khandasari sugar is obtained by boiling the clarified sugar cane juice quickly to a required consistency to induce crystallization of sugar. The crystals are recovered by centrifugation and dried.

Sugar is manufactured from sugar cane juice in three different forms, raw sugar, refined sugar and white sugar. The sugar cane juice obtained by crushing the cane is dark green in colour and turbid. It is mildly acidic with a pH 5.0–5.4 and the sucrose content is 10–18%. The coarse suspended impurities are filtered off through fine screens. Milk of lime (calcium hydroxide) is added to the juice and heated to boiling to neutralize and precipitate the organic acids and phosphate as well as to flocculate colloidal impurities. Lime treatment makes the juice mildly alkaline (pH ~9) and thereby prevents inversion of sucrose. The hot juice is clarified to yield a yellow brown liquid with about 15% sugar content which is then evaporated in multiple-effect evaporators to a syrup containing 75–85% sugar. Raw sugar crystallizes out from the syrup held in single effect vacuum pans. The crystals are separated from the massecuite (mixture of crystals and mother liquor or molasses) by centrifugation. The crystals are light brown in colour with an adhering film of molasses and contain 96–97% sugar. Raw sugar is refined by dissolving it in water and decolourizing the solution with animal charcoal. Refined sugar crystallizes out in multiple-effect vacuum pans, which are separated from the syrup and dried in revolving drums or granulators. The size of the crystals is important and hence the crystals are screened into granulated sugars of different coarseness. In white sugar manufacture, the lime mixed cane juice is treated with sulphur dioxide (sulphitation) or carbon dioxide (carbonation) to remove the excess of calcium, and is then clarified to yield a light coloured syrup which on crystallization in multiple-effect vacuum pans yields white sugar crystals. The recovery of sugar is about 10–11% of weight of sugar cane.

25.1.2 Sugar Beet

Beet sugar is obtained from sugar beet. The beet is cleaned, sliced into thin shreds and extracted with hot water under mildly alkaline conditions. The thin extract is treated with lime, carbonated and quickly filtered to remove the precipitated impurities. The clear liquid is passed through ion-exchange resins to remove ionic impurities which retard the process of crystallization of sugar, and also to partially bleach the syrup. The syrup is then evaporated in multiple effect evaporators with temperatures controlled below 135°C to yield a thick syrup. Raw sugar crystals are obtained from the thick syrup. The raw sugar crystals are purified by affination process in which the crystals are mixed with the thick syrup to form a thick paste or magma. The magma is heated in a mixer to reduce its viscosity and centrifuged in basket centrifuges. The crystals, which remain in the basket, are spray washed with a small volume of water or

steamed to solubilize only the outer surface of the crystals. Such cleaned sugar is called *affinade* or affinated sugar. The affination syrup is recycled for further crystallization. The affinade undergoes further purification to give white sugar by the process of raffination. In this step the affinade is solubilized to a clear liquid and evaporated under reduced pressure to cystallize the raffinade or granulated sugar crystals of highest purity. Table sugar or granulated sugar is made from fine sized grains of *raffinade*, and cubes or tablets are made by forming a mixture of sugar crystals and white sugar syrup under pressure followed by drying. Powdered sugar for icing of confectionery, cakes, pastries and bakery products is made by pulverizing granulated sugar with or without the addition of edible starch. Starch addition prevents caking by absorbing the moisture. *Diamond sugar* is a decorative sugar in small rectangular crystals used in confectionery and beetle nuts.

25.2 SWEETENERS

These are manufactured mainly from any starch source such as wheat, maize or corn, and are liquefied in the presence of enzymes. The liquid then undergoes saccharification after it is cooled to about 60°C or so. The various types of sweeteners are discussed in detail in the following sections.

25.2.1 Starch Hydrolysates

Starch syrup (glucose or maltose syrup), dried starch syrup, glucose (dextrose) and high fructose syrup are some of the sweeteners derived from starch degradation. Starch saccharification is carried out by either acidic or enzymatic hydrolysis under controlled processing conditions to yield starch hydrolysates with different composition to suit the diversified requirements. The industrial uses include manufacture of soft caramel candies, fondants, chewing gum, ice cream, alcoholic beverages and soft drinks, canning and processing of fruits and in the baking industry. The progress of hydrolysis is measured by a Dextrose Equivalent (DE) in which the specific chemical reducing power of the mixture of monomers, oligomers and polymers is measured relative to that of pure glucose or dextrose; the latter is deemed to have a value of 100. The enzyme α-amylase from *Bacillus subtilis* or *B.licheniformis* is commonly used at pH 6.5 and temperatures of 70–100°C. Maltose syrup consisting mostly of maltose, maltotriose and small amounts of glucose is obtained by starch hydrolysis using amylase from *Aspergillus oryzae*. Dried starch syrup with a moisture content of 3–4%, dextran (50%), maltose (30%) and glucose (20%) is obtained by spray drying starch hydrolysates. It is readily soluble in water and diluted alcohol and is used in sausage production as a red colour enhancer.

25.2.2 High Fructose Corn Syrup

The commercial value of *high fructose corn syrup* (HFCS) is based on the increased sweetness of the HFCS compared to the starting material glucose obtained from starch. On a sweetness comparison basis, glucose is only about 70% as sweet as sucrose whereas fructose is more sweet compared to sucrose to the extent of 20–60% depending conditions used. The abundant supplies of cheap corn in USA, as well as efficient processing techniques and valuable by-products such as corn oil and protein provide the basis for the low cost manufacture of HFCS. Typical syrups made from starch contain a mixture of glucose, fructose and other saccharides. The equilibrium limitation of the enzymatic conversion of glucose to fructose, as well as sweetness and manufacturing cost restrictions, led to the development of fructose syrups containing 42% fructose on a dry basis to replace liquid sucrose in many food applications.

The first stage in HFCS production is the hydrolysis of polymeric corn starch containing about 74% amylopectin and 26% amylose to monomeric glucose. The DE value rises from almost zero for starch during hydrolysis and in commercial practice, the hydrolysis of starch is stopped between 42 and 63 DE to give products which have those combinations of osmolarity, and hygroscopicity which are suitable for a range of commercial applications.

In the later stages of HFCS production, large scale chromatographic techniques are used to produce syrups with various proportions of fructose and glucose. The degree of isomerization (DI) is the percentage of sugars, dry basis, by weight as fructose. Thus a 42 DI 'Regular' HFCS contains 42% fructose, 52% glucose and 6% higher sugars. It is common practice to incorporate the DI value in the product description, e.g. HFCS-55.

Ultra high fructose corn syrup (UHFCS) is generally restricted to those syrups having a DI value greater than 50. Sometimes *ultra high fructose glucose syrup* (UHFGS) or second generation HFCS or (IG) are used instead of HFCS.

Manufacturing process. Three main stages involving three different enzymes may be identified in the manufacturing process of HFCS. These include: (i) liquefaction of starch, (ii) saccharification and (iii) isomerization. A flow chart of the process is given in Figure 25.1.

(i) *Liquefaction of starch (dextrinization).* Corn, waxy maize (high in amylopectin), a high amylose hybrid maize or wheat may be used as starch source. Liquefaction involves cooking starch to disrupt the granules leading to dissolution of starch. In a typical liquefaction process, starch slurry containing about 33% dry solids is liquefied with a bacterial α-amylase (*B. amyloliquefaciens*) at temperatures ranging from 80–110°C. The liquefaction process is continuous with residence times ranging from

2 to 4 hours. Sometimes jet cooking at 140°C in the absence of enzyme is used as a pretreatment step. The α-amylase commonly used has a calcium requirement (200–500 ppm) and the pH of the starch slurry is adjusted to pH 6–7 with calcium hydroxide.

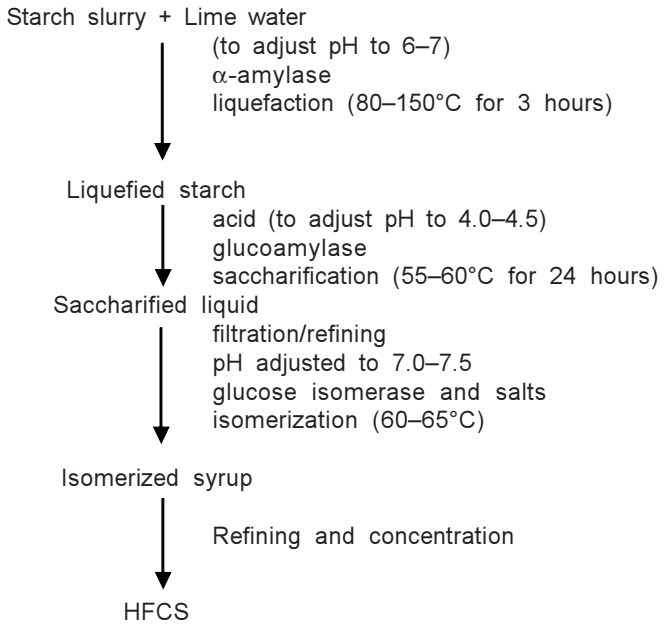


Figure 25.1 Flow chart for production of HFCS.

(ii) *Saccharification.* The liquefied starch is flash cooled to about 60°C and saccharified by treatment with a fungal glucoamylase (*Aspergillus sp. or Rhizopus sp.*). Conditions for the saccharification are 55–60°C, pH 4.0–4.5 and a holding time of 24–90 hours depending upon the amount of glucoamylase used and production scheduling. The saccharified starch, contains 94–96% dextrose and is decolourized with carbon and deionized with strong acid and weak base ion-exchange resins. The deionization step is necessary to remove most of the calcium ions which would inactivate the isomerase enzyme in the subsequent stage.

(iii) *Isomerisation.* The refined dextrose liquor is reconcentrated or blended to a dry solids level of 40–50%. After concentration, the liquor is pumped to a make-up tank where various salts are added. Magnesium ions are required for optimum catalytic activity for all commercially used isomerases with the exception of isomerase from *L.brevis* which requires manganese. Bisulphite or iron may be added to enhance the activity or stability of the isomerase. The pH of the liquor is adjusted to 7.0–8.5 with NaOH. Batch or continuous isomerisation using soluble or immobilized isomerase may be carried out. The dextrose content of the substrate at

this point determines the monosaccharide (dextrose + fructose) content of the finished fructose syrup.

25.2.3 Glucose (Dextrose)

Starch from corn, potatoes or wheat is saccharified enzymatically by α-amylase and/or microbial amyloglucosidase. The enzyme from *Aspergillus niger* hydrolyses starch at pH 4.5 and 60°C to a product with 95% glucose. The syrup is purified and evaporated to crystallize glucose as α-D-glucose monohydrate. Drying in a stream of warm air or crystallization from ethanol, methanol or glacial acetic acid gives the anhydrous form. Glucose is used as an invigorating and strengthening agent in many nourishing formulations and medicines and also a red colour enhancer of meat and frying sausages.

25.2.4 Fructose

Fructose is obtained by acid hydrolysis of inulin, a natural polymer of fructose found in tubers. Fructose is sweeter than sucrose and is used as a sugar substitute for diabetics.

25.2.5 Lactose

Lactose is produced from whey concentrates. The concentrate is adjusted to pH 4.7 and heated with steam to 95°C to remove milk albumins by thermal denaturation. The filtered deproteinated fluid is concentrated in multiple effect evaporators to separate salts. The desalted concentrate yields a yellow raw lactose which is raffinated by solubilization, filtration and crystallization. The snow white α-D-lactose monohydrate is pulverized in a pin mill and separated in a centrifugal classifier. β-lactose which is more soluble, sweet and easily digestible compared to the α-lactose, is obtained by heating lactose solution to 93.5°C and discharging the crystals into a vacuum drum dryer. β-lactose finds use as a nutrient for children, filler or diluter in medicinal preparations and tablets and ingredient of nutrient solutions used in microbial production of antibiotics.

25.2.6 Sugar Alcohols

Sorbitol is a hygroscopic alcohol half as sweet as sucrose and is manufactured by catalytic hydrogenation of glucose. It finds use as a sweetener for diabetics, in food canning as a softener, and humectant (moisture retention) in candies. Sorbose is obtained by oxidation of sorbitol by *Acetobacter xylium* as an intermediate product for commercial

production of ascorbic acid. Sorbose also finds use as sucrose substitute for diabetics and as an ingredient with negligible carcinogenicity in low calorie foods. Xylitol is obtained by catalytic hydrogenation of xylose which is obtained by hydrolysis of hemicelluloses. Xylitol is used as sugar substitute for diabetics and as a low carcinogenic ingredient in the production of chewing gum. Mannitol, obtained from fructose hydrogenation, has a laxative effect and is used in chewing gum manufacture. Maltitol, which is sweeter than maltose, finds its use in the manufacture of sweet commodities. It is obtained by hydrogenation of starch hydrolysate as a non-fermentable and less carcinogenic product.

25.2.7 Maple Syrup

Maple syrup with a sugar content of about 65% is obtained by evaporating the sap of maple tree (*Acer saccharum*). The sap from the tree has no flavour but develops a special flavour during evaporation. Maple syrup is used as a sweetener and as a flavouring agent in pancakes, waffles and in baked products.

25.3 HONEY

Honey is produced by honeybees which suck up the nectar and honeydew from flowers and other sweet saps of plants and store the nectar in their honey sac or pouch. Honey production starts immediately in the bee's pouch by the addition of substances secreted by the bee. The mixture is given to worker bees in the hive to deposit it into six-sided cells of the honeycomb. The nectar is converted into honey in the honeycomb through several stages of water evaporation, thickening, acid and enzymic hydrolysis of sugar to invert sugar, isomerization of glucose to fructose and assimilation of proteins, acids, minerals, vitamins and aroma substances. When the water content decreases to about 16–18% the cells are closed with a wax lid and ripening continues with enzyme invertase hydrolysis of sucrose and synthesis of new sugars.

Comb honey is honey present in freshly built closed combs devoid of brood combs (young virgin combs). Honey is extracted (extracted honey) from the comb by a honey extractor which is a centrifuge operated at slightly elevated temperatures. Gentle warming to 40°C facilitates the release of honey from the combs. Pressed honey is collected by pressing brood-free honey combs in a hydraulic press at room temperature. Strained honey is obtained by gentle heating of pulped or unpulped brood-free honey combs followed by pressing. Beetle honey used for feeding honey bees is obtained pulping honey combs which include brood combs.

Based on its end use, honey is classified as honey for domestic use of highest purity and bakery honey of not so high purity used as sugar

substitute in baking industry. Honey is also classified on the basis of its geographical origin as German, Hungarian, Californian, Chilean, Canadian, Yemeni, etc. The flavour and colour of honey depends on the origin of nectar from flowers and plants. Thus flower honey is white, greenish yellow or brownish coloured thick, transparent liquid obtained from nectars of acacia, sweet and white clovers, alfalfa, rape, buckwheat, etc. Flower honey has a characteristic sweet and highly aromatic flavour which is dependent on the flavour substances in the nectar collected by the bees. Honeydew honey from pine, spruce or leaf honeydew is less sweet and darkly coloured with terpene like odour and flavour. It solidifies with difficulty.

Honey is marketed as a liquid or as a semisolid creamed honey and contains about 38% fructose, 31% glucose and 2% sucrose. It also contains enzymes such as α- and β-amylases (diastase), glucose oxidase, catalase and acid phosphatase. Invertase and diastase activities together with hydroxymethyl furfural content are used for assessing whether or not honey was heated, and also for detection of adulteration with caramel. Honey also contains free amino acids (about 100 mg/100 g solids), particularly proline, proteins and organic acids such as gluconic acid together with small amounts of acetic, butyric, lactic, citric, succinic, formic, maleic, malic and oxalic acids. Aroma substances found in honey include about 120 volatile compounds such as esters, aldehydes, ketones and alcohols. Poisonous honey contains toxic substances and comes from nectars of rhododendron species, some species of mad berries, certain evergreen shrubs and honeydew exudates of grasshoppers. Honey has water retaining capacity and hence cakes and candies containing honey remain moist for a longer period.

25.4 CONFECTIONERY

25.4.1 Chocolate Confectionery

Chocolate is made from non-alkalized cocoa liquor by mixing with sucrose, cocoa butter and aroma substances including milk solids, nuts, coffee paste, etc. The ingredients are processed through several steps to yield a final product 'which melts in the mouth but not in hand'. The various processing steps include: (i) mixing, (ii) refining, (iii) ripening, (iv) conching and (v) tempering and molding.

(i) *Mixing.* Cocoa liquor, crystalline sugar, cocoa butter and milk powder in the case of milk chocolates are intensely mixed in a melangeur or paster to form a homogeneous, coarse chocolate paste.

(ii) *Refining.* Single or multiple refining rollers with provision for controlling temperature are used to disintegrate the chocolate paste into a smooth-textured mass of fine particles. The refined mass has a fat content of about 23–28% and the particle size in range of 35–75 μm.

(iii) *Ripening*. The refined chocolate mass is a dry powder at room temperature with a harsh, sour flavour. Storing the mixture in warm chambers at 45–50°C for about 24 hours ripens it. The chocolate mass attains a dough consistency and is used for making commercial chocolates and for baking. For making high quality chocolates the ripened chocolate mass is further processed through conching and tempering steps.

(iv) *Conching*. This is a mixing process in which the ripened chocolate mass is mixed, ground and kneaded to produce chocolate mass with a fine flavour, desired texture and smoothness. Conching is performed in round or oblong conche pots with rotary or roller conches in two stages. In the first stage of conching the chocolate mass is heated at 80°C for 6–12 hours. Moisture and a portion of volatile compounds such as methanol, ethanol, isopropanol, isobutanol, isopentanol, acetaldehyde, acetone, diacetyl and ethyl acetate are expelled and the fat is uniformly distributed so that each cocoa particle is coated with a film of fat. The temperature is allowed to rise at this stage and aroma substances such as pyrazines may be lost. In the second and final stage of conching, the mass is liquified by adding cocoa butter and thus homogenized. Lecithin is added to reduce the viscosity of the mass and give required fluidity with less of cocoa butter and homogenization is continued till the mass attains the desired texture and flavour.

(v) *Tempering and molding*. The conched chocolate mass on cooling without proper control will yield a product with granular texture and poor colour and blotchy appearance. Hence controlled cooling of the conched mass with tempering is necessary to obtain chocolate tablets or chocolate covered confectionery with good texture, colour and smooth/glossy appearance and stability. The tempering step involves an initial cooling of the molten chocolate from about 50°C to 18°C within 10 minutes with constant stirring. The mass is kept at this temperature for another 10 minutes to facilitate the formation of stable β-form of cocoa butter. The temperature is once again increased to 30–31°C within 5 minutes and delivered to warm molds made of metal or plastic. The filled molds pass over a vibrating shaker to allow the escape of entrapped air and then passed through a cooling zone. Slow cooling allows the formation of abundant fat crystals of small size with high melting points. A homogeneous solid with fine, crystalline and heat-stable fat structure having a glossy appearance and the desired melting properties is formed by this tempering process. The mass hardens finally at 10°C and the finished product falls out from the mold.

The finished chocolate contains at least 40% cocoa liquor or a blend of liquor and cocoa butter and up to 60% sugar. A variety of chocolates are marketed with different ingredient compositions such as

baking chocolate (cocoa — 33%, cocoa butter — 5–7% and sugar — 50–60%),

chocolate for coating (cocoa—35–60%, cocoa butter—15% and sugar—35–50%),

milk cream chocolate (cocoa—10–20%, cocoa butter—10–22%, skim milk powder—10–15%, butter fat—5–10% and sugar 35–60%),

whole milk chocolate (cocoa—10–30%, cocoa butter—12–20%, skim milk powder—10–22%, butter fat—3–7% and sugar—3–60%)

and icings (cocoa—32–65%, cocoa butter—5–25% and sugar—25–50%).

Cocoa and chocolate products require careful storage condition of dry (55–65% humidity), cool (10–12°C), well aerated space protected from light and odorous substances. Improper storing results in the formation of sugar bloom causing tiny sugar particles to deposit on the surface which dissolves in moist conditions and forms larger crystals resulting in a greying matte surface. A fat bloom develops with fluctuating temperatures and temperature higher than 30°C. The liquid separates out and after repeated congealing, forms large white spots.

25.4.2 Sugar Confectionery

Sugar confectionery includes both crystalline and amorphous types made from boiled sugar syrup. The temperature of boiling sugar solution, the ingredients used to aid crystallization or inhibit crystallization and the method of handling of the supercooled sugar solution determine the crystalline or amorphous nature of the end product. Crystalline confectionery or candies have a smooth texture, amenable for cutting with knife and easily chewable. These include fondants and fudges. Amorphous candies have a heterogeneous soft structure and break into pieces rather than be cut with a knife. Examples of amorphous candies include toffee, brittles and caramel. Caramel is the softest of amorphous candies but is amenable for cutting.

Crystalline confectionery is made by adding ingredients such as invert sugar, glucose or corn syrup, which aid the formation of fine sugar crystals from sugar syrup. The candy mixture is then concentrated by boiling (112–115°C) to achieve the desired consistency. The concentrated product is then poured onto a flat surface, allowed to cool to about 40°C and beaten continuously to incorporate air to get a creamy stiff mass. Ripening of the creamy mass over 24 hours soften the candy sufficiently to give the final product with soft yet firm and smooth texture. Fondants are used in the preparation of mints, coating fruit and nut mixtures that are moulded and sliced and as cream centres in chocolate confectionery. Fudges contain butter and milk which interfere with crystallization and corn sugar as ingredients. The sugar solution is boiled (117–118°C) and fat and essence are added. Subsequent handling of the boiled sugar solution is similar to that used in fondant making.

Amorphous confectionery is made by preventing crystallization of the sugar either by cooking the sugar solution at higher temperatures (120–150°C) and allowing the product to harden quickly or by adding large amounts of ingredients which inhibit crystallization, such as corn syrup, fats or concentrated milk products. Caramelization occurs due to thermal degradation of the sugar and the brown colour develops due to both caramelization as well as Maillard reaction. Brittles are made by melting sugar and caramelized sugar. Toffees are made from sugar solution with added lemon juice or vinegar. Spongy candies such as marshmallows and gum drops are made using gelatin as an ingredient.

Questions

1. Give an outline of the manufacture of sugar.
2. How is jaggery obtained?
3. What are starch hydrolysates? How are they obtained?
4. Give an account of the steps involved in the manufacture of HFCS.
5. What are sugar alcohols? What is their use?
6. How is honey obtained?
7. Write an account on the manufacture of chocolate confectionery.

CHAPTER

26

Salt and Spices

26.1 SALT

Salt occupies a special position as an ingredient to enhance the taste, palatability and flavour of foods. It is also a preservative of a variety of food products when used in large amounts. The need for maintaining a constant level of sodium and chloride ions in plasma and extracellular fluids is an important factor in the human diet. The daily requirement of common salt is about 5 g and excessive intake is also detrimental to health. Cooking salt is almost entirely sodium chloride with other salts such as magnesium and calcium chlorides, and magnesium, calcium and sodium sulphates constituting about 2–2.5%. Other trace elements are also found to be present in the cooking salt. The moisture content is about 3%. Salt occurs naturally as rock salt, in seawater (3–3.7%) or salt springs.

Salt production depends on the source. Rock salt is mined, crushed and finely ground. Brine (~4%) from salt springs is evaporated directly or pre-concentrated to about 20% sodium chloride in cascade solar evaporation units and then under vacuum. During crystallization, additives such as calcium or magnesium carbonate (0.25–2.0%) prevent lumping of the salt. The addition of potassium ferrocyanide (20 ppm) modifies the crystallization pattern of sodium chloride during evaporation to dendrite form. The dendrite crystals have low density, smaller volume and low tendency to agglomerate. In tropical places, sea water is concentrated in shallow flat basins by sun, heat and wind until it crystallizes.

Iodized salt containing 5 mg of sodium or potassium or calcium iodide per kg of cooking salt is used to cater to the need of iodine in our diet to prevent goitre, a disease of the thyroid gland. Nitrite salt containing 0.5% of sodium nitrite in common salt is used for pickling or dry curing of meat. Low sodium diet required by some patients to avoid excess sodium intake involves the use of substitutes for common salt in the form

of flavouring agents. The substitutes marketed as diet salts include potassium, calcium and magnesium salts of adipic, succinic, glutamic, carbonic, lactic, tartaric and citric acids or monopotassium phosphate, adipic and glutamic acids and potassium sulphate.

26.2 SPICES

Spices (and condiments) are plant products which contribute distinctive flavour and aroma to foods. A variety of spices are used as flavouring agents and for seasoning of food. Spices are parts of plants such as buds, flowers, fruits, barks, roots or seeds obtained from a variety of plants. Spices may be classified into several groups taking into consideration their origin and the characteristic flavour or aroma. Accordingly, the same spice may be classified into more than one group: (i) aromatic fruits like black pepper, paprika or red pepper, vanilla, chillies, dill, juniper berries, coriander, anise, fenugreek, celery, cumin, fennel and caraway; (ii) aromatic seeds such as cardamom, nutmeg and mustard; (iii) Aromatic barks including cinnamon and cassia; (iv) aromatic flower such as clove; (v) rhizomes such as ginger and turmeric; (vi) leaves such as parsley, marjoram, rosemary and sage; (vii) pungent spices such as pepper, ginger, chillies and mustard; (viii) phenolic spices such as clove, eugenol and allspice and (ix) coloured spices such as saffron, turmeric and paprika.

The characteristic flavour and aroma of spices are due to steam volatile essential or volatile oils and organic solvent soluble oleoresins. The main constituents of essential oils include mono- and sesqui-terpenes, phenols and phenolethers, aromatic hydrocarbons, alkaloids and pyrazines. Eugenol, carvacrol, thymol, estragole, anethole, safrole and myristicin are some of the important phenols and phenolethers. Aromatic hydrocarbons such as p-cymene, 1,3,8-menthatriene and 1-methyl-4-isopropenylbenzene also occur in spices. A variety of mechanisms are responsible for the generation of the characteristic flavour and aroma compounds in spices. Thus biosynthesis of cinnamaldehyde, the main constituent of cinnamon bark and of eugenol and safrole originates from phenylalanine. Terpene oxidation produces some aromatic hydrocarbons in spices. For example, 1-methyl-4-isopropenylbenzene is derived from 1,3,8-menthatriene, and (+)ar-curcumene is derived from zingiberene or β-sesquiphellandrene. Character impact compounds include cinnamaldehyde in cinnamon, anethole in anise, d-carvone in caraway, eugenol in cloves, 1-methyl-4-isopropenylbenzene and 1,3,8-menthatriene in parsley leaves, turmerone and ar-turmerone in turmeric.

Spices such as mustard (and horseradish) contain glucosinolates in the cells. After the rupture of the cells, thioglucosidase enzymes act on glucosinolates yielding isothiocyantes such as allyl isothiocyanate from sinigrin, and p-hydroxybenzyl isothiocyanate from sinalbin, which are responsible for the pungent burning taste of the mustard and horseradish.

The aroma of capsicum pepper is mainly due to pyrazines, particularly 2-isobutyl-3-methoxypyrazine. The hot, burning pungent flavour of paprika, pepper and ginger are due to alkaloids such as piperine, piperanine, piperylin, ginerol, shogaol, zingerone, capsaicin and its derivatives.

Spices such as paprika, saffron and curcuma also contain pigments which find use as food colourants. Paprika contains carotenoid pigments with capsanthin as the main compound. Curcuma contains curcurmin as the pigment. Antioxidants, mainly the cyclic diterpene diphenols such as carnosolic acid and carnosol, are also found in the extracts of spices such as sage and rosemary.

26.2.1 Harvesting and Processing of Spices

Black pepper, considered to be the black gold of India and the king of spices, is obtained by harvesting the spikes of fully mature fruits when they start yellowing. The berries are separated from the spikes by rubbing, threshing or trampling, and are blanched by immersing them in boiling water to inactivate the native enzymes. These are then sun dried for a few days. The dried berries become dark brown to black in colour and when completely dry, the outer skin is shrivelled to yield a shiny black product. The spice value of pepper is due to the presence of alkaloids (5%), pungent constituents and pepper oil (2–3%) containing 95% of terpene hydrocarbons and 4% oxygenated terpenes, which give rise to the characteristic odour of the oil. *Green pepper* with a growing demand for garnishing meat dishes, is obtained by harvesting immature pepper and is dehydrated or canned in brine or vinegar.

Chillies in unripe condition are green and the ripe ones are red in colour. The green chillies are used fresh while the nearly ripe fruits are picked and cured in heaps for 2–3 days and then sun dried for long-term storage.

Ginger is the underground stem or rhizome of the zingiberous herb. The crop is dug in stages depending upon the type of ginger required. Green ginger, the tender rhizome used for pickling, is harvested during the fifth month of the crop. Dry ginger is harvested eight months after sowing, peeled to remove the epidermal layer, washed and sun dried. Ginger oil (2–3%) obtained by steam distillation of the rhizome is responsible for the characteristic aroma and odour. The oil contains sesquiterpenes as the major constituent (65%) and oxygenated terpenes (17%) and monoterpenes (4%). The active principle of pungency is mainly based upon the oleresin consisting of oxymethyl phenols (gingerol, shogaol and other related compounds).

Cardamom is the dried ripe fruit with flavouring and medicinal characteristics. The fully ripe fruits are harvested and sun dried. The pale green to yellow capsules get bleached naturally during drying to give a straw yellow product. Bleaching may also be carried out by exposure to

sulphur dioxide or steeping in a dilute solution of sulphurous acid. Volatile oils (5–10%) present in the seeds are the major contributor to the spice value. Cardamom oil, which is obtained by steam distillation of the seeds or whole fruits, contains terpenes (terpeniol, terpinene, limonene, sabinine and cineol).

Turmeric rhizomes are cured by boiling in lime water or soda bicarbonate solution. The well-cooked rhizomes are sun dried, cleaned and polished in a rotary drum to yield a finished brittle product with an yellow colour. Essential oil (2%) contains sesquiterpene ketones (55%) and tertiary alcohol (9–10%). The oleoresin contains all the aroma and flavour constituents of turmeric. Turmeric contains curcumin as the yellow pigment and antioxidant phenolic substances. Turmeric finds use as the flavouring and colouring agent and also as an important ingredient in medicines.

Mustard is an important oilseed crop. The dried whole seeds find use as a preservative and seasoning agent.

Tamarind is the ripe fruit pulp from which seeds have been removed and sun dried. It is rich in acids in the free and combined forms and is used as an acidulant in culinary preparations.

Cloves are flower buds which are harvested and dried. The buds contain about 17% essential oils (mainly eugenol) which contribute to their spice value. Clove oil also finds use as antiseptic and antimicrobial cure for toothache.

Mint (pudina) is aromatic herb and an important source of peppermint oil and menthol.

Saffron is a colouring spice which contains crocin as the pigment. It finds use as an ingredient in cosmetics and merdicines.

Garlic has a strong flavour due to sulphur containing allicin. It has antibacterial and anti-insecticidal activity. It is believed to decrease the cholesterol content in blood serum and reduce hypertension.

Asafoetida is the gum that oozes from the rhizome of Ferula species. The strong flavour is due to steam volatile sulphur containing organic compounds.

Coriander, cumin, fenugreek, allspice and mace and nutmeg are some other spices used in food preparations for their flavour.

26.2.2 Spice Powders and Extracts

Spice powders and extracts or concentrates are finding increasing acceptance among consumers. Spices are ground into powder using a cryogenic mill and stored in the absence of air at temperatures less than 20°C and a relative humidity less than 60% to enhance the shelf life. Ground spices generally lose their aroma and absorb aroma from other substances. Essential oils are obtained by steam distillation of spices at controlled pressure and temperature in order to minimize thermal

decomposition of the aroma compounds. The extracted oil is separated from the aqueous layer, clarified and stored. Oleoresins are obtained from raw spices with a low essential oil content. In such cases, the aroma compounds may be lost due to solubility in water on steam distillation and hence these are recovered by solvent extraction. Solvents such as hexane, methylene chloride, acetone, ethanol or edible oils and fats may be used. Supercritical fluid extraction using carbon dioxide as solvent is yet another important method for extracting essential oils and resins as solvent removal or recovery from the extract is quite easy. Extraction is usually followed by chromatographic or counter-current separation to isolate the desired aroma fractions.

Questions

1. How is salt manufactured?
2. What are spices? How are they classified?
3. Write a note on different spices used in food preparations.

CHAPTER

27

Food Quality

Quality of foods and food products may be defined as the degree of excellence of the various characteristics that influence consumer acceptance as well as consumer safety. The selection of a particular food by a discerning consumer is made by the judgment of all the physical senses, i.e. sight, tough, smell, taste and hearing. However, the consumer cannot easily judge the nutritional qualities of a food and the presence or absence of pathogens, contaminants, toxins and chemical additives. Food processing and preservation aims at meeting the requirements of both consumer acceptance as well as safety. Evaluation of food quality for consumer acceptance is based on the use of sensory organs or by the use of appropriate instruments. Consumer safety requires the evaluation of food quality with respect to nutritional quality, hygienic condition and keeping (storage) quality.

27.1 SENSORY EVALUATION OF FOOD QUALITY

Sensory evaluation of foods may be carried out by human operators or by automated instruments. Sensory evaluation includes quality check on the basis of three basic factors: (i) appearance, (ii) texture and (iii) flavour. These factors determine the organoleptic properties of food.

27.1.1 Appearance Factors

The quality of a food item may simply be judged from its appearance when it is placed in front of a consumer. For example, the presence of 'eyes' in Swiss cheese is indicative of acceptable flavour and texture obtained by proper bacterial fermentation while eye formation in Cheddar cheese is considered a defect. Similarly, the appearance of foam at the top in a cup of tea or a glass of wine is a quality defect. A slight turbidity or cloudiness in orange juice is acceptable but not in apple juice, which must be crystal clear.

Physical factors such as size, shape, freedom from defect/damaged surface, type and extent of damaged parts and optical properties such as colour, gloss and transparency and the consistency of the product in different batches/packages are also appearance factors that are indicative of quality. The appearance factors are highly useful and practised in quality evaluation of plant foods including raw materials such as whole fruits and vegetables, as well as in processed plant and animal foods.

Sorting and grading operations using size, shape, colour and transparency for quality evaluation and separation of foods and food products into different quality categories have already been discussed in Chapter 13. The consistency of a viscous or semisolid food product such as tomato sauce, sugar syrup, chocolate syrup, ketchup or honey may be determined by the resistance of the food product to flow using different types of viscometers.

27.1.2 Textural Factors

These include various physical factors such softness, hardness, firmness, juiciness, grittiness or chewiness felt by the consumer when he handles the food with fingers (hand feel) or with the tongue, teeth or palate (mouth feel). Any deviation from the expected texture is said to be a quality defect. The texture of foods changes due to aging, improper processing or storage. Thus fresh fruits and vegetables become soggy due to over-ripening, and bread and cakes become stiff due to staling. Texture is an important factor in deciding the consumer acceptance of a food. In fact, quality of a food is mainly judged by its freshness/ripeness/maturity or proper processing. For example, crispness of potato chips, the firmness and crunchiness of apples, a springy bread, compressibility and shearability of steak between teeth, juiciness of fruits such as melons, non-sticky but quick melting softness of chocolates and chewiness of chewing gum are indicative of prime quality of the food.

Measuring the resistance offered by the food to an applied force may be related to evaluation of textural quality. Thus compressive force used on a fruit to squeeze out juice may be related to succulence while squeezing of bread may be related to springy and spongy texture indicative of freshness of the food. Shearing moves one part of the food to slide past another and is related to the chewiness of the chewing gum. A combination of compressive and shear forces are helpful in determining the tenderness of peas. The texture of muffins may be related to the tensile force, which tears or pulls apart the food. The toughness or tenderness of meat is related to the yielding of meat while chewing with a composite of compressive, shear and tensile forces. The firmness of a gel is measured by penetration test in which the force required to move a fixed plunger over a fixed distance through a food material is determined.

It is possible to use non-destructive testing of food samples by correlating colour with texture or use of sound waves. The ripeness of fruits and vegetables as measured by colour measurement may be used as an indication of an acceptable texture, as colour and texture development to the optimum level occur simultaneously in such foods. The degree of 'eye' formation in Swiss cheese during ripening is tested by using sound waves, which are absorbed to different extents depending on the firmness of the product.

27.1.3 Flavour Factors

Evaluation of flavour factor is highly subjective and depends on the discriminating ability of the consumer as flavour includes the sense of smell as well as the sense of taste as experienced by a consumer. People differ in their sensitivity to different odours and tastes as much as in their preferences for various types of foods. In addition, consumers are influenced to some extent on the appearance, colour and texture of the food while evaluating the flavour characteristics. Flavour characteristics may be evaluated by instrumental methods as well by panels of judges. The nature and concentration of volatile aroma compounds may be determined by gas chromatography. Chemical properties and physical properties such as refractive index, electrical conductivity and pH may be used to determine specific ingredients, which contribute to the specific taste of the food. Panels of judges consisting of trained individuals are used for evaluating foods, particularly grading of tea, coffee and wine.

27.2 QUALITY FACTORS FOR CONSUMER SAFETY

The quality factors, which may not be apparent even to a discriminating consumer, include nutritional quality, sanitary quality and keeping quality.

27.2.1 Nutritional Quality

Nutritional quality of a food or food product may be evaluated by specific analysis for essential nutrients such as proteins, vitamins, minerals and other nutrients. Chemical and instrumental methods are used for evaluating the nutritional quality. Animal feeding tests are also used for evaluating foods from new sources.

27.2.2 Sanitary Quality

Evaluation of sanitary quality of food is based on determining the

microbial load. It is not sufficient to test merely the final product. Reduction of microbial load and prevention of recontamination right from the raw material stage through various processing steps to the final product is absolutely essential. It is in this context that the practice of hazard analysis and critical control points (HACCP) is crucial for good manufacturing practice in a food processing plant.

Hazard analysis and critical control points (HACCP). This technique is used to analyze potential hazards in an operation, identifying where these may occur and how much these are critical to consumer safety. It also establishes control systems that focus on the prevention of such hazards rather than relying on end-product testing. HACCP is applied in food-chain from primary producer to final consumer. Potentially hazardous foods (PHFs) include sea foods, livestock and poulty meat, dairy products such as cheese and butter, cooked vegetables, preserved fruit products, soft drinks and beverages, cereals and cereal products, nuts and nut products. Potential hazards for such foods arise from pathogenic bacteria, viruses and fungi, toxic chemicals, industrial effluents, and physical matter like fibers, particles, etc. These hazards can occur during all stages of manufacture, from raw materials to the consumption stage. Appropriate action is taken to ensure that areas identified as critical control points (CCPs) are kept under control and not allowed to endanger the items produced.

The successful application of HACCP requires the full commitment and involvement of the management and the workforce and a multidisciplinary approach such as plant design as per good manufacturing practice (GMP), pollution control, insect and pest control, hygiene and sanitation (HASSOP—Hygiene and Standard Sanitation Operating Procedures) and trained personnel. The HACCP system consists of the following principles:

(i) Conducting a hazard analysis with help of product description and process flow charts.

(ii) Identifying the CCPs.

(iii) Determining critical limits for each CCP.

(iv) Establishing a monitoring system for the control of CCP.

(v) Establishing the corrective action to be taken when monitoring indicates that a particular CCP is not under control.

(vi) Establishing procedures for verification to confirm that the HACCP system is working effectively.

(vii) Documenting all procedures and records appropriate to these principles and their application.

(viii) Subjecting to internal audit, third party audit and certification.

(ix) Training of personnel.

The application of HACCP is compatible with the implementation of quality management systems such as ISO 9000 series. HACCP certification against IS 15000 for providing an assurance for food safety through HACCP system is becoming imperative in food industry.

27.2.3 Keeping Quality of Foods

The storage stability or keeping quality of foods may be determined by accelerated tests simulating the conditions under which the product is expected to be handled and stored before it reaches the consumer.

27.3 FOOD SAFETY STANDARDS

Codex Alimentarius Commission is an international body under the joint sponsorship of Food and Agriculture Organization (FAO) and the World Health Organization (WHO), setting standards containing requirements for food products aimed at ensuring a sound, wholesome food product, free from adulteration, correctly labelled and presented to the consumer. Codex is responsible for the formulation of internationally recognized food standards for all principal foods whether raw, semi-processed or processed. It also includes provisions for ensuring hygienic and nutritional qualities of food, microbiological norms, nature of food additives, identification and elimination of contaminants, methods of sampling and analysis, and labelling and presentation.

A preventive approach to the safety of foods is more effective than testing or inspection of processed foods at the final stage. Good manufacturing practice (GMP) assures food safety through vigilant measures at the source, product design and process control. The various steps involved in good manufacturing practice include the (i) safety of raw materials from contamination by pesticides, toxins, microbes, agrochemicals and animal drugs, (ii) the judicious use of food additives and (iii) application of hygienic practices in handling, processing, packaging, labelling, distribution, sale, preparation and use of the food product.

Questions

1. What is sensory evaluation of food quality?
2. How is food quality evaluated with respect to consumer safety?
3. What is hazard analysis? How is it important in food processing?

References

Belitz, H.D. and Grosch, W., *Food Chemistry,* Springer Verlag, 1987.

BIOTOL Series (Biotechnology by open learning), *Biotechnological Innovations in Food Processing,* Butterworth-Heinemann, Oxford, 1991.

Birch, G.G., Camerron, A.G. and Spencer, M., *Food Science*, 3rd ed., Pergamon Press, Oxford, 1986.

Birch, G.G., Brennan, J.G. and Parker, K.J., *The Sensory Properties of Foods*, Applied Science Pub., London, 1977.

Brennan, J.G., Butters, J.R., Cowell, N.D. and Lilly, A.E.V., *Food Engineering Operations,* 2nd ed., Applied Science Pub. Ltd., London, 1979.

Charley, H., *Food Science,* John Wiley and Sons Inc., New York, 1982.

Coultate, T.P., *Food—The Chemistry of Its Components*, 2nd ed., Royal Society of Chemistry, London, 1992.

Desrosier, N.W. and Desrosier, J.N., *The Technology of Food Preservation*, 4th ed., AVI Pub. Inc. Westport, 1977.

Fennema, O.R. (Ed.), *Principles of Food Science,* Part I, *Food Chemistry,* Marcel Dekker, New York, 1976.

Fox, B.A. and Cameron, A.G., *Food Science, Nutrition and Health*, 5th ed., Edward Arnold, London, 1989.

Frazier, W.C. and Westhoff, D.C., *Food Microbiology,* 3rd ed., Tata McGraw-Hill, Delhi, 1978.

Geankoplis, C.J., *Transport Processes and Unit Operations*, 3rd ed., Prentice-Hall of India, New Delhi, 1997.

Hudson, B.J.F. (Ed.), *Development in Food Proteins,* Vol. I, Applied Science Pub., London, 1982.

Jay, J.M., *Modern Food Microbiology*, 3rd ed., CBS Pub., Delhi, 1987.

Maguer, M.Le and Jelen, P. (Eds.), *Food Engineering and Process Applications*—Vol. 2, *Unit Operations,* Elsevier Applied Science, New York, 1986.

Marton, I.D. and Macbod, A.J., *Food Flavours,* Elesvier Scientific Pub. Co., New York, 1982.

Peterson, M.S. and Johnson, A.H. (Eds.), *Encyclopedia of Food Science,* AVI Pub. Co., Westport, 1978.

Potter, N.N., *Food Science*, 3rd ed., AVI Pub. Co. Inc., Westport, 1978.

Robinson, D.S., *Food—Biochemistry and Nutritional Value,* Longman Scientific and Technical, London, 1987.

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Food Processing and Preservation

B. Sivasankar

Food science and technology is an interdisciplinary subject involving topics from chemistry, microbiology, chemical engineering and process technology. These topics need an interactive approach in order to comprehend the complexities involved in food processing and preservation. This book provides a thorough understanding of all major aspects of food processing with an emphasis on the microorganisms associated with food, before going into the problems of large-scale production and preservation of foodstuffs.

Written in a style that is accessible to students, the text introduces the important aspects of food science, such as functional role of the nutrients, the changes that the nutrients undergo during processing and preservation, and the chemical reactions responsible for spoiling various food materials as well as maintaining the organoleptic properties of foods. The use of various chemical additives during food processing is exhaustively discussed in a separate chapter.
In addition, the text presents a detailed discussion on microorganisms, which is followed by the unit operations practised in food industry for processing and preservation involving the use of high and low temperatures and radiation. Production of various types of food products including alcoholic beverages, spices and confectionery items are also discussed.

Intended as a textbook for undergraduate students of science and engineering, the study would also benefit the postgraduate students offering courses in food science as well as professionals and researchers.

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